



BASE24[®]

Integrated Server Transaction Security Manual

© 1998 by Applied Communications, Inc.

All rights reserved. All information contained in this manual is confidential and proprietary to Applied Communications, Inc. and has been delivered for use pursuant to license. No part of this manual may be photocopied, electronically transferred, modified, or reproduced in any manner that is contrary to license without the prior written consent of Applied Communications, Inc.

BASE24, BASE24-atm, BASE24-pos, and BASE24-teller are registered trademarks, and BASE24 Remote Banking, BASE24-telebanking, and BASE24-billpay are trademarks of Applied Communications, Inc.

All other brand or product names are trademarks or registered trademarks of their respective companies or organizations.

Print Record

11/1998, Version 2 (Update)

08/1997, Version 1 (New)

Contents

Preface.	xv
Conventions Used in this Manual.	xxiii
Section 1	
Introduction.	1-1
Definition of Terms	1-2
External Processors.	1-2
Remote Banking Endpoint Devices	1-2
Primary Account Number (PAN)	1-3
SYSGEN Utility and DSC Utility	1-3
Remote Banking Products	1-4
Transaction Security Processing Software	1-6
Integrated Authorization Server Process	1-6
Interface Master Process.	1-6
Integrated Endpoint Process	1-7
ISO Host Interface Process.	1-7
Data Encryption and Decryption	1-8
DES Algorithm	1-9
Keys	1-9
Encryption Security	1-10
Translation	1-11
PIN Management	1-12
PIN Encryption, Decryption, and Translation	1-13
PIN Verification	1-13
PIN Change.	1-14
Message Authentication	1-15
Security Modules	1-17
Check Digits	1-18

Introduction *continued*

Key Management	1-20
ASMcom Utility	1-20
RSMcom Utility	1-21
Supercrypt Utility	1-22
Dynamic Key Management	1-22

Section 2

Transaction Processing and PINs.	2-1
PIN Encryption	2-2
PIN Blocks	2-3
PIN/PAN PIN Blocks	2-3
PIN/PAD PIN Blocks	2-7
Racal-Guardata Host Security Module Requirements	2-7
Internal Messages and Transaction Data	2-10

Section 3

Remote Banking Device PIN Encryption Support	3-1
Integrated Authorization Server Process	3-2
Remote Banking Device Keys	3-3
Inbound Encryption Keys	3-3
Intermediate Keys	3-3
Previous Keys	3-5
PIN Encryption Options	3-6
Incoming Transaction Processing	3-8
Clear PINs	3-8
Encrypted PINs	3-8
Setting Up the BASE24 Database	3-10
ISF Screen 1	3-11
ISF Screen 2	3-13

Section 4

External Processor PIN Encryption Support	4-1
ISO Host Interface Process	4-2
External Processor Keys	4-3
Inbound Encryption Keys	4-3
Outbound Encryption Keys	4-4
Intermediate Keys	4-4
Previous Keys	4-5
PIN Key Exchange Keys	4-6
Dynamic Key Management	4-6
PIN Encryption Options	4-7
BASE24 Carries Clear PINs; External Processor Carries Clear PINs	4-9
Processing Outgoing Messages	4-9
Processing Incoming Messages	4-9
Setting Up the BASE24 Database	4-9
BASE24 Carries Clear PINs; External Processor Carries Encrypted PINs	4-11
Processing Outgoing Messages	4-11
Processing Incoming Messages	4-13
Setting Up the BASE24 Database	4-15
BASE24 Carries Encrypted PINs; External Processor Carries Clear PINs	4-24
Processing Outgoing Messages	4-24
Processing Incoming Messages	4-26
Setting Up the BASE24 Database	4-27
BASE24 and External Processor Carry Encrypted PINs	4-32
Processing Outgoing Messages	4-32
Processing Incoming Messages	4-34
Setting Up the BASE24 Database	4-35

Section 5

Message Authentication Support	5-1
BASE24 Remote Banking Message Authentication	5-2
Transaction Flow	5-3
Hardware and Software Authentication	5-4

Message Authentication Support *continued*

Authentication Inputs	5-5
MAC Key	5-5
Data to be Authenticated.	5-5
Location and Length of MAC.	5-6
Authentication Algorithm.	5-6
BASE24 Remote Banking Internal Messages	5-7

Section 6**PIN Verification Support. 6-1**

BASE24 Remote Banking PIN Verification	6-2
Institution PIN Verification Parameters	6-3
KEYA Screen 1	6-3
KEYA Screens 2 Through 5.	6-4
IDF Screen 2	6-4
IDF Screen 4	6-7
IDF Screen 40	6-10
Customer PIN Verification Parameters	6-12
CSTT Screen 1	6-12
PIN Verification By Authorization Level.	6-14
BASE24 Remote Banking Internal Messages	6-15
BASE24 Remote Banking PIN Verification Processing	6-16
Local PIN Verification	6-18
PIN Change and PIN Select Options	6-19
PIN Change Configuration Requirements	6-20
PIN Change Processing	6-21
PINs at Customer Service Representative Terminals.	6-23

Section 7**IBM 3624-Format PIN Verification Support. 7-1**

IBM 3624-Format PIN Verification	7-2
PIN Verification Inputs.	7-3
Validation Data	7-3
PIN Verification Key	7-4

PIN Verification Inputs *continued*

Decimalization Table	7-5
PIN Offset	7-6
PIN	7-7
Setting Up IBM 3624-Format PIN Verification	7-8
Hardware Module Requirements	7-8
Institution Definition File (IDF) Settings	7-8
Customer Table (CSTT) Settings	7-15
Key Authorization File (KEYA) Settings	7-17

Section 8**Diebold PIN Verification Support 8-1**

Diebold PIN Verification	8-2
PIN Verification Inputs	8-3
Diebold Number Table	8-3
Customer ID	8-4
Algorithm Number	8-4
PIN Offset	8-5
PIN	8-5
Setting Up Diebold PIN Verification	8-7
Hardware Module Requirements	8-7
Institution Definition File (IDF) Settings	8-8
Customer Table (CSTT) Settings	8-15
Key Authorization File (KEYA) Settings	8-17

Section 9**Identikkey PIN Verification Support 9-1**

Identikkey PIN Verification	9-2
PIN Verification Inputs	9-3
Customer ID	9-3
PIN	9-4
Bank ID	9-4
Compare Indicator	9-5
PIN Verification Number (PVN)	9-5

Identikey PIN Verification Support *continued*

Setting Up Identikey PIN Verification	9-6
Hardware Module Requirements	9-6
Institution Definition File (IDF) Settings	9-6
Customer Table (CSTT) Settings	9-13
Key Authorization File (KEYA) Settings.....	9-15

Section 10**Visa PVV PIN Verification Support 10-1**

Visa PVV PIN Verification	10-2
PIN Verification Inputs.....	10-3
Customer ID	10-3
PIN	10-3
PIN Verification Key (PVK) Pair.....	10-4
PIN Verification Key Indicator (PVKI)	10-5
PIN Verification Value (PVV)	10-5
Setting Up Visa PVV PIN Verification	10-7
Hardware Module Requirements	10-7
Institution Definition File (IDF) Settings	10-8
Customer Table (CSTT) Settings	10-14
Key Authorization File (KEYA) Settings.....	10-17

Section 11**Remote Banking Dynamic Key Management..... 11-1**

Dynamic Key Management	11-2
Inbound PIN Encryption Keys	11-2
Key Exchange Keys	11-3
Key Management Processors	11-4
Key Management Messages	11-5
Processing Key Management Messages.....	11-6
External ISO Messages.....	11-7
Cryptographic Service Message	11-9
Key Distribution Environment	11-9
Message Classes	11-9
Message Fields and Subfields.....	11-9
Message Formats	11-10

Cryptographic Service Message *continued*

Message Length	11-13
Required Fields	11-14
New Key Message Flow Diagram	11-15
Processing Summary	11-16
RBSI Dynamic Key Management Thresholds	11-19
Time Threshold	11-20
Transaction Threshold	11-20
Setting Up the BASE24 Database	11-23
Interface Security File Settings	11-23
ITS Interface Configuration File Settings	11-27

Section 12

External Processor Dynamic Key Management	12-1
Dynamic Key Management	12-2
Working Keys	12-2
Key Exchange Keys	12-3
Key Management Processors	12-4
Key Management Messages	12-6
Processing Key Management Messages	12-7
Network Management Messages	12-9
Cryptographic Service Message	12-12
Key Distribution Environment	12-12
Message Classes	12-12
Message Fields and Subfields	12-12
Message Formats	12-13
Message Length	12-17
Message Flow Diagrams	12-19
Dynamic Key Management Thresholds	12-22
Time Threshold	12-23
Transaction Thresholds	12-23
Setting Up the BASE24 Database	12-28
Key File Settings	12-28
External Message File Settings	12-42
Host Configuration File Settings	12-43

Section 13

Security Module Support	13-1
BASE24 Remote Banking Security Module Usage	13-2
Supported Security Modules.....	13-4
Number of Security Modules	13-4
Security Device Failures and Timeouts	13-8
Configuring Security Modules	13-9

Appendix A

File and Table Maintenance Screens	A-1
Customer Table (CSTT) Screen 1	A-2
Host Configuration File (HCF) Screen 1	A-3
Institution Definition File (IDF) Screen 2	A-4
Institution Definition File (IDF) Screen 4	A-5
Institution Definition File (IDF) Screen 40	A-6
Interface Security File (ISF) Screen 1	A-7
Interface Security File (ISF) Screen 2	A-8
Key Authorization File (KEYA) Screen 1	A-9
Key Authorization File (KEYA) Screen 2	A-10
Key Authorization File (KEYA) Screen 3	A-11
Key Authorization File (KEYA) Screen 4	A-12
Key Authorization File (KEYA) Screen 5	A-13
Key File (KEYF) Screen 1	A-14
Key File (KEYF) Screen 2	A-15
Key File (KEYF) Screen 3	A-16
VRU Configuration Data (VCD) Screen 1	A-17
VRU Configuration Data (VCD) Screen 2	A-18
VRU Configuration Data (VCD) Screen 4	A-19
VRU Configuration Data (VCD) Screen 18	A-20

Appendix B

BASE24 Supercrypt Utility	B-1
Using the Supercrypt Utility	B-2
Starting the Supercrypt Utility	B-2
Exiting the Supercrypt Utility	B-2
Accessing the Supercrypt Utility	B-3
Error Messages	B-3

Appendix C

ASMcom Utility	C-1
BASE24 Keys	C-3
Translating Keys	C-4
Changing Master File Keys	C-4
Variants	C-4
Using the ASMcom Utility	C-6
Starting the ASMcom Utility	C-6
Specialized Use of the Return Key	C-7
ASMcom Utility Variables	C-8
Main Menu	C-9
Validate/Generate Check Digits	C-10
Translate Security Keys	C-14
Translate KEK-Encrypted PIN Working Key for Local Storage	C-18
Translate KEK-Encrypted MAC Working Key for Local Storage	C-19
Alter Security Module Device Name	C-20
Alter LCONF Location	C-21
Alter Summary Report Output Location	C-22
Help Function	C-22
Status Messages	C-22
Exiting the ASMcom Utility	C-23
Reports	C-24
Check Digit Generation Report	C-25
Check Digit Validation Report	C-26
Key Translation Report	C-27

Appendix D

RSMcom Utility	D-1
BASE24 Keys	D-3
Key Pairs	D-4
Changing Local Master Key Pairs	D-4
Changing the Diebold Number Table Key Pair	D-5
Using the RSMcom Utility	D-6
Starting the RSMcom Utility	D-6
Specialized Use of the Return Key	D-7
Documentation of the RSMcom Utility Variables	D-8
Main Menu	D-9
Generate Clear Intermediate Working Key	D-10
Translate PIN Working Key from ZMK to LMK Encryption	D-11
Translate MAC Working Key from ZMK to LMK Encryption	D-13
Validate/Generate Check Digits	D-14
Translate Security Keys	D-18
Alter Security Module Device Name	D-25
Alter LCONF Location	D-25
Alter Summary Report Output Location	D-26
Upload Diebold Number Table to KEYA	D-26
Help Function	D-28
Status Messages	D-28
Exiting the RSMcom Utility	D-28
Reports	D-29
Check Digit Generation Report	D-30
Check Digit Validation Report	D-31
Key Translation Report	D-32
DNT Upload Summary Report	D-33

Appendix E

Atalla Security Modules	E-1
Security Processing Support	E-2
Configuring Atalla Security Devices	E-4
Atalla Security Modules	E-4
Atalla A5000 Network Security Processors	E-6
Atalla A6000 Network Security Processors	E-7

Configuring Atalla Security Devices *continued*

Atalla A7000 Network Security Processors	E-9
Atalla A10000E Network Security Processors	E-10
Boxcar Interface Module	E-13
Logical Network Configuration File (LCONF)	E-15
LCONF Assigns	E-15
LCONF Params.	E-17
Atalla Security Device Support	E-24
Preparing Data for the BASE24 Files.	E-24
ASM and NSP Online Functions Supported by BASE24	E-26
Security Module Capabilities	E-28

Appendix F

Racal-Guardata Host Security Modules	F-1
Security Processing Support	F-2
Configuring Racal-Guardata Security Devices.	F-3
Data Communications Options	F-3
Asynchronous Emulation (ASYNCH)	F-3
SDLC Emulation.	F-4
Ethernet (TCP/IP)	F-5
Logical Network Configuration File (LCONF)	F-9
LCONF Assigns	F-9
LCONF Params.	F-11
Racal-Guardata Host Security Module Support	F-16
Preparing Data for the BASE24 Files.	F-16
RSM Online Functions Supported by BASE24	F-20
Index	Index-1
Index by Field Name.	Index-7

Applied Communications, Inc.

Preface

The ***BASE24 Integrated Server Transaction Security Manual*** describes the support provided by BASE24 Remote Banking products for transaction security including PIN encryption, PIN verification, message authentication, and dynamic key management, along with the database settings required to implement this support.

The manual also describes how hardware security modules are integrated into BASE24 Remote Banking processing and the duties they can assume as a part of that processing.

Audience

This manual is intended for the system administrators and operational staff who determine how the BASE24 Remote Banking system is to handle transaction security and who set up the BASE24 database.

Prerequisites

This manual discusses a number of general topics relating to transaction security. These discussions have been included primarily to provide a stronger insight into the way BASE24 Remote Banking products handle transaction security. Persons using this manual should be familiar with the BASE24 Remote Banking products, their basic functions, terminology, and the BASE24 files they use.

The manual is not intended as a substitute for the industry literature relating to transaction security. Individuals making decisions on BASE24 Remote Banking transaction security processing should be familiar with the industry literature available for the type of transaction security processing being performed on their BASE24 Remote Banking system.

Refer to the following ANSI and ISO standards for additional information about transaction security:

- The ANSI X3.92 standard, ***Data Encryption Algorithm Standard***
- The 1982 ANSI X9.8 standard, ***American National Standard for Personal Identification Number (PIN) Management and Security***
- The 1986 ANSI X9.9 standard, ***Financial Institution Message Authentication (Wholesale)***
- The 1985 ANSI X9.17 standard, ***Financial Institution Key Management (Wholesale)***
- The 1986 ANSI X9.19 standard, ***Financial Institution Retail Message Authentication***
- The 1987 ISO 8583 standard, ***Bank Card Originated Messages—Interchange Message Specification***
- The 1993 ISO 8583 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specifications***
- The ISO 8732 standard, ***Banking—Key Management (Wholesale)***
- The ISO 9564 standard, ***Banking—Personal Identification Number Management and Security***

Also important is an understanding of the basic functionality of the security modules to be integrated with the BASE24 Remote Banking system. Security modules are all integrated into the BASE24 Remote Banking system in a similar fashion; however, the individual functions of each module and the ways in which these functions are accessed vary by vendor and module. Thus, users of this manual should have a working knowledge of the security modules they plan to use, especially in the area of key generation and management.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The **BASE24 Remote Banking Transaction Processing Manual** provides information regarding the processing of transactions by the BASE24-telebanking and BASE24-billpay products.
- The **BASE24 Base Files Maintenance Manual** provides data entry screens and field descriptions for base files.
- The **BASE24 Core Files and Tables Maintenance Manual** provides detailed descriptions and valid values for configuring BASE24 Remote Banking files and tables.
- The **BASE24 ISO Host Interface Manual** provides information regarding the BASE24 ISO Host Interface external message structure.
- The **BASE24 External Message Manual** describes the BASE24 External Message. Configured using the External Message File (EMF), it is the ISO-based message used to communicate with hosts.
- The **BASE24 Text Command Reference Manual** provides descriptions and syntax for BASE24 send or deliver process and interprocess commands.
- The **BASE24 Logical Network Configuration File Manual** contains additional information regarding BASE24 assigns and params in the Logical Network Configuration File (LCONF).

This manual references the **NET24 IEP Objects Reference Manual** for more information on the context management functions performed by the Integrated Endpoint process and the following informal document available from ACI for detailed information on configuring and starting the Remote Application Manager process: **Module Specification - Remote Application Manager**.

This manual also contains references to the following Tandem publications:

- The Tandem **Himalaya S-Series Hardware Installation Guide** provides information on adding a 3861 Ethernet 4 ServerNet Adapter (E4SA) controller.
- The Tandem **TLAM Configuration and Management Manual** provides information on adding an Ethernet controller and line.

Software

This manual documents standard transaction security processing for BASE24-telebanking release 1.3 and BASE24-billpay release 1.3. All transaction security processing for BASE24-billpay is provided in BASE24-telebanking software.

Customer-specific code modifications (that is, RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The ***BASE24 Remote Banking Transaction Security Manual*** is organized by security function (for example, PIN verification, message authentication). When configuring a process, it is important that all applicable security functions are included in the setup procedure.

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to transaction security, including overviews of key management, PIN encryption, message authentication, PIN verification, PIN change, and their support by BASE24 Remote Banking products.

Section 2, “Transaction Processing and PINs,” summarizes how BASE24 Remote Banking products use PINs in transaction processing in conjunction with the remote banking devices and external processors to which they are connected. This section also includes a discussion of PIN blocks and their role in PIN encryption.

Section 3, “Remote Banking Device PIN Encryption Support,” describes how BASE24 Remote Banking products support PIN encryption between remote banking devices and the BASE24 Remote Banking system using the Integrated Authorization Server process.

Section 4, “External Processor PIN Encryption Support,” describes how BASE24 Remote Banking products support PIN encryption between external processors and the BASE24 Remote Banking system using the ISO Host Interface process.

Section 5, “Message Authentication Support,” summarizes how the ISO Host Interface process manages message authentication in conjunction with various external processors on behalf of BASE24 Remote Banking products.

Section 6, “PIN Verification Support,” describes how BASE24 Remote Banking products support PIN verification. The basic options available, the fields used in the BASE24 Remote Banking internal messages for PIN verification, and descriptions of the processing performed are presented in this section.

Section 7, “IBM 3624-Format PIN Verification Support,” describes BASE24 Remote Banking support for the IBM 3624-format PIN verification method, including how the BASE24 database must be set up to allow BASE24 Remote Banking to verify PINs under this method of verification.

Section 8, “Diebold PIN Verification Support,” describes BASE24 Remote Banking support for the Diebold PIN verification method, including how the BASE24 database must be set up to allow BASE24 Remote Banking to verify PINs under this method of verification.

Section 9, “Identikey PIN Verification Support,” describes BASE24 Remote Banking support for the Identikey PIN verification method, including how the BASE24 database must be set up to allow BASE24 Remote Banking to verify PINs under this method of verification.

Section 10, “Visa PVV PIN Verification Support,” describes BASE24 Remote Banking support for the Visa PVV PIN verification method, including how the BASE24 database must be set up to allow BASE24 Remote Banking to verify PINs under this method of verification.

Section 11, “Dynamic Key Management,” describes the dynamic key management options provided by the ISO Host Interface process on behalf of BASE24 Remote Banking. It also describes the Cryptographic Service Message (CSM) element in network management messages used for dynamic key management.

Section 12, “Security Module Support,” describes the usage of security modules with BASE24 Remote Banking and the functions of each that are supported.

Appendix A, “File and Table Maintenance Screens,” contains illustrations of the screens used to perform the file and table maintenance procedures described throughout this manual.

Appendix B, “BASE24 Supercrypt Utility,” describes the BASE24 Supercrypt utility, a program that is required for encrypting inbound and outbound keys for entry into the BASE24 database when BASE24 Remote Banking performs external message PIN encryption in software.

Appendix C, “BASE24 ASMcom Utility,” describes the BASE24 ASMcom utility, a program used to interact with an Atalla Security Module (ASM) or Atalla Network Security Processor (NSP) for translating inbound and outbound keys for entry into the BASE24 database.

Appendix D, “BASE24 RSMcom Utility,” describes the BASE24 RSMcom utility, a program used to interact with a Racal-Guardata Host Security Module for translating intermediate, inbound, and outbound keys for entry into the BASE24 database.

Appendix E, “Atalla Security Modules,” describes how Atalla security modules must be configured to work with BASE24 Remote Banking. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 Remote Banking and the information a user requires for configuring the modules for use by BASE24 Remote Banking.

Appendix F, “Racal-Guardata Host Security Modules,” describes how Racal-Guardata Host security modules must be configured to work with BASE24 Remote Banking. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 Remote Banking and the information a user requires for configuring the modules for use by BASE24 Remote Banking.

Readers can use the index by field name to locate information about a particular screen field.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, TB-AE000-13 for the ***BASE24 Integrated Server Transaction Security Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 11/1998 for November, 1998) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

Applied Communications, Inc.

Conventions Used in this Manual

This section explains how command syntax notation is documented in this manual.

Syntax Notation

Syntax descriptions in this manual use the following symbols and conventions to denote how commands must be entered:

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
<i>lowercase italics</i>	Identify variables that you must provide.
Brackets []	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar ().
Vertical bar	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.

Notation	Meaning
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Section 1

Introduction

The transaction security requirements of any organization can vary greatly according to the environment in which the organization is operating. An organization often requires an integrated system of software, industry hardware, and manual procedures to properly implement transaction security.

BASE24 Remote Banking transaction security includes several components that work together:

- PIN encryption ensures that PIN information is maintained in a confidential manner.
- PIN verification ensures that the true customer is performing the transaction.
- Message authentication ensures that data remains tamper-free while passing between the BASE24 Remote Banking system and an external processor.

This section includes overviews of each of these transaction security components and how they are supported by BASE24 Remote Banking products. It also provides a definition of terms used throughout the manual, an introduction to the security modules supported by BASE24 Remote Banking products, the use of check digits with hardware-encrypted keys, and key management tools provided with BASE24 Remote Banking products.

Definition of Terms

The following definitions are provided to give the reader a clear understanding of specific terms as they are used in this manual.

External Processors

This manual uses the term *external processor* to describe a transaction processing entity that is external to a BASE24 Remote Banking system. External processors can include the following:

- A data processing center (DPC), which is the BASE24 designation for host computers connected online to a BASE24 system.
- A co-network, which is another BASE24 system or another transaction processing network connected online to a BASE24 system.

When this manual is referring to processing performed by one of these specific entities, that entity is referred to by the name shown in the list above. If the processing can be performed by any of these entities (DPC or co-network), the term *external processor* is used.

Remote Banking Endpoint Devices

This manual uses the term *remote banking endpoint device* to describe any device that acquires BASE24 Remote Banking transactions using the Remote Banking Standard Interface or the Customer Service Interface. The Customer Service Interface supports Pathway-based CSR terminals only. The Remote Banking Standard Interface supports any of the following devices: interactive voice response system (IVR), call center, screen phone, personal computer, third-party processor, kiosk, personal digital assistant, or the Internet. When this manual is referring to processing performed by one of these specific devices, that device is referred to by the name shown in the list above. If the processing can be performed by any of these devices, the term *remote banking endpoint device* is used.

If the processing can be performed only by a device using a particular interface standard, the term *<interface>-compliant device* is used, where *<interface>* is an acronym for the interface standard (e.g., RBSI for the Remote Banking Standard Interface). For example, dynamic key management is only supported for RBSI-compliant devices.

Primary Account Number (PAN)

This manual at times uses the term *primary account number (PAN)* instead of the term customer ID. PAN is simply another term for a customer ID.

SYSGEN Utility and DSC Utility

This manual frequently references the terms *SYSGEN utility* and *DSC utility*. These terms refer to the Tandem System Generation (SYSGEN) and the Tandem Dynamic System Configuration (DSC) utilities.

Remote Banking Products

The BASE24 Remote Banking product family currently includes two products—BASE24-telebanking and BASE24-billpay. The BASE24-telebanking product provides online banking services (for example, balance inquiries, check clearance inquiries, and immediate transfers between two accounts) to customers using remote devices (for example, a touch-tone telephone or personal computer) or calling a customer service representative (CSR) or branch employee. The BASE24-billpay product provides automated payment and scheduled transfer services (for example, immediate, future, and recurring payments, future and recurring transfers, and associated inquiries) to customers using remote devices or calling a CSR or branch employee. The BASE24-billpay product provides files that can be used as input to programs that generate checks and ACH output for payments; however, the BASE24-billpay product does not currently include these programs. The BASE24-billpay product also provides facilities for signing up new customers and accessing customer information.

Optional Enhanced Telebanking product modules provide financial institutions and service providers the ability to support all BASE24-telebanking transactions as well as the subset of banking transactions that are provided with the BASE24-billpay product.

Enhanced Telebanking provides for the scheduling of one-time future transfers and recurring transfers. The product modules monitor future and recurring transfers, and on the scheduled transaction date, automatically initiate the transfer transaction. When Enhanced Telebanking initiates a transfer, BASE24 routes the transaction to the appropriate authorizing entity for authorization. Various reports are available directly related to these transfer transactions.

Enhanced Telebanking also supports customer and account maintenance transactions, financial institution and financial institution branch list inquiry transactions, loan application transactions, and miscellaneous request transactions.

The BASE24 Remote Banking software also provides the three transaction security functions described on the previous page (that is, PIN encryption, PIN verification, and message authentication) that are used to support the services provided by BASE24-telebanking and BASE24-billpay.

As described in the ***BASE24 Remote Banking Transaction Processing Manual***, BASE24-telebanking allows for a wide range of software and hardware options geared toward the individual needs of institutions.

Because the BASE24-billpay product relies on the BASE24-telebanking software for transaction security and financial transaction processing, the BASE24-billpay product cannot be installed without the BASE24-telebanking product. The BASE24-telebanking product can be installed without the BASE24-billpay product; however, scheduled transfer capabilities and customer signup functions are available only if both products are installed.

This manual documents transaction security processing for the BASE24-telebanking and BASE24-billpay products. Throughout this manual, references are made to BASE24, BASE24 Remote Banking, BASE24-telebanking, and BASE24-billpay, as follows:

- BASE24 includes BASE24-telebanking and BASE24-billpay, as well as other BASE24 products.
- BASE24 Remote Banking includes BASE24-telebanking and BASE24-billpay.
- BASE24-telebanking indicates the BASE24-telebanking product with or without the BASE24-billpay product. For example, the BASE24-telebanking Integrated Authorization process performs PIN verification for both BASE24-telebanking transactions and BASE24-billpay transactions.
- BASE24-billpay indicates the BASE24-billpay product only.

Transaction Security Processing Software

BASE24 Remote Banking products use four processes for transaction security processing: the Integrated Authorization Server process, the Interface Master process, the Integrated Endpoint process, and the ISO Host Interface process.

The distinction between these processes is greater for BASE24 Remote Banking products than for other BASE24 products. The ISO Host Interface process obtains the information needed for PIN encryption, PIN translation, message authentication, and dynamic key management from the Key File (KEYF). The Integrated Authorization Server and Interface Master processes obtain the information needed for PIN encryption, PIN translation, and dynamic key management from a separate file. For other BASE24 products, all processes use the Key File (KEYF) for this information. The Integrated Authorization Server and Interface Master process also use different security-related Logical Network Configuration File (LCONF) assigns and params than other BASE24 processes use.

Integrated Authorization Server Process

The Integrated Authorization Server process encrypts, decrypts, and translates PINs received from all remote banking devices that it controls. PINs received from a customer service representative (CSR) terminal are always in the clear. The process also handles all PIN verification performed by BASE24 Remote Banking products, regardless of transaction source.

The Integrated Authorization Server process calls the Interface Master process to generate new keys, formats the external response, and sends the external response to the Integrated Endpoint process.

The Interface Security File (ISF) provides the information and parameters required by the Integrated Authorization Server process for PIN encryption and translation. The Key Authorization File (KEYA) provides the information and parameters required for PIN verification. Message authentication is not currently supported between BASE24 and a remote banking device.

Interface Master Process

The Interface Master process maintains dynamic key management counters and generates new keys for remote banking devices that use the Remote Banking Standard Interface (that is, RBSI-compliant devices).

The Interface Security File (ISF) and ITS Interface Configuration File (IFCF) provide the information and parameters required by the Interface Master process to perform dynamic key management processing.

Integrated Endpoint Process

The Integrated Endpoint process plays a minor role when dynamic key management processing is performed. When the Integrated Authorization Server process sends a response containing new key data to the Integrated Endpoint process, the Integrated Endpoint process stores context. For responses without new key data, context is not stored.

When a response sent to an RBSI-compliant device contains new key data, the process routes the subsequent network management message received from the device to the Interface Master process. If the context times out before this message is received, the Integrated Endpoint process returns the context to the Interface Master process.

For detailed information on context processing performed by the Integrated Endpoint process, refer to the *NET24 IEP Objects Reference Manual*.

ISO Host Interface Process

The ISO Host Interface process encrypts, decrypts, and translates PINs received from a host or being sent to a host. It also handles message authentication and dynamic key management between BASE24 and a host.

The Key File (KEYF) provides the information and parameters required by the ISO Host Interface process for PIN encryption and translation, message authentication, and dynamic key management.

Data Encryption and Decryption

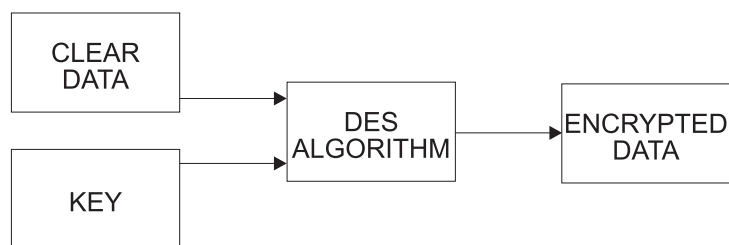
Data encryption and decryption are the means by which sensitive data such as a PIN can be rendered unintelligible for storage in a database or transmission over communications lines and then subsequently returned to its original form when necessary.

The support provided by BASE24 Remote Banking for data encryption and decryption is based on the use of the Data Encryption Standard (DES) algorithm, a data encryption algorithm widely used throughout the financial industry for various cryptographic functions.

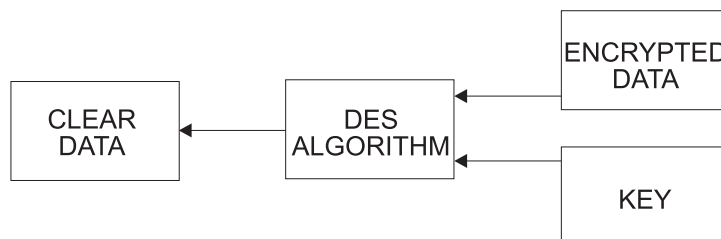
DES encryption starts with a data item in its original, or clear, form. Then, together with a second data item called a key, that clear data is used as input to the DES algorithm, which yields a new, encrypted value bearing no outward resemblance to the original.

DES decryption is the reverse of the encryption operation. It starts with the encrypted data and the same key that was used in the encryption process. In this case, the DES algorithm is applied in reverse to the two data items, bringing the unintelligible data back to its original, or clear, form.

As shown in the following diagram, in data encryption, the clear data is used as input to the DES algorithm, along with an encryption key.



In data decryption, the encrypted data is used as input to the DES algorithm, along with the same key that was used to encrypt the data originally. This is illustrated in the following diagram.



As described on the following pages, there are three components of DES encryption and decryption: the DES algorithm, a key, and the data to be encrypted or decrypted.

DES Algorithm

The DES algorithm is a set of mathematical operations widely used throughout the financial industry for data encryption and decryption. It is also known as the data encryption algorithm (DEA), which is approved by the International Organization for Standardization (ISO) for use with ISO standards related to PIN security and message authentication. As stated previously, it is the DEA on which BASE24 Remote Banking cryptographic support is based.

The DES algorithm is available royalty-free for public use. The algorithm operations are not documented in this manual. For information on the actual workings of the DES algorithm, refer to the ANSI X3.92 standard, ***Data Encryption Algorithm Standard***.

Keys

DES encryption requires what is called a key. A key is a sequence of numbers used as one of the two variables input into the DES algorithm. Its role in the encryption and decryption process is major.

DES encryption and decryption require two input data blocks. From these two values a third is computed. When encrypting a block of data, the inputs are the clear version of the data to be encrypted and the encryption key. The result of the operation is the encrypted form of the clear data. When decrypting a block of data,

the inputs are the encrypted version of the data and the same encryption key that was used to encrypt the data. The output in this case is the clear form of the encrypted data.

In both cases, the encryption key is one of the two given blocks of data which allows for derivation of the third.

Encryption Security

The security of DES encryption and decryption is dependent on keeping the encryption keys confidential.

Since the DES algorithm is widely used, if an encryption key becomes compromised, the data encrypted with it becomes vulnerable to decryption by unauthorized parties.

For encrypted data stored on a database, security is fairly straightforward. The encryption key can be maintained and kept secure by the party in control of the database. When security modules are used for data encryption and decryption, the key itself is often composed of parts, with each part known only to one person. Under this method, when a key must be placed in the security module, each person enters the part known to them separately, such that no one person has knowledge of the entire key.

File maintenance screens are used to place encryption keys in the BASE24 database. Each key is entered in its entirety; therefore, the option of maintaining key information in parts is not available when entering keys in the BASE24 database.

For encrypted data transmitted between two parties or networks, encryption key security becomes slightly more involved because both parties must know the encryption key and, at the same time, maintain its secrecy.

Translation

Another data security concept that is closely related to encryption and decryption is translation. In situations involving more than one data format or key (for example, a remote banking device that acquires a transaction uses one key and the external processor that authorizes the transaction uses another key), the data must be translated as necessary so that both entities can accurately read the data. This differs from encryption and decryption because the data is in encrypted form before and after translation without appearing in clear form, as shown below.

Process	Input Data	Output Data
Encryption	Clear	Encrypted
Decryption	Encrypted	Clear
Translation	Encrypted	Encrypted

PIN Management

In an electronic banking environment, transactions can be initiated from several types of originators, which are also known as acquiring endpoints. An acquiring endpoint can be a customer-operated device such as a telephone or personal computer or a customer service representative terminal. PIN verification can be performed by the Integrated Authorization Server process or as a prescreening step for transactions authorized by a host.

PIN is an acronym for personal identification number, which is a number assigned to a customer to allow the customer to initiate BASE24 Remote Banking transactions.

Because of the major role that PINs play in the authorization of transactions, all parties involved in the processing of transactions are obliged to maintain PIN confidentiality.

If a transaction is accompanied by the customer's PIN, the authorizer of the transaction can assume that the transaction is being made by the true customer or on the authority of the true customer.

However, if a transaction is accompanied by a PIN other than the customer's, the authorizer of the transaction must assume that the transaction is being attempted without the consent of the true customer and therefore must decline the transaction.

Not only must customers keep their PINs secret, but their confidentiality must also be ensured as they travel from the point at which a transaction is originated to the point at which the transaction is authorized—an electronic journey that can span multiple processors and can cross numerous communications links.

Because of the relative ease with which data can be tapped while being transmitted over various commercial communications lines or discerned during processing, PINs and other sensitive data items are often encrypted to guard against unauthorized detection. This encryption, while effective in disguising data, is itself only viable if certain aspects of the encryption are kept confidential.

The problem of preserving PIN confidentiality is further compounded by the fact that the PIN must ultimately be verified in an electronic medium, requiring that an authorizer be able to electronically determine the validity of a PIN entered in a transaction. This not only requires that the authorizer be able to deal with an encrypted version of the PIN, but also requires that keys used to encrypt the PIN be kept confidential to ensure that PINs are not generated in an unauthorized manner.

There are a number of methods of PIN verification used throughout the financial industry. Many approach PIN verification in different ways. However, most are based on a derivation of certain items of control information that can be kept on file for a customer, in lieu of the actual PIN, and then mathematically linked to the customer's PIN for verification at the time of a transaction.

A number of industry-wide standards and practices have been developed to effectively guard PINs while traveling through various electronic media and during verification. BASE24 Remote Banking allows for integrating many of these standards and practices in day-to-day processing, and provides various options to its users for tailoring PIN security to their own situations and needs. BASE24 Remote Banking PIN support can be divided into three areas, each having its own distinct set of options:

- Encryption, decryption, and translation
- Verification
- Change

PIN Encryption, Decryption, and Translation

BASE24 Remote Banking PIN encryption and decryption duties are handled by the Integrated Authorization Server process that sends transactions to and receives transactions from remote banking devices and by the ISO Host Interface process that sends transactions to and receives transactions from external processors. When a transaction is received from outside the BASE24 Remote Banking system, the PIN may have to be encrypted, decrypted, or translated from one encrypted form to another, depending on how the BASE24 Remote Banking system and remote banking device or external processor are set up to handle PINs.

BASE24 Remote Banking support of PIN encryption is described in sections 2, 3, and 4.

PIN Verification

While PINs can be verified outside of the BASE24 Remote Banking system by an authorizing host, PIN verification can also be performed within the BASE24 Remote Banking system by the Integrated Authorization Server process or locally at the remote banking endpoint device.

BASE24 Remote Banking supports four methods of PIN verification:

- IBM 3624-format PIN verification
- Diebold PIN verification
- Identikay PIN verification
- Visa PIN verification value (PVV) verification

Each of these methods has its own unique set of requirements and options. The mechanics of BASE24 Remote Banking PIN verification are described in section 6.

PIN Change

The BASE24 Remote Banking transaction set also includes PIN change transactions for use by the customer and customer service representatives.

To aid in preserving PIN confidentiality, BASE24 Remote Banking can be configured to force the customer to perform a PIN change transaction prior to performing any other BASE24 Remote Banking transaction. An institution can also configure their system to restrict its customer service representatives from changing a PIN when the customer is not present.

Message Authentication

In a remote banking environment, data can pass between the BASE24 Remote Banking system and an external processor electronically. The integrity of this data is a key to the success of all of these systems. Message authentication is a method used to monitor the integrity of data flowing between the BASE24 Remote Banking system and an external processor.

Note: Message authentication is not supported between the BASE24 Remote Banking system and a remote banking device.

The message authentication performed in BASE24 Remote Banking is based on the use of the authentication algorithm defined in the ANSI X9.9 standard, ***Financial Institution Message Authentication (Wholesale)*** and the ANSI X9.17 standard, ***Financial Institution Key Management (Wholesale)***. This algorithm is used by the ISO Host Interface process.

The ISO Host Interface process generates and verifies values called message authentication codes (MACs) between it and a host in Atalla or Racal-Guardata hardware or in software. The Integrated Authorization Server process does not support message authentication between it and a remote banking endpoint device or customer service representative terminal.

The illustration on the following page demonstrates message authentication. Like PIN encryption, message authentication also uses keys. While PIN encryption uses keys to encrypt and decrypt data, message authentication uses keys to encrypt the same data twice and compare the results. The MAC is part of the message that passes between a transaction originator and a transaction authorizer. The originator uses the key from its database and the information that is already in the message to generate a MAC and then adds the MAC to the message before sending it. The transaction recipient verifies the message by generating a second MAC based on the same message information used by the transaction originator, then comparing the MAC it generates with the MAC contained in the message. If the two MACs agree, the message can be accepted because it has not changed between the originator and the recipient. If the two MACs do not agree, the message should not be accepted because it has changed in some way between the originator and the recipient. A message can change as the result of tampering or because of communications difficulties. The MACs also do not agree if the keys are out of synchronization (that is, the originator and recipient are not using the same key).



As illustrated above, before a message is sent, the message authentication key and selected fields from the message are used as input to the authentication algorithm to generate a MAC that is then carried in the message.

MACs can be generated and verified in messages moving between the BASE24 Remote Banking system and external processors. BASE24 Remote Banking support of message authentication is described in section 5.

Security Modules

A number of vendors offer hardware modules that can be set up to perform PIN protection and verification, as well as other security-related functions. BASE24 Remote Banking currently supports three types of security modules from two vendors:

- Atalla Security Module (ASM)
- Atalla Network Security Processor (NSP)
- Racal-Guardata Host Security Module (RSM)

The functions within these modules are proprietary. However, most of the principles of PIN encryption, PIN verification, and message authentication remain the same regardless of whether they are performed in hardware or software. The difference is that instead of performing the transaction security processing in software, BASE24 Remote Banking sends information to the security module and accepts the results.

The advantage of using a security module is that instead of BASE24 Remote Banking having the keys in the clear, which is a requirement for software encryption, the keys themselves can be encrypted under a key known only to the security module. Thus, BASE24 Remote Banking is only given access to encrypted versions of the keys; the security module itself is responsible for decrypting the keys it is given prior to processing.

For general information on the functions supported by BASE24 Remote Banking for each type of security module and how security modules are used in the system, refer to section 13. For information on how devices must be configured for use with BASE24 Remote Banking systems, refer to either appendix E, for information regarding Atalla Security Modules and Atalla Network Security Processors, or appendix F, for information regarding Racal-Guardata Host Security Modules.

Check Digits

Throughout the BASE24 database, many of the fields that can contain a hardware-encrypted key are accompanied by an optional four-character check digits field. For example, screen 2 of the Key File (KEYF) displays eight fields containing keys used for PIN encryption or translation and message authentication. A check digits field appears next to each of these key fields.

When calculating an encrypted value, also known as a cryptogram, to be used as a key, security modules also assign check digits. These check digits can be used to verify the cryptogram was entered or generated correctly in the BASE24 database. BASE24 provides two utilities that assist in this verification: the BASE24 Atalla Security Module Communications (ASMcom) utility and the BASE24 Racal Security Module Communications (RSMcom) utility. These utilities are described later in this section.

The ASMcom and RSMcom utilities have two ways that check digits can be used for cryptogram verification:

- Enter only the cryptogram in the field on the BASE24 data entry screen.

When you use the Generate Check Digits option of the ASMcom or RSMcom utility, the utility sends the cryptogram to the security module, obtains from the security module a check digit value, and updates the BASE24 database with this check digit value.

This check digit value will then be displayed in the check digits field on the data entry screen the next time you display the screen. If the cryptogram was entered correctly, these check digits will match the check digits originally obtained from the security module.

- Enter both the cryptogram and the check digits on the BASE24 data entry screen.

When you use the Validate Check Digits option of the ASMcom or RSMcom utility, the utility obtains from the security module the check digits for the cryptogram entered on the screen, compares the value returned from the security module with the check digits already entered, and indicates on the Check Digit Validation report if the check digits were validated.

Note: If the security module assigns check digits that are greater than four characters in length, the left-most (most significant) four digits generated by the security module should be entered on the BASE24 data entry screen.

In addition to check digits generation and verification, the ISO Host Interface process uses the check digits in message authentication performed by the Atalla Network Security Processor (NSP). When the MAC received with a message from an external processor and the MAC generated by the ISO Host Interface process do not match, the ISO Host Interface process compares the check digits in the BASE24 database with the check digits returned by the NSP to determine the response code that the BASE24 Remote Banking system should return.

If the BASE24 and NSP check digits match, the ISO Host Interface process returns an invalid MAC response code indicating the message cannot be authenticated because the data has been modified. If the BASE24 and NSP check digits do not match, the BASE24 Remote Banking system returns a key synchronization error response code indicating that the message could not be authenticated because the MAC key check digits in the BASE24 database do not match the check digits returned by the NSP.

A mismatch of the two check digits is known as a key synchronization error and indicates that new keys are necessary. The Racal-Guardata Host Security Module (RSM) does not return check digits to the BASE24 Remote Banking system with the MAC, so the RSM always returns an invalid MAC response code when the MAC received with a message and the MAC generated by the ISO Host Interface process do not match.

Key Management

Keys play an important role in every transaction security process, whether it is data encryption, decryption, or translation; PIN verification; PIN change; message authentication; or security device operation. BASE24 provides the tools listed below to assist in managing the keys used in these processes.

- Atalla security module communications (ASMcom) utility
- Racal security module communications (RSMcom) utility
- Supercrypt utility
- Dynamic key management

ASMcom Utility

ACI provides the ASMcom utility for communicating with the Atalla Security Module (ASM) and the Atalla Network Security Processor (NSP). In addition to the check digits generation and verification described earlier, the ASMcom utility performs key translation.

Note: The ASMcom utility does not currently modify the Interface Security File (ISF) in any way. For the ISF, only procedures in which you must copy down the ASMcom utility output and manually enter it, can be used with the ISF.

When security modules are used, working keys (for example, inbound encryption keys and outbound encryption keys) are in the clear only within the security module. The external processor and the BASE24 system administrator use cryptograms (that is, encrypted versions) of the working keys. The external processor uses a key exchange key (KEK) to encrypt the cryptogram of a working key before providing the cryptogram to the BASE24 system administrator. The BASE24 system administrator uses the master file key (MFK) of the ASM or NSP to encrypt the cryptogram of a working key.

The BASE24 system administrator uses the ASMcom utility with the security module to translate working keys from encryption under the KEK to encryption under the MFK of the security module. The ASMcom utility also uses the security module to translate keys in the BASE24 database from encryption under one security module MFK to encryption under a new security module MFK.

Generating check digits and translating keys from encryption under one MFK to encryption under another MFK result in the modification of security-related information in the BASE24 database. These ASMcom utility functions also

require that the network be shut down while they are performed, so the use of this utility by anyone other than qualified, trusted network security staff is discouraged, and should not be attempted without a full understanding of the processing involved.

Appendix C provides a more detailed description of this processing, as well as step-by-step operating procedures for the ASMcom utility.

RSMcom Utility

ACI provides the RSMcom utility for communicating with the Racal-Guardata Host Security Module (RSM). In addition to the check digits generation and verification described earlier, the RSMcom utility uses the RSM to generate clear intermediate keys, upload an encrypted Diebold number table (DNT) to the BASE24 Remote Banking system for storage, and perform key translation.

Note: The RSMcom utility currently does not modify the Interface Security File (ISF) in any way. For the ISF, only procedures in which you must copy down the RSMcom utility output and manually enter it, can be used with the ISF.

When security modules are used, working keys (for example, inbound encryption keys and outbound encryption keys) are in the clear only within the security module. The external processor and the BASE24 system administrator use cryptograms (that is, encrypted versions) of the working keys. The external processor uses a zone master key (ZMK) to encrypt the cryptogram of a working key before providing the cryptogram to the BASE24 system administrator. The BASE24 system administrator uses the local master key (LMK) of the RSM to encrypt the cryptogram of a working key.

The BASE24 system administrator uses the RSMcom utility with the security module to translate working keys from encryption under the ZMK to encryption under the LMK of the security module. The RSMcom utility also uses the security module to automatically translate keys in the BASE24 database from encryption under one security module LMK to encryption under a new security module LMK.

Generating check digits, translating keys from encryption under one LMK to encryption under another LMK, and uploading the DNT result in the modification of security-related information in the BASE24 database. These RSMcom utility functions also require that the network be shut down while they are performed, so the use of this utility by anyone other than qualified, trusted network security staff is discouraged, and should not be attempted without a full understanding of the processing involved.

Appendix D provides a more detailed description of this processing, as well as step-by-step operating procedures for the RSMcom utility.

Supercrypt Utility

ACI provides the Supercrypt utility for encrypting the inbound and outbound PIN encryption working keys when the external processor is encrypting PINs and the BASE24 Remote Banking system is carrying them in the clear.

Appendix B provides the step-by-step operating procedures for the Supercrypt utility.

Dynamic Key Management

Integrated Authorization Server processes and ISO Host Interface processes use working keys to protect the PINs moving through the BASE24 network. ISO Host Interface processes also use working keys to protect MACs moving between the BASE24 Remote Banking system and an external processor. Working keys should be changed regularly to maintain proper security. The monitoring and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management.

Dynamic key management requires the use of security modules to perform PIN encryption and message authentication. It is not supported when these functions are performed in software. Message authentication is only supported between BASE24 and an external processor; it is not supported between BASE24 and an RBSI-compliant device.

Dynamic key management support is currently provided between external processors and the ISO Host Interface process, and between RBSI-compliant devices and the Integrated Authorization Server process, Interface Master process, and Integrated Endpoint process. Dynamic key management is not currently supported between the BASE24 and CSR terminals or between BASE24 and EBMS-compliant devices.

Dynamic key management allows BASE24 or an external processor to automatically change working keys based on a number of factors. For dynamic key management between BASE24 and an RBSI-compliant device, only BASE24 can automatically change working keys. Thresholds can be established based on time, number of transactions, number of errors, number of consecutive errors, or any combination of these factors. Operator commands are also available for

changing working keys as required. In order for working keys to be changed automatically, another key, known as the key exchange key (KEK), is used to encrypt the working keys for distribution.

Dynamic key management provided by the Integrated Authorization Server process, Interface Master process, Integrated Endpoint process, and ISO Host Interface process is based on the ANSI X9.17 standard, ***Financial Institution Key Management (Wholesale)*** and the ISO 8732 standard, ***Banking—Key Management (Wholesale)***.

BASE24 Remote Banking support of dynamic key management between RBSI-compliant devices and the Integrated Authorization Server process, Interface Master process, and Integrated Endpoint process is described in section 11.

BASE24 Remote Banking support of dynamic key management between external processors and the ISO Host Interface process is described in section 12.

Applied Communications, Inc.

Section 2

Transaction Processing and PINs

This section provides an overview of BASE24 Remote Banking transaction processing and an introduction to the PIN encryption support provided by BASE24 Remote Banking.

PIN Encryption

BASE24 Remote Banking provides PIN encryption support in the Integrated Authorization Server and ISO Host Interface processes. Encryption refers to the process of rendering PINs unintelligible for transmission, and decryption refers to subsequently returning PINs to their original form. The whole process includes encrypting, decrypting, and translating PINs.

Within the BASE24 Remote Banking system, PINs travel in the clear, encrypted under a secure key within a hardware security module, or encrypted using the DES algorithm within software.

When a security module is being used by BASE24 Remote Banking for PIN encryption, PINs are never in the clear within BASE24. They are passed from process to process in their encrypted forms.

When a security module is not being used for PIN encryption, BASE24 Remote Banking decrypts PINs upon entry to the system, and the PINs remain in the clear until they are verified or sent to an external system.

Within BASE24 Remote Banking, transactions can be sent between the BASE24 Remote Banking system and a remote banking endpoint device and can be linked to a number of different external processors, all of which can have their own types of PIN encryption and decryption requirements. BASE24 Remote Banking must provide for a flexible interface between its own PIN handling, which is performed by the Integrated Authorization Server process, and the PIN handling of external processing entities.

BASE24 Remote Banking utilizes the ISO Host Interface process to communicate with external processors. The ISO Host Interface process assumes responsibility for the PIN encryption between the BASE24 Remote Banking system and the respective external processors.

PIN Blocks

During encryption, a PIN is not simply fed into the encryption routines as is. Rather, it is presented to the routines in what is called a PIN block.

A PIN block is a 16-digit series of hexadecimal characters, of which the PIN is a part.

The DES algorithm accepts only 8-byte or 16-hexadecimal-digit blocks of data for input, and since PINs can be from 4 to 12 digits in length, the PIN block type defines what to do with the gap between the PIN length and DES input requirements.

Thus, when encrypting a PIN, the routines are really encrypting a clear PIN block. Likewise, when decrypting a PIN, what is really being decrypted is an encrypted PIN block. This is true, regardless of whether the encryption or decryption is being performed in software by BASE24 Remote Banking or by a hardware module.

BASE24 Remote Banking supports two types of PIN blocks—PIN/PAN PIN blocks and PIN/PAD PIN blocks—both of which are commonly used throughout the industry.

PIN/PAN PIN Blocks

The ISO Host Interface process supports three different PIN/PAN PIN blocks when communicating with external processors, including one PIN/PAN PIN block commonly known as the ANSI PIN block. The ANSI PIN block consists of the 12 right-most digits, excluding the check digit, and is the standard PIN block prescribed by the ANSI X9.8 standard, ***American National Standard for Personal Identification Number (PIN) Management and Security***. ANSI PIN blocks also are called Format 0 PIN blocks by the ISO 9564 standard, ***Banking—Personal Identification Number Management and Security***.

The Integrated Authorization Server process supports only the ANSI PIN block when communicating with a remote banking device.

Building PIN/PAN PIN blocks is a multistage process. Two strings—a PIN string and a PAN string—must first be created. BASE24 Remote Banking uses the PIN, the PIN length, and the pad character F to create a PIN string. BASE24 Remote

Banking then uses a subset of the PAN to create a PAN string. Finally, these strings are combined using an exclusive-OR operation to obtain the final PIN/PAN PIN block.

To build a PIN/PAN PIN block, the Integrated Authorization Server process or ISO Host Interface process performs the following steps:

1. Builds a 16-character PIN string, as follows:
 - a. Places a zero in the first, left-most position.
 - b. Places a hexadecimal representation of the clear PIN length in the second position. (For example, if a PIN is 10 digits in length, the length is specified by the hexadecimal value A.)
 - c. Places the clear PIN in the string, starting in the third position.
 - d. Pads (fills in) the remaining positions to the right with the hexadecimal value F.

Example: A PIN of 8051 (4 digits in length) would result in a PIN string of 048051FFFFFFFF.

2. Builds a 16-character PAN string, as follows:
 - a. Zero-fills the first four positions, starting from the left.
 - b. Selects 12 digits from the PAN of the account involved in the transaction being processed.

The ANSI standard uses the right-most 12 digits of the PAN, excluding the check digit. Therefore, the Integrated Authorization Server process always uses these 12 digits when forming PIN/PAN PIN blocks to use within the BASE24 Remote Banking system when external processors are not involved.

When building PIN blocks for external processors, the ISO Host Interface process uses the value in the ANSI PAN FORMAT field on Key File (KEYF) screen 1 to determine the 12 digits used by the external processor to form the PIN block. Valid values are as follows:

- 0 = Use the 12 right-most digits, excluding the check digit (ANSI standard).
- 1 = Use the 12 right-most digits, including the check digit.
- 2 = Use the 12 left-most digits.

If the PAN does not contain enough significant digits to provide the 12 digits specified by the ANSI standard or the value in the ANSI PAN FORMAT field on KEYF screen 1, the Integrated Authorization Server process or the ISO Host Interface process use leading zeros to reformat the PAN used in creating the PAN string.

- c. Places the selected 12 digits from the PAN in the right-most positions of the PAN string.

Example: Using the 12 right-most digits, excluding the check digit, a PAN of 1040000533921277526 results in a PAN string of 0000053392127752.

3. Combines the PIN string and PAN string through an exclusive-OR operation, as follows:
 - a. Converts each hexadecimal digit in the PIN and PAN strings into its binary representation. As an example, a hexadecimal 2 has a binary representation of 0010 and a hexadecimal A has a binary representation of 1010.
 - b. Combines, using the exclusive-OR operation, the binary representation (four bits) for the first digit in the PIN string with the binary representation (four bits) for the first digit in the PAN string so that the result is still a binary representation. This exclusive-OR operation is also known as modulo 2 addition. For each bit, if the values are the same (both 0 or both 1), the resulting bit is 0, and if the values are different (one is 0 and the other is 1), the resulting bit is 1. As an example, 0010 plus 1010 equals 1000.

The Integrated Authorization Server process or ISO Host Interface process then repeats this process for each digit of the two strings.

- c. Converts the binary representations obtained in step b back to their hexadecimal equivalents. As an example, the binary representation of 1000 converts to a hexadecimal 8.

The following is an example of the creation of a PIN/PAN PIN block using the 12 right-most PAN digits, excluding the check digit:

INPUTS:

PIN: 8 0 5 1

PAN: 1 0 4 0 0 0 0 5 3 3 9 2 1 2 7 7 5 2 6

STRINGS:

PIN STRING: 0 4 8 0 5 1 F F F F F F F F F F

PAN STRING: 0 0 0 0 0 5 3 3 9 2 1 2 7 7 5 2

RESULTING PIN BLOCK: 0 4 8 0 5 4 C C 6 D E D 8 8 A D

PIN/PAD PIN Blocks

PIN/PAD PIN blocks are another form of PIN block that can be used in PIN encryption.

The PIN and a hexadecimal pad character of A through F are used to build a PIN/PAD PIN block.

To build a PIN/PAD PIN block, the Integrated Authorization Server or ISO Host Interface process perform the following steps:

1. Places the PIN in the left-most positions.
2. Pads each of the remaining positions after the PIN with the hexadecimal pad character. For the ISO Host Interface process, this character is specified in the PIN PAD CHARACTER field on KEYF screen 1. For the Integrated Authorization Server process, this character is specified in the PIN PAD CHARACTER field on Interface Security File (ISF) screen 1.

Note: The Racal-Guardata security module restricts the pad character to hexadecimal F. For other security module pad character requirements, refer to the vendor documentation for the security module being used.

The following is an example of the creation of a PIN/PAD PIN block:

INPUTS:

PIN: 8 0 5 1

PAD CHARACTER: F

RESULTING PIN BLOCK: 8 0 5 1 F F F F F F F F F F F F

Racal-Guardata Host Security Module Requirements

Special requirements exist for PIN blocks in configurations using the Racal-Guardata Host Security Module (RSM). These requirements affect translations between two PIN/PAN PIN blocks, the IBM 3624-format PIN verification method, and the Visa PVV PIN verification method.

PIN Translations

PIN translation using an RSM requires that the same PAN digits used to create the input PIN block in a PIN/PAN-to-PIN/PAN translation are also used to create the output PIN block. This requirement does not apply to translations between a PIN/PAN PIN block and a PIN/PAD PIN block.

As an example, the ISO Host Interface process receives a transaction from a host and the transaction has a PIN/PAN PIN block created using a PAN string containing the 12 right-most digits, excluding the check digit. In this case, the RSM can translate the PIN block into a PIN/PAD PIN block or another PIN/PAN PIN block that uses a PAN string containing the 12 right-most digits, excluding the check digit. The RSM cannot translate the PIN block into another PIN/PAN PIN block using a PAN string containing the 12 left-most digits or one using a PAN string containing the 12 right-most digits, including the check digit.

Refer to appendix F for the specific BASE24 file maintenance settings to meet this requirement.

IBM 3624-Format PIN Verification

IBM 3624-format PIN verification using an RSM requires that the validation data include the five right-most positions of the PAN, excluding the check digit. Therefore, the PAN VERIFY OFFSET and PAN VERIFY LENGTH fields in the Key Authorization File (KEYA) must be set appropriately to capture this portion of the PAN.

Refer to section 7 for the BASE24 file maintenance settings for this verification method.

Visa PVV PIN Verification

Visa PIN verification value (PVV) method of PIN verification using an RSM requires that external processors use PIN/PAD PIN blocks or ANSI X9.8 standard PIN/PAN PIN blocks in the messages sent to BASE24 Remote Banking. The Visa PVV PIN verification method uses the right-most digits of the PAN excluding the check digit. The ANSI X9.8 standard PIN/PAN PIN block uses the 12 right-most digits of the PAN excluding the check digit. The RSM returns a PIN block error if the external processors use a PIN/PAN PIN block other than the ANSI X9.8 standard.

This requirement does not affect any BASE24 file maintenance settings. The external processor must be instructed to use either PIN/PAD PIN blocks or ANSI X9.8 PIN/PAN PIN blocks.

For more information on the file maintenance settings for the Visa PVV PIN method of verification, refer to section 10.

Internal Messages and Transaction Data

Transactions move through BASE24 Remote Banking systems in the form of standard internal messages or internal transaction data. Standard internal messages are used to communicate between the Integrated Authorization Server process and other processes such as the ISO Host Interface process and Billpay Server process. Internal transaction data is used to communicate within the Integrated Authorization Server process. All information relating to PIN encryption, and the key under which the PIN is encrypted, is carried in the standard internal message or internal transaction data. When a transaction is passed between the Integrated Authorization Server process and another process, it is converted from internal transaction data to standard internal message format. This conversion is reversed when a transaction is passed from another process to the Integrated Authorization Server process.

When a message is received from a remote banking device, the Integrated Authorization Server process converts that message from the native format of the device to an Internal Transaction Data (ITD) format it recognizes. The Integrated Authorization Server process then performs whatever actions are required on the PIN.

When a message is sent to an external processor, the Integrated Authorization Server process converts the ITD to a BASE24-telebanking Standard Internal Message (BSTM) format recognizable to the ISO Host Interface process. The ISO Host Interface process, which is responsible for creating the external message, determines the form the PIN is in from the fields in the internal message and performs any necessary actions (decryption, encryption, or translation) based on the current form of the PIN and the needs of the external entity.

For a complete list and descriptions of all BSTM and ITD fields, refer to appendix A of the ***BASE24 Remote Banking Transaction Processing Manual***.

Section 3

Remote Banking Device PIN Encryption Support

This section presents a detailed discussion of the PIN encryption support provided by the Integrated Authorization Server process.

In reviewing the information presented in this section, note that the term *remote banking device* can refer to any remote banking endpoint device or customer service terminal as defined in the “Definition of Terms” discussion in section 1.

The Integrated Authorization Server process uses external messages that are based on the 1993 ISO 8583 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification***.

Note: The Integrated Authorization Server process and the ISO Host Interface process both use the ISO 8583 standard; however, there are differences such as message layout and valid field values used by the two processes because the ISO Host Interface process uses the 1987 ISO 8583 standard, ***Bank Card Originated Messages—Interchange Message Specifications***, instead of the 1993 ISO 8583 standard.

Integrated Authorization Server Process

In the Remote Banking system, the task of PIN encryption, decryption, and translation on behalf of remote banking devices is assigned to the Integrated Authorization Server process.

When the BASE24 Remote Banking system receives an external message containing a PIN from a remote banking device, the PIN can be in clear or encrypted format. The message is routed to the Integrated Authorization Server process which is tasked with processing the message, including PIN processing.

Note: If dynamic key management is used for an RBSI-compliant device, the PIN in the external message must be in hardware-encrypted format. Dynamic key management processing is not available for remote banking endpoint devices using the Customer Service Interface. For detailed information on dynamic key management for RBSI-compliant devices, refer to section 11.

When the Integrated Authorization Server process receives the message, it uses the Interface Security File extended memory table to perform any required encryption, decryption, or translation as dictated by the interface instance for the remote banking device. The steps required are determined by the PIN block type, either PIN/PAN PIN block or PIN/PAD PIN block, as specified in the table. The Interface Security File extended memory table is built from the Interface Security File (ISF).

PIN encryption data for messages acquired from a remote banking device is configured on Interface Security File (ISF) screens 1 and 2.

Remote Banking Device Keys

The Integrated Authorization Server process uses inbound encryption keys as well as intermediate keys. Inbound encryption keys are used to handle PINs between the BASE24 Remote Banking system and a remote banking device. Intermediate keys are keys for which both the clear and encrypted versions are available to the BASE24 Remote Banking system (the encrypted version being encrypted under the key of a security module).

The keys required by BASE24 Remote Banking must be physically placed in the BASE24 database through file maintenance, except for the inbound PIN encryption key which can also be placed in the BASE24 database as a result of dynamic key management processing.

Inbound Encryption Keys

An inbound encryption key is the working key BASE24 Remote Banking uses for dealing with incoming PINs from a remote banking device. If a PIN is encrypted coming from the remote banking device, this is the key under which it is encrypted. If a PIN is in the clear coming from the remote banking device and must be encrypted by BASE24 Remote Banking, it is encrypted in software using the intermediate software key described below under “Intermediate Keys.”

When encryption is performed by a security module, the value entered in the BASE24 database is a cryptogram (that is, encrypted version) of the inbound encryption key. This cryptogram is obtained using the Atalla master file key (MFK) or the Racal-Guardata local master key (LMK).

Intermediate Keys

BASE24 Remote Banking uses intermediate keys to decrypt PINs that are encrypted under keys managed by a security module. BASE24 Remote Banking also uses intermediate keys to take PINs that are received in the clear and encrypt them under a key that is managed by a security module. The clear version of an intermediate key is known as an intermediate software key, while the hardware-encrypted version of an intermediate key is known as an intermediate hardware key.

Intermediate Software Keys

Intermediate software keys are used to encrypt PINs when a clear PIN is received from a remote banking device and it needs to be processed by a security module in encrypted form. Intermediate software keys are used to decrypt PINs when an encrypted PIN is received from a remote banking device and it needs to be processed in software in clear form.

Intermediate Hardware Keys

Intermediate hardware keys are used to translate PINs when a clear PIN is received from a remote banking device and it needs to be processed by a security module in encrypted form. Intermediate hardware keys are used to translate PINs when an encrypted PIN is received from a remote banking device and it needs to be processed in software in clear form.

Intermediate Key Processing

When a PIN that has been encrypted in hardware by a remote banking device must be brought into the clear for internal processing, BASE24 Remote Banking can have the security module reencrypt the PIN under the intermediate key using the encrypted version, allowing BASE24 Remote Banking to decrypt the PIN in software using the clear version of the key. This is accomplished through the following steps:

1. The PIN is translated from being encrypted under the current inbound encryption key to being encrypted under the corresponding hardware intermediate key from the Interface Security File extended memory table.
2. The PIN is decrypted using the corresponding software intermediate key from the Interface Security File extended memory table.

When a PIN that is received in the clear must be encrypted for processing by a security module, BASE24 Remote Banking can encrypt the PIN in software using the clear version of the key, allowing the security module to translate the PIN using the encrypted version of the key.

Previous Keys

If both a current and previous inbound encryption key exists, the Integrated Authorization Server process sends both encryption keys to a security module to verify or translate a PIN before allowing the transaction to be declined due to a sanity check error or PIN block error (Racal-Guardata security modules only). The Integrated Authorization Server process tries the current key first, and if a sanity check or PIN block error occurs, tries to use the previous key.

If a PIN length error occurs using the current key, the transaction is denied. In this case, the security module does not attempt to use the previous key because the security module was able to decrypt enough of the PIN block using the current key to determine that the PIN length found at the beginning of the block was invalid. This means that the current key was valid; using the previous key will not change the error condition.

PIN Encryption Options

BASE24 Remote Banking can use several PIN encryption options internally and externally for PIN conversion. External PIN encryption options dictate how the PIN is encrypted in external messages. Internal PIN encryption options dictate how the PIN is encrypted in internal messages.

External PIN encryption options include the following:

- Clear PIN processing. All PINs carried in the external message passed through the BASE24 Remote Banking system are in the clear.
- Hardware-encrypted PIN processing. A security module is accessed for PIN processing. All PINs carried in the external message passed through the BASE24 Remote Banking system are hardware-encrypted under secure keys.

This is the only option available when using dynamic key management for endpoints using the Remote Banking Standard Interface (i.e., RBSI-compliant devices). For detailed information on dynamic key management, refer to section 11.

- Software-encrypted PIN processing. The software uses DES PIN processing. All PINs carried in the external message passing through the BASE24 Remote Banking system are encrypted in software, using the DES algorithm. No security module is in use.

Internal PIN encryption options include the following:

- Clear PIN processing. All PINs carried in the ITD or BSTM within the BASE24 Remote Banking system are in the clear.
- Hardware-encrypted PIN processing. A security module is accessed for PIN processing. All PINs carried in the ITD or BSTM within the BASE24 Remote Banking system are hardware-encrypted under secure keys.
- Software-encrypted PIN processing. The software uses DES PIN processing. All PINs carried in the ITD or BSTM within the BASE24 Remote Banking system are encrypted in software, using the DES algorithm. No security module is in use.

The PIN processing conversion combinations resulting from the PIN encryption options offered by BASE24 Remote Banking are as follows:

External PIN Encryption	Internal PIN Encryption
Clear	Clear
Clear	Hardware encrypted
Clear	Software encrypted
Hardware encrypted	Clear
Hardware encrypted	Hardware encrypted
Hardware encrypted	Software encrypted
Software encrypted	Clear
Software encrypted	Software encrypted

Note: Software security can be used instead of a security module for encrypting PINs in the external messages when internal messages carry PINs encrypted under a secure key. However, ACI recommends hardware security for the following reasons:

- Security modules that are already in place for internal message PIN management can also be used for external message PIN management.
- A higher level of security exists because keys are controlled within security modules and intermediate keys are not required.

BASE24 Remote Banking allows for each of these combinations, and each combination requires its own file settings in the BASE24 database. This section discusses each of these combinations and describes how the BASE24 database must be set up for each.

When a transaction is received from a remote banking device, BASE24 Remote Banking may be required to move the PIN from one type of encryption to another for use internally by the system.

Incoming Transaction Processing

BASE24 Remote Banking acquires transactions from endpoints such as interactive voice response systems (IVRs), personal computers, and customer service representative (CSR) terminals. The following are examples of the processing that can take place when the BASE24 Remote Banking system acquires a transaction.

Note: PINs acquired from a CSR terminal are always in the clear.

Clear PINs

In the case of an incoming message in which the PIN is carried in the clear, the Integrated Authorization Server process performs the following PIN processing:

1. Accesses the Interface Security File extended memory table to determine what actions are required for this transaction.
2. Accepts the PIN sent by the remote banking device as the clear PIN for the transaction.
3. Performs any PIN translation required. PIN translation is required only if the clear PIN needs to be encrypted.
4. Updates the PIN information in the ITD.

The message then follows the normal transaction processing as described in the ***BASE24 Remote Banking Transaction Processing Manual***.

Encrypted PINs

In the case of an incoming message in which the PIN is carried in encrypted format, the Integrated Authorization Server process performs the following PIN processing:

1. Accesses the Interface Security File extended memory table to determine what actions are required for this transaction.
2. Performs any PIN translation required. PIN translation is required only if the PIN needs to be decrypted. If both the external and internal PINs are hardware encrypted, no translation takes place.
3. Updates the PIN information in the ITD.

The message then follows the normal transaction processing as described in the ***BASE24 Remote Banking Transaction Processing Manual***.

Setting Up the BASE24 Database

The Interface Security File (ISF) contains the information and parameters required for PIN encryption and PIN translation with the BASE24 Remote Banking software. The ISF contains information for both issuer and acquirer interfaces; however, only acquirer interfaces are currently supported by BASE24 Remote Banking within an Integrated Authorization Server process.

The ISF contains one record for each record in the ITS Interface Configuration File (IFCF). The IFCF contains one record for all customer service representative (CSR) terminals and one record for each group of up to 90 remote banking stations. Having one ISF record for each IFCF record enables an institution to configure security parameters for the CSR terminals and each group of remote banking stations. The VRU Configuration Data (VCD) screens are used to maintain the IFCF.

Note: The ISO Host Interface described in section 4 uses the Key File (KEYF) instead of the ISF for transaction security parameters. Institutions using the BASE24 Remote Banking product and authorizing transactions on a host must define transaction security parameters in both the KEYF and the ISF.

ISF screens use the following acronyms:

Acronym	Description
KAM*	Key used to authenticate message data.
KDT*	Key used to encrypt data for transport. This key also is used with a PIN transport key (KPT) to encrypt PINs for transport.
KKT-AM*	Key used to encrypt message authentication keys (KAMs) for transport.
KKT-DT*	Key used to encrypt data transport keys (KDTs) for transport.
KKT-PT	Key used to encrypt PIN transport keys (KPTs) for transport.
KPT	Key to encrypt PINs for transport.
VVK	Verification value (check digits) associated with a key.

* These keys are not currently used.

ISF Screen 1

Interface Security File (ISF) screen 1 contains PIN encryption parameters, message authentication parameters, and key transport key information. The following fields on this screen are pertinent to transaction security functions performed by the Integrated Authorization Server process. Any fields appearing on this screen that are not described here are not currently used by the Integrated Authorization Server process. For detailed descriptions of these fields and a screen illustration, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

INTERFACE NAME — The name of the interface instance using the information in this ISF record. An interface instance provides a means of dividing interface processing according to specific business relationships and purposes. Interface processing is divided according to groups of stations, with up to 90 stations per each interface instance. For example, a separate interface instance could be used for each financial institution using the interface. One interface instance is used to define interface processing for all customer service representative (CSR) terminals. The value in this field must match the value in the INTERFACE NAME field on VRU Configuration Data (VCD) screen 1 for the same interface instance.

EXTRN PIN ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external entity. The default value is H (Hardware PIN encryption). Valid values are as follows:

H = Hardware PIN encryption
N = None (clear PINs)
S = Software PIN encryption

This field should always be set to a value of N for the CSR terminals interface instance.

A description of the code entered is displayed immediately to the right of the value in the EXTRN PIN ENCRYPT TYPE field.

INTRN PIN ENCRYPT TYPE — Specifies the type of PIN encryption used by BASE24 products. The default value is H (Hardware PIN encryption). Valid values are as follows:

H = Hardware PIN encryption
N = None (clear PINs)
S = Software PIN encryption

If the value in the EXTRN PIN ENCRYPT TYPE field is S, the value in this field must be N or S. If the value in the EXTRN PIN ENCRYPT TYPE field is H or N, the value in this field can be H, N, or S.

A description of the code entered is displayed immediately to the right of the value in the INTRN PIN ENCRYPT TYPE field.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in external messages. The default value is 1 (ANSI (PIN/PAN) PIN block). Valid values are as follows:

- 0 = Clear PINs
- 1 = ANSI (PIN/PAN) PIN block
- 3 = PIN/PAD PIN block

If the value in the EXTRN PIN ENCRYPT TYPE field is N, the value in this field must be 0. If the value in the EXTRN PIN ENCRYPT TYPE field is H or S, the value in this field must be 1 or 3.

A description of the code entered is displayed immediately to the right of the value in the PIN BLOCK FORMAT field.

PIN PAD CHARACTER — If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the pad character used in the formation of the external message PIN block. Valid values are A through F. The default value is F.

PIN ENCRYPT LEVEL — A code identifying the level of encryption used for PINs exchanged between an external entity and the BASE24 interface. The valid value is 1 (single).

OLD KEY TIMER LIMIT — The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the interface when dynamic key management processing is configured. The keys in the KPT1 and KPT2 fields on ISF screen 2, KAM1 and KAM2 fields on ISF screen 3, and KDT1 and KDT2 fields on ISF screen 4 permit information for two keys. The values in the CURRENT INDEX fields on each screen identify whether key number 1 or key number 2 is in use.

At the time a new key is placed in service, the other key becomes the old key. For example, if inbound PIN transport key (KPT) number 1 is currently in use when the BASE24 interface receives a new key, the new key is stored as KPT number 2.

KPT number 2 becomes the current key and KPT number 1 becomes the old key. (If KPT number 2 is the current key, it becomes the old key and the new key is stored as KPT number 1.)

When processing a transaction, the Integrated Authorization Server process uses the new key for PIN processing. If the security device used by the Integrated Authorization Server process detects an error because the external entity is still using the old key, it can repeat the PIN processing using the old key before the Integrated Authorization Server process declines the transaction. The value in this field determines how long the old key is available. If the old key is not available, a second PIN processing attempt is not made. The default value is 0. Valid values are as follows:

- 0 = The old key is not cleared.
- 1–99999 = The maximum length of time, in seconds, that an old key can be used before being cleared.

ISF Screen 2

ISF screen 2 contains PIN transport key information. The following fields on this screen are pertinent to transaction security functions performed by the Integrated Authorization Server process. Any fields appearing on this screen that are not described here are not currently used by the Integrated Authorization Server process. This includes all of the fields that appear under the OUTBOUND KEYS label on the screen. For detailed descriptions of these fields and a screen illustration, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

PIN TRANSPORT KEY INFORMATION

The following fields contain information for keys used to transport PINs between a remote banking device and the BASE24 Remote Banking system.

Intermediate keys are utilized whenever a PIN that has been encrypted under a secure key (hardware PIN encryption) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

INTERM SFTWR KEY — The clear version of the intermediate key. The key is encrypted in software for storage. The value in this field is used to encrypt a PIN that is received in the clear from a remote banking device. The value in this field is

used to decrypt a PIN that is received in encrypted format from a remote banking device. The default is set to zeros. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

INTERM HDWR KEY — The hardware-encrypted intermediate key. The value in this field is used to translate a PIN that is received in the clear or in encrypted format from a remote banking device. The default is set to zeros. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

VVK — The verification value associated with the value in the INTERM HDWR KEY field. The default is set to zeros. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is, transactions sent by the remote banking device to the Integrated Authorization Server process).

If external PIN management is performed in hardware, security module-encrypted keys are entered in these fields. If external PIN management is performed in software, clear keys are entered in these fields. However, they are encrypted in software for storage. If external PIN management is performed in the clear, keys entered in these fields are not used.

KPT1 — The key utilized by the external entity to encrypt PINs in transactions sent to the Integrated Authorization Server process. The Integrated Authorization Server process uses the key in this field when the CURRENT INDEX field contains a value of 0. The default is set to zeros. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

VVK1 — The verification value associated with the value in the KPT1 field. The default is set to zeros. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

KPT2 — The key utilized by the external entity to encrypt PINs in transactions sent to the Integrated Authorization Server process. The Integrated Authorization Server process uses the key in this field when the CURRENT INDEX field

contains a value of 1. The default is set to zeros. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

VVK2 — The verification value associated with the value in the KPT2 field. The default is set to zeros. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

CURRENT INDEX — Identifies the key the interface is currently using. The default value is 0. Valid values are as follows:

- 0 = The value in the KPT1 field
- 1 = The value in the KPT2 field

Applied Communications, Inc.

Section 4

External Processor PIN Encryption Support

This section presents a detailed discussion of the PIN encryption support provided by the ISO Host Interface process in BASE24 Remote Banking systems.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC) or co-network as defined in the “Definition of Terms” discussion in section 1.

ISO Host Interface Process

The ISO Host Interface process handles PIN encryption between BASE24 Remote Banking systems and their respective external processors. The following documentation includes the PIN encryption options provided by the ISO Host Interface process. The options selected for the ISO Host Interface process are based on a combination of interface process capabilities and user preferences.

The ISO Host Interface process provides external messages that are based on the 1987 ISO 8583 standard, ***Bank Card Originated Messages—Interchange Message Specifications***. The ISO Host Interface process does not use the ANSI-based message format used by some BASE24 products. For more information on the ISO Host Interface process, refer to the ***BASE24 ISO Host Interface Manual*** and ***BASE24 External Message Manual***.

Note: The ISO Host Interface process and the Integrated Authorization Server process both use the ISO 8583 standard; however, there are differences such as message layout and valid field values used by the two processes because the Integrated Authorization Server process uses the 1993 ISO 8583 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification***, instead of the 1987 ISO 8583 standard.

External Processor Keys

BASE24 identifies five types of keys for use in handling PINs between BASE24 Remote Banking systems and an external processor:

- Inbound encryption keys
- Intermediate keys
- Outbound encryption keys
- PIN key exchange keys
- Previous inbound and outbound keys

In most cases, the keys required by BASE24 Remote Banking must be physically placed in the BASE24 database through file maintenance. BASE24 utility programs are provided to assist in the preparation of this information. The key fields in the database can be left with their default values (zeros) if keys are provided by an automatic key change when the interface process is started; however, the PIN key exchange key, if used, must be entered manually.

The ISO Host Interface process offers dynamic key management for PIN encryption keys. Refer to section 12 for additional information about dynamic key management support for external processors.

Inbound Encryption Keys

An inbound encryption key is the working key BASE24 uses for dealing with incoming PINs from an external processor. If a PIN is encrypted coming from the external processor, this is the key under which it is encrypted. If a PIN is in the clear coming from the external processor and must be encrypted by BASE24, this is the key under which it is to be encrypted.

When encryption is performed by a security module, the value entered in the BASE24 database is a cryptogram (that is, encrypted version) of the inbound encryption key. This cryptogram is obtained using the Atalla master file key (MFK) or the Racal-Guardata local master key (LMK). If the external processor provides a cryptogram of the key, this cryptogram is encrypted under the PIN key exchange key (KEK). The security module translates the inbound encryption key from encryption under the KEK to encryption under the MFK or LMK. The clear version of the inbound encryption key does not appear in the BASE24 database.

Outbound Encryption Keys

An outbound encryption key is the working key under which a PIN being sent to an external processor must be encrypted, if encryption is required by the external processor.

When encryption is performed by a security module, the value entered in the BASE24 database is a cryptogram (that is, encrypted version) of the outbound encryption key. This cryptogram is obtained using the Atalla MFK or the Racal-Guardata LMK. If the external processor provides a cryptogram of the key, this cryptogram is encrypted under the KEK. The security module translates the outbound encryption key from encryption under the KEK to encryption under the MFK or LMK. The clear version of the outbound encryption key does not appear in the BASE24 database.

Intermediate Keys

An intermediate key is a key for which both the clear and encrypted versions are available to the BASE24 Remote Banking system (the encrypted version being encrypted under the key of a security module).

BASE24 uses intermediate keys to decrypt PINs that are encrypted under keys managed by a security module. BASE24 also uses intermediate keys to take PINs that are received in the clear and encrypt them under a key that is managed by a security module.

When an encrypted PIN must be brought into the clear for processing or for sending to an external processor, BASE24 can have the security module reencrypt the PIN under the intermediate key using the encrypted version, allowing BASE24 to decrypt the PIN in software using the clear version of the key.

When a PIN that is received in the clear must be encrypted for processing by a security module, BASE24 can encrypt the PIN in software using the clear version of the key, allowing the security module to translate the PIN using the encrypted version of the key.

Previous Keys

After a key change has taken place between BASE24 and an external processor, PINs can be sent between BASE24 and the external processor encrypted under the new key or the previous key. Because the external processor does not specify the key under which a PIN is encrypted, the ISO Host Interface process saves the previous key and places it in the BASE24-telebanking Standard Internal Message (BSTM) along with the new key.

When translating PINs, the ISO Host Interface process sends both encryption keys to a security module before allowing a transaction to be declined due to a sanity check error or PIN block error. The interface process tries the new key first, and if a sanity check error or PIN block error occurs, tries to use the previous key. The interface process can translate PINs in the following situations:

- In outgoing messages, when internal messages carry encrypted PINs and external messages carry clear PINs.
- In outgoing messages, when both internal and external messages carry encrypted PINs.
- In incoming messages, when internal messages carry clear PINs and external messages carry encrypted PINs.

Some interface processes use the CLEAR OLD KEY TIMER VALUE field in the Key File (KEYF) to control how long the previous key can be used. This field is accessed using KEYF screen 3.

Note: Only certain external processors support the use of a previous key.

PIN Key Exchange Keys

A PIN key exchange key (KEK) is the key BASE24 uses to encrypt or decrypt outbound PIN encryption keys when the BASE24 Remote Banking system and the external processor use dynamic key management to automatically change the inbound and outbound PIN encryption keys.

A KEK can be single length (16 hexadecimal characters) or double length (32 hexadecimal characters). Using double-length KEKs provides additional security by increasing the difficulty of using trial and error to obtain the clear value of the working keys. The KEKs used for encrypting PIN encryption working keys and message authentication working keys both must be the same length, whether single or double. An Atalla Network Security Processor is required when double-length KEKs are used.

Dynamic Key Management

Dynamic key management allows a BASE24 Remote Banking system and an external processor to change working keys based on time, number of transactions, number of errors, number of consecutive errors, or any combination of these factors. The ISO Host Interface process currently supports dynamic key management.

Dynamic key management requires the use of security modules and is not supported in BASE24 software. Supported hardware modules include the following:

- Atalla Security Module
- Atalla Network Security Processor
- Racal-Guardata Host Security Module

For detailed information regarding hardware support, refer to the appropriate security module vendor manual. Refer to section 12 for additional information on dynamic key management support for external processors.

PIN Encryption Options

An external processor can use the following three PIN encryption options:

- Clear PIN processing. All PINs carried in the external message passed between BASE24 and the external processor are in the clear.
- Encrypted PINs: security module accessed for PIN processing. All PINs carried in the external message passed between BASE24 and the external processor are encrypted under secure keys.
- Encrypted PINs: software DES PIN processing. All PINs carried in the external message passed between BASE24 and the external processor are encrypted using software, using the DES algorithm. No security module is in use.

BASE24 can use the following two PIN encryption options internally:

- Clear PIN processing. For BASE24 Remote Banking systems, all PINs carried in the BASE24-telebanking Standard Internal Message (BSTM) are in the clear.
- Encrypted PINs: security module accessed for PIN processing. For BASE24 Remote Banking systems, all PINs carried in the BASE24-telebanking Standard Internal Message (BSTM) are encrypted under secure keys.

The three external processor options and the two BASE24 options offer the following combinations:

1. Internal and external messages carry clear PINs.
2. Internal messages carry clear PINs and external messages carry PINs encrypted in software or under secure keys.
3. Internal messages carry PINs encrypted under secure keys and external messages carry clear PINs.
4. Internal and external messages carry PINs encrypted under secure keys.

Note: Software security can be used instead of a security module for encrypting PINs in the external messages when internal messages carry PINs encrypted under a secure key. However, ACI recommends hardware security for the following reasons:

- Security modules that are already in place for internal message PIN management can also be used for external message PIN management.

- A higher level of security exists because keys are controlled within security modules and intermediate keys are not required.

BASE24 allows for each of these combinations, and each combination requires its own file settings in the BASE24 database. This section discusses each of these combinations and describes how the BASE24 database must be set up for each.

When a transaction is received from an external processor, BASE24 may be required to move the PIN from one type of encryption to another for use internally by the system. Similarly, when a transaction is sent to an external processor, BASE24 may be required to move the PIN from one type of encryption to another prior to sending it.

BASE24 Carries Clear PINs; External Processor Carries Clear PINs

If PINs are carried in clear form by an external processor, no special PIN handling is required.

Processing Outgoing Messages

The ISO Host Interface process sends the clear PIN to the external processor.

Processing Incoming Messages

In the case of incoming messages, the ISO Host Interface process accepts the PIN sent by the external processor as the clear PIN for the transaction.

Setting Up the BASE24 Database

Fields on screen 1 of the Key File (KEYF) record for each external processor are used when the external processor carries PINs in clear form. Fields pertain to inbound and outbound keys. Fields on KEYF screen 3 deal exclusively with dynamic key management. Screens 2 and 3 are not used when all PINs are carried in clear form. Refer to the ***BASE24 Base Files Maintenance Manual*** for detailed information on KEYF screen 1 and its fields.

The following fields on KEYF screen 1 must be set as indicated when both the external processor and BASE24 carry PINs in clear form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear by the external processor.

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear by BASE24.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages. This field must be set to a value of 0 (clear PINs) because PINs are carried in the clear, meaning PIN blocks are not used by the external processor or BASE24.

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message. This field must be set to a value of 0 (12 right-most digits, excluding the check digit) even though PIN blocks are not used by the external processor or BASE24.

PIN PAD CHARACTER — Specifies the pad character used in the formation of the PIN block in the external message. This field must be set to a value of F even though PIN blocks are not used by the external processor or BASE24.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 are the same or different. The value in this field is used when PIN or MAC keys are exchanged automatically.

- If message authentication is not being performed, this field must be set to the default value of 1.
- If message authentication is being performed using the same key for inbound and outbound messages, this field must be set to a value of 1.
- If message authentication is being performed using different keys for inbound and outbound messages, this field must be set to a value of 2.

KEY LENGTH — Identifies whether single- or double-length key exchange keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If message authentication is not being performed, this field must be set to the default value of 1.
- If single-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 2.

BASE24 Carries Clear PINs; External Processor Carries Encrypted PINs

If PINs are carried in clear form by BASE24, but are carried in encrypted form by the external processor, BASE24 PIN processing is carried out as described below.

PINs are carried in clear form in the BASE24-telebanking Standard Internal Message (BSTM) and either software or hardware security can be used for encrypting PINs in the external messages between BASE24 and the external processor.

Processing Outgoing Messages

In the case of outgoing messages, BASE24 must encrypt PINs prior to sending them to the external processor. The ISO Host Interface process performs as follows:

1. Creates a PIN block using the clear PIN.
 - If the PIN BLOCK FORMAT field on KEYF screen 1 is set to a value of 1 (ANSI PIN block), BASE24 creates a PIN/PAN PIN block using the appropriate characters from the customer's PAN, as specified in the ANSI PAN FORMAT field in the KEYF.
 - If the PIN BLOCK FORMAT field on KEYF screen 1 is set to a value of 3 (PIN/PAD PIN block), BASE24 creates a PIN/PAD PIN block using the appropriate pad character specified in the PIN PAD CHARACTER field in the KEYF.

Note: If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), a security module must be brought into use. In this case, the interface process formats a PIN/PAD PIN block, regardless of the setting in the PIN BLOCK FORMAT field on KEYF screen 1, because the PIN block is an intermediate PIN block that will be translated in step 5 to the format required by the external processor.

2. Determines the outbound encryption key to use to encrypt the PIN.
 - If the OUTBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key in the OUTBOUND KEYS: PIN KEY1 field on KEYF screen 2 is used.
 - If the OUTBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key in the OUTBOUND KEYS: PIN KEY2 field on KEYF screen 2 is used.
3. Determines the encryption type and continues accordingly.
 - If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), processing continues with step 4.
 - If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), the ISO Host Interface process encrypts the PIN block in software using the DES algorithm and the outbound encryption key selected in step 2. The interface process then sends the encrypted PIN block to the external processor.
4. Encrypts the PIN block in software under the clear intermediate key taken from the INTERMEDIATE KEYS: CLEAR field on KEYF screen 1.
5. Requests that the security module translate the PIN block for sending to the external processor.

The ISO Host Interface process provides the security module with the following information:

- The encrypted PIN block.
- The encrypted version of the intermediate key taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1.
- The encrypted version of the outbound encryption key selected in step 2.
- The type of PIN block required by the external processor, specified in the PIN BLOCK FORMAT field on KEYF screen 1.
- The PAN digits specified by the ANSI PAN FORMAT field or the PIN PAD CHARACTER field on KEYF screen 1, depending on the type of PIN block required by the external processor.

The security module performs the following steps:

- a. Decrypts the intermediate key and the outbound encryption key.
- b. Decrypts the PIN block using the decrypted intermediate key.
- c. Extracts the clear PIN from the PIN block.
- d. Creates a PIN block of the type required by the external processor.

- e. Encrypts the PIN block using the decrypted outbound encryption key.
- f. Returns the encrypted PIN block to the interface process.

The interface process then sends the encrypted PIN block to the external processor.

Processing Incoming Messages

In the case of incoming messages, the ISO Host Interface process must decrypt PINs upon arrival from the external processor. The ISO Host Interface process performs the following steps:

1. Determines the inbound encryption key to use for decrypting the PIN.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2 is used first and the encryption key in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2, if one exists, is used as the previous key in step 4.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2 is used and the encryption key in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2, if one exists, is used as the previous key in step 4.
2. Determines the encryption type and continues accordingly.
 - If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), the ISO Host Interface process must have the PIN translated with an intermediate key prior to decrypting it. Processing continues with step 3.
 - If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), the ISO Host Interface process decrypts the PIN block in software using the DES algorithm and the inbound encryption key selected in step 1. Processing then continues with step 6.

3. Requests that the security module translate the PIN block for decryption by BASE24.

The ISO Host Interface process provides the security module with the following information:

- The encrypted PIN block taken from the external message.
- The encrypted version of the key under which the PIN is currently encrypted. This is the inbound key selected in step 1.
- The encrypted version of the intermediate key taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1.
- The type of PIN block used by the external processor, specified in the PIN BLOCK FORMAT field on KEYF screen 1.
- The PAN digits specified by the ANSI PAN FORMAT field or the PIN PAD CHARACTER field on KEYF screen 1, depending on the type of PIN block used by the external processor.

The security module then performs the following steps:

- a. Decrypts the inbound encryption key and the intermediate key.
- b. Decrypts the PIN block using the decrypted inbound encryption key.
- c. Extracts the clear PIN from the PIN block.
- d. Creates a PIN/PAN PIN block using the clear PIN.

A PIN/PAN PIN block with a PAN string containing the 12 right-most digits, excluding the check digit, is always requested by the ISO Host Interface process when the PIN is being translated to encryption under the intermediate key.

- e. Encrypts the PIN/PAN PIN block using the decrypted intermediate key.
- f. Returns the encrypted PIN block to the interface process.

4. Repeats step 3 with the previous inbound encryption key, if necessary.
 - If the security module successfully translated the PIN block in step 3 using the current key identified in step 1, processing continues with step 5.
 - If the security module returned a sanity check error or PIN block error in step 3, the interface process repeats step 3 using the previous key identified in step 1. If the security module successfully translates the PIN block, processing continues with step 5.

- If the security module returns a sanity check error or PIN block error when repeating step 3 with the previous key, the interface process declines the transaction.
 - If no previous key exists (that is, the previous key field contains zeros or blanks), the interface process declines the transaction.
5. Decrypts the PIN block using the clear version of the intermediate key.
- Since the PIN block is now encrypted under the intermediate key, the ISO Host Interface process can use the clear version of the key to decrypt the PIN block in software.
- The clear version of the intermediate key is taken from the INTERMEDIATE KEYS: CLEAR field on KEYF screen 1.
6. Extracts the clear PIN from the PIN block and places it in the PIN field of the internal message.
7. Computes the size of the clear PIN and notes the size in the PIN-LGTH field of the internal message.

Setting Up the BASE24 Database

Fields on screens 1 and 2 of the Key File (KEYF) record for each external processor are used when BASE24 carries PINs in clear form and the external processor carries PINs in encrypted form. Fields on KEYF screen 3 deal exclusively with dynamic key management and are described in section 12. Refer to the ***BASE24 Base Files Maintenance Manual*** for detailed information on KEYF screens 1 and 2 and their fields.

KEYF Screen 1

The following fields on KEYF screen 1 must be set as indicated when BASE24 carries PINs in clear form and the external processor carries PINs in encrypted form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor.

- If a security module is being used, this field must be set to a value of 1 (security module PIN management).
- If PINs are encrypted by software for the external processor, this field must be set to a value of 2 (software DES PIN management).

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear in the BASE24-telebanking Standard Internal Message (BSTM).

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages.

- If PINs are encrypted for or by the external processor using a PIN/PAN PIN block, this field must be set to a value of 1 (ANSI PIN block).
- If PINs are encrypted for or by the external processor using a PIN/PAD PIN block, this field must be set to a value of 3 (PIN/PAD PIN block).

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message.

- If PIN/PAN PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 1), this field must specify the PAN digits to use for forming a PIN/PAN PIN block in external messages.
- If PIN/PAD PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 3), this field must be set to a value of 0.
- If a Racal-Guardata Host Security Module is used, this field must be set to a value of 0 (12 right-most PAN digits, excluding check digit) to match the PAN string used by BASE24 in internal messages.

PIN PAD CHARACTER — Specifies the pad character used in the formation of the PIN block in the external message.

- If PIN/PAD PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 3), this field must specify the pad character to use for forming a PIN/PAD PIN block in external messages.
- If PIN/PAN PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 1), this field must be set to a value of F.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 are the same. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If the same key is used for inbound and outbound messages, this field must be set to a value of 1.

- If different keys are used for inbound and outbound messages, this field must be set to a value of 2.

KEY LENGTH — Identifies whether single- or double-length key exchange keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. The PIN KEK and MAC KEK must be the same length. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If single-length KEKs are being used for key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for key exchanges, this field must be set to a value of 2.

INTERMEDIATE KEYS

The following three fields are used whenever a PIN that has been encrypted under a secure key (security module PIN encryption) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

CLEAR — Specifies the clear version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

- If the ENCRYPT TYPE field is set to a value of 1 (security module PIN management), this field must contain the clear version of the intermediate key.
- If the ENCRYPT TYPE field is set to a value of 2 (software DES PIN management), this field must be set to zeros.

CHECK DIGITS — Specifies the check digits associated with the clear version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the clear version of the intermediate key.

ENCRYPTED — Specifies the encrypted version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

- If the ENCRYPT TYPE field is set to a value of 1 (security module PIN management), this field must contain the encrypted version of the intermediate key.
- If the ENCRYPT TYPE field is set to a value of 2 (software DES PIN management), this field must be set to zeros.

EXCHANGE KEYS

The following two fields are used to create keys that BASE24 uses when PIN keys are automatically exchanged between the ISO Host Interface process and the host.

PIN KEY — Identifies the security module-encrypted form of the key exchange key (KEK) used when inbound and outbound PIN keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. There are two sets of 16 hexadecimal characters. The first set is used for single-length keys and both sets are used for double-length keys. Valid values for each character in this field are 0 through 9 and A through F. Both sets must contain 16 valid characters. If the value in the KEY LENGTH field is 1 (single-length keys), the second set of characters must be zeros. Refer to section 12 for additional information about dynamic key management.

CHECK DIGITS — Specifies the check digits associated with the PIN KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the PIN KEK.

KEYF Screen 2

The following fields on KEYF screen 2 must be set as indicated when BASE24 carries PINs in clear form and the external processor carries PINs in encrypted form.

PIN KEY INFORMATION

The PIN KEY INFORMATION section of KEYF screen 2 contains two groups of fields, OUTBOUND KEYS and INBOUND KEYS, each containing six fields.

OUTBOUND KEYS

The following six fields pertain to PIN encryption and PIN translation.

PIN KEY1 — Identifies the first key to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

CHECK DIGITS1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Indicates a second key to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the outbound PIN key has been changed by dynamic key management processing. This counter is incremented whenever the outbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The ISO Host Interface process increments the counter automatically and attempts to resynchronize it with the external processor.

INBOUND KEYS

The following six fields contain the PIN keys used for inbound transactions (that is transactions sent by the external processor to BASE24).

PIN KEY1 — Identifies the first key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS 1— Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Identifies a second key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the inbound PIN key has been changed by dynamic key management processing. This counter is incremented whenever the inbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The ISO Host Interface process increments the counter automatically and attempts to resynchronize it with the external processor.

BASE24 Carries Encrypted PINs; External Processor Carries Clear PINs

If PINs are carried in encrypted form by BASE24 and are carried in clear form by the external processor, BASE24 PIN processing is carried out as described below.

Security modules must be used to perform internal message PIN management when PINs are carried in encrypted form in the BSTM. PINs in the external messages between BASE24 and the external processor are carried in the clear.

Processing Outgoing Messages

BASE24 must decrypt PINs prior to sending them to the external processor. The ISO Host Interface process performs the following steps:

1. Requests that the security module translate the PIN block for decryption by BASE24.

The interface process provides the security module with the following information:

- The encrypted PIN block taken from the PIN field in the internal message.
- The type of PIN block used by BASE24, taken from the PIN Data field in the internal message.
- The PAN digits specified by the ANSI offset value in the internal message, or the PIN pad character value from the internal message, depending on the type of PIN block used.
- The encrypted version of the key under which the PIN block is currently encrypted, taken from the PIN Data field in the internal message.
- The encrypted version of the intermediate key, taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1.

The security module performs the following steps:

- a. Decrypts the intermediate key and the key under which the PIN block is encrypted.
- b. Decrypts the PIN block using the decrypted version of the key used to encrypt the PIN block.
- c. Extracts the clear PIN from the PIN block.

- d. Creates a PIN/PAN PIN block using the clear PIN.

A PIN/PAN PIN block with a PAN string containing the 12 right-most digits, excluding the check digit, is always requested by BASE24 when the PIN is being translated to encryption under the intermediate key.
 - e. Encrypts the PIN/PAN PIN block using the decrypted intermediate key.
 - f. Returns the encrypted PIN block to the interface process.
2. Repeats step 1 with the previous encryption key, if necessary.
 - If the security module successfully translated the PIN block in step 1 using the key taken from the PIN key field in the internal message, processing continues with step 3.
 - If the security module returned a sanity check error or PIN block error in step 1, the interface process repeats step 1 using the previous encryption key, taken from the previous key field in the internal message. If the security module successfully translates the PIN block, processing continues with step 3.
 - If the security module returns a sanity check error or PIN block error when repeating step 1 with the previous key, the interface process declines the transaction.
 - If no previous key exists (that is, the previous key field in the internal message contains zeros or blanks), the interface process declines the transaction.
 3. Decrypts the PIN block using the clear version of the intermediate key.

Since the PIN block has been encrypted under the intermediate key, BASE24 can use the clear version of the key to decrypt the PIN block in software.

The clear version of the intermediate key is taken from the INTERMEDIATE KEYS: CLEAR field on KEYF screen 1.
 4. Extracts the PIN from the clear PIN/PAN PIN block.

Processing Incoming Messages

In the case of incoming messages, BASE24 must encrypt PINs upon arrival from the external processor. The ISO Host Interface process performs the following steps:

1. Creates a PIN/PAN PIN block using the right-most 12 digits of the PAN, excluding the check digit.
2. Encrypts the PIN block using the DES algorithm and the clear intermediate key taken from the INTERMEDIATE KEYS: CLEAR field on KEYF screen 1.
3. Determines the inbound encryption key to use to encrypt the PIN.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2 is used.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2 is used.
4. Encrypts the clear PIN using the intermediate key.
5. Requests that the security module translate the PIN block to encryption under the inbound encryption key.

The interface process provides the security module with the following information:

- The encrypted PIN block.
- The encrypted version of the inbound encryption key selected in step 3.
- The encrypted version of the intermediate key taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1.
- The type of PIN block created by BASE24 in step 1.
- The PAN digits used for the PIN block created in step 1.

The security module then performs the following steps:

- a. Decrypts the inbound encryption key and the intermediate key.
- b. Decrypts the PIN block using the decrypted intermediate key.
- c. Encrypts the PIN block using the decrypted inbound encryption key.
- d. Returns the encrypted PIN block to the interface process.

6. Places the encrypted PIN block in the PIN field of the internal message, and sets the PIN-LGTH field in the internal message to a value of 16.
7. Sets the PIN Data field in the internal message to indicate the PIN is encrypted and that a PIN/PAN PIN block was used.
8. Sets the PIN key and previous key fields in the internal message.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2 is the PIN key and the encryption key in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2 is the previous key.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2 is the PIN key and the encryption key in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2 is the previous key.

Setting Up the BASE24 Database

Fields on screens 1 and 2 of the Key File (KEYF) record for each external processor are used when BASE24 carries PINs in encrypted form and the external processor carries PINs in clear form. Fields on KEYF screen 3 deal exclusively with dynamic key management and are described in section 12. Refer to the ***BASE24 Base Files Maintenance Manual*** for detailed information on KEYF screens 1 and 2 and their fields.

KEYF Screen 1

The following fields on KEYF screen 1 must be set as indicated when BASE24 carries PINs in encrypted form and the external processor carries PINs in clear form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear by the external processor.

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 1 (security module PIN management) to indicate that PINs are carried in encrypted form by BASE24.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages. This field must be set to a value of 1 (ANSI PIN block) to identify the format of the PIN block returned to the interface process by the security module.

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message. This field must be set to a value of 0 (12 right-most digits, excluding check digit) to identify the PAN digits carried in the PIN block returned to the interface process by the security module.

PIN PAD CHARACTER — Specifies the pad character used in the formation of the PIN block in the external message. This field must be set to a value of F.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 are combined or separate. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If message authentication is not being performed, this field must be set to the default value of 1.
- If message authentication is being performed using the same key for inbound and outbound messages, this field must be set to a value of 1.
- If message authentication is being performed using different keys for inbound and outbound messages, this field must be set to a value of 2.

KEY LENGTH — Identifies whether single- or double-length key exchange keys (KEKs) are being used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If message authentication is not being performed, this field must be set to the default value of 1.
- If single-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 2.

INTERMEDIATE KEYS

The following three fields are used when a PIN that has been encrypted under a secure key (security module PIN encryption) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

CLEAR — Identifies the clear version of the intermediate key to be used for translating PINs for software decryption (outbound) or software encryption (inbound). Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS — Specifies the check digits associated with the clear version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the clear intermediate key.

ENCRYPTED — Identifies the encrypted version of the intermediate key to be used for translating PINs for software decryption (outbound) or software encryption (inbound). Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

KEYF Screen 2

The following fields on KEYF screen 2 must be set as indicated when BASE24 carries PINs in encrypted form and the external processor carries PINs in clear form.

PIN KEY INFORMATION

The PIN KEY INFORMATION section of KEYF screen 2 contains two groups of fields, OUTBOUND KEYS and INBOUND KEYS, each containing six fields.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is transactions sent by the external processor to BASE24).

PIN KEY1 — Identifies the first key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Identifies the second key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

BASE24 and External Processor Carry Encrypted PINs

If PINs are carried in encrypted form by both BASE24 and the external processor, BASE24 PIN processing is carried out as described below.

Security modules must be used to perform internal message PIN management when PINs are carried in encrypted form in the BSTM. While either software or hardware security can be used for encrypting PINs in the external messages between BASE24 and the external processor, ACI recommends hardware security for the following reasons:

- Security modules that are already in place for internal message PIN management can also be used for external message PIN management.
- A higher level of security exists because keys are controlled within security modules and intermediate keys are not required.

The processing descriptions and database settings that follow are based on security modules being used to perform PIN management in external messages as well as internal messages.

Processing Outgoing Messages

In the case of outgoing messages, BASE24 must translate PINs prior to sending them to the external processor.

Note: In situations where BASE24 has received a message containing an encrypted PIN from a host or device and is routing the message to an external processor that expects an encrypted PIN, the PIN block carried in the internal message at this point has not been modified from the way it was received from the device or first host.

To translate PINs, the ISO Host Interface process performs the following steps:

1. Determines the encryption key to use to encrypt the PIN for sending to the external processor.
 - If the OUTBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key in the OUTBOUND KEYS: PIN KEY1 field on KEYF screen 2 is used.

- If the OUTBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key in the OUTBOUND KEYS: PIN KEY2 field on KEYF screen 2 is used.
2. Requests that a security module translate the PIN block to encryption under the outbound encryption key.

The interface process provides the security module with the following:

- The encrypted PIN block taken from the PIN field in the internal message.
- The encrypted version of the key under which the PIN is currently encrypted, taken from the PIN Data field in the internal message.
- The type of PIN block used by BASE24, taken from the PIN Data field in the internal message.
- The PIN pad character value from the internal message.
- The encrypted outbound encryption key selected in step 1.
- The type of PIN block required by the external processor taken from the PIN BLOCK FORMAT field on KEYF screen 1.

Note: Some security modules require that the same PAN digits used to create the input PIN block in a PIN/PAN-to-PIN/PAN translation are also used to create the output PIN block. This requirement does not apply to translations between a PIN/PAN PIN block and a PIN/PAD PIN block. The Racal-Guardata Host Security Module has this requirement. The Atalla Security Module (ASM) also has this requirement; however the requirement can be eliminated if the ASM includes support for command 33. The Atalla Network Security Processor does not have this requirement.

The security module then performs the following steps:

- a. Decrypts the outbound key and the key under which the PIN block is encrypted.
- b. Decrypts the PIN block using the decrypted version of the key under which the PIN block is encrypted.
- c. Extracts the clear PIN from the clear PIN block.
- d. Creates a new PIN block of the type required by the external processor.
- e. Encrypts the PIN block using the decrypted outbound key.
- f. Returns the encrypted PIN block to the interface process.

3. Repeats step 2 with the previous encryption key, if necessary.
 - If the security module successfully translated the PIN block in step 2. using the key taken from the PIN Data field in the internal message, the PIN block is ready for sending to the external processor.
 - If the security module returned a sanity check error or PIN block error in step 2, the interface process repeats step 2 using the previous encryption key, taken from the previous key field in the internal message. If the security module successfully translates the PIN block, the PIN block is ready for sending to the external processor.
 - If the security module returns a sanity check error or PIN block error when repeating step 2 with the previous key, the interface process declines the transaction.
 - If no previous key exists (that is, the previous key field in the internal message contains zeros or blanks), the interface process declines the transaction.

Processing Incoming Messages

In the case of incoming messages, no translation is required by BASE24. The ISO Host Interface process performs the following steps:

1. Moves the PIN block from the external message into the PIN field of the internal message and sets the PIN-LGTH field in the internal message to a value of 16.
2. Sets the PIN Data and previous key fields in the internal message.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2 is the PIN key and the encryption key in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2 is the previous key.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2 is the PIN key and the encryption key in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2 is the previous key.
3. Sets the PIN block information in the internal message, based upon the value of the PIN BLOCK FORMAT field on KEYF screen 1.

- If the PIN BLOCK FORMAT field on KEYF screen 1 is set to a value of 1 (ANSI PIN block), the interface process sets the PIN format field in the internal message to indicate a PIN/PAN PIN block was used by the external processor.

Note: If the external PIN block is an ANSI PIN block, but the 12 right-most digits from the PAN, excluding the check digit, were not used to create the PIN block, then the external PIN must be translated from its current PIN block to an encrypted PIN block using the 12 right-most digits from the PAN, excluding the check digit.

- If the PIN BLOCK FORMAT field on KEYF screen 1 is set to a value of 3 (PIN/PAD PIN block), the interface process sets the PIN format field in the internal message to indicate a PIN/PAD PIN block was used by the external processor. It also sets the PIN pad character field in the internal message from the PIN PAD CHARACTER field on KEYF screen 1.

Setting Up the BASE24 Database

Fields on screens 1 and 2 of the Key File (KEYF) record for each external processor are used when both the external processor and BASE24 carry their PINs in encrypted form. Fields on KEYF screen 3 deal exclusively with dynamic key management and are described in section 12. Refer to the ***BASE24 Base Files Maintenance Manual*** for detailed information on KEYF screens 1 and 2 and their fields.

KEYF Screen 1

The following fields on KEYF screen 1 must be set as indicated when BASE24 and the external processor carry PINs in encrypted form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor. This field must be set to a value of 1 (security module PIN management).

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 1 (security module PIN management) to indicate that PINs are carried in encrypted form by BASE24.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages.

- If PINs are encrypted by or for the external processor using a PIN/PAN PIN block, this field must be set to a value of 1 (ANSI PIN block).
- If PINs are encrypted by or for the external processor using a PIN/PAD PIN block, this field must be set to a value of 3 (PIN/PAD PIN block).

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message.

- If PIN/PAN PIN blocks are used for the external processor (PIN BLOCK FORMAT field on KEYF screen 1 contains a value of 1), this field must specify the PAN digits to use for forming a PIN/PAN PIN block.
- If PIN/PAD PIN blocks are used for the external processor (PIN BLOCK FORMAT field on KEYF screen 1 contains a value of 3), this field must be set to a value of 0.

The Racal-Guardata Host Security Module, if used, requires that the PAN string used to create the input PIN block in a PIN/PAN to PIN/PAN translation is also used to create the output PIN block.

PIN PAD CHARACTER — Specifies the pad character used in the formation of the PIN block in the external message.

- If PIN/PAD PIN blocks are used for the external processor, this field must specify the pad character to use for forming a PIN/PAD PIN block.
- If PIN/PAN PIN blocks are used for the external processor, this field must be set to a value of F.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 are combined or separate. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If the same key is used for inbound and outbound messages, this field must be set to a value of 1.
- If different keys are used for inbound and outbound messages, this field must be set to a value of 2.

KEY LENGTH — Identifies whether single- or double-length key exchange keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. The PIN KEK and MAC KEK must be the same length. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If single-length KEKs are being used for key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for key exchanges, this field must be set to a value of 2.

EXCHANGE KEYS

The following two fields must be set as indicated to allow PIN keys to be exchanged between the ISO Host Interface process and a host in encrypted form.

PIN KEY — Identifies the security module-encrypted form of the key exchange key (KEK) used when inbound and outbound PIN keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. There are two sets of 16 hexadecimal characters. The first set is used for single-length keys and both sets are used for double-length keys. Valid values for each character in this field are 0 through 9 and A through F. Both sets must contain 16 valid characters. If the value in the KEY LENGTH field is 1 (single-length keys), the second set of characters must be zeros. Refer to section 12 for additional information about dynamic key management.

CHECK DIGITS — Identifies the check digits associated with the PIN KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the PIN KEK.

KEYF Screen 2

The following fields on KEYF screen 2 must be set as indicated when BASE24 and the external processor carry PINs in clear form.

PIN KEY INFORMATION

The PIN KEY INFORMATION section of KEYF screen 2 contains two groups of fields, OUTBOUND KEYS and INBOUND KEYS, each containing six fields.

OUTBOUND KEYS

The following six fields pertain to PIN encryption and PIN translation.

PIN KEY1 — Identifies the first key to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

CHECK DIGITS1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Indicates a second key to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the outbound PIN key has been changed by dynamic key management processing. This counter is incremented whenever the outbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The ISO Host Interface process increments the counter automatically and attempts to resynchronize it with the external processor.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is transactions sent by the external processor to BASE24).

PIN KEY1 — Identifies the first key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS1— Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Identifies a second key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix B.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the outbound PIN key has been changed by dynamic key management processing. This counter is incremented whenever the outbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The ISO Host Interface process increments the counter automatically and attempts to resynchronize it with the external processor.

Section 5

Message Authentication Support

This section provides an introduction to the message authentication support provided in the ISO Host Interface process for BASE24 Remote Banking.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC) or co-network as defined in the “Definition of Terms” discussion in section 1.

BASE24 Remote Banking Message Authentication

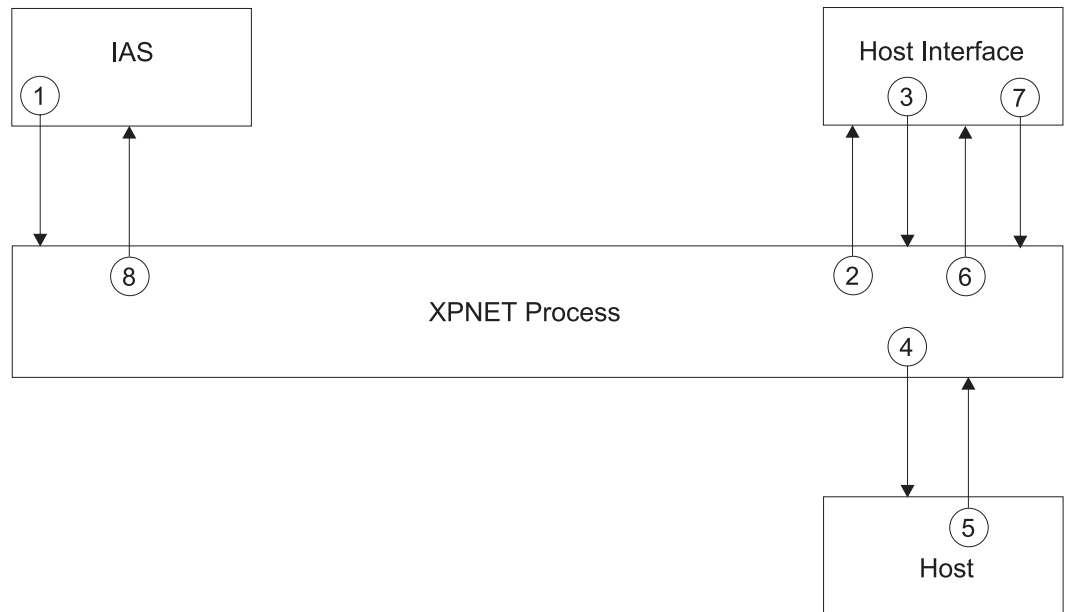
BASE24 Remote Banking provides message authentication support in the ISO Host Interface process. Message authentication is utilized to ensure transaction messages are received exactly as sent by the legitimate originator. It is designed to protect messages against accidental or deliberate alteration. The Integrated Authorization Server process does not support message authentication between it and a remote banking endpoint device or customer service representative terminal at the present time.

The message authentication code (MAC) is generated by the originator of the message, based upon message elements identified in advance by the originator and recipient, and included with the message. To verify a MAC it receives, the recipient computes a MAC based upon the same message elements, then compares the computed MAC and the MAC received with the message. If the two MACs match, the message is valid and has not been altered. However, if the two MACs do not match, the message cannot be validated. This can be because alterations may have occurred in the message or because the originator and recipient are not using the same keys to compute the MAC.

Message authentication occurs between an external processor and an ISO Host Interface process. The MAC is not carried in the BASE24-telebanking Standard Internal Message or in Internal Transaction Data. For example, message authentication on a transaction originating at a device for authorization at an external processor does not occur until it reaches the ISO Host Interface process, which generates a MAC value in either Racal-Guardata or Atalla hardware or in software. The transaction is then sent on to the external processor for authentication and processing.

Transaction Flow

The following transaction flow diagram illustrates message authentication for a transaction originating from the Integrated Authorization Server process and authorized by a host. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process

The steps illustrated in the diagram are described below.

1. The Integrated Authorization Server process determines that the request must be sent to an issuer host for authorization, creates a BSTM, and passes the message to the XPNET process for delivery to the ISO Host Interface process.
2. The XPNET process delivers the request to the ISO Host Interface process.
3. The ISO Host Interface process determines the appropriate host from the DPC-NUM field in the BSTM; creates an external message from the data included in the BSTM; and generates a MAC and includes it in the external message. The ISO Host Interface process passes the message to the XPNET process for delivery to the host.
4. The XPNET process passes the external message to the host to be verified and processed.

5. The host validates the MAC, processes the transaction, and returns a message either approving or declining the transaction to the XPNET process. The host generates a new MAC based on information in the response and includes the new MAC with the response.
6. The XPNET process delivers the response message to the ISO Host Interface process.
7. The ISO Host Interface process validates the MAC received in the response, updates the BSTM from the external message using only the information that was changed by the host, and passes the message to the XPNET process for return to the Integrated Authorization Server process. The BSTM does not contain the MAC.
8. The XPNET process delivers the response message from the host to the Integrated Authorization Server process to continue internal processing of the transaction.

For more information regarding BASE24 Remote Banking financial message handling and transaction flows refer to the ***BASE24 ISO Host Interface Manual***.

Hardware and Software Authentication

The ISO Host Interface process supports message authentication in both hardware and software. The ISO Host Interface process uses the value from the MAC ENCRYPT TYPE field on KEYF screen 1 to determine the type of message authentication that is to be performed. Valid values for this field are as follows:

- 0 = No MAC processing
- 1 = Security module MAC processing
- 2 = Software MAC processing (not currently supported)

This field must be set to the value 1 if any of the MAC key timer fields on KEYF screen 3 are set to a nonzero value. This field or the ENCRYPT TYPE field must be set to a value 1 if the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 contains a nonzero value or the KEY PROCESSING TYPE field on KEYF screen 3 contains a value other than N (N indicates no dynamic key management is to be performed).

Hardware message authentication can be performed with the Atalla Network Security Processor (NSP) and the Racal-Guardata Security Module.

Authentication Inputs

The MAC resulting from generation or verification depends on several variables. All information used to generate a MAC initially must match the information used to verify the MAC when it is received. Parameters for these computations include the following:

- MAC key
- Data to be authenticated
- Location and length of the MAC
- Authentication algorithm

MAC Key

The key used to verify the MAC must match the key used to generate the MAC initially. The key used by an external processor and the key used by the ISO Host Interface process must be the same. The key must contain only hexadecimal characters.

Check digits associated with the MAC key are entered with the MAC key in the Key File (KEYF). New check digits are calculated each time the MAC key is changed. These check digits help verify the accuracy of the key itself and are used each time a MAC is generated or verified to ensure that the security module is using the proper key. BASE24 Remote Banking returns a MAC failure response when a MAC cannot be verified and the check digits returned by the security module match the check digits in the KEYF. BASE24 Remote Banking returns a KMAC (MAC key) synchronization error when a MAC cannot be verified and the check digits returned by the security module do not match the check digits in the KEYF.

Data to be Authenticated

The data to be authenticated can be an entire message or selected fields from a message, depending on configuration options provided by the external processor and the security device. The message originator and recipient must use the same data arranged in the same order when generating and verifying MACs. Data must also be in the same character set (that is, ASCII or EBCDIC) when generating and verifying MACs.

Note: When using the Atalla Network Security Processor (NSP), the length of the data being authenticated cannot exceed 4096 bytes. When using the Racal-Guardata Security Module, the length of the data being authenticated cannot exceed 400 bytes.

Location and Length of MAC

The MAC generated by the transaction originator is expected to be located in the final 64 bits of the message of the ISO-based external message exchanged between an external processor and an interface process.

In an ISO message, the MAC can be in data element P-64 or in data element S-128, depending on whether any secondary data elements are in the message. Data element P-64 contains the MAC when the message contains only primary data elements (P-1 through P-63). Data element S-128 contains the MAC if the message contains any of the other secondary data elements (S-65 through S-127). The MAC can be in data element P-64 or data element S-128, but not both.

According to the 1986 ANSI X9.9 standard, ***Financial Institution Message Authentication (Wholesale)***, the MAC can be 32, 48, or 64 bits in length. BASE24 Remote Banking supports only the 32-bit, left-justified MAC. The MAC is represented as eight hexadecimal characters.

Authentication Algorithm

The authentication algorithm is the formula used to obtain the MAC from the authentication data and MAC key. BASE24 Remote Banking uses the authentication algorithm defined in the 1986 ANSI X9.9 standard, ***Financial Institution Message Authentication (Wholesale)*** when generating or verifying MACs. The ANSI-standard authentication algorithm is used in all message authentication performed with security modules. It is also used when message authentication is performed in software by the ISO Host Interface process.

BASE24 Remote Banking Internal Messages

Transactions move through BASE24 Remote Banking systems in the form of standard internal messages, which are passed among the BASE24 Remote Banking processes.

When a message is received from an external processor, the ISO Host Interface process converts the external message to a BASE24-telebanking Standard Internal Message (BSTM) format recognizable to BASE24 Remote Banking processes. When the ISO Host Interface process sends a message to an external processor, it converts the message from BSTM format to the external message format of the external processor to which the message is sent.

Message authentication only occurs on messages exchanged between the ISO Host Interface process and an external processor. Message authentication does not occur on messages exchanged between the Integrated Authorization Server process and a remote banking device.

No fields specific to message authentication are carried in the BSTM, although the following field in the BSTM is set accordingly by the ISO Host Interface to indicate the outcome of message authentication processing for incoming messages.

Failure Flag — Indicates whether this is a failed message. Valid values are as follows:

0 = No failure
1 = Failure

Applied Communications, Inc.

Section 6

PIN Verification Support

This section presents an introduction to the PIN verification support provided by BASE24 Remote Banking.

Several file maintenance screens are discussed in this section. These screens are illustrated in appendix A.

BASE24 Remote Banking PIN Verification

PIN verification, that is, the authentication of a customer by a transaction authorizer, can be performed for transactions to be authorized by the Integrated Authorization Server process or as a prescreening step for transactions authorized by a host. PIN verification can also be performed locally at the remote banking endpoint device, allowing PINs to be verified at a remote banking endpoint device before the transaction is forwarded to the BASE24 Remote Banking system for authorization or to an authorizing host. BASE24 Remote Banking also supports customer PIN selection and PIN change options.

PIN verification confirms that the PIN entered with a transaction is the actual PIN of the customer authorized to use the customer ID involved in the transaction. When transactions are received by the Integrated Authorization Server process, part of the authorization procedure can include verification of the PIN.

If PINs are to be verified by the Integrated Authorization Server process, they can be verified in software or by a security module. For verifying PINs in software, the PIN and all of the data required for verification must be in the clear or software-encrypted so that they can be used by the Integrated Authorization Server process. For verifying PINs using a security module, the PIN is carried in the system encrypted under a key known only to the security module.

The following documentation includes the PIN verification options provided by BASE24 Remote Banking. Information includes basic configuration options for the Integrated Authorization Server process PIN verification option, local PIN verification option, PIN selection, and PIN change options. Also included are BASE24-telebanking Standard Internal Message (BSTM) field descriptions and an overview of the procedures performed within the Integrated Authorization Server process.

Institution PIN Verification Parameters

PIN verification by the Integrated Authorization Server process requires a record for each institution performing PIN verification in the following files:

- Key Authorization File (KEYA)
- Institution Definition File (IDF)

The following are descriptions of the screens and fields required for PIN verification to be performed by the Integrated Authorization Server process.

For detailed information on IDF and KEYA screens and fields, refer to the ***BASE24 Base Files Maintenance Manual***.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the IBM 3624-format, Diebold, Identikey, or Visa PVV PIN verification method.

GRP — The code used in the IDF to identify this KEYA record. If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with the institution's FIID in this field.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. Valid values are as follows:

- 01 = IBM 3624-format PIN verification on screen 2
- 02 = Diebold PIN verification on screen 3
- 03 = Identikey PIN verification on screen 4
- 04 = Visa PVV PIN verification on screen 5

KEYA Screens 2 Through 5

BASE24 Remote Banking supports several PIN verification methods. The KEYA screens required for maintaining the KEYA depend on the PIN verification method used, as shown in the following table. Field descriptions for each of these screens are provided in other sections, as shown in the following table:

PIN Verification Method	Screen	Section
IBM 3624-format	2	7
Diebold	3	8
Identikey	4	9
Visa PVV	5	10

IDF Screen 2

IDF screen 2 enables an institution to set up the customer processing parameters. BASE24 Remote Banking uses some of the parameters on IDF screen 2 even though BASE24 Remote Banking does not require cards to initiate transactions. In particular, the PIN CHECK TYPE field is used to determine the PIN verification method to be used by the institution. The following fields on IDF screen 2 pertain to PIN verification processing by BASE24 Remote Banking.

PIN — Specifies whether the PIN entered by the customer is checked during transaction screening. The Integrated Authorization Server process uses this field with authorization level 1 (online) and authorization level 3 (online/offline) only, since prescreening checks are made only for these authorization levels. Valid values are as follows:

Y = Yes, check the customer's PIN; if invalid, decline the request and do not send the request to the host.

N = No, do not check the customer's PIN before sending the request to the host if the host is online.

The default value is N.

MAX PIN TRIES — The code indicates the number of times that a customer can enter an incorrect PIN. The Integrated Authorization Server process uses the Customer Table (CSTT) to accumulate this information for a customer regardless of authorization method. The default value is 1.

BAD PIN ACTION — The Integrated Authorization Server process does not use this code because no card is involved in a BASE24 Remote Banking transaction. The default value is 0.

POFST/PVV LOC — A code specifying the location of the IBM 3624-format or Diebold PIN verification method PIN offset, the Visa PVV PIN verification method PIN Verification Value (PVV), or the Identikey PIN verification method PIN Verification Number (PVN). Valid values are as follows:

- 0 = No PIN offset or PVN. A value of 0000 is used if an offset is required for the verification method. This value is not valid for Visa PVV.
- 1 = PIN offset, PVV, or PVN is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the offset.
- 2 = PIN offset, PVV, or PVN is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the offset.

A description of the code entered is displayed to the right of the POFST/PVV LOC field. The default value is 0.

PIN CHECK TYPE — A code indicating the PIN verification method used by the institution. Valid values are as follows:

- 00 = No verification
- 01 = IBM 3624-format
- 02 = Diebold
- 03 = Identikey
- 04 = Visa PVV

A description of the code entered is displayed to the right of the PIN CHECK TYPE field. The default value is 00.

ALGO NUMBER LOC — A code specifying the location of the algorithm number. Currently, the algorithm number is required only for the Diebold PIN verification method. When other PIN verification methods are used, this field should contain a 0. Valid values are as follows:

- 0 = Algorithm number is not required for the PIN verification method.
- 1 = Algorithm number is located in the Key Authorization File (KEYA).
- 2 = Algorithm number is located in the Key Authorization File (KEYA).

A description of the code entered is displayed to the right of the ALGO NUMBER LOC field. The default value is 0.

CARDHOLDER PIN SELECT — The Integrated Authorization Server process does not use this code. For BASE24 Remote Banking, initial PIN selection is controlled at the customer-level in the Customer Table (CSTT).

PIN TRIES RESET OPTION — A code indicating how the BAD PIN COUNT field on CSTT screen 1 is to be reset for customers of an institution.

The Integrated Authorization Server process keeps track of the number of bad PIN tries for a customer according to the authorization parameters established by an institution. This allows bad PIN tries to be accumulated over a period of time, and institutions can then choose to decline authorization of a transaction for a customer if that customer has had an excessive number of incorrect PIN tries. The PIN tries limit is located in the MAX PIN TRIES field on IDF screen 2.

BASE24 products offer two methods for clearing accumulated bad PIN tries. The first allows for clearing the accumulated bad PIN tries when the customer enters a correct PIN. The second method allows for clearing the accumulated bad PIN tries at the end of each usage accumulation period. This field controls how these clearance methods are to be applied for an institution's customers.

The bad PIN tries in the CSTT are not automatically cleared at the end of each usage accumulation period. Institutions using the CSTT can choose—using this field—to have their CSTT bad PIN tries automatically cleared with the rest of their totals at the end of each usage accumulation period, when a correct PIN is entered, or both. The bad PIN tries in the CSTT can also be reset by refreshing the file.

BASE24 Remote Banking uses customer processing dates in the BASE24-telebanking segment of the IDF to track usage accumulation periods. The default value for this field is 0. Valid values are as follows:

- 0 = Reset the bad PIN tries at the end of the usage accumulation period, but not when a correct PIN is entered.
- 1 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.
- 3 = Reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 4 = Reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.

For more information on setting up CSTT screen 1, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

IDF Screen 4

IDF screen 4 enables an institution to specify its withdrawal period parameters. BASE24 Remote Banking uses parameters on IDF screen 4 for PIN tries only. The fields on IDF screen 4 that pertain to PIN verification are described below.

WITHDRAWAL PERIOD PARAMETERS

The values in the following fields specify the withdrawal period. BASE24 Remote Banking uses these parameters for PIN tries only. The Integrated Authorization Server process maintains customer PIN tries in the Customer Table (CSTT).

BEGINNING DATE — The starting date (YYMMDD) of the current usage accumulation period for all customers belonging to this institution.

The date in this field is placed in the CSTT when usage accumulation fields in that table have been cleared. This date is placed in the LAST RESET DATE field on CSTT screen 1.

BASE24 products update this field automatically at the beginning of a new withdrawal period. See the description of the NEXT BEGINNING DATE field on this screen for more information.

If the value in the PERIOD LENGTH field on this screen is 83, 84, 85, or 86, the day in this field must be between 01 and 28.

Note: BASE24 Remote Banking uses the CURRENT PERIODIC USAGE BEGIN DATE and CURRENT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

NEXT BEGINNING DATE — The starting date (YYMMDD) of the next usage accumulation period.

When the date in the CURRENT BUSINESS DATE field on IDF screen 40 is greater than or equal to the value in this field, the value in this field is moved to the BEGINNING DATE field on this screen. A new value for this field is then computed by BASE24 products and placed in the field.

Note: BASE24 Remote Banking uses the NEXT PERIODIC USAGE BEGIN DATE and NEXT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

PERIOD LENGTH — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

The value in this field is referenced only if the value in the WORK DAY CODE field on this screen is set to 0, indicating that the usage accumulation period length should be specified by this field. If the value in the WORK DAY CODE field is set to 1, 2, or 3, the value in this field must be set to 0.

If the value in this field is 83, 84, 85, or 86, the day in the BEGINNING DATE field must be between 01 and 28. Valid values are as follows:

- 0 = Usage accumulation period is specified by the value in the WORK DAY CODE field.
- 1–79 = Number of days in the usage accumulation period.
- 80 = Usage accumulation period is one week (7 days).
- 81 = Usage accumulation period is two weeks (14 days).
- 82 = Usage accumulation period begins on the first and 15th of each month.
- 83 = Usage accumulation period is one month.
- 84 = Usage accumulation period is three months.
- 85 = Usage accumulation period is six months.
- 86 = Usage accumulation period is one year.

Note: BASE24 Remote Banking uses the PERIODIC USAGE LENGTH and CYCLIC USAGE LENGTH fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 1. A description of the code entered is displayed to the right of the PERIOD LENGTH field.

WORK DAY CODE — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

If codes 1, 2, or 3 are used in this field, the value in the PERIOD LENGTH field on this screen must be set to the value 0. If the value in the PERIOD LENGTH field on this screen is to be used to specify the usage accumulation period length, then the value in this field must be set to 0.

Valid values are as follows:

- 0 = Use the value in the PERIOD LENGTH field; weekends and holidays are not taken into account with this code.
- 1 = Clear usage accumulation fields daily, except for weekends and specified holidays.
- 2 = Clear usage accumulation fields daily, except for Sundays and specified holidays.
- 3 = Clear usage accumulation fields daily, except for Saturdays and specified holidays.

Holidays are specified in the HOLIDAYS fields on this screen.

Note: BASE24 Remote Banking uses the PERIODIC WORK DAY and CYCLIC WORK DAY fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 0. A description of the code entered is displayed to the right of the WORK DAY CODE field.

HOLIDAYS — A maximum of twenty dates (YYMMDD) defining the legitimate holidays for the institution.

The values in these fields are used if the value in any of the fields listed below is set to 1, 2, or 3. Any of these values indicates that usage accumulation fields are not cleared on the dates specified by the HOLIDAYS fields.

- WORK DAY CODE field on IDF screen 4 (this screen)
- PERIODIC WORK DAY field on IDF screen 41
- CYCLIC WORK DAY field on IDF screen 41

Blank fields can not be left between dates; however, unused fields should be left blank.

Example: 961124 961225 970101 970116 970213
 970529 970704 970904 971009 971111
 971123 971225 980101 _____

IDF Screen 40

IDF screen 40 contains BASE24 Remote Banking processing information. The fields on IDF screen 40 that pertain to PIN verification include the following:

PIN VERIFICATION GROUP — The value used by the Integrated Authorization Server process to select the Key Authorization File (KEYA) record when verifying PINs for customers of this financial institution. The value in this field is matched with the value in the GRP field on KEYA screen 1. The default value is the value in the FIID field.

PIN REQUIRED — A code identifying whether a PIN is required with a transaction that is entered at a customer service representative (CSR) terminal. The default value is Y. Valid values are as follows:

Y = Yes, a PIN is required.

N = No, a PIN is not required.

O = CSR terminal operator can override a transaction that is entered without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.

PIN CHANGE ALLOWED — A code identifying whether a PIN change transaction can be performed at a customer service representative (CSR) terminal. The default value is N. Valid values are as follows:

Y = Yes, PIN change transactions are allowed.

N = No, PIN change transactions are not allowed.

Customer PIN Verification Parameters

Customer PIN information is maintained in the Customer Table (CSTT).

CSTT Screen 1

CSTT screen 1 contains authorization parameters and usage accumulation information for BASE24 Remote Banking customers. The fields used on CSTT screen 1 that pertain to PIN verification include the following:

CUSTOMER ID — The unique identification number assigned to this CSTT row. An unlimited number of individual accounts and persons (each identified by a unique personal ID) can be associated with one customer ID number.

A customer ID number can also be known as a primary account number (PAN). Other BASE24 products use the PAN to identify information on a plastic card. If customers have already been issued plastic cards, existing PANs could be conveniently used as customer IDs; however, the PAN used by BASE24 Remote Banking does not have to be associated with a plastic card.

FIID — The FIID of the financial institution for this CSTT row. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution and to secure file record and table row access by financial institution. The value entered in this field should match the value of the FIID in the Institution Definition File (IDF).

PIN VERIFICATION DIGITS — The PIN offset value, PIN verification number (PVN), or the PIN verification value (PVV) used in PIN verification processing. The information in this field depends on the PIN verification parameters specified in the PIN CHECK TYPE and POFST/PVV LOC fields on Institution Definition File (IDF) screen 2. This field is required if the PIN verification digits are to be maintained in the CSTT.

If one zero is entered, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry. There is no default value.

LAST UPDATE — The date and time (YYYYMMDDhhmmss) the PIN VERIFICATION DIGITS field was last updated.

PIN VERIFICATION KEY INDEX — Identifies the location of the PIN verification value (PVV) when the Visa PVV PIN verification method is specified in the PIN CHECK TYPE field on Institution Definition File (IDF) screen 2. Valid values are 1 through 6 for the Visa PVV verification method and blank for other PIN verification methods. There is no default value.

BAD PIN COUNT — The number of times an incorrect PIN has been entered for this CSTT row during the current usage accumulation period.

Each institution defines the maximum number of incorrect PIN tries allowed for the customer in the MAX PIN TRIES field on Institution Definition File (IDF) screen 2. When this number of incorrect PIN tries is exceeded, subsequent transactions for the customer are rejected according to the setting in the MAX PIN TRIES RESET OPTION field on IDF screen 2.

PIN CHANGE REQUIRED — A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. If the institution wants to define a customer before the initial PIN has been selected, this field must be set to the value S. When this field is set to the value S or Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. The default value is N. Valid values are as follows:

Y = Yes, a PIN change is required. The customer must change their current PIN; the only transaction allowed is the PIN change transaction.

N = No, a PIN change is not required; the current PIN can be used.

S = Customer Select. The customer must select their initial PIN; the only transaction allowed is the PIN change transaction.

LAST RESET DATE — The date and time (YYYYMMDDhhmmss) the BAD PIN COUNT field was last cleared. The BAD PIN COUNT field is cleared when the customer performs the first transaction of a business day or enters a correct PIN, based on the value in the PIN TRIES RESET OPTION field on Institution Definition File (IDF) screen 2.

A transaction is considered to be the first one of the day when the date in this field is less than the date in the BEGINNING DATE field on IDF screen 4. After the BAD PIN COUNT field is cleared, the date in this field is replaced with the date from the BEGINNING DATE field on IDF screen 4.

PIN Verification By Authorization Level

There are three authorization levels available through BASE24 Remote Banking:

Level 1 (online)

Level 2 (offline)

Level 3 (online/offline)

Under authorization level 1 (online), transactions must be sent to a host for authorization. Under this authorization level, the Integrated Authorization Server process can be configured to perform PIN verification as a part of its transaction screening actions by setting the PIN field on IDF screen 2 to a value of Y and the PIN CHECK TYPE field on IDF screen 2 to a value in the range of 01 through 04. This means that the Integrated Authorization Server process verifies the customer PIN prior to sending the transaction to the host and declines the transaction if the customer's PIN is incorrect.

Under authorization level 2 (offline), transactions are not authorized by a host, but by BASE24 Remote Banking exclusively. In this case, the value of the PIN field is not used. The PIN CHECK TYPE field determines if and how PINs are verified.

Under authorization level 3 (online/offline), transactions must be sent to a host for authorization but can be subsequently authorized by BASE24 Remote Banking if the host is unavailable. Under this authorization level, BASE24 Remote Banking can be configured to perform PIN verification as a part of its transaction screening actions by setting the PIN field to a value of Y and the PIN CHECK TYPE field to a value in the range of 01 through 04. In this case, the Integrated Authorization Server process verifies the customer's PIN prior to sending the transaction to the host and declines the transaction if the customer's PIN is incorrect. Subsequently, if the transaction is sent to the host and the host is unavailable, the Integrated Authorization Server process stands in to authorize the transaction. BASE24 Remote Banking then automatically verifies the PIN if it did not verify the PIN prior to sending the transaction to the host.

The BASE24 Remote Banking authorization level is set in the AUTH LEVEL field on Institution Routing Configuration File (IRCF) screen 1.

BASE24 Remote Banking Internal Messages

All information required for transaction processing relating to the PIN is contained in internal messages. The BASE24-telebanking Standard Internal Message (BSTM) is exchanged between processes in a BASE24 Remote Banking system. Internal Transaction Data (ITD) fields are used within the Integrated Authorization Server process of a BASE24 Remote Banking system. Transactions are converted between these two formats as needed when processed internally by the BASE24 Remote Banking system. The ITD is enveloped within BSTM messages.

The ITD fields that carry PIN information are PIN data, PIN verification digits, and PIN change required. For a complete discussion of these ITD fields and the BSTM format used by BASE24 Remote Banking, refer to appendix A of the *BASE24 Remote Banking Transaction Processing Manual*.

BASE24 Remote Banking PIN Verification Processing

In BASE24 Remote Banking several processes are called to perform the PIN verification actions during its processing of a transaction. PIN verification actions constitute only a portion of the overall transaction processing performed in processing a transaction.

The steps summarized here are intended to give a general view of PIN verification. For specific information on how each PIN verification method operates, refer to sections 7 through 10.

PIN verification processing is performed while the transaction is in the control of the Integrated Authorization Server process.

Prior to verifying the PIN, the Integrated Authorization Server process performs the following actions:

- Verifies that the transaction is allowed for the specified customer using information in the customer extended memory table, verifies that the customer is defined in the Customer Table (CSTT), and identifies the account numbers involved in the transaction by reading the Customer/Account Relation Table (CACT).
- Retrieves the PIN verification digits, PVK index, PIN change required flag, and bad PIN count from the customer extended memory table.
- Locates the appropriate entry in the Institution Definition File extended memory table for this transaction and includes a response that includes the values from the BAD PIN LIMIT field, BAD PIN TRIES RESET OPTION field, and other processing information. It also provides the prescreening information and Customer Service Interface information from the IDF.

During PIN verification processing, the Integrated Authorization Server process performs the following:

1. Determines the FIID, PIN offset, bad PIN count, PIN verify flag, PIN change flag, and PIN verification key index.
2. Retrieves the PIN information from the ITD.
3. Formats a security information request that includes the FIID associated with the customer.
4. Retrieves the security information from the ITD.

5. Extracts the specified fields from the ITD.
6. Locates the appropriate entry in the Institution Definition File extended memory table and formats a response that includes the PIN verification group, bad PIN limit, bad PIN reset option, and PIN offset location.
7. Retrieves the required PIN verification data from the Key Authorization File extended memory table.
8. Verifies the PIN by performing the following steps:
 - a. Determines the bad PIN limit.
 - b. Determines the bad PIN count.
 - c. Determines the bad PIN reset options.
 - d. Verifies the PIN in software or sends a request to a security module to verify the PIN. The PIN is either accepted as valid or rejected.
 - e. Updates the PIN information in the ITD accordingly.

The transaction then continues processing within the Integrated Authorization Server process according to the institution parameters for that type of transaction.

Local PIN Verification

If the remote banking endpoint device has the functional capability, the customer PIN can be verified by the remote banking endpoint device prior to entering the BASE24 Remote Banking system. The remote banking endpoint device does not place the PIN in the request message when the PIN is verified locally.

In BASE24 Remote Banking, VRU Configuration Data (VCD) screens are used to configure the Remote Banking Standard Interface and the Customer Service Interface. These interfaces are responsible for communicating with remote banking endpoint devices, endpoint devices managed by the Visa Interactive Gateway Server, and customer service representative (CSR) terminals, respectively. An interactive voice response (IVR) system represents just one type of remote banking endpoint device. A remote banking endpoint device could also be a screen phone or a personal computer.

The value in the LOCAL PIN VERIFICATION field on VCD screen 18 identifies whether a remote banking endpoint device can perform local PIN verification. Valid values are as follows:

Y = Yes, local PIN verification is supported.

N = No, local PIN verification is not supported.

If the LOCAL PIN VERIFICATION field is set to the value Y, then BASE24 Remote Banking uses the ITD point of service (ITD-PT-SVC) field in the BSTM to indicate that the PIN may be verified by the remote banking endpoint device. The information in this field of the internal message is based on the 1993 ISO 8583 standard, ***Financial Transaction Card Originated Messages - Interchanges Message Specification standard***.

The ITD-PT-SVC field contains a series of codes from subfields intended to identify terminal capability, terminal environment, and presentation of security data. It is used to indicate specific conditions that are present at the time a transaction takes place at the point of service or when the transaction is initiated.

For transactions being verified by the remote banking endpoint device, the PIN-CAPTR-CAP (PIN capture capability) subfield in the ITD-PT-SVC field indicates the length of PIN that the acquirer endpoint is capable of capturing. A value of S indicates the PIN has been verified. This subfield can be set by either the remote banking device or the Integrated Authorization Server process.

In the case of a remote banking endpoint device verifying the transaction PIN locally, the PIN-CAPTR-CAP subfield would be set to the value of S and no further PIN verification would be performed for that transaction.

PIN Change and PIN Select Options

BASE24 Remote Banking gives institutions the ability to allow their customers to select or change their PINs. The PIN select option allows customers to select their first PINs. The PIN change option allows customers to change their existing PINs.

The PIN select option differs from the PIN change option in the following ways:

- PIN selection is one step in a BASE24 Remote Banking transaction while PIN change is a stand-alone BASE24 Remote Banking transaction.
- PIN selection is part of the first customer transaction only, while the PIN change option is a transaction that a customer can perform at any time after the account has been opened.

The PIN select and PIN change options also have several similarities:

- Both options can be used on the same customer ID.
- Both options allow PIN processing to be performed in software or in a security module.
- Neither option can be performed locally at a remote banking endpoint device because with local PIN processing, the PIN verification digits are not transmitted with the transaction to the BASE24 Remote Banking system.

PIN Change Configuration Requirements

The PIN change option allows a customer to change the existing PINs. When this option is used, BASE24 Remote Banking uses the existing PIN to verify the customer and then allows the customer to enter a new PIN twice, computes the corresponding PIN verification digits, and places the PIN verification digits in the Customer Table. From that point on, BASE24 Remote Banking will recognize the new PIN entered by the customer as the customer's valid PIN.

The following BASE24 Remote Banking configuration requirements apply when the PIN change transaction is offered:

- If PIN verification is performed in software, the PIN verification method must be IBM 3624-format, Diebold, or Visa PVV.
- If PIN verification is performed using Atalla Network Security Processor (NSP) security modules, all four PIN verification methods are available.
- If PIN verification is performed using Racal-Guardata security modules, all PIN verification methods except Identikey are available.
- The Atalla Security Module (ASM) does not support PIN change transactions.
- PIN verification digits must be carried in the Customer Table (CSTT).

The following field located on CSTT screen 1 must contain specific values when the PIN change transaction is offered.

PIN CHANGE REQUIRED — A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. If the institution wants to define a customer before the initial PIN has been selected, this field must be set to the value S. When this field is set to the value S or Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. The default value is N. Valid values are as follows:

Y = Yes, a PIN change is required. The customer must change their current PIN; the only transaction allowed is the PIN change transaction.

N = No, a PIN change is not required; the current PIN can be used.

S = Customer Select. The customer must select their initial PIN; the only transaction allowed is the PIN change transaction.

PIN Change Processing

A PIN change transaction changes the PIN that is assigned to a customer ID. A customer starts a PIN change transaction by correctly entering the current PIN. This establishes that the customer has been granted security access to use the customer ID. Next, the customer enters the new PIN and the PIN is changed. At some remote banking devices, the customer must enter the new PIN twice to verify that it was typed in accurately. The customer must then use the new PIN to initiate future transactions for that customer ID.

A PIN change transaction is submitted in one of two ways, either by the customer at a remote banking endpoint device or by a customer service representative at a customer service representative terminal on behalf of a customer.

The PIN change transaction processing is the same whether initiated because the customer chooses to change their PIN or because the customer's financial institution forces the customer to change their PIN.

When a customer selects their initial PIN, the processing is similar to that of other PIN change transactions except that the BASE24 Remote Banking system does not expect a current PIN to be included in the transaction request.

The steps for PIN change processing are the same as previously described for PIN verification processing, with the following additional processing.

1. As part of the transaction PIN analysis as described in step 8 for PIN verification processing, the Integrated Authorization Server process checks the PIN-CHNG-REQ field in the ITD, to determine if this transaction is a PIN change request. This flag is set to a value of Y when the PIN CHANGE REQUIRED field on CSTT screen 1 contains a value of Y or S, indicating that a PIN change or PIN select transaction is required, respectively.
2. If the transaction is initiated by a customer service representative at a customer service representative terminal, the following requirements must be met:
 - If the PIN REQUIRED field on IDF screen 40 contains the value Y, the transaction must be entered with a PIN.
 - If the PIN REQUIRED field on IDF screen 40 contains the value O, the transaction can be entered with or without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.

- The PIN CHANGE ALLOWED field on IDF screen 40 must contain the value Y, indicating that PIN change transactions are allowed from CSR terminals.
3. The Integrated Authorization Server process completes the following:
 - Validates the old PIN.
 - Determines that this is a PIN change transaction.
 - Determines the new PIN entered either by the customer or the customer service operator.
 - Performs a comparison between the NEW PIN 1 and NEW PIN 2 fields if the new PIN is entered twice for verification purposes. This is required by some remote banking endpoint devices. Other devices can perform this verification before sending the transaction to the BASE24 Remote Banking system, in which case BASE24 Remote Banking assumes that the NEW PIN 1 field contains the accurate new PIN.
 - Sends a PIN change request to a hardware security module or performs it in software.
 - Obtains new PIN verification digits (PVD).
 4. The Integrated Authorization Server process formats a response to the PIN verification request containing the new PVD.
 5. The Integrated Authorization Server process updates the Customer Table (CSTT) with the new PVD.

The transaction then continues processing within the Integrated Authorization Server process according to the institution parameters for that type of transaction.

PINs at Customer Service Representative Terminals

BASE24 Remote Banking allows an institution to determine whether a valid PIN is required when a customer service representative enters a transaction on behalf of a customer. It also permits an institution to restrict its customer service representatives from changing a PIN when the customer is not present. The following fields located on IDF screen 40 contain parameters for PINs with transactions performed at a customer service representative terminal.

For more information on setting up the database and IDF screen 40, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN REQUIRED — A code identifying whether a PIN is required with a transaction that is entered at a customer service representative (CSR) terminal. The default value is Y. Valid values are as follows:

Y = Yes, a PIN is required.

N = No, a PIN is not required.

O = CSR terminal operator can override a transaction that is entered without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.

PIN CHANGE ALLOWED — A code identifying whether a PIN change transaction can be performed at a customer service representative (CSR) terminal. The default value is N. Valid values are as follows:

Y = Yes, PIN change transactions are allowed.

N = No, PIN change transactions are not allowed.

Applied Communications, Inc.

Section 7

IBM 3624-Format PIN Verification Support

This section describes Integrated Authorization Server process support of the IBM 3624-format PIN verification method, including how the BASE24 database must be set up to allow the Integrated Authorization Server process to verify PINs under this method of verification.

The Integrated Authorization Server process allows for implementation of this PIN verification method in software or in a security module.

IBM 3624-Format PIN Verification

IBM 3624-format PIN verification is based on the use of a PIN verification key, a decimalization table, and a PIN offset. IBM 3624-format PIN verification can be performed in software or using a hardware module linked to the processor on which the BASE24 Remote Banking system is running.

At the time a customer relationship is established, the institution generates a unique PIN and PIN offset combination for each customer, using validation data taken from the customer ID. If the PIN is institution-assigned or chosen by the customer, the PIN offset is generated to correspond to the PIN; if the PIN offset is a set value, the PIN is generated to correspond to the PIN offset. The PIN offset is then placed in the BASE24 database.

The following is a summary of the steps taken to verify a PIN using the IBM 3624-format PIN verification method in software or hardware. If a security module is being used, however, it must be given additional information to decrypt the PIN block and extract a clear PIN for verification and also to decrypt the PIN verification key.

The verifying entity performs the following steps:

1. Creates a validation data item using the customer ID.
2. Encrypts the validation data using the PIN verification key.
3. Decimalizes the encrypted validation data using the decimalization table.
4. Creates an intermediate PIN by computing the length of the customer's PIN offset and taking that same number of characters from the left of the decimalized validation data.
5. Adds the customer's PIN offset (using the modulo 10 technique) to the intermediate PIN to derive a PIN check number.
6. Compares the PIN check number to the PIN entered by the customer. If they match, the customer entered the PIN correctly. If they do not match, the customer entered the PIN incorrectly.

PIN Verification Inputs

IBM 3624-format PIN verification requires the following information as input. Some of these items vary from customer to customer, while others are constant for an institution.

- Validation data
- PIN verification key
- Decimalization table
- PIN offset
- PIN

These input items are described separately within this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Validation Data

Validation data, as applied to IBM 3624-format PIN verification, is a 16-digit string of data unique to the customer. This data is used in the IBM 3624-format PIN verification routines for generating a PIN.

The basis for creating validation data is the customer ID, also known as the primary account number (PAN), associated with the customer. The PAN equates to the customer ID stored in the BASE24 database.

To derive validation data for use with IBM 3624-format PIN verification, three values are required in addition to the PAN: the PAN offset, the PAN length and a PAN pad character.

The PAN offset value and length are used to extract a string of characters from the customer ID. The PAN pad character is then used to pad that string of characters out to the 16 digits required for the validation data.

To illustrate, if the PAN verification offset is 3 and the PAN verification length is 10, the Integrated Authorization Server process would skip to position 4 of the customer ID and, starting with that position, pick up 10 consecutive characters of the account number. Then, the Integrated Authorization Server process would add six of the PAN pad characters to the right of the account digits to expand the number out to 16 positions. The PAN must be at least as long as the PAN verification offset plus the PAN verification length.

An example of this follows, using a PAN pad character of F.

CUSTOMER ID:	5	0	1	2	7	2	1	3	6	7	2	5	3	3	1	1
OFFSET:	1	2	3													
				↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			
CUSTOMER ID:				2	7	2	1	3	6	7	2	5	3			
VALIDATION DATA:				2	7	2	1	3	6	7	2	5	3	F	F	F

To perform the above operation, the Integrated Authorization Server process requires that the PAN verification offset, PAN verification length, and PAN pad character be specified in the Key Authorization File (KEYA). These values are specified in the PAN VERIFY OFFSET, PAN VERIFY LENGTH, and PAN PAD CHARACTER fields on KEYA screen 2, respectively.

Note: If the value in the PAN VERIFY LENGTH field is set to 16, no pad characters are used. The 16 digits are taken from the customer ID and used as the validation data.

PIN Verification Key

IBM 3624-format PIN verification requires that the validation data be encrypted for processing. The PIN verification key is the key used to encrypt that validation data. In this case, encryption is used to generate a different sequence of digits and to tie that string of digits to a secret value, the key.

There are two versions of the PIN verification key that can be required, depending on whether the verification is being done exclusively in software by the Integrated Authorization Server process or by a hardware module.

In the case of verification performed by the Integrated Authorization Server process, a clear key is required for the process to encrypt the validation data.

In the case of verification performed by a hardware module, an encrypted key (encrypted by or for the hardware module) is required that can be passed to the module for encrypting the validation data.

If some verification is done in software and some in hardware, both a clear PIN verification key and an encrypted PIN verification key are required.

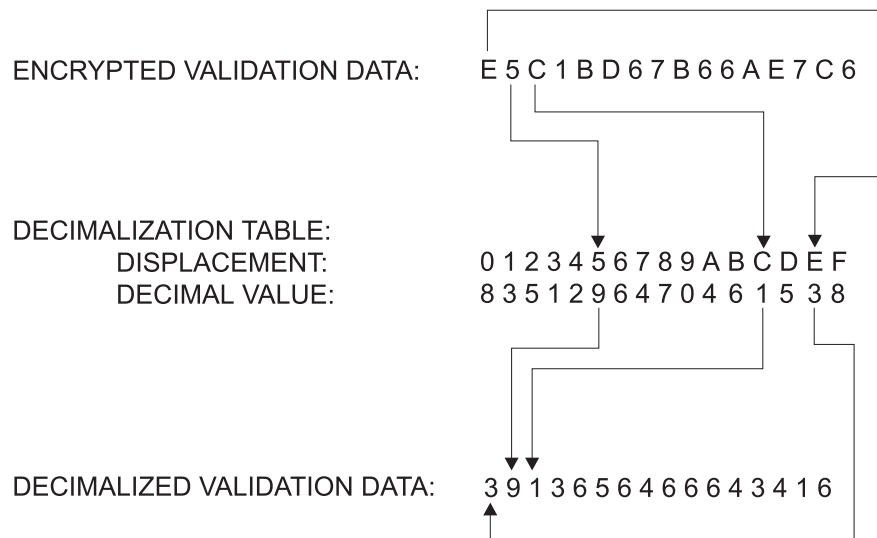
The clear PIN verification key must be specified in the CLEAR KEY field on KEYA screen 2. The encrypted PIN verification key must be specified in the ENCRYPTED KEY field on KEYA screen 2.

Decimalization Table

Because encryption of the validation data used in IBM 3624-format PIN verification yields hexadecimal characters, the encrypted version of the validation data must be decimalized prior to further processing. When data is decimalized, all of the hexadecimal characters are converted to decimal characters.

The technique used to decimalize the encrypted validation data requires a decimalization table, which is a random series of 16 digits selected by the BASE24 system administrator. These digits must be in the range of 0 through 9.

To decimalize encrypted validation data, each character in the data is mapped to its equivalent position in the table, and the character in that position of the table is then substituted for the one in the validation data. The 16 table positions are numbered 0 through 15. An example is shown below:



In the example shown, the 5 in the validation data is replaced with a 9 because the sixth position of the decimalization table contains a 9. Likewise, each of the Cs in the validation data (the hexadecimal equivalent for the decimal number 12) is replaced with a 1 because the thirteenth position of the decimalization table contains a 1. Each position in the validation data is converted to a decimal value in this manner.

The decimalization table to be used by the Integrated Authorization Server process must be placed in the DECIMALIZATION TABLE field on KEYA screen 2. There are no requirements for the decimalization table except that all 16 characters be numeric (0 through 9).

PIN Offset

Once the validation data has been decimalized, the left-most digits are taken to form an intermediate PIN. The number of digits to take is determined by the length of the customer's PIN offset. The PIN offset is located in the Customer Table (CSTT). An example is shown below:

DECIMALIZED VALIDATION DATA: 3 9 1 3 6 5 6 4 6 6 6 4 3 4 1 6

PIN OFFSET: 8 7 6 5 (Length of 4)

INTERMEDIATE PIN: 3 9 1 3

The PIN offset then is the value which when added (using the modulo 10 technique) to the intermediate PIN yields the PIN check number to be matched against the PIN entered by the customer. Modulo 10 addition is a technique for adding numbers without carryover. For example, under this technique, 8 plus 4 would be 2, rather than 12, because the carried 1 is disregarded. Similarly, 8888 plus 4444 would be 2222 rather than 13332 because all of the carries are disregarded. An example of this is shown below:

INTERMEDIATE PIN: 3 9 1 3

PIN OFFSET (added Mod-10): 8 7 6 5

PIN CHECK NUMBER: 1 6 7 8

The Integrated Authorization Server process allows institutions to specify how PIN offsets are to be obtained for verification using the PIN OFFSET LOC field on Institution Definition File (IDF) screen 2. The following options are available:

- A standard PIN offset value of 0000 can be used for all verification actions.
- The PIN offset can be taken from the Customer Table (CSTT). In this case, the PIN offset can be up to 12 digits in length.

PIN

In IBM 3624-format PIN verification, the PIN entered by the customer is compared to the PIN check number computed by the verification routines.

If PINs are verified by software, they must be in the clear when it comes time to verify them.

If PINs are verified by a hardware module, they must be passed to the module in encrypted form, along with the key used to encrypt them. The module then decrypts the PIN for comparing to the PIN check number it computes.

BASE24 Remote Banking allows customers to change their existing PINs. The PIN offset must already exist in the CSTT before this transaction is performed. The PIN change transaction can be performed with IBM 3624-format PIN verification in hardware or software; however, only Atalla Network Security Processors (NSPs) and Racal-Guardata Host Security Modules support this transaction. The Atalla Security Module (ASM) does not support the PIN change transaction. Upon receiving a PIN change transaction, the Integrated Authorization Server process or the security module computes an appropriate PIN offset based on the PIN entered by the customer and uses it to replace the existing offset in the CSTT. From that point on, the new PIN will be the only one allowed.

PINs can be from four to 12 digits in length. However, the length of the PIN offset is the minimum required length of the PIN. For example, if the PIN offset is four digits, the PIN must be at least four digits as well. The PIN can be longer than the PIN offset; however, if the PIN is longer than the PIN offset, only the right-most digits of the PIN will be used in the verification.

Setting Up IBM 3624-Format PIN Verification

To support IBM 3624-format PIN verification, the Integrated Authorization Server process requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to structure its customer IDs, how it wishes verification to be handled, and whether or not it plans to use a hardware module for performing verification.

The remainder of this section discusses the Integrated Authorization Server process-related requirements for IBM 3624-format PIN verification. The discussion is divided into two parts:

- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Hardware Module Requirements

IBM 3624-format PIN verification can be performed by the following hardware modules supported by the Integrated Authorization Server process:

- Atalla Security Module (ASM)
- Atalla Network Security Processor (NSP)
- Racal-Guardata Host Security Module (RSM)

For instructions on configuring these devices, refer to section 13, appendix E and appendix F.

Note: IBM 3624-format PIN verification using an RSM requires that the validation data include the five right-most positions of the PAN, excluding the check digit. Therefore, the PAN VERIFY OFFSET and PAN VERIFY LENGTH fields in the KEYA must be set appropriately to capture this portion of the PAN.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the IBM 3624-format PIN verification method. Appendix A contains illustrations of IDF screens 2, 4, and 40.

For more information on IDF screens 2, 4, and 40, refer to the ***BASE24 Base Files Maintenance Manual***.

IDF Screen 2

IDF screen 2 enables an institution to set parameters for its customers. The following fields must be completed when using the IBM 3624-format PIN verification method.

PIN — Specifies whether the customer's entered PIN is checked during transaction screening. The Integrated Authorization Server process uses this field with authorization level 1 (online) and authorization level 3 (online/offline) only, since prescreening checks are made only for these authorization levels. The default value is N. Valid values are as follows:

Y = Yes, check the customer's PIN; if invalid, decline the request and do not send the request to the host.

N = No, do not check the customer's PIN before sending the request to the host if the host is online.

MAX PIN TRIES — The code indicates the number of times that a customer can enter an incorrect PIN. The Integrated Authorization Server process uses the Customer Table (CSTT) to accumulate this information for a customer regardless of authorization method. The default value is 1.

POFST/PVV LOC — A code specifying the location of the PIN offset. Valid values are as follows:

0 = No PIN offset. A value of 0000 is used if an offset is required for the verification method.

1 = PIN offset is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the offset.

2 = PIN offset is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the offset.

A description of the code entered is displayed to the right of the POFST/PVV LOC field. The default value is 0.

PIN CHECK TYPE — A code indicating the PIN verification method used by the institution. The valid value for IBM 3624-format PIN verification is 01.

A description of the code entered is displayed to the right of the PIN CHECK TYPE field. The default value is 00 (no verification).

ALGO NUMBER LOC — A code specifying the location of the algorithm number. This field should contain a value of 0 for the IBM 3624-format PIN verification method.

A description of the code entered is displayed to the right of the ALGO NUMBER LOC field. The default value is 0.

PIN TRIES RESET OPTION — A code indicating how the BAD PIN COUNT field on CSTT screen 1 is to be reset for the customers of an institution.

The Integrated Authorization Server process keeps track of the number of bad PIN tries for a customer according to the authorization parameters established by an institution. This allows bad PIN tries to be accumulated over a period of time, and institutions can then choose to decline authorization of a transaction for a customer if that customer has had an excessive number of incorrect PIN tries. The PIN tries limit is located in the MAX PIN TRIES field on IDF screen 2 (this screen).

BASE24 products offer two methods for clearing accumulated bad PIN tries. The first allows for clearing the accumulated bad PIN tries when the customer enters a correct PIN. The second method allows for clearing the accumulated bad PIN tries at the end of each usage accumulation period. This field controls how these clearance methods are to be applied for an institution's customers.

The bad PIN tries in the CSTT are not automatically cleared at the end of each usage accumulation period. Institutions using the CSTT can choose—using this field—to have their CSTT bad PIN tries automatically cleared with the rest of their totals at the end of each usage accumulation period, when a correct PIN is entered, or both. The bad PIN tries in the CSTT can also be reset by refreshing the file.

The Integrated Authorization Server process uses customer processing dates in the product-specific segments of the IDF to track usage accumulation periods. The default value for this field is 0. Valid values are as follows:

0 = Reset the bad PIN tries at the end of the usage accumulation period, but not when a correct PIN is entered.

- 1 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.
- 3 = Reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 4 = Reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.

IDF Screen 4

IDF screen 4 enables an institution to specify its withdrawal period parameters. BASE24 Remote Banking uses parameters on IDF screen 4 for PIN tries only. Descriptions of the fields on IDF screen 4 that pertain to PIN verification are as follows:

WITHDRAWAL PERIOD PARAMETERS

The values in the following fields specify the withdrawal period. The Integrated Authorization Server process uses these parameters for PIN tries only. The process maintains customer PIN tries in the Customer Table (CSTT).

BEGINNING DATE — The starting date (YYMMDD) of the current usage accumulation period for all customers belonging to this institution.

This date is placed in the LAST RESET DATE field on CSTT screen 1.

BASE24 Remote Banking updates this field automatically at the beginning of a new withdrawal period. See the description of the NEXT BEGINNING DATE field on this screen for more information.

If the value in the PERIOD LENGTH field on this screen is 83, 84, 85, or 86, the day in this field must be between 01 and 28.

Note: BASE24 Remote Banking uses the CURRENT PERIODIC USAGE BEGIN DATE and CURRENT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

NEXT BEGINNING DATE — The starting date (YYMMDD) of the next usage accumulation period.

When the date in the CURRENT BUSINESS DATE field on IDF screen 40 is greater than or equal to the value in this field, the value in this field is moved to the BEGINNING DATE field on this screen. A new value for this field is then computed by BASE24 Remote Banking and placed in the field.

Note: BASE24 Remote Banking uses the NEXT PERIODIC USAGE BEGIN DATE and NEXT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

PERIOD LENGTH — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

The value in this field is referenced only if the value in the WORK DAY CODE field on this screen is set to 0, indicating that the usage accumulation period length should be specified by this field. If the value in the WORK DAY CODE field is set to 1, 2, or 3, the value in this field must be set to 0.

If the value in this field is 83, 84, 85, or 86, the day in the BEGINNING DATE field must be between 01 and 28. Valid values are as follows:

- 0 = Usage accumulation period is specified by the value in the WORK DAY CODE field.
- 1–79 = Number of days in the usage accumulation period.
- 80 = Usage accumulation period is one week (7 days).
- 81 = Usage accumulation period is two weeks (14 days).
- 82 = Usage accumulation period begins on the first and 15th of each month.
- 83 = Usage accumulation period is one month.
- 84 = Usage accumulation period is three months.
- 85 = Usage accumulation period is six months.
- 86 = Usage accumulation period is one year.

Note: BASE24 Remote Banking uses the PERIODIC USAGE LENGTH and CYCLIC USAGE LENGTH fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 1. A description of the code entered is displayed to the right of the PERIOD LENGTH field.

WORK DAY CODE — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

If codes 1, 2, or 3 are used in this field, the value in the PERIOD LENGTH field on this screen must be set to the value 0. If the value in the PERIOD LENGTH field on this screen is to be used to specify the usage accumulation period length, then the value in this field must be set to 0. Valid values are as follows:

- 0 = Use the value in the PERIOD LENGTH field; weekends and holidays are not taken into account with this code.
- 1 = Clear usage accumulation fields daily, except for weekends and specified holidays.
- 2 = Clear usage accumulation fields daily, except for Sundays and specified holidays.
- 3 = Clear usage accumulation fields daily, except for Saturdays and specified holidays.

Holidays are specified in the HOLIDAYS fields on this screen.

Note: BASE24 Remote Banking uses the PERIODIC WORK DAY and CYCLIC WORK DAY fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 0. A description of the code entered is displayed to the right of the WORK DAY CODE field.

HOLIDAYS — A maximum of twenty dates (YYMMDD) defining the legitimate holidays for the institution.

The values in these fields are used if the value in any of the fields listed below is set to 1, 2, or 3. Any of these values indicates that usage accumulation fields are not cleared on the dates specified by the HOLIDAYS fields.

- WORK DAY CODE field on IDF screen 4 (this screen)
- PERIODIC WORK DAY field on IDF screen 41
- CYCLIC WORK DAY field on IDF screen 41

Blank fields can not be left between dates; however, unused fields should be left blank.

Example: 961124 961225 970101 970116 970213
 970529 970704 970904 971009 971111
 971123 971225 980101 _____

IDF Screen 40

IDF screen 40 contains BASE24 Remote Banking processing information. The fields used on IDF screen 40 for the IBM 3624-format PIN verification method include the following:

TELEBANKING IDF INFORMATION

The following field contains BASE24 Remote Banking processing information.

PIN VERIFICATION GROUP — The value used by the Integrated Authorization Server process to select the Key Authorization File (KEYA) record when verifying PINs for customers of this financial institution. The value in this field is matched with the value in the GRP field on KEYA screen 1. The default value is the value in the FIID field.

CUSTOMER SERVICE INTERFACE CONTROL PARAMETERS

The following fields contain parameters used to process transactions initiated at customer service representative (CSR) terminals.

PIN REQUIRED — A code identifying whether a PIN is required with a transaction that is entered at a customer service representative (CSR) terminal. The default value is Y. Valid values are as follows:

Y = Yes, a PIN is required.

N = No, a PIN is not required.

O = CSR terminal operator can override a transaction that is entered without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.

PIN CHANGE ALLOWED — A code identifying whether a PIN change transaction can be performed at a customer service representative (CSR) terminal. The default value is N. Valid values are as follows:

Y = Yes, PIN change transactions are allowed.

N = No, PIN change transactions are not allowed.

Customer Table (CSTT) Settings

Customer Table (CSTT) screen 1 contains authorization parameters and usage accumulation information for BASE24 Remote Banking customers. The fields used on CSTT screen 1 include the following.

For more information regarding setting up CSTT screen 1, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

CUSTOMER ID — The unique identification number assigned to this CSTT row. An unlimited number of individual accounts and persons (each identified by a unique personal ID) can be associated with one customer ID number.

A customer ID number can also be known as a primary account number (PAN). Other BASE24 products use the PAN to identify information on a plastic card. If customers have already been issued plastic cards, existing PANs could be conveniently used as customer IDs; however, the PAN used by BASE24 Remote Banking does not have to be associated with a plastic card.

The customer ID is used as input to create validation data for the IBM 3624-format PIN verification method.

PIN VERIFICATION DIGITS — The PIN offset value used in PIN verification processing. The information in this field depends on the PIN verification parameters specified in the PIN CHECK TYPE and POFST/PVV LOC fields on Institution Definition File (IDF) screen 2. This field is required if the PIN verification digits are to be maintained in the CSTT.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

If one zero is entered, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry. There is no default value.

LAST UPDATE — The date and time (YYYYMMDDhhmmss) the PIN VERIFICATION DIGITS field was last updated.

PIN VERIFICATION KEY INDEX — This field is not used for the IBM 3624-format PIN verification method. There is no default value.

BAD PIN COUNT — The number of times an incorrect PIN has been entered for this CSTT row during the current usage accumulation period.

Each institution defines the maximum number of incorrect PIN tries allowed for the customer in the MAX PIN TRIES field on Institution Definition File (IDF) screen 2. When this number of incorrect PIN tries is exceeded, subsequent transactions for the customer are rejected according to the setting in the PIN TRIES RESET OPTION field on IDF screen 2.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN CHANGE REQUIRED — A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. If the institution wants to define a customer before the initial PIN has been selected, this field must be set to the value S. When this field is set to the value S or Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. The default value is N. Valid values are as follows:

Y = Yes, a PIN change is required. The customer must change their current PIN; the only transaction allowed is the PIN change transaction.

N = No, a PIN change is not required; the current PIN can be used.

S = Customer select. The customer must select their initial PIN; the only transaction allowed is the PIN change transaction.

LAST RESET DATE — The date and time (YYYYMMDDhhmmss) the BAD PIN COUNT field was last cleared. The BAD PIN COUNT field is cleared when the customer performs the first transaction of a business day or enters a correct PIN, based on the value in the PIN TRIES RESET OPTION field on Institution Definition File (IDF) screen 2.

A transaction is considered to be the first one of the day when the date in this field is less than the date in the BEGINNING DATE field on IDF screen 4. After the BAD PIN COUNT field is cleared, the date in this field is replaced with the date from the BEGINNING DATE field on IDF screen 4.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID group using the IBM 3624-format PIN verification method. Appendix A contains illustrations of all KEYA screens.

For more information on KEYA screens, refer to the ***BASE24 Base Files Maintenance Manual***.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the IBM 3624-format PIN verification method.

GRP — The code used in the IDF to identify this record. If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with the institution's FIID in this field.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. When using the IBM 3624-format PIN verification method, the valid value for this field is 01.

KEYA Screen 2

KEYA screen 2 contains IBM 3624-format PIN verification information. The following fields must be completed when using the IBM 3624-format PIN verification method.

DECIMALIZATION TABLE — The decimalization table to be used by the Integrated Authorization Server process. This table must be 16 characters in length. Valid values for each character in this field are 0 through 9. For protection in the database, this table is scrambled for storage in the file by the KEYA file server process.

PAN VERIFY OFFSET — The offset to the starting position in the PAN for extracting validation data. Position 00 identifies the first digit in the PAN.

CLEAR KEY — The clear PIN verification key used when PIN verification is performed without a security module.

- If verification is, at least some of the time, to be performed without a security module, this field must contain a clear PIN verification key. This key must be 16 hexadecimal characters. Valid values for each character in this field are 0 through 9 and A through F. For protection in the database, this key is scrambled for storage by the KEYA file server process.
- If PIN verification is to be performed by a hardware module exclusively, this field is not used.

PAN VERIFY LENGTH — The length of the data (number of digits) to be extracted from the PAN for the formation of validation data. Valid values are 04 through 16.

ENCRYPTED KEY — The encrypted PIN verification key used when PIN verification is performed by a hardware security module.

- If verification is, at least some of the time, to be performed by a hardware module, this field must contain an encrypted PIN verification key. The PIN verification key must be encrypted by or for the specific security module being used, such that the module is able to decrypt the key when it is provided by the Integrated Authorization Server process. The encrypted key must be 16 characters in length. Valid values for each character in this field are 0 through 9 and A through F.
- If PIN verification is to be performed by the Integrated Authorization Server process exclusively, this field is not used.

PAN PAD CHARACTER — The character to use for padding the validation data out to 16 digits. This can be 0 through 9 or A through F.

CHECK DIGITS — The check digits associated with the encrypted PIN verification key used when PIN verification is performed by a hardware security module. Valid values for each character in this field are 0 through 9 and A through F. A value of 0000 in this field indicates check digits are not being used with PIN verification keys.

Applied Communications, Inc.

Section 8

Diebold PIN Verification Support

This section describes the Integrated Authorization Server process support of the Diebold PIN verification method, including how the BASE24 database must be set up to allow the Integrated Authorization Server process to verify PINs under this method of verification.

The Integrated Authorization Server process allows for implementation of this PIN verification method in software or a security module.

Diebold PIN Verification

Diebold PIN verification is based on the use of a Diebold number table, an algorithm number, and a PIN offset to verify a customer's PIN. Diebold PIN verification can be performed in software or using a hardware module linked to a processor on which the BASE24 Remote Banking system is running; however, an institution cannot perform this verification method in both software and hardware. Each institution using this verification method must determine whether to use software or hardware.

At the time the customer relationship is established, the institution generates a PIN offset for each customer ID, using the Diebold number table, an algorithm number, and other account-specific data. The PIN offset is then placed in the BASE24 Remote Banking database.

The following is a summary of the steps taken to verify a PIN using Diebold PIN verification in software or hardware.

The verifying entity performs the following steps:

1. Processes each digit of the customer ID against the Diebold number table, using the selected algorithm number as an index into the table.
2. Processes the right-most four digits against the Diebold number table a second time to derive an intermediate PIN.
3. Subtracts the customer's PIN offset from the intermediate PIN (using the modulo 10 technique).
4. Compares the result from step 3 with the PIN entered by the customer. If they match, the customer entered the PIN correctly. If they do not match, the customer entered the PIN incorrectly.

PIN Verification Inputs

Diebold PIN verification requires the following information as input. Some of these items vary from customer to customer, while others are constant for the institution.

- Diebold number table (DNT)
- Customer ID
- Algorithm number
- PIN offset
- PIN

These input items are described separately within this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Diebold Number Table

A Diebold number table (DNT) is a random set of 256 nonrepeating values used in Diebold PIN verification.

The Integrated Authorization Server process requires access to the institution's Diebold number table in order to perform PIN verification in software. The table must be placed in the Key Authorization File (KEYA) using KEYA screen 3.

Each value in the Diebold number table consists of a pair of hexadecimal characters (for example, AA, A1, 9F, etc.). There are 256 possible combinations of hexadecimal pairs, and these combinations can be scrambled to derive a unique set of table entries, subject to the following rules:

- There must be 256 values in the table.
- No hexadecimal pair can be repeated.
- The first character of the first hexadecimal pair of the table must be a C. Thus, the first hexadecimal pair can be any one of the 16 combinations from C0 to CF.

Note: This table functions in much the same way as a DES key; therefore, it must be kept confidential.

Customer ID

The customer ID, also known as the primary account number (PAN), is used to identify each customer relationship. It must be carried in the BASE24 Remote Banking database and is used as input to Diebold PIN verification.

Under Diebold PIN verification, each digit of the PAN is processed individually against the Diebold number table (regardless of the length of the PAN), starting from the right and moving left.

Each digit is processed against the table seven times. The right-most four digits of the PAN are then processed against the table again to derive an intermediate PIN.

Algorithm Number

An algorithm number is the two-digit number used as a starting index in the Diebold number table for calculations on the PAN. An algorithm number can be any value from 00 to 99. Like the Diebold number table, the algorithm number can be chosen randomly and it must be kept confidential. The confidentiality of the DNT and algorithm number enforce the security of the processing.

Institutions must specify how algorithm numbers are to be handled by the Integrated Authorization Server process using the ALGO NUMBER LOC field on Institution Definition File (IDF) screen 2. For Diebold PIN verification the algorithm number must be taken from the KEYA. In this case, a single algorithm number is used for all verification actions.

PIN Offset

A PIN offset is subtracted (using the modulo 10 technique) from the intermediate PIN to determine a computed PIN. The computed PIN is then compared to the PIN entered by the customer. Modulo 10 subtraction is a technique for subtracting numbers without borrowing. For example, under this technique, 2 minus 8 would be 4: 8 cannot be subtracted from 2; therefore, the 2 would automatically be assumed to have a value of 12. Similarly, 3333 minus 8888 would be 5555, because each number is treated separately. An example of this is shown below:

INTERMEDIATE PIN:	2 9 9 2
PIN OFFSET (subtracted Mod-10):	8 7 6 5
COMPARISON NUMBER:	4 2 3 7

BASE24 Remote Banking allows institutions to specify how PIN offsets are to be obtained for verification. Under Diebold PIN verification, PIN offsets must be four digits in length. The following options are available:

- A standard PIN offset value of 0000 can be used for all verification actions.
- The PIN offset can be taken from the PIN VERIFICATION DIGITS field on Customer Table (CSTT) screen 1. In this case, the PIN offset can be up to 12 digits in length.

PIN

In Diebold PIN verification, the PIN entered by the customer is compared to the PIN computed by the verification routines.

If PINs are verified by software, they must be in the clear when it comes time to verify them.

If PINs are verified by a hardware module, they must be passed to the module in encrypted form, along with the key used to encrypt them. The module then decrypts the PIN for comparing to the PIN it computes.

BASE24 Remote Banking also allows for customers to change their existing PINs. The PIN offset must already exist in the CSTT before this transaction is performed. The PIN change transaction can be performed with Diebold PIN verification in hardware or software; however, only Atalla Network Security Processors (NSPs) and Racal-Guardata Host Security Modules support this

transaction. The Atalla Security Module (ASM) does not support the PIN change transaction. Upon receiving a PIN change transaction, the Integrated Authorization Server process or the hardware security module computes an appropriate PIN offset based on the PIN entered by the customer and uses it to replace the existing offset in the CSTT. From that point on, the new PIN will be the only one allowed.

Under this form of PIN verification, PINs must be four digits in length.

Setting Up Diebold PIN Verification

To support Diebold PIN verification, the Integrated Authorization Server process requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to structure its customer IDs, how it wants verification to be handled, and whether or not it plans to use a hardware module for performing verification.

The remainder of this section discusses the Integrated Authorization Server process-related requirements for Diebold PIN verification. The discussion is divided into two parts:

- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Hardware Module Requirements

Diebold PIN verification can be performed by the following hardware modules supported by BASE24 Remote Banking:

- Atalla Security Module (ASM)
- Atalla Network Security Processor (NSP)
- Racal-Guardata Host Security Module (RSM)

For instructions on configuring these devices, refer to section 13, appendix E, and appendix F.

An institution cannot perform Diebold PIN verification in both software and hardware.

ASM or NSP support of Diebold PIN verification requires that a clear Diebold number table (DNT) be entered in the DIEBOLD TABLE field on KEYA screen 3. It also requires that a key be provided, in clear and encrypted form, to encrypt the DNT for sending to the ASM or NSP. The clear version of the key must be placed in the CLEAR KEY field on KEYA screen 3. The encrypted version of the key must be placed in the ENCRYPTED KEY field on KEYA screen 3. During initialization, the Integrated Authorization Server process uses the clear key for encrypting the DNT and sends the encrypted version of the key to the ASM or NSP along with the encrypted DNT to allow the ASM or NSP to decrypt the DNT.

RSM support of Diebold PIN verification requires only that the DNT be accessible to the RSM. The DIEBOLD TABLE field on KEYA screen 3 can contain all zeros or an encrypted version of the DNT returned from the RSM. When Diebold PIN verification is specified, the Integrated Authorization Server process, during initialization, calls a utility and passes the DNT from the KEYA to the RSM. If the DNT from the KEYA contains all zeros, the utility asks the RSM to verify that the RSM already has a valid DNT. If the DNT in the KEYA contains values other than zero, the utility loads the KEYA DNT to the RSM. The DNT in the RSM can be uploaded to the KEYA using the BASE24 RSMcom utility.

In ASM, NSP, and RSM support, a relative location for the DNT in the security module must be specified in the NUMBER TABLE RELATIVE LOCATION field on KEYA screen 3. This allows for keeping the DNTs, which could be different for each institution, from overlaying each other in the security module.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the Diebold PIN verification method. For detailed information on IDF screens 2, 4, and 40, refer to the ***BASE24 Base Files Maintenance Manual***.

IDF Screen 2

IDF screen 2 enables an institution to set parameters for its customers. The following fields must be completed when using the Diebold PIN verification method.

PIN — Specifies whether the customer's entered PIN is checked during transaction screening. The Integrated Authorization Server process uses this field with authorization level 1 (online) and authorization level 3 (online/offline) only, since prescreening checks are made only for these authorization levels. The default value is N. Valid values are as follows:

Y = Yes, check the customer's PIN; if invalid, decline the request and do not send the request to the host.

N = No, do not check the customer's PIN before sending the request to the host if the host is online.

MAX PIN TRIES — The code indicates the number of times that a customer can enter an incorrect PIN. The Integrated Authorization Server process uses the Customer Table (CSTT) to accumulate this information for a customer regardless of authorization method. The default value is 1.

POFST/PVV LOC — A code specifying the location of the Diebold PIN verification method PIN offset. Valid values are as follows:

- 0 = No PIN offset. A value of 0000 is used if an offset is required for the verification method.
- 1 = PIN offset is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the offset.
- 2 = PIN offset is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the offset.

A description of the code entered is displayed to the right of the POFST/PVV LOC field. The default value is 0.

PIN CHECK TYPE — A code indicating the PIN verification method used by the institution. The valid value for Diebold PIN verification is 02.

A description of the code entered is displayed to the right of the PIN CHECK TYPE field. The default value is 00 (no verification).

ALGO NUMBER LOC — A code specifying the location of the algorithm number. Valid values for the Diebold PIN verification method are as follows:

- 1 = Algorithm number is located in the Key Authorization File (KEYA).
- 2 = Algorithm number is located in the Key Authorization File (KEYA).

A description of the code entered is displayed to the right of the ALGO NUMBER LOC field. The default value is 0, which is not valid for the Diebold PIN verification method.

PIN TRIES RESET OPTION — A code indicating how the BAD PIN COUNT field on CSTT screen 1 is to be reset for customers of an institution.

The Integrated Authorization Server process keeps track of the number of bad PIN tries for a customer according to the authorization parameters established by an institution. This allows bad PIN tries to be accumulated over a period of time, and

institutions can then choose to decline authorization of a transaction for a customer if that customer has had an excessive number of incorrect PIN tries. The PIN tries limit is located in the MAX PIN TRIES field on IDF screen 2 (this screen).

BASE24 products offer two methods for clearing accumulated bad PIN tries. The first allows for clearing the accumulated bad PIN tries when the customer enters a correct PIN. The second method allows for clearing the accumulated bad PIN tries at the end of each usage accumulation period. This field controls how these clearance methods are to be applied for an institution's customers.

The bad PIN tries in the CSTT are not automatically cleared at the end of each usage accumulation period. Institutions using the CSTT can choose—using this field—to have their CSTT bad PIN tries automatically cleared with the rest of their totals at the end of each usage accumulation period, when a correct PIN is entered, or both. The bad PIN tries in the CSTT can also be reset by refreshing the file.

BASE24 Remote Banking uses customer processing dates in the product-specific segments of the IDF to track usage accumulation periods. The default value for this field is 0. Valid values are as follows:

- 0 = Reset the bad PIN tries at the end of the usage accumulation period, but not when a correct PIN is entered.
- 1 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.
- 3 = Reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 4 = Reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.

IDF Screen 4

IDF screen 4 enables an institution to specify its withdrawal period parameters. BASE24 Remote Banking uses parameters on IDF screen 4 for PIN tries only. Descriptions of the fields on IDF screen 4 that pertain to PIN verification are as follows:

WITHDRAWAL PERIOD PARAMETERS

The values in the following fields specify the withdrawal period. BASE24 Remote Banking uses these parameters for PIN tries only. The Integrated Authorization Server process maintains customer PIN tries in the Customer Table (CSTT).

BEGINNING DATE — The starting date (YYMMDD) of the current usage accumulation period for all customers belonging to this institution.

This date is placed in the LAST RESET DATE field on CSTT screen 1.

BASE24 products update this field automatically at the beginning of a new withdrawal period. See the description of the NEXT BEGINNING DATE field on this screen for more information.

If the value in the PERIOD LENGTH field on this screen is 83, 84, 85, or 86, the day in this field must be between 01 and 28.

Note: BASE24 Remote Banking uses the CURRENT PERIODIC USAGE BEGIN DATE and CURRENT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

NEXT BEGINNING DATE — The starting date (YYMMDD) of the next usage accumulation period.

When the date in the CURRENT BUSINESS DATE field on IDF screen 40 is greater than or equal to the value in this field, the value in this field is moved to the BEGINNING DATE field on this screen. A new value for this field is then computed by BASE24 products and placed in the field.

Note: BASE24 Remote Banking uses the NEXT PERIODIC USAGE BEGIN DATE and NEXT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

PERIOD LENGTH — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

The value in this field is referenced only if the value in the WORK DAY CODE field on this screen is set to 0, indicating that the usage accumulation period length should be specified by this field. If the value in the WORK DAY CODE field is set to 1, 2, or 3, the value in this field must be set to 0.

If the value in this field is 83, 84, 85, or 86, the day in the BEGINNING DATE field must be between 01 and 28. Valid values are as follows:

- 0 = Usage accumulation period is specified by the value in the WORK DAY CODE field.
- 1–79 = Number of days in the usage accumulation period.
- 80 = Usage accumulation period is one week (7 days).
- 81 = Usage accumulation period is two weeks (14 days).
- 82 = Usage accumulation period begins on the first and 15th of each month.
- 83 = Usage accumulation period is one month.
- 84 = Usage accumulation period is three months.
- 85 = Usage accumulation period is six months.
- 86 = Usage accumulation period is one year.

Note: BASE24 Remote Banking uses the PERIODIC USAGE LENGTH and CYCLIC USAGE LENGTH fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 1. A description of the code entered is displayed to the right of the PERIOD LENGTH field.

WORK DAY CODE — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

If codes 1, 2, or 3 are used in this field, the value in the PERIOD LENGTH field on this screen must be set to the value 0. If the value in the PERIOD LENGTH field on this screen is to be used to specify the usage accumulation period length, then the value in this field must be set to 0. Valid values are as follows:

- 0 = Use the value in the PERIOD LENGTH field; weekends and holidays are not taken into account with this code.

- 1 = Clear usage accumulation fields daily, except for weekends and specified holidays.
- 2 = Clear usage accumulation fields daily, except for Sundays and specified holidays.
- 3 = Clear usage accumulation fields daily, except for Saturdays and specified holidays.

Holidays are specified in the HOLIDAYS fields on this screen.

Note: BASE24 Remote Banking product uses the PERIODIC WORK DAY and CYCLIC WORK DAY fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 0. A description of the code entered is displayed to the right of the WORK DAY CODE field.

HOLIDAYS — A maximum of twenty dates (YYMMDD) defining the legitimate holidays for the institution.

The values in these fields are used if the value in any of the fields listed below is set to 1, 2, or 3. Any of these values indicates that usage accumulation fields are not cleared on the dates specified by the HOLIDAYS fields.

- WORK DAY CODE field on IDF screen 4 (this screen)
- PERIODIC WORK DAY field on IDF screen 41
- CYCLIC WORK DAY field on IDF screen 41

Blank fields can not be left between dates; however, unused fields should be left blank.

Example:

961124	961225	970101	970116	970213
970529	970704	970904	971009	971111
971123	971225	980101	_____	_____
_____	_____	_____	_____	_____

IDF Screen 40

IDF screen 40 contains BASE24 Remote Banking processing information. The fields used on IDF screen 40 in support of the Diebold PIN Verification method include the following:

TELEBANKING IDF INFORMATION

The following field contains BASE24 Remote Banking processing information.

PIN VERIFICATION GROUP — The value used by the Integrated Authorization Server process to select the Key Authorization File (KEYA) record when verifying PINs for customers of this financial institution. The value in this field is matched with the value in the GRP field on KEYA screen 1. The default value is the value in the FIID field.

CUSTOMER SERVICE INTERFACE CONTROL PARAMETERS

The following fields contain parameters used to process transactions initiated at customer service representative (CSR) terminals.

PIN REQUIRED — A code identifying whether a PIN is required with a transaction that is entered at a customer service representative (CSR) terminal. The default value is Y. Valid values are as follows:

Y = Yes, a PIN is required.

N = No, a PIN is not required.

O = CSR terminal operator can override a transaction that is entered without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.

PIN CHANGE ALLOWED — A code identifying whether a PIN change transaction can be performed at a customer service representative (CSR) terminal. The default value is N. Valid values are as follows:

Y = Yes, PIN change transactions are allowed.

N = No, PIN change transactions are not allowed.

For more information on setting up the database and IDF screen 2, refer to the ***BASE24 Base Files Maintenance Manual***.

Customer Table (CSTT) Settings

Customer Table (CSTT) screen 1 contains authorization parameters and usage accumulation information for BASE24 Remote Banking customers. The fields used on CSTT screen 1 include the following:

CUSTOMER ID — The unique identification number assigned to this CSTT row. An unlimited number of individual accounts and persons (each identified by a unique personal ID) can be associated with one customer ID number.

A customer ID number can also be known as a primary account number (PAN). Other BASE24 products use the PAN to identify information on a plastic card. If customers have already been issued plastic cards, existing PANs could be conveniently used as customer IDs; however, the PAN used by BASE24 Remote Banking does not have to be associated with a plastic card.

The customer ID is used as input for Diebold PIN verification.

PIN VERIFICATION DIGITS — The PIN offset value used in PIN verification processing. The information in this field depends on the PIN verification parameters specified in the PIN CHECK TYPE and POFST/PVV LOC fields on Institution Definition File (IDF) screen 2. This field is required if the PIN verification digits are to be maintained in the CSTT.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

If one zero is entered, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry. There is no default value.

LAST UPDATE — The date and time (YYYYMMDDhhmmss) the PIN VERIFICATION DIGITS field was last updated.

PIN VERIFICATION KEY INDEX — This field is not used for the Diebold PIN verification method. There is no default value.

BAD PIN COUNT — The number of times an incorrect PIN has been entered for this CSTT row during the current usage accumulation period.

Each institution defines the maximum number of incorrect PIN tries allowed for the customer in the MAX PIN TRIES field on Institution Definition File (IDF) screen 2. When this number of incorrect PIN tries is exceeded, subsequent transactions for the customer are rejected according to the setting in the PIN TRIES RESET OPTION field on IDF screen 2.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN CHANGE REQUIRED — A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. If the institution wants to define a customer before the initial PIN has been selected, this field must be set to the value S. When this field is set to the value S or Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. The default value is N. Valid values are as follows:

Y = Yes, a PIN change is required. The customer must change their current PIN; the only transaction allowed is the PIN change transaction.

N = No, a PIN change is not required; the current PIN can be used.

S = Customer Select. The customer must select their initial PIN; the only transaction allowed is the PIN change transaction.

LAST RESET DATE — The date and time (YYYYMMDDhhmmss) the BAD PIN COUNT field was last cleared. The BAD PIN COUNT field is cleared when the customer performs the first transaction of a business day or enters a correct PIN, based on the value in the PIN TRIES RESET OPTION field on Institution Definition File (IDF) screen 2.

A transaction is considered to be the first one of the day when the date in this field is less than the date in the BEGINNING DATE field on IDF screen 4. After the BAD PIN COUNT field is cleared, the date in this field is replaced with the date from the BEGINNING DATE field on IDF screen 4.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID group using the Diebold PIN verification method. For detailed information on KEYA screens and fields, refer to the ***BASE24 Base Files Maintenance Manual***.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the Diebold PIN verification method.

GRP — The code used in the IDF to identify this record. If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with the institution's FIID in this field.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. When using the Diebold PIN verification method, the valid value for this field is 02.

KEYA Screen 3

KEYA screen 3 contains Diebold PIN verification information. The following fields must be completed when using the Diebold PIN verification method.

NUMBER TABLE RELATIVE LOCATION — Specifies the relative location of the Diebold number table (DNT) in the security device if PIN verification is to be performed by a hardware module.

When the security device contains one DNT, the value entered in this field is 1. When the security device contains multiple DNTs, the relative location of the DNT used with this KEYA record is entered in this field. The actual value is calculated by the ASMcom or RSMcom utility.

ALGO NUMBER — Specifies the two-digit algorithm number for the Diebold PIN verification method. This value is used as the starting index into the Diebold number table. Valid values are hexadecimal characters 00 through 99. The first position allows values of 0 through 9; the second position allows values of 0 through 9 or A through F, up to the allowed limit of 99.

CLEAR KEY — The clear key used for encrypting the Diebold number table (DNT).

- If Diebold PIN verification is to be performed by an Atalla Security Module (ASM) or an Atalla Network Security Processor (NSP), the DNT must be passed to it in encrypted form. This field must contain the clear key used by the Integrated Authorization Server process to encrypt the DNT. This key must be 16 hexadecimal characters. Valid values for each character in this field are 0 through 9 and A through F. For protection in the database, this key is scrambled for storage by the KEYA file server process.
- If the Racal-Guardata Host Security Modules (RSMs) are being used, this field is not required and must be set to zeros. The DNT is encrypted by the security module before it is uploaded to the BASE24 Remote Banking system. It is stored in encrypted form and sent back to the RSM in encrypted form as well.

ENCRYPTED KEY — The encrypted version of the key used for encrypting the Diebold number table (DNT).

- If Diebold PIN verification is to be performed by an ASM or an NSP, the DNT must be passed to it in encrypted form. This field must contain the encrypted version of the key in the CLEAR KEY field.

This key must be encrypted by or for the specific security module being used, such that the module is able to decrypt the key when it is provided by the Integrated Authorization Server process. Once the key is decrypted, the module uses the clear version for decrypting the DNT. This value must be 16 hexadecimal characters.

- If the RSM is used because the DNT, if present, is already encrypted by a local master key (LMK) pair, this field is not required.

DIEBOLD TABLE — The DNT to be used by the Integrated Authorization Server process. The table must consist of 256 hexadecimal number pairs. For protection in the database, this table is scrambled for storage by the KEYA file server process.

The DNT should be in the clear for Atalla security modules, encrypted for Racal-Guardata security modules, or contain zeros. Refer to the topic “Hardware Module Requirements” earlier in this section for additional information.

Applied Communications, Inc.

Section 9

Identikey PIN Verification Support

This section describes the Integrated Authorization Server process support of the Identikey PIN verification method, including how the BASE24 database must be set up to allow the Integrated Authorization Server process to verify PINs under this method of verification.

The Integrated Authorization Server process only allows for implementation of this type of PIN verification using Atalla hardware modules.

Identikey PIN Verification

Identikey PIN verification is based on the use of the Atalla Identikey algorithm, which is proprietary to Atalla. Therefore, the Integrated Authorization Server process does not support this type of PIN verification in software.

Under this verification type, at the time the customer relationship is established, the institution generates a PIN verification number for each customer ID according to the specifications provided by Atalla. This PIN verification number is then placed in the BASE24 Remote Banking database.

To verify a customer's PIN in a transaction, the Integrated Authorization Server process passes this PIN verification number to an Atalla module, along with the PIN entered by the customer and other parameters set by the institution. The Atalla module then uses this information to verify the PIN and notify the Integrated Authorization Server process of the outcome.

PIN Verification Inputs

Identikey PIN verification requires the following information as input. Note that some of these items change from customer to customer, while others are constant for the institution.

- Customer ID
- PIN
- Bank ID
- Compare indicator
- PIN verification number (PVN)

These input items are described separately in this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Customer ID

The customer ID, also known as the primary account number (PAN), is used to identify each customer relationship. It must be carried in the BASE24 Remote Banking database and is used as input to Identikey PIN verification.

The Integrated Authorization Server process must derive the portion of the PAN to be sent as input to the Atalla module. To do this, it requires two other values: an offset and a length.

The offset denotes the starting point in the PAN for extracting data, and the length indicates the number of characters to take from the PAN. For example, if the PAN offset is 3 and the length is 10, the Integrated Authorization Server process skips to position 4 of the PAN and, starting with that position, picks up 10 consecutive characters of the PAN.

The offset and length must be specified in the PARTIAL PAN OFFSET and PARTIAL PAN LENGTH fields respectively on Key Authorization File (KEYA) screen 4.

PIN

In Identikey PIN verification, the customer-entered PIN must be sent to the Atalla module along with the other information for PIN verification.

If PINs are to be verified by an Atalla Security Module (ASM) or Atalla Network Security Processor (NSP), they can be in the clear or encrypted. If they are presented to the module in encrypted form, the key used to encrypt them must also be presented.

BASE24 Remote Banking allows for customers to change their existing PINs. The PIN verification number (PVN) must already exist in the Customer Table (CSTT) before this transaction is performed. Only Atalla Network Security Processors (NSPs) support this transaction. The Atalla Security Module (ASM) does not support the PIN change transaction. Upon receiving a PIN change transaction, the NSP computes an appropriate PVN based on the PIN entered by the customer and uses it to replace the existing PVN in the CSTT. From that point on, the new PIN will be the only one allowed.

Customers cannot select their own PINs as part of the first transaction with Identikey PIN verification.

Bank ID

A Bank ID is a two-, six-, or eight-digit value associated with the institution for use by the Identikey algorithm. It can be a two-digit backward index, a six-digit ISO number, or an eight-digit routing and transit number.

The use of this value is prescribed by the Identikey algorithm. The Integrated Authorization Server process passes it to the hardware module. In order for the Integrated Authorization Server process to pass it to the module, it must be specified in the INSTITUTION IDENTIFIER field on KEYA screen 4.

Compare Indicator

The compare indicator is a value indicating which portion of the Identikey number derived by the hardware module should be compared to the customer's PIN verification number.

Like the Bank ID, the function of the compare indicator is prescribed by the Identikey algorithm. The Integrated Authorization Server process gets the compare indicator value from the KEYA and passes it to the module. Therefore, the value must be specified in the COMPARE INDICATOR field on KEYA screen 4.

PIN Verification Number (PVN)

A PIN verification number (PVN) is a computed value, unique to each customer ID, created using a portion of the PAN, the customer's PIN, and a set of other values prescribed by the Identikey algorithm.

The Integrated Authorization Server process passes this value to the hardware module.

PVNs are stored in the Customer Table (CSTT) on a BASE24 Remote Banking system. This is specified using the POFST/PVV LOC field on Institution Definition File (IDF) screen 2.

PVNs can be four, six, or eight digits in length. The Integrated Authorization Server process uses all of the digits placed for PVNs stored in the CSTT.

Setting Up Identikey PIN Verification

To support Identikey PIN verification, the Integrated Authorization Server process requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to structure its accounts and how it wishes verification to be handled.

The remainder of this section discusses the Integrated Authorization Server process-related requirements for Identikey PIN verification. The discussion is divided into two parts:

- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Hardware Module Requirements

Identikey PIN verification can be performed by the following hardware modules supported by the Integrated Authorization Server process:

- Atalla Security Module
- Atalla Network Security Processor

For instructions on configuring these devices, refer to section 13 and appendix E.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the Identikey PIN verification method. Appendix A contains illustrations of IDF screens 2, 4, and 40.

For more information on IDF screens and fields, refer to the ***BASE24 Base Files Maintenance Manual***.

IDF Screen 2

IDF screen 2 enables an institution to set parameters for its customers. The following fields must be completed when using the Identikey PIN verification method.

PIN — Specifies whether the customer's entered PIN is checked during transaction screening. The Integrated Authorization Server process uses this field with authorization level 1 (online) and authorization level 3 (online/offline) only, since prescreening checks are made only for these authorization levels. The default value is N. Valid values are as follows:

Y = Yes, check the customer's PIN; if invalid, decline the request and do not send the request to the host.

N = No, do not check the customer's PIN before sending the request to the host if the host is online.

MAX PIN TRIES — The code indicates the number of times that a customer can enter an incorrect PIN. The Integrated Authorization Server process uses the Customer Table (CSTT) to accumulate this information for a customer regardless of authorization method. The default value is 1.

POFST/PVV LOC — A code specifying the location of the Identikey PIN verification method PIN Verification Number (PVN). Valid values are as follows:

0 = No PVN. A value of 0000 is used if a PVN is required for the verification method.

1 = PVN is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the PVN.

2 = PVN is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the PVN.

A description of the code entered is displayed to the right of the POFST/PVV LOC field. The default value is 0.

PIN CHECK TYPE — A code indicating the PIN verification method used by the institution. The valid value for the Identikey PIN verification method is 03.

A description of the code entered is displayed to the right of the PIN CHECK TYPE field. The default value is 00 (no verification).

ALGO NUMBER LOC — A code specifying the location of the algorithm number. This field should contain a value of 0 for the Identikey PIN verification method.

A description of the code entered is displayed to the right of the ALGO NUMBER LOC field. The default value is 0.

PIN TRIES RESET OPTION — A code indicating how the BAD PIN COUNT field on CSTT screen 1 is to be reset for customers of an institution.

The Integrated Authorization Server process keeps track of the number of bad PIN tries for a customer according to the authorization parameters established by an institution. This allows bad PIN tries to be accumulated over a period of time, and institutions can then choose to decline authorization of a transaction for a customer if that customer has had an excessive number of incorrect PIN tries. The PIN tries limit is located in the MAX PIN TRIES field on IDF screen 2 (this screen).

BASE24 products offer two methods for clearing accumulated bad PIN tries. The first allows for clearing the accumulated bad PIN tries when the customer enters a correct PIN. The second method allows for clearing the accumulated bad PIN tries at the end of each usage accumulation period. This field controls how these clearance methods are to be applied for an institution's customers.

The bad PIN tries in the CSTT are not automatically cleared at the end of each usage accumulation period. Institutions using the CSTT can choose—using this field—to have their CSTT bad PIN tries automatically cleared with the rest of their totals at the end of each usage accumulation period, when a correct PIN is entered, or both. The bad PIN tries in the CSTT can also be reset by refreshing the file.

BASE24 Remote Banking uses customer processing dates in the product-specific segments of the IDF to track usage accumulation periods. The default value for this field is 0. Valid values are as follows:

- 0 = Reset the bad PIN tries at the end of the usage accumulation period, but not when a correct PIN is entered.
- 1 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.
- 3 = Reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.

- 4 = Reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.

IDF Screen 4

IDF screen 4 enables an institution to specify its withdrawal period parameters. BASE24 Remote Banking uses parameters on IDF screen 4 for PIN tries only. Descriptions of the fields on IDF screen 4 that pertain to PIN verification are as follows:

WITHDRAWAL PERIOD PARAMETERS

The values in the following fields specify the withdrawal period. BASE24 Remote Banking uses these parameters for PIN tries only. The Integrated Authorization Server process maintains customer PIN tries in the Customer Table (CSTT).

BEGINNING DATE — The starting date (YYMMDD) of the current usage accumulation period for all customers belonging to this institution.

This date is placed in the LAST RESET DATE field on CSTT screen 1.

BASE24 Remote Banking updates this field automatically at the beginning of a new withdrawal period. See the description of the NEXT BEGINNING DATE field on this screen for more information.

If the value in the PERIOD LENGTH field on this screen is 83, 84, 85, or 86, the day in this field must be between 01 and 28.

Note: BASE24 Remote Banking uses the CURRENT PERIODIC USAGE BEGIN DATE and CURRENT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

NEXT BEGINNING DATE — The starting date (YYMMDD) of the next usage accumulation period.

When the date in the CURRENT BUSINESS DATE field on IDF screen 40 is greater than or equal to the value in this field, the value in this field is moved to the BEGINNING DATE field on this screen. A new value for this field is then computed by BASE24 Remote Banking and placed in the field.

Note: BASE24 Remote Banking product uses the NEXT PERIODIC USAGE BEGIN DATE and NEXT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

PERIOD LENGTH — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

The value in this field is referenced only if the value in the WORK DAY CODE field on this screen is set to 0, indicating that the usage accumulation period length should be specified by this field. If the value in the WORK DAY CODE field is set to 1, 2, or 3, the value in this field must be set to 0.

If the value in this field is 83, 84, 85, or 86, the day in the BEGINNING DATE field must be between 01 and 28. Valid values are as follows:

- 0 = Usage accumulation period is specified by the value in the WORK DAY CODE field.
- 1–79 = Number of days in the usage accumulation period.
- 80 = Usage accumulation period is one week (7 days).
- 81 = Usage accumulation period is two weeks (14 days).
- 82 = Usage accumulation period begins on the first and 15th of each month.
- 83 = Usage accumulation period is one month.
- 84 = Usage accumulation period is three months.
- 85 = Usage accumulation period is six months.
- 86 = Usage accumulation period is one year.

Note: BASE24 Remote Banking uses the PERIODIC USAGE LENGTH and CYCLIC USAGE LENGTH fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 1. A description of the code entered is displayed to the right of the PERIOD LENGTH field.

WORK DAY CODE — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

If codes 1, 2, or 3 are used in this field, the value in the PERIOD LENGTH field on this screen must be set to the value 0. If the value in the PERIOD LENGTH field on this screen is to be used to specify the usage accumulation period length, then the value in this field must be set to 0. Valid values are as follows:

- 0 = Use the value in the PERIOD LENGTH field; weekends and holidays are not taken into account with this code.
- 1 = Clear usage accumulation fields daily, except for weekends and specified holidays.
- 2 = Clear usage accumulation fields daily, except for Sundays and specified holidays.
- 3 = Clear usage accumulation fields daily, except for Saturdays and specified holidays.

Holidays are specified in the HOLIDAYS fields on this screen.

Note: BASE24 Remote Banking uses the PERIODIC WORK DAY and CYCLIC WORK DAY fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 0. A description of the code entered is displayed to the right of the WORK DAY CODE field.

HOLIDAYS — A maximum of twenty dates (YYMMDD) defining the legitimate holidays for the institution.

The values in these fields are used if the value in any of the fields listed below is set to 1, 2, or 3. Any of these values indicates that usage accumulation fields are not cleared on the dates specified by the HOLIDAYS fields.

- WORK DAY CODE field on IDF screen 4 (this screen)
- PERIODIC WORK DAY field on IDF screen 41
- CYCLIC WORK DAY field on IDF screen 41

Blank fields can not be left between dates; however, unused fields should be left blank.

Example: 961124 961225 970101 970116 970213
 970529 970704 970904 971009 971111
 971123 971225 980101 _____

IDF Screen 40

IDF screen 40 contains BASE24 Remote Banking processing information. The fields used on IDF screen 40 include the following:

TELEBANKING IDF INFORMATION

The following field contains BASE24 Remote Banking processing information.

PIN VERIFICATION GROUP — The value used by the Integrated Authorization Server process to select the Key Authorization File (KEYA) record when verifying PINs for customers of this financial institution. The value in this field is matched with the value in the GRP field on KEYA screen 1. The default value is the value in the FIID field.

CUSTOMER SERVICE INTERFACE CONTROL PARAMETERS

The following fields contain parameters used to process transactions initiated at customer service representative (CSR) terminals.

PIN REQUIRED — A code identifying whether a PIN is required with a transaction that is entered at a customer service representative (CSR) terminal. The default value is Y. Valid values are as follows:

Y = Yes, a PIN is required.

N = No, a PIN is not required.

O = CSR terminal operator can override a transaction that is entered without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.

PIN CHANGE ALLOWED — A code identifying whether a PIN change transaction can be performed at a customer service representative (CSR) terminal. The default value is N. Valid values are as follows:

Y = Yes, PIN change transactions are allowed.

N = No, PIN change transactions are not allowed.

Customer Table (CSTT) Settings

Customer Table (CSTT) screen 1 contains authorization parameters and usage accumulation information for BASE24 Remote Banking customers. The fields used on CSTT screen 1 include the following.

For detailed information on CSTT screen 1 and its fields, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

CUSTOMER ID — The unique identification number assigned to this CSTT row. An unlimited number of individual accounts and persons (each identified by a unique personal ID) can be associated with one customer ID number.

A customer ID number can also be known as a primary account number (PAN). Other BASE24 products use the PAN to identify information on a plastic card. If customers have already been issued plastic cards, existing PANs could be conveniently used as customer IDs; however, the PAN used by BASE24 Remote Banking does not have to be associated with a plastic card.

The customer ID is used as input into the Identikey PIN verification method.

PIN VERIFICATION DIGITS — The PIN verification number (PVN) used in PIN verification processing. The information in this field depends on the PIN verification parameters specified in the PIN CHECK TYPE and POFST/PVV LOC fields on Institution Definition File (IDF) screen 2. This field is required if the PIN verification digits are to be maintained in the CSTT.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

If one zero is entered, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry. There is no default value.

LAST UPDATE — The date and time (YYYYMMDDhhmmss) the PIN VERIFICATION DIGITS field was last updated.

PIN VERIFICATION KEY INDEX — This field is not used for the Identikey PIN verification method.

BAD PIN COUNT — The number of times an incorrect PIN has been entered for this CSTT row during the current usage accumulation period.

Each institution defines the maximum number of incorrect PIN tries allowed for the customer in the MAX PIN TRIES field on Institution Definition File (IDF) screen 2. When this number of incorrect PIN tries is exceeded, subsequent transactions for the customer are rejected according to the setting in the PIN TRIES RESET OPTION field on IDF screen 2.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN CHANGE REQUIRED — A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. When this field is set to the value Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. The default value is N. Valid values are as follows:

Y = Yes, a PIN change is required. The customer must change their current PIN; the only transaction allowed is the PIN change transaction.

N = No, a PIN change is not required; the current PIN can be used.

Note: The value S (customer select) is not valid for the Identikey PIN verification method.

LAST RESET DATE — The date and time (YYYYMMDDhhmmss) the BAD PIN COUNT field was last cleared. The BAD PIN COUNT field is cleared when the customer performs the first transaction of a business day or enters a correct PIN, based on the value in the PIN TRIES RESET OPTION field on Institution Definition File (IDF) screen 2.

A transaction is considered to be the first one of the day when the date in this field is less than the date in the BEGINNING DATE field on IDF screen 4. After the BAD PIN COUNT field is cleared, the date in this field is replaced with the date from the BEGINNING DATE field on IDF screen 4.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID group using the Identikey PIN verification method. Appendix A contains illustrations of all KEYA screens.

For more information on KEYA screens and fields, refer to the ***BASE24 Base Files Maintenance Manual***.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the Identikey PIN verification method.

GRP — The code used in the IDF to identify this record. If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with the institution's FIID in this field.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. When using the Identikey PIN verification method, the valid value for this field is 03.

KEYA Screen 4

KEYA screen 4 contains Identikey PIN verification information. The following fields must be completed when using the Identikey PIN verification method.

INSTITUTION IDENTIFIER — Contains the two-, six-, or eight-character bank ID used by the Identikey algorithm to identify the institution. If the bank ID is less than eight characters in length, it must be left-justified with no embedded spaces.

ID LENGTH — Specifies the length of the bank ID. The Integrated Authorization Server process computes this value automatically based on the length of the data entered in the INSTITUTION IDENTIFIER field.

PARTIAL PAN OFFSET — The offset to the starting position in the PAN for extracting the partial PAN. Position 00 identifies the first digit in the PAN.

PARTIAL PAN LENGTH — The length of the data (that is, number of digits) to be extracted from the PAN for the formation of the partial PAN. Valid values are 4 through 19.

COMPARE INDICATOR — The portion of the Identikey number to be compared to the customer's PVN. Valid values are as follows:

- L = Left-most four digits of the Identikey number.
- M = Middle four digits of the Identikey number.
- R = Right-most four digits of the Identikey number.

The Identikey number is created by the hardware module during PIN verification. Customer PVNs can be four, six, or eight digits in length; however, this value is only used if PVNs are four digits in length.

Section 10

Visa PVV PIN Verification Support

This section describes the Integrated Authorization Server process support of the Visa PVV PIN verification method, including how the BASE24 database must be set up to allow the Integrated Authorization Server process to verify PINs under this method of verification.

The Integrated Authorization Server process allows for implementation of this PIN verification method in software or a security module.

Visa PVV PIN Verification

The Visa PVV PIN verification method is based on using a PIN verification value (PVV) for verifying a customer's PIN. Visa PVV PIN verification can be performed in software or using a hardware module linked to a processor on which the BASE24 Remote Banking system is running.

At the time the customer relationship is established, the institution generates a control PVV for each customer ID using the standard DES algorithm, a pair of PIN verification keys, the PIN, and other customer-specific data. This PVV is then placed in the BASE24 database.

The following is a summary of the steps taken to verify a PIN using Visa PVV PIN verification in software or hardware.

The verification entity performs the following steps:

1. Creates a 16-digit value using 11 digits from the PAN, the 1-digit PIN verification key indicator, and 4 digits from the PIN.
2. Encrypts the 16-digit value using the DES algorithm and the first key of the PIN verification key (PVK) pair.
3. Performs a decrypt function on the result from step 2 using the DES algorithm and the second key of the PVK pair.
4. Performs an encrypt function on the result from step 3 using the DES algorithm and the first key of the PVK pair.
5. Extracts a 4-digit transaction PVV from the result from step 4.

To do this, the result is scanned—left to right—and the first 4 digits found with a value equal to or less than 9 are taken. The result from step 4 is in hexadecimal representation.

If fewer than 4 digits are found with a value equal to or less than 9, any additional values needed to complete the 4 digits are derived by rescanning the result from step 4, taking only those digits greater than 9 and subtracting 10 from each until the 4 digits have been obtained.

6. Compares the transaction PVV with the control PVV. If they match, the customer entered the PIN correctly. If they do not match, the customer entered the PIN incorrectly.

PIN Verification Inputs

Visa PVV PIN verification requires the following information as input. Note that some of these items change from customer to customer, while others are constant for the institution.

- Customer ID
- PIN
- PIN verification key (PVK) pair
- PIN verification key indicator (PVKI)
- PIN verification value (PVV)

These input items are described separately in this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Customer ID

The customer ID, also known as the primary account number (PAN), is used to identify each customer relationship. It must be carried in the BASE24 Remote Banking database and is used as input to Visa PVV PIN verification.

Visa PVV PIN verification uses the right-most 11 digits of the customer's PAN, not including the check digit (last digit), as input to create the PIN verification value. An example is shown below:

PAN:	1	2	3	4	5	6	7	8	9	9	8	7	6	5	4	7
PAN DIGITS USED:					5	6	7	8	9	9	8	7	6	5	4	

PIN

In Visa PVV PIN verification, the PIN entered by the customer is used in the computation of a transaction PVV.

If the PIN is four digits in length, the entire PIN is used in the computations. If the PIN is longer than four digits, only the left-most four digits are used in the computations.

For PIN verification performed by the Integrated Authorization Server process in software, the PIN must be in the clear.

For PIN verification performed by a hardware module, the PIN must be passed to the module in encrypted form, along with the key used to encrypt it. The hardware module then decrypts the PIN for use in generating a transaction PVV.

BASE24 Remote Banking allows for customers to change their existing PINs. The PIN verification value (PVV) must already exist in the Customer Table (CSTT) before this transaction is performed. Visa PVV PIN verification can be performed in hardware or software; however, only Atalla Network Security Processors (NSPs) and Racal-Guardata Host Security Modules (RSMs) support this transaction. The PIN change transaction is not supported when Visa PVV PIN verification is performed using the Atalla Security Module (ASM). Upon receiving a PIN change transaction, the Integrated Authorization Server process or the hardware security module computes an appropriate PVV based on the PIN entered by the customer and uses it to replace the existing PVV in the CSTT. From that point on, the new PIN is the only one allowed.

PIN Verification Key (PVK) Pair

Two PIN verification keys are required in Visa PVV PIN verification. These keys, called the PVK pair, are used as input into the DES algorithm to alternately encrypt and decrypt transaction data in the computation of a PIN verification value. For information on encryption and decryption and the DES algorithm, refer to section 2.

An institution can use the same PVK pair to generate PIN verification values for all of its customer IDs; however, the Integrated Authorization Server process allows for use of up to six different pairs. As with any DES-based operation, the security of the operation is dependent on keeping the keys confidential. Therefore, by using multiple sets of keys, the risk can be divided in case a pair of keys is compromised. In this situation, only those customer IDs for which PIN verification values were created with the compromised keys need to be reissued.

The PVK pairs used to create the institution's PIN verification values must be specified in the Key Authorization File (KEYA).

If the Integrated Authorization Server process performs PIN verification in software, clear versions of the keys must be entered in the CLEAR field on KEYA screen 5.

If a hardware module performs PIN verification, encrypted versions of the keys must be entered in the ENCRYPTED field on KEYA screen 5. In this case, these keys must be encrypted such that they can be decrypted by the hardware module.

PIN Verification Key Indicator (PVKI)

Because multiple pairs of PIN verification keys can be used, an indicator is required to specify the set of keys used to create the PIN verification value for each customer ID.

This indicator, called the PIN verification key indicator (PVKI), is a one-digit number associated with a customer ID identifying the pair of keys used to create the control PVV for that customer ID. Since there are up to six different sets of keys possible, PVKI values can be from 1 to 6.

The PVKI is carried in the PIN VERIFICATION KEY INDEX field on Customer Table (CSTT) screen 1.

Besides noting the PVK pair to be used, the PVKI is also used as part of the actual input to the DES algorithm. This input must be 16 characters in length and is composed of 11 digits from the PAN, the one-digit PVKI, and four digits from the entered PIN. An example is shown below:

PAN DIGITS:	5 6 7 8	9 9 8 7	6 5 4
PVKI:	2		
PIN:	4 5 5 0		
DES ALGORITHM INPUT:	5 6 7 8	9 9 8 7	6 5 4 2 4 5 5 0

PIN Verification Value (PVV)

A PIN verification value (PVV) is a computed four-digit value created using the PAN, the customer's PIN, and a specific set of encryption and decryption keys as input to the standard DES algorithm.

The basis for Visa PVV PIN verification is comparing a transaction PVV against a control PVV. The control PVV is created when the customer relationship is established; the transaction PVV is created as the PIN is being verified. Thus, it is the control PVV that is required as input to Visa PVV PIN verification.

The control PVV can be stored in the PIN VERIFICATION DIGITS field on Customer Table (CSTT) screen 1. This is specified using the POFST/PVV LOC field on IDF screen 2.

Setting Up Visa PVV PIN Verification

To support Visa PVV PIN verification, the Integrated Authorization Server process requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to structure its customer IDs, how it wishes verification to be handled, and whether or not it plans to use a hardware module for performing verification.

The remainder of this section discusses the Integrated Authorization Server process-related requirements for Visa PVV PIN verification. The discussion is divided into two parts:

- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Hardware Module Requirements

Visa PVV PIN verification can be performed by the following hardware modules supported by the Integrated Authorization Server process:

- Atalla Security Module (ASM)
- Atalla Network Security Processor (NSP)
- Racal-Guardata Host Security Module (RSM)

For instructions on configuring these devices, refer to section 13, appendix E, and appendix F.

Note: Visa PVV PIN verification using an RSM requires that external processors use PIN/PAD PIN blocks or the ANSI X9.8 standard PIN/PAN PIN blocks in the messages sent to the BASE24 Remote Banking system. The Visa PVV PIN verification method uses the right-most digits of the PAN excluding the check digit (last digit). The ANSI X9.8 standard PIN/PAN PIN block uses the 12 right-most digits of the PAN excluding the check digit. The RSM returns a PIN block error if the external processors use a PIN/PAN PIN block other than the ANSI X9.8 standard.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the Visa PVV PIN verification method. Appendix A contains illustrations of IDF screens 2, 4, and 40.

For more information on IDF screens and fields, refer to the ***BASE24 Base Files Maintenance Manual***.

IDF Screen 2

IDF screen 2 enables an institution to set up parameters for its customers. The following fields must be completed when using the Visa PVV PIN verification method.

PIN — Specifies whether the customer's entered PIN is checked during transaction screening. The Integrated Authorization Server process uses this field with authorization level 1 (online) and authorization level 3 (online/offline) only, since prescreening checks are made only for these authorization levels. The default value is N. Valid values are as follows:

Y = Yes, check the customer's PIN; if invalid, decline the request and do not send the request to the host.

N = No, do not check the customer's PIN before sending the request to the host if the host is online.

MAX PIN TRIES — The code indicates the number of times that a customer can enter an incorrect PIN. The Integrated Authorization Server process uses the Customer Table (CSTT) to accumulate this information for a customer regardless of authorization method. The default value is 1.

POFST/PVV LOC — A code specifying the location of the Visa PVV PIN verification method PIN Verification Value (PVV). Valid values are as follows:

- 1 = PVV is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the PVV.
- 2 = PVV is in the Customer Table (CSTT). The PIN VERIFICATION DIGITS field on CSTT screen 1 contains the PVV.

A description of the code entered is displayed to the right of the POFST/PVV LOC field. The default value is 0.

PIN CHECK TYPE — A code indicating the PIN verification method used by the institution. The valid value for Visa PVV PIN verification is 04.

A description of the code entered is displayed to the right of the PIN CHECK TYPE field. The default value is 00 (no verification).

ALGO NUMBER LOC — A code specifying the location of the algorithm number. This field should contain a value of 0 for the Visa PVV PIN verification method.

A description of the code entered is displayed to the right of the ALGO NUMBER LOC field. The default value is 0.

PIN TRIES RESET OPTION — A code indicating how the BAD PIN COUNT field on CSTT screen 1 is to be reset for customers of an institution.

The Integrated Authorization Server process keeps track of the number of bad PIN tries for a customer according to the authorization parameters established by an institution. This allows bad PIN tries to be accumulated over a period of time, and institutions can then choose to decline authorization of a transaction for a customer if that customer has had an excessive number of incorrect PIN tries. The PIN tries limit is located in the MAX PIN TRIES field on IDF screen 2 (this screen).

BASE24 products offer two methods for clearing accumulated bad PIN tries. The first allows for clearing the accumulated bad PIN tries when the customer enters a correct PIN. The second method allows for clearing the accumulated bad PIN tries at the end of each usage accumulation period. This field controls how these clearance methods are to be applied for an institution's customers.

The bad PIN tries in the CSTT are not automatically cleared at the end of each usage accumulation period. Institutions using the CSTT can choose—using this field—to have their CSTT bad PIN tries automatically cleared with the rest of their totals at the end of each usage accumulation period, when a correct PIN is entered, or both. The bad PIN tries in the CSTT can also be reset by refreshing the file.

The Integrated Authorization Server process uses customer processing dates in the product-specific segments of the IDF to track usage accumulation periods. The default value for this field is 0. Valid values are as follows:

- 0 = Reset the bad PIN tries at the end of the usage accumulation period, but not when a correct PIN is entered.
- 1 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 = Reset the bad PIN tries at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.
- 3 = Reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 4 = Reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.

IDF Screen 4

IDF screen 4 enables an institution to specify its withdrawal period parameters. BASE24 Remote Banking uses parameters on IDF screen 4 for PIN tries only. Descriptions of the fields on IDF screen 4 that pertain to PIN verification are as follows:

WITHDRAWAL PERIOD PARAMETERS

The values in the following fields specify the withdrawal period. The Integrated Authorization Server process uses these parameters for PIN tries only. The process maintains customer PIN tries in the Customer Table (CSTT).

BEGINNING DATE — The starting date (YYMMDD) of the current usage accumulation period for all customers belonging to this institution.

This date is placed in the LAST RESET DATE field on CSTT screen 1.

BASE24 Remote Banking updates this field automatically at the beginning of a new withdrawal period. See the description of the NEXT BEGINNING DATE field on this screen for more information.

If the value in the PERIOD LENGTH field on this screen is 83, 84, 85, or 86, the day in this field must be between 01 and 28.

Note: BASE24 Remote Banking product uses the CURRENT PERIODIC USAGE BEGIN DATE and CURRENT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

NEXT BEGINNING DATE — The starting date (YYMMDD) of the next usage accumulation period.

When the date in the CURRENT BUSINESS DATE field on IDF screen 40 is greater than or equal to the value in this field, the value in this field is moved to the BEGINNING DATE field on this screen. A new value for this field is then computed by BASE24 products and placed in the field.

Note: BASE24 Remote Banking product uses the NEXT PERIODIC USAGE BEGIN DATE and NEXT CYCLIC USAGE BEGIN DATE fields on IDF screen 41 in place of this field for transfer and payment usage periods.

PERIOD LENGTH — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

The value in this field is referenced only if the value in the WORK DAY CODE field on this screen is set to 0, indicating that the usage accumulation period length should be specified by this field. If the value in the WORK DAY CODE field is set to 1, 2, or 3, the value in this field must be set to 0.

If the value in this field is 83, 84, 85, or 86, the day in the BEGINNING DATE field must be between 01 and 28. Valid values are as follows:

- 0 = Usage accumulation period is specified by the value in the WORK DAY CODE field.
- 1–79 = Number of days in the usage accumulation period.
- 80 = Usage accumulation period is one week (7 days).
- 81 = Usage accumulation period is two weeks (14 days).
- 82 = Usage accumulation period begins on the first and 15th of each month.
- 83 = Usage accumulation period is one month.

- 84 = Usage accumulation period is three months.
- 85 = Usage accumulation period is six months.
- 86 = Usage accumulation period is one year.

Note: The Integrated Authorization Server process uses the PERIODIC USAGE LENGTH and CYCLIC USAGE LENGTH fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 1. A description of the code entered is displayed to the right of the PERIOD LENGTH field.

WORK DAY CODE — A code defining the institution's usage accumulation period length. The length of a usage accumulation period defines how long customer usage data in the Customer Table (CSTT) is allowed to accumulate before it is cleared.

If codes 1, 2, or 3 are used in this field, the value in the PERIOD LENGTH field on this screen must be set to the value 0. If the value in the PERIOD LENGTH field on this screen is to be used to specify the usage accumulation period length, then the value in this field must be set to 0.

Valid values are as follows:

- 0 = Use the value in the PERIOD LENGTH field; weekends and holidays are not taken into account with this code.
- 1 = Clear usage accumulation fields daily, except for weekends and specified holidays.
- 2 = Clear usage accumulation fields daily, except for Sundays and specified holidays.
- 3 = Clear usage accumulation fields daily, except for Saturdays and specified holidays.

Holidays are specified in the HOLIDAYS fields on this screen.

Note: The Integrated Authorization Server process uses the PERIODIC WORK DAY and CYCLIC WORK DAY fields on IDF screen 41 in place of this field for transfer and payment usage periods.

The default value is 0. A description of the code entered is displayed to the right of the WORK DAY CODE field.

HOLIDAYS — A maximum of twenty dates (YYMMDD) defining the legitimate holidays for the institution.

The values in these fields are used if the value in any of the fields listed below is set to 1, 2, or 3. Any of these values indicates that usage accumulation fields are not cleared on the dates specified by the HOLIDAYS fields.

- WORK DAY CODE field on IDF screen 4 (this screen)
- PERIODIC WORK DAY field on IDF screen 41
- CYCLIC WORK DAY field on IDF screen 41

Blank fields can not be left between dates; however, unused fields should be left blank.

Example: 961124 961225 970101 970116 970213
 970529 970704 970904 971009 971111
 971123 971225 980101 _____

IDF Screen 40

IDF screen 40 contains BASE24 Remote Banking processing information. The fields used on IDF screen 40 include the following:

TELEBANKING IDF INFORMATION

The following field contains BASE24 Remote Banking processing information.

PIN VERIFICATION GROUP — The value used by the Integrated Authorization Server process to select the Key Authorization File (KEYA) record when verifying PINs for customers of this financial institution. The value in this field is matched with the value in the GRP field on KEYA screen 1. The default value is the value in the FIID field.

CUSTOMER SERVICE INTERFACE CONTROL PARAMETERS

The following fields contain parameters used to process transactions initiated at customer service representative (CSR) terminals.

PIN REQUIRED — A code identifying whether a PIN is required with a transaction that is entered at a customer service representative (CSR) terminal. The default value is Y. Valid values are as follows:

Y = Yes, a PIN is required.

N = No, a PIN is not required.

O = CSR terminal operator can override a transaction that is entered without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.

PIN CHANGE ALLOWED — A code identifying whether a PIN change transaction can be performed at a customer service representative (CSR) terminal. The default value is N. Valid values are as follows:

Y = Yes, PIN change transactions are allowed.

N = No, PIN change transactions are not allowed.

Customer Table (CSTT) Settings

Customer Table (CSTT) screen 1 contains authorization parameters and usage accumulation information for BASE24 Remote Banking customers. The fields used on CSTT screen 1 include the following.

For more information regarding setting up CSTT screen 1, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

CUSTOMER ID — The unique identification number assigned to this CSTT row. An unlimited number of individual accounts and persons (each identified by a unique personal ID) can be associated with one customer ID number.

A customer ID number can also be known as a primary account number (PAN). Other BASE24 products use the PAN to identify information on a plastic card. If customers have already been issued plastic cards, existing PANs could be conveniently used as customer IDs; however, the PAN used by BASE24 Remote Banking product does not have to be associated with a plastic card.

The customer ID is used as input into the Visa PVV PIN verification method.

PIN VERIFICATION DIGITS — The PIN verification value (PVV) used in PIN verification processing. The information in this field depends on the PIN verification parameters specified in the PIN CHECK TYPE and POFST/PVV LOC fields on Institution Definition File (IDF) screen 2. This field is required if the PIN verification digits are to be maintained in the CSTT.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

If one zero is entered, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry. There is no default value.

LAST UPDATE — The date and time (YYYYMMDDhhmmss) the PIN VERIFICATION DIGITS field was last updated.

PIN VERIFICATION KEY INDEX — Identifies the location of the PIN verification value (PVV). Valid values are 1 through 6, and there is no default value.

BAD PIN COUNT — The number of times an incorrect PIN has been entered for this CSTT row during the current usage accumulation period.

Each institution defines the maximum number of incorrect PIN tries allowed for the customer in the MAX PIN TRIES field on Institution Definition File (IDF) screen 2. When this number of incorrect PIN tries is exceeded, subsequent transactions for the customer are rejected according to the setting in the PIN TRIES RESET OPTION field on IDF screen 2.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN CHANGE REQUIRED — A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. If the institution wants to define a customer before the initial PIN has been selected, this field must be set to the value S. When this field is set to the value S or Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. The default value is N. Valid values are as follows:

Y = Yes, a PIN change is required. The customer must change their current PIN; the only transaction allowed is the PIN change transaction.

N = No, a PIN change is not required; the current PIN can be used.

S = Customer Select. The customer must select their initial PIN; the only transaction allowed is the PIN change transaction.

LAST RESET DATE — The date and time (YYYYMMDDhhmmss) the BAD PIN COUNT field was last cleared. The BAD PIN COUNT field is cleared when the customer performs the first transaction of a business day or enters a correct PIN, based on the value in the PIN TRIES RESET OPTION field on Institution Definition File (IDF) screen 2.

A transaction is considered to be the first one of the day when the date in this field is less than the date in the BEGINNING DATE field on IDF screen 4. After the BAD PIN COUNT field is cleared, the date in this field is replaced with the date from the BEGINNING DATE field on IDF screen 4.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID group using the Visa PVV PIN verification method. Appendix A contains illustrations of all KEYA screens.

For more information on KEYA screens and fields, refer to the ***BASE24 Base Files Maintenance Manual***.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the Visa PVV PIN verification method.

GRP — The code used in the IDF to identify this record. If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with the institution's FIID in this field.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. For the Integrated Authorization Server process, this field must contain asterisks (*****) because cards are not used.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. When using the Visa PVV PIN verification method, the valid value for this field is 04.

KEYA Screen 5

KEYA screen 5 contains Visa PVV PIN verification information. The following fields must be completed when using the Visa PVV PIN verification method.

Visa PVV Keys

The following fields contain data for the Visa PVV PIN verification method.

CLEAR — The clear PIN verification keys. At least one clear or encrypted PVK pair must contain a value other than 16 zeros. The fields are labeled 1 through 6. The PVKI contains a value in the range of 1 through 6 to identify which PVK pair should be used.

- If PIN verification is to be performed without a security module, these fields must contain at least one pair of clear PIN verification keys. Up to six pairs can be specified. Each key must be 16 hexadecimal characters. Valid values for each character in the fields making up the PVK pairs are 0 through 9 and A through F. For protection in the database, all of these keys are scrambled for storage by the KEYA file server process.
- If PIN verification is to be performed by a hardware module exclusively, these fields are not used.

ENCRYPT — The encrypted PIN verification keys. At least one clear or encrypted PVK pair must contain a value other than 16 zeros. The fields are labeled 1 through 6. The PVKI contains a value in the range of 1 through 6 to identify which PVK pair should be used.

- If PIN verification is to be performed by a hardware module, these fields must contain at least one pair of encrypted PIN verification keys. Up to six pairs can be specified. Each key must be encrypted by or for the specific security module being used, such that the module is able to decrypt the key when it is provided by the Integrated Authorization Server process. The encrypted keys entered on the screen must be 16 hexadecimal characters. Valid values for each character in the fields making up the PVK pairs are 0 through 9 and A through F.
- If PIN verification is to be performed by the Integrated Authorization Server process exclusively, these fields are not used.

CHECK DIGITS — The check digits corresponding to the value in the ENCRYPT field located directly above this field on the screen. Valid values for each character in the check digits fields are 0 through 9 and A through F. A value of 0000 in this field indicates check digits are not being used with the key in the ENCRYPT field.

Applied Communications, Inc.

Section 11

Remote Banking Dynamic Key Management

This section describes the dynamic key management options provided by the Integrated Authorization Server process, Interface Master process, and Integrated Endpoint process for RBSI-compliant devices.

Note: Dynamic key management is not supported between BASE24 Remote Banking and EBMS-compliant devices or CSR terminals.

The information presented includes the database requirements and describes the network management messages for dynamic key management, including the information contained in the Cryptographic Service Message (CSM) element of the ISO external messages.

The Integrated Authorization Server process, Interface Master process, and Integrated Endpoint process use external messages that are based on the 1993 ISO 8583 standard, *Financial Transaction Card Originated Messages—Interchange Message Specification*.

Note: The above processes and the ISO Host Interface process both use the ISO 8583 standard; however, there are differences such as message layout and valid field values between the two because the ISO Host Interface process uses the 1987 ISO 8583 standard, *Bank Card Originated Messages—Interchange Message Specifications*, instead of the 1993 ISO 8583 standard.

Dynamic Key Management

The Integrated Authorization Server process uses working keys to protect the transaction data moving through the BASE24 Remote Banking network. The Interface Master process generates new working keys when keys need to be changed. The Integrated Endpoint process stores context when a response sent to an RBSI-compliant device contains a new working key. The installation and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management.

Dynamic key management allows BASE24 Remote Banking to automatically change working keys based on a number of factors. Thresholds can be established based on time, number of transactions, number of PIN translation errors, number of consecutive PIN translation errors, number of key synchronization errors, or any combination of these factors. A working key can be changed automatically when any of the defined thresholds is surpassed. Working keys also can be changed manually at any time using the KPT NEW server control command.

Note: The Integrated Authorization Server process, Interface Master process, and Integrated Endpoint process support just one type of working key—inbound PIN encryption keys. Message authentication keys and data transport keys are not currently used.

Dynamic key management provided by the Integrated Authorization Server process, Interface Master process, and Integrated Endpoint process is based on the ANSI X9.17 standard, *Financial Institution Key Management (Wholesale)* and the ISO 8732 standard, *Banking—Key Management (Wholesale)*.

Dynamic key management requires the use of security modules to generate new PIN encryption keys. This function is not supported in software by BASE24 Remote Banking. In addition, dynamic key management is optional for RBSI-compliant devices that send encrypted PINs to BASE24 Remote Banking and is not required for RBSI-compliant devices that always send PINs in the clear to BASE24 Remote Banking or perform local PIN verification.

Inbound PIN Encryption Keys

If a PIN is encrypted coming from an RBSI-compliant device, this is the key under which it is encrypted. This key can also be referred to as a PIN transport key or KPT.

Key Exchange Keys

Dynamic key management requires key exchange keys (KEKs), also referred to as key transport keys, in addition to the inbound PIN encryption keys. Inbound PIN encryption keys must be encrypted before being exchanged between BASE24 Remote Banking and the RBSI-compliant device. One KEK is used to encrypt inbound PIN encryption working keys. This KEK must be manually exchanged and entered in the BASE24 Remote Banking database before dynamic key management can occur.

Atalla uses the term *key exchange key* (KEK) to describe the keys used for exchanging PIN keys. Racal-Guardata uses the term *zone master key* (ZMK) to describe the key used for exchanging PIN keys. References to the key exchange key (KEK) in BASE24 documentation include both the Atalla and Racal-Guardata keys.

Key Management Processors

Key management involves two parties: BASE24 Remote Banking and the RBSI-compliant device. Each party has specific key management responsibilities.

The processor type assigned to each party identifies its responsibilities when performing key management, whether key management is performed manually or automatically. There are two possible processor types, each of which is described below.

Master Key Processor — The master key processor controls all key generation. When one key management party is the master, the other party must be the slave. For BASE24 Remote Banking, only the master can generate a key. Keys are never generated in response to a request from the slave. In addition, BASE24 Remote Banking always functions as the master key processor.

Slave Key Processor — The slave key processor has no responsibility for key generation. When one key management party is the slave, the other party must be the master. For BASE24 Remote Banking, the slave cannot generate a key or request the master to generate a new key. The slave can, however, accept or reject a key generated by the master. RBSI-compliant devices must always function as slave key processors.

Key Management Messages

Key management messages make up a subset of authorization response (1110 or 1210) messages and network management (0800) messages. There are three key management messages, each of which is described below. Two of these messages contain Cryptographic Service Messages (CSMs), which are described later in this section.

Note: The new key message can be generated automatically as the result of thresholds being surpassed. Thresholds are described in the topic “Dynamic Key Management Thresholds” presented later in this section. In addition, this message can be initiated by an operator issuing the KPT NEW command from the Server Control screen. The syntax for this command is described in the ***BASE24 Remote Banking Transaction Processing Manual***. The Server Control screen is described in the ***NET24 ITS Kernel Files Maintenance Manual***.

New Key Message — A new key message carries a new inbound PIN encryption key from BASE24 Remote Banking to the RBSI-compliant device. The RBSI-compliant device verifies the new key and returns a network management request message to BASE24 Remote Banking indicating that the new key has been implemented or that an error was encountered when verifying the key. If the new key was found to be valid, the device starts using it when it sends this request.

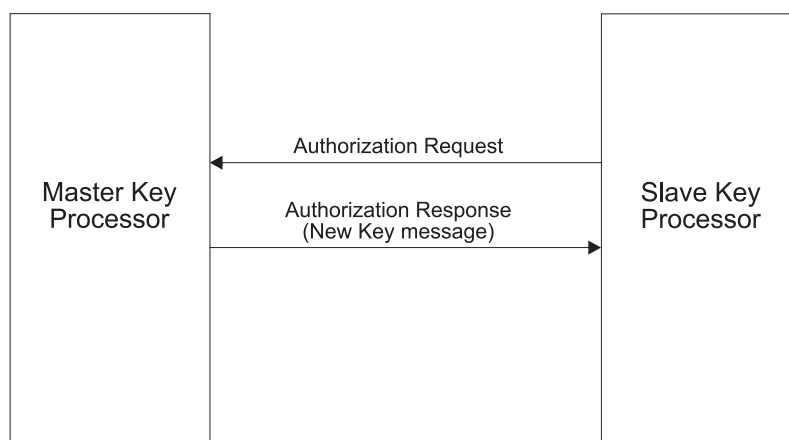
BASE24 Remote Banking sends the new key message to an RBSI-compliant device when it has determined that the key needs to be changed. This can occur as a result of a threshold being reached or in response to receipt of a KPT NEW command (the KPT NEW command has the same effect as passing a threshold). In either case, BASE24 Remote Banking sends the new key message in response to the next authorization request (financial or nonfinancial) received from the device that contains a PIN. The new key message is carried in the authorization response message and uses the Key Service Message, which is a CSM. Using an authorization response message to send new keys prevents the RBSI-compliant device from receiving unsolicited messages from BASE24 Remote Banking.

Network Management Request Message — This message is sent from the RBSI-compliant device to BASE24 Remote Banking and indicates whether the new inbound PIN key received in a new key message was determined to be valid. If the new key was determined to be valid, this message contains a Response Service Message in the request message. If the new key was determined to be invalid, this message contains an Error Service Message in the request message. Both the Response Service Message and Error Service Message are CSMs.

Network Management Response Message — This message is sent from BASE24 Remote Banking to the RBSI-compliant device in response to a network management request message. This message contains no CSMs and simply echos back the function code received in the network management request message.

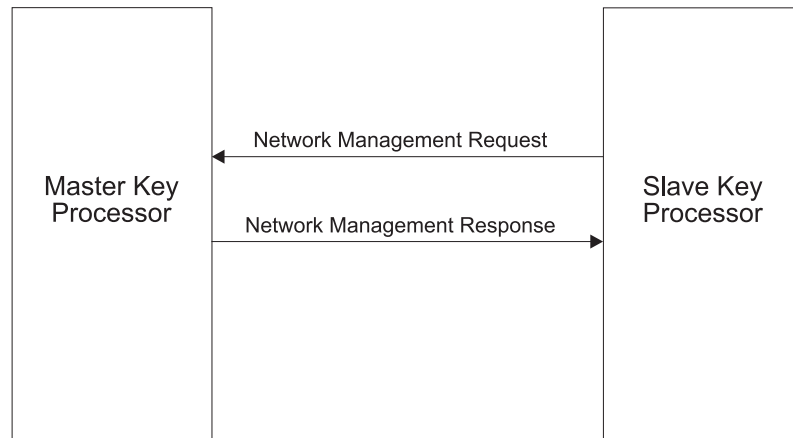
Processing Key Management Messages

The processing for a new key message is demonstrated in the following example. In this case, BASE24 Remote Banking (master key processor) determines that a new inbound PIN encryption key is required for an RBSI-compliant device. The new key is sent in response to the next authorization request received from the device (slave key processor) that contains a PIN.



When the RBSI-compliant device receives the new key message, it verifies the new inbound PIN encryption key, generates a network management request message, and sends the message to BASE24 Remote Banking. In this example, the new key

is valid, so the request message contains a Response Service Message. BASE24 Remote Banking responds by sending a network management response message back to the RBSI-compliant device.



The RBSI-compliant device starts using the new inbound PIN encryption key when it sends the network management request with the RSM. BASE24 Remote Banking maintains the old and new keys until it receives the network management request with the RSM, at which time, it starts using the new key. The old key remains available until the clear old key timer expires. At this time, the Interface Master process deletes the old key from the database.

External ISO Messages

All key management messages are ISO messages exchanged between BASE24 and the RBSI-compliant device. The message type for each of these ISO messages is provided below in parentheses following the descriptive message name.

New key messages are always sent in response to authorization request (1100), authorization repeat (1101), financial request (1200), or financial repeat (1201) messages. The authorization response message (1110) is used in response to nonfinancial authorization requests. The financial response message (1210) is used in response to financial authorization requests.

The network management request (1804) or network management repeat (1805) messages are sent in response to a new key message. The network management response (1814) is sent in response to an 1804 or 1805 message.

The following table describes the ISO message elements used to carry the data required for key management messages. These data elements are used in 1110, 1210, 1804, 1805, and 1814 key management messages. For more information on ISO messages, refer to the ***BASE24 Remote Banking Standard Interface Support Manual***.

Data Element	Description
System Trace Audit Number (P-11)	The system trace audit number is a number assigned by a message initiator that uniquely identifies a transaction within a 24 hour time period. The same system trace audit number provided in an 1110 or 1210 response containing a new key message must be provided in the Original Data Elements (P-56) bit included in an 1804 or 1805 network management request or repeat message.
Function Code (P-24)	All 1804, 1805, and 1814 key management messages must contain a function code of 811 in this data element.
Original Data Elements (P-56)	Each 1804 or 1805 key management message must include the system trace audit number received in the 1110 or 1210 response that prompted the 1804 or 1805 message.
Key Management Data (S-96)	Contains the Cryptographic Service Message (CSM), which indicates the key management function to be performed. In 1110 and 1210 response messages, this data element contains the Key Service Message (KSM) when a new inbound PIN encryption key is included. In 1804 and 1805 network management request messages, this data element contains either a Response Service Message (RSM) or Error Service Message (ESM), depending on whether the new inbound PIN encryption key was successfully verified.

Cryptographic Service Message

The ANSI X9.17 standard, *Financial Institution Key Management (Wholesale)*, establishes standards for key management. This standard also defines the Cryptographic Service Message (CSM) used for moving key management data between processors when keys are distributed. CSM information is contained in the Key Management Data data element (S-96) of the ISO external message.

Key Distribution Environment

BASE24 Remote Banking and RBSI-compliant devices use the point-to-point key distribution environment. In this environment, two parties share a key encryption key so that further PIN keys can be exchanged and one or both of the parties has the capability to generate or otherwise acquire keys. For BASE24 Remote Banking, only the Interface Master process can generate keys, and the RBSI-compliant device can only acquire keys.

Message Classes

BASE24 Remote Banking and RBSI-compliant devices support the following classes of CSMs used in point-to-point environments:

- Key Service Message (KSM)
- Response Service Message (RSM)
- Error Service Message (ESM)

The Request Service Initiation Message (RSI) is not used because RBSI-compliant devices cannot request keys from BASE24 Remote Banking and BASE24 Remote Banking cannot request keys from RBSI-compliant devices.

Message Fields and Subfields

Each CSM class contains several fields, with many of the fields containing subfields. Some fields and subfields are mandatory while others are optional. Because of this flexibility, each field begins with a unique 2- or 3-character identifier and ends with a blank. (The only exception to this rule is the VVK field, which does not end with a blank.) The CSM itself always begins with the literal CSM, and the message contents are always carried between a pair of parentheses.

The fields shown in the following table are the only ones that appear in the CSMs created by BASE24 Remote Banking. There are four CSM formats because the ESM class can have two formats, depending on the type of error being reported. Each of the CSM formats is described in greater detail, following the table.

CSM Class	KSM	RSM	ESM	
			With Counts*	Without Counts
CSM Fields	MCL RCV ORG KPTR CTR VVK	MCL RCV ORG	MCL RCV ORG CTP CTR ERF	MCL RCV ORG CTR ERF

* Although counts can be used to synchronize parties in key exchanges, BASE24 expects RBSI-compliant devices to always accept or adopt the count provided by BASE24 in Key Service Messages. The BASE24 count is obtained from the KEY COUNTER field on ISF screen 2.

When BASE24 Remote Banking receives an RSM or ESM generated by an RBSI-compliant device, it searches only for the fields shown in the preceding table and the MCL field is expected to be the first field.

Message Formats

BASE24 Remote Banking dynamic key management processing uses the CSM to move key information between BASE24 Remote Banking and the RBSI-compliant device in two of the three key management messages described earlier in this section: new key messages and network management request messages. The network management response message does not use the CSM because it only echoes the function code received in a network management request message back to the RBSI-compliant device.

The following describes and presents examples of the BASE24 Remote Banking supported CSM classes:

Key Service Message (KSM) — Used when a new inbound PIN encryption key is passed between BASE24 and the RBSI-compliant device.

An example of a CSM containing the KSM format is shown below:

CSM(**MCL**/KSM**RCV**/1234567890123456**ORG**/1234567890123456**KPTR**/1234567890123456**CTR**/12345678901234**VVK**/1234)

The information in this example is actually a continuous line. It has been broken here due to space constraints. Each field in the message is preceded by one of the following identifiers. The identifiers are shown in boldface type. The blanks identified by the character **b** in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

- MCL/ = Message class (KSM)
Field Length: 3 alphabetic characters
- RCV/ = Receiver of the message taken from the RECEIVING ID field on VCD screen 4. This field can contain blanks.
Field Length: 4–16 alphanumeric characters
- ORG/ = Originator of the message taken from the ORIGINATING ID field on VCD screen 4. This field can contain blanks.
Field Length: 4–16 alphanumeric characters
- KPTR/ = New key value, generated by the security module for a new key request.
Field Length: 16 hexadecimal characters
- CTR/ = Hexadecimal key counter taken from the INBOUND KEYS: KEY COUNTER field on ISF screen 2. Valid values for this field are 0–FFFFFF and leading zeroes are suppressed.
Field Length: 1–14 hexadecimal or blank characters
- VVK/ = Key verification value taken from the INBOUND KEYS: VVK1 or VVK2 field on ISF screen 2.
Field Length: 4 hexadecimal characters

Response Service Message (RSM) — Used to respond to a KSM that is processed successfully. An example of a CSM containing the RSM format is shown below:

CSM(**MCL**/RSM**RCV**/1234567890123456**ORG**/1234567890123456**b**)

Each field in the message is preceded by one of the following identifiers. The identifiers are shown in boldface type. The blanks identified by the character / in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

MCL/ = Message class (RSM)
 Field Length: 3 alphabetic characters

RCV/ = Receiver of the message
 Field Length: 16 alphanumeric or blank characters

ORG/ = Originator of the message
 Field Length: 16 alphanumeric or blank characters

Error Service Message (ESM) — Used to respond to a Key Service Message (KSM) that cannot be processed successfully. The ESM can have two formats, depending on the type of error being reported. The expected key count (CTP field) is included only when the key count received from the message originator (CTR field) does not match the expected count. The CTR field is always included in the ESM. The error code (ERF field) in the examples below can contain only alphabetic characters. Numbers are used to demonstrate field length.

Note: BASE24 expects the RBSI-compliant device to always accept or adopt the count provided by BASE24 in the Key Service Message (KSM). Therefore, key validation by the RBSI-compliant device should never fail due to a count synchronization error and the first format below should never be used, even though it is valid format.

An example of a CSM containing the ESM format with the expected key counter (CTP field) is shown below:

```
CSM(MCL/ESM|RCV/1234567890123456|ORG/1234567890123456|'  
CTP/12345678901234|CTR/12345678901234|ERF/1234567890123456| )
```

An example of a CSM containing the ESM format without the expected key counter is shown below:

```
CSM(MCL/ESM|RCV/1234567890123456|ORG/1234567890123456|'  
CTR/12345678901234|ERF/1234567890123456| )
```

The information in each of these examples is actually a continuous line. It has been broken here due to space constraints. Each field in the messages is preceded by one of the following identifiers. The identifiers are shown in boldface type.

The blanks identified by the character / in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

- MCL/ = Message class (ESM)
Field Length: 3 alphabetic characters
- RCV/ = Receiver of the message
Field Length: 4–16 alphanumeric characters
- ORG/ = Originator of the message
Field Length: 4–16 alphanumeric characters
- CTP/ = Hexadecimal key count expected by the receiver of the KSM. This identifier and field are only included in the message when error code P is returned.
Field Length: 1–14 hexadecimal characters
- CTR/ = Hexadecimal key count originally sent in the KSM by the message originator.
Field Length: 1–14 hexadecimal characters
- ERF/ = The following codes identify the errors that were detected during processing.
C = Cannot process (general error)
F = Format error
H = Invalid receiver ID or originator ID
P = The value in the CTP field of the KSM does not match the expected count (BASE24 Remote Banking does not expect to ever receive this code)
Field Length: 1-16 alphabetic characters

Message Length

The length of each field in the CSM is determined by the data it carries. The message format field descriptions presented earlier in this section include the range of possible lengths for each field. The total length for each message must account for the CSM literal, message parentheses, field identifiers, and field separators. The maximum lengths for each CSM are as follows: KSM (104 bytes), ESM with the expected key counts (114 bytes), ESM without the expected key counts (95 bytes), RSM (55 bytes).

Note: Fixed-length ISO messages are not supported between BASE24 and a RBSI-compliant device.

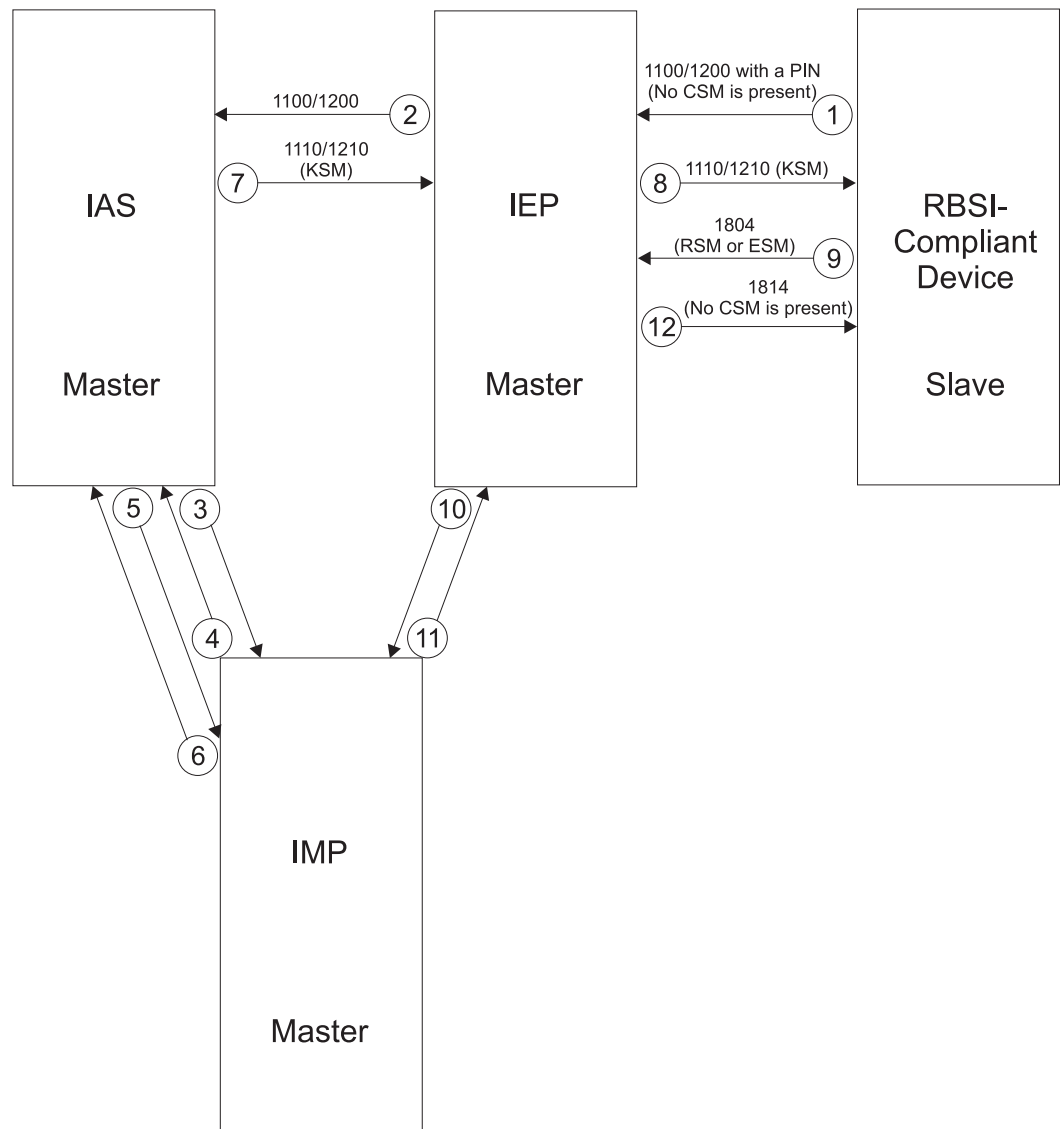
Required Fields

All fields are required except for the expected key count field (CTP) in the ESM. This key count field is conditional in the ESM, depending on whether the error code field (ERF) contains a value of P (counts do not match). This key count field is not included when the error code is a value other than P.

New Key Message Flow Diagram

The following diagram illustrates how messages containing CSM formats are used when a new key is passed from BASE24 Remote Banking to an RBSI-compliant device. A key to the abbreviations follows the illustration.

Note: This message flow does not show interactions with the XPNET process or object-level interactions. For a detailed message flow including these interactions, refer to the ***BASE24 Remote Banking Transaction Processing Manual***.



Key

CSM	=	Cryptographic Service Message
ESM	=	Error Service Message
IAS	=	Integrated Authorization Server process
IEP	=	Integrated Endpoint process
IMP	=	Interface Master process
KSM	=	Key Service Message
RBSI	=	Remote Banking Standard Interface
RSM	=	Response Service Message

Processing Summary

The new key messages are processed as follows:

1. The RBSI-compliant device sends a financial or nonfinancial authorization request containing a PIN to BASE24 Remote Banking.
2. The Integrated Endpoint process performs message-based routing and routes the message to the Integrated Authorization Server process.
3. The Integrated Authorization Server process calls the Interface Master process to perform station validation.
4. The Interface Master process validates the station on which the request was received and returns this information to the Integrated Authorization Server process.

No dynamic key management counters are increased and no new key data is returned at this time.

5. The Integrated Authorization Server process sends one or more commands, depending on the configuration of the ISF, to a security module to decrypt the PIN and verify it. If a sanity check or PIN length error occurs, the process sets the appropriate action code. Otherwise, the process completes authorization of the transaction. For the purpose of this message flow, assume that a sanity check or PIN length error occurred. The Integrated Authorization Server process calls the Interface Master process with another station validation request to increase the appropriate dynamic key management threshold.

Note: The Integrated Authorization Server sends this validation request only if dynamic key management is enabled for the interface instance and the transaction included a PIN.

6. The Interface Master process updates the appropriate dynamic key management counts in the Interface extended memory table and compares the counts to their corresponding limits. For the purpose of this message flow, assume that one of limits has been reached or that a KPT NEW command was previously received for the interface instance. The Interface Master process requests a security module to generate a new inbound PIN encryption key. The new key is stored in the Interface Security File (ISF) for the interface instance, but the process does not toggle the current key index at this time. The process updates the key counter in the ISF and returns the new key data to the Integrated Authorization Server process in the station validation response.
7. The Integrated Authorization Server process logs the transaction to the ITS Transaction Log File (ITLF), however, the new key data is not included. The process then formats an external message authorization response containing a Key Service Message (KSM) and sends it to the Integrated Endpoint process. The Integrated Authorization Server process requests that the Integrated Endpoint process store context for the response and specifies that the subsequent network management message (message type 1804) or context that times out should be routed to the Interface Master process.
8. The Integrated Endpoint process stores context, sets the network management message timer from the Interface Configuration File (ICF), and sends the response with the KSM to the RBSI-compliant device.
9. The RBSI-compliant device validates the KSM and sends a network management message (message type 1804) back to BASE24 Remote Banking. This request message contains an RSM when no errors are detected in the KSM or an ESM when errors are found in the KSM. If the new inbound PIN encryption key is found to be valid, the device starts using the new key after it sends the RSM. For the purpose of this message flow, assume the message contains an RSM.
10. The Integrated Endpoint process receives the 1804 message within the network management message time limit, retrieves context, and sends the message and context to the Interface Master process.
11. The Interface Master process determines that the new key was accepted by the RBSI-compliant device, toggles the current key index in the ISF to start using the new key, and sends a network management response message (message type 1814) to the Integrated Endpoint process.

If the 1804 request message had indicated an error, the Interface Master process would generate an event message indicating the error that occurred. The process would continue to use the old key and would again attempt to generate a new key and send it in the KSM of a response to the next authorization request containing a PIN received from the device.

12. The Integrated Endpoint process sends the 1814 network management response to the RBSI-compliant device.

RBSI Dynamic Key Management Thresholds

Remote Banking Standard Interface (RBSI) dynamic key management supports five types of thresholds for changing inbound PIN encryption keys. One or more of the thresholds can be established for inbound PIN encryption keys. These thresholds are set on VRU Configuration Data (VCD) screen 4.

A flag indicating that dynamic key management will be performed with BASE24 acting as the master must be set, before dynamic key management can occur (either through threshold checking or by operator entry of the KPT NEW command). This flag is set in the KEY MGMT TYPE field on VCD screen 1.

If threshold checking is to be used, at least one threshold must be set. When the first threshold is reached for an interface instance, an internal flag is set to indicate that the key needs to be changed. A new inbound PIN encryption key is generated and sent to the RBSI-compliant device in response to the next authorization request received from the device that contains a PIN.

RBSI dynamic key management thresholds are based on the following information:

- Length of time
- Number of transactions
- Number of key synchronization errors
- Number of errors
- Number of consecutive errors

The length-of-time threshold uses a timer to track the length of time that has elapsed since the last key exchange. The remaining thresholds use counters to track the number of transactions or errors that have occurred since the last key exchange. The circumstances that cause a timer or counter to be increased or reset depend on the threshold being monitored. These thresholds are described below and on the following pages. The timer and counts are maintained on an interface instance basis in a table internal to the Interface Master process that is built from the Interface extended memory table.

Time Threshold

The length-of-time threshold is based solely on the length of time that has elapsed since the timer was set or reset. The timer is reset differently, depending on whether it is used by itself or in combination with other threshold limits.

When the length-of-time threshold is used by itself, it is reset only when the timer expires, thereby creating a series of windows of equal length. When the timer expires, BASE24 Remote Banking sets an internal flag to change the key for the interface instance. When the next authorization request containing a PIN is received from the RBSI-compliant device, the key is changed and returned in the authorization response.

When the length-of-time threshold is used in combination with other thresholds, it is reset (along with all of the other threshold counts) when any one of the other thresholds is reached or when the timer expires, whichever occurs first.

The timer is reset in the following situations:

- When a KPT NEW command is issued for the device's interface instance.
- When the Interface Master process is initialized or the Interface Master Services object in the process receives an LCONF Warmboot command.
- When the timer is used by itself, it is reset when the timer expires.
- When the timer is used in combination with other thresholds, it is reset when any one of the other thresholds is reached or when the timer expires, whichever occurs first.

Transaction Threshold

Each of the transaction thresholds is based on the number of transactions containing a PIN that have occurred since its corresponding transaction counter was set. Each time a counter is increased, it is compared to its threshold. When the counter meets or exceeds its threshold, BASE24 Remote Banking sets an internal flag to change the inbound PIN encryption key for the interface instance. When the next authorization request containing a PIN is received from the RBSI-compliant device for the interface instance, the key is changed and returned in the authorization response.

Number of Transactions

The number of transactions counter is increased each time an external message is received from an RBSI-compliant device containing a hardware-encrypted PIN.

The transaction counter is reset in the following situations:

- When a KPT NEW command is issued for the device's interface instance.
- When the Interface Master process is initialized.
- When any of the thresholds configured for the device's interface instance are reached.

Number of Errors

The number of errors counter is increased each time a PIN length or sanity check error occurs while attempting to process a hardware-encrypted PIN contained in the external message received from an RBSI-compliant device.

The error counter is reset in the following situations:

- When a KPT NEW command is issued for the device's interface instance.
- When the Interface Master process is initialized.
- When any of the thresholds configured for the device's interface instance are reached.

Number of Consecutive Errors

The number of consecutive errors counter is increased each time a PIN length or sanity check error occurs while attempting to process a hardware-encrypted PIN contained in the external message received from an RBSI-compliant device.

The consecutive error counter is reset in the following situations:

- When a KPT NEW command is issued for the device's interface instance.
- When the Interface Master process is initialized.

- When any of the thresholds configured for the device's interface instance are reached.
- When the inbound PIN encryption key is used successfully in processing a transaction.

Number of Key Synchronization Errors

The number of key synchronization errors counter is increased each time a sanity check error occurs while attempting to process a hardware-encrypted PIN contained in the external message received from an RBSI-compliant device.

The number of key synchronization errors counter is reset in the following situations:

- When a KPT NEW command is issued for the device's interface instance.
- When the Interface Master process is initialized.
- When any of the thresholds configured for the device's interface instance are reached.

Setting Up the BASE24 Database

Settings in the Interface Security File (ISF) and ITS Interface Configuration File (IFCF) must be considered when using dynamic key management. The IFCF is maintained using the VRU Configuration Data (VCD) screens. Appendix A contains illustrations of the ISF and VCD screens presented in this section.

For more information on ISF and VCD screens, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

Interface Security File Settings

The Interface Security File (ISF) contains the information and parameters required by integrated transaction services (ITS) interfaces for PIN encryption, PIN translation, message authentication, and dynamic key management.

Note: The message authentication fields in the ISF are not currently used.

ISF Screen 1

Interface Security File (ISF) screen 1 contains PIN encryption parameters and key transport key information. The following fields are required when using dynamic key management.

EXTRN PIN ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external entity.

- If dynamic key management is used, this field must be set to a value of H (hardware PIN encryption).
- If dynamic key management is not used for PIN keys, this field can be set to any valid value.

INTRN PIN ENCRYPT TYPE — Specifies the type of PIN encryption used by BASE24 products. This field can be set to any value (H, S, or N) when dynamic key management is used.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages.

- If dynamic key management is used, which requires the use of security modules for PIN encryption, this field must be set to a value of 1 (ANSI PIN block) or 3 (PIN/PAD PIN block).
- If dynamic key management is not used for PIN keys, this field can be set to any valid value.

PIN PAD CHARACTER — If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the pad character used in the formation of the external message PIN block. Valid values are A through F.

PIN ENCRYPT LEVEL — A code identifying the level of encryption used for PINs exchanged between a RBSI-compliant device and BASE24. The valid value is 1 (single).

OLD KEY TIMER LIMIT — The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the Interface Master process. The keys in the INBOUND KEYS: KPT1 and KPT2 fields on ISF screen 2 permit information for two inbound PIN encryption keys. The values in the INBOUND KEYS: CURRENT INDEX field on ISF screen 2 identifies whether key number 1 or key number 2 is in use.

At the time a new key is placed in service, the other key becomes the old key. For example, if inbound PIN transport key (KPT) number 1 is currently in use when the Interface Master process generates a new key, the new key is stored as KPT number 2. KPT number 2 becomes the current key and KPT number 1 becomes the old key only after the Interface Master process has received notice from the RBSI-compliant device that the new key was successfully verified. (If KPT number 2 is the current key, it becomes the old key and the new key is stored as KPT number 1.)

When processing a transaction, the Integrated Authorization Server process uses the new key for PIN processing. If the security module used by the Integrated Authorization Server process detects an error because the external entity is still using the old key, it can repeat the PIN processing using the old key before it declines the transaction. The value in this field determines how long the old key is available. If the old key is not available, a second PIN processing attempt is not made.

A value of 0 in this field indicates the the old key is not cleared until the next time the inbound PIN encryption key is changed.

KEY TRANSPORT KEY INFORMATION

When PIN keys are transported between BASE24 Remote Banking and the RBSI-compliant device, they are encrypted under this key for the external entity. The key transport key must be manually exchanged between the two entities.

KEY LENGTH — A code identifying whether single- or double-length key transport keys (KKTs) are being used with this interface. This field can be set to a value of 1 (single-length key transport keys) or 2 (double-length key transport keys) for dynamic key management.

KKT-PT — The security module-encrypted form of the key used to encrypt PIN transport keys (KPTs). There are two sets of 16 hexadecimal characters. The first set is used for single-length keys and both sets are used for double-length keys. Valid values for each position in this field are 0 through 9 and A through F. Both sets must contain 16 valid characters. If the value in the KEY LENGTH field is 1 (single-length key transport keys), the second set of characters must be spaces.

VVK — The verification value associated with the value in the KKT-PT field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

VARIANT — A code that specifies the key variant to be applied to the KKT-PT key when translating PIN transport keys (KPTs). The same key variant must be used by the RBSI-compliant device.

ISF Screen 2

ISF screen 2 contains PIN transport key information. Descriptions of the fields required for dynamic key management are as follows.

PIN TRANSPORT KEY INFORMATION

The following fields contain information for keys used to transport PINs between an external entity and BASE24.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is, transactions sent by the RBSI-compliant device to the Integrated Authorization Server process).

Because external PIN management must be performed in hardware for dynamic key management, only security module-encrypted keys are entered in these fields.

KPT1 — The key utilized by the RBSI-compliant device to encrypt PINs in transactions sent to BASE24. The Integrated Authorization Server process uses the key in this field when the CURRENT INDEX field contains a value of 0. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

VVK1 — The verification value associated with the value in the KPT1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

KPT2 — The key utilized by the RBSI-compliant device to encrypt PINs in transactions sent to BASE24. The Integrated Authorization Server process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

VVK2 — The verification value associated with the value in the KPT2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

CURRENT INDEX — Identifies the key the interface is currently using. Valid values are as follows:

- 0 = Value in the KPT1 field
- 1 = Value in the KPT2 field

The Interface Master process changes the value in this field only after it has sent a new inbound PIN encryption key to the RBSI-compliant device, and the RBSI-compliant device has sent an RSM to the process indicating that the new key was successfully verified.

KEY COUNTER — The number of times the key has been changed by the Interface Master process for dynamic key management processing.

This counter is updated whenever a new inbound PIN encryption key is generated by a security module and stored in the ISF. This value should require adjustment using file maintenance only if the external entity expects a value other than zero when dynamic key management is established. Valid values for each position in this field are 0 through 9 and A through F. This field must contain six valid characters.

When BASE24 Remote Banking sends a new inbound PIN encryption key to an RBSI-compliant device, it expects the RBSI-compliant device to accept or adopt the key counter value from this field.

ITS Interface Configuration File Settings

The ITS Interface Configuration File (IFCF) contains configuration information for RBSI-compliant devices, including dynamic key management thresholds.

VCD Screen 1

VCD screen 1 defines the processing options associated with a particular interface instance. The following fields on VCD screen 1 should be considered when configuring dynamic key management for an interface instance.

NMM ENABLED — A code indicating whether BASE24 products can send network management messages to this interface instance. Valid values are as follows:

- Y = Yes, BASE24 products can send network management messages to this interface instance.
- N = No, BASE24 products cannot send network management messages to this interface instance.

This field must be set to a value of Y when dynamic key management is used. Network management messages are used in dynamic key management processing.

KEY MGMT TYPE — A code identifying the type of key management processing this interface instance can perform. Valid values are as follows:

- M = Dynamic key management with the BASE24 product as the master.
- N = No dynamic key management.

This field must be set to a value of M to enable dynamic key management for an interface instance.

IMP SYMBOLIC NAME — The symbolic name of the Interface Master process that performs dynamic key management for this RBSI-compliant device.

VCD Screen 2

VCD screen 2 defines additional processing options associated with an interface instance. The following field on VCD screen 2 should be considered when configuring dynamic key management for an interface instance.

NETWORK MANAGEMENT — The number of seconds that a BASE24 product waits for a network management message (message type 1804) after transmitting an authorization response containing a new key to an RBSI-compliant device. Network management messages include dynamic key management messages. This field must be set to a reasonable non-zero value for dynamic key management. Valid values are 0 through 99999.

VCD Screen 4

VCD screen 4 defines the thresholds for dynamic key management. This screen appears only when the KEY MGMT TYPE field on VCD screen 1 contains a valid value other than N.

Three types of keys are displayed on VCD screen 4. These keys are identified by the following abbreviations:

KAM = This key is used to authenticate message data.

KDT = This key is used to encrypt data. This key also is used with a PIN transport key (KPT) to encrypt PINs for transport.

KPT = This key is used to encrypt PINs.

Only the KPT keys are used for dynamic key management. The following fields on VCD screen 4 should be considered when configuring dynamic key management for an interface instance.

LIMITS

The following fields are used to set the dynamic key management parameters for this interface instance. Keys can be changed based on time intervals, transaction activity, the number of errors since the last key change, or the number of consecutive errors. These limits can be used individually or in any combination.

At least one PIN transport key (KPT) limit on this screen must be set to a nonzero value for the interface instance using this record to initiate dynamic key management for PINs, unless you are initiating key changes using the KPT NEW command only.

The counters and timers for the key are reset when any one of the configured count or time limits is reached, the KPT NEW command is received for the interface instance, or the Interface Master process is initialized. The timer limit is also reset when the Interface Master Services object in the process receives an LCONF Warmboot command.

Each limit can be set for PIN transport keys (KPTs) only. A value of 0 in a field indicates that the limit is not used.

KEY SYNC ERRORS — The number of sanity check errors allowed before BASE24 Remote Banking performs a key change. Sanity check errors can occur when decrypting the PIN block. Valid values are as follows:

- 0 = This parameter is not used.
- 1–9999 = The maximum number of errors.

TIMER VALUE — The maximum length of time, in minutes, that a key should be used before it is changed. Valid values are 0 through 99999 minutes.

TRANSACTIONS — The maximum number of transactions containing a PIN that can be performed with the current key before it must be changed. Valid values are as follows:

- 0 = This parameter is not used.
- 1–99999 = The maximum number of transactions.

ERRORS — The maximum number of translation errors that can occur with the current key before it must be changed. Translation errors include both PIN length errors and sanity check errors. Valid values are as follows:

- 0 = This parameter is not used.
- 1–9999 = The maximum number of errors.

CONSECUTIVE ERRORS — The maximum number of consecutive translation errors that can occur with the current key. Translation errors include both PIN length errors and sanity check errors. Valid values are as follows:

- 0 = This parameter is not used.
- 1–9999 = The maximum number of errors.

Changing a key or using it successfully resets the counter associated with this limit.

KEY SYNC ERROR ACTION — This field is not used by BASE24 Remote Banking for dynamic key management. The action is always to perform a key change.

KPT CONSECUTIVE ERROR ACTION — This field is not used by BASE24 Remote Banking for dynamic key management. The action is always to perform a key change.

NUMBER OF KEYS — A code indicating whether the inbound and outbound keys are combined or separate. Valid values are as follows:

- 1 = Combined. The inbound and outbound PIN transport keys (KPTs) are the same, the inbound and outbound message authentication keys (KAMs) are the same, and the inbound and outbound data transport keys (KDTs) are the same.
- 2 = Separate. The inbound and outbound KPTs are different, the inbound and outbound KAMs are different, and the inbound and outbound KDTs are different.

This field must be set to a value of 1 for dynamic key management.

ORIGINATING ID — Indicates the sender of dynamic key management messages.

The values in the RECEIVING ID and ORIGINATING ID fields need to be agreed upon by the sender and receiver. The sender's originating ID is the receiver's receiving ID and the sender's receiving ID is the receiver's originating ID. For example, if you have agreed on the IDs AAAA for the remote banking endpoint and BBBB for the BASE24 transaction processing system, this field would contain the value BBBB (BASE24 system's originating ID) and the RECEIVING ID field would contain the value AAAA. These values do not need to match any specific values in the BASE24 database.

BASE24 products place the originating ID and receiving ID in the dynamic key management messages that they send to a remote banking endpoint. These IDs can provide the remote banking endpoint another way of checking that a message has been routed properly. For example, assume you use the IDs AAAA and BBBB described above. A message from the BASE24 system to the remote banking endpoint has an originating ID of BBBB and a receiving ID of AAAA. The remote banking endpoint or host can check the receiving ID, accepting the

message only if it contains a receiving ID of AAAA. The message returned from the remote banking endpoint has an originating ID of AAAA and a receiving ID of BBBB; however, BASE24 products do not verify these IDs in the messages that they receive from the remote banking endpoint.

RECEIVING ID — Indicates the receiver of dynamic key management messages.

Refer to the ORIGINATING ID field description for an example of how this field is used.

Applied Communications, Inc.

Section 12

External Processor Dynamic Key Management

This section describes the dynamic key management options provided by the ISO Host Interface process.

The information presented includes the database requirements and describes the network management messages for dynamic key management, including the information contained in the Cryptographic Service Message (CSM) element of the ISO external messages.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC) or co-network as defined in the “Definition of Terms” discussion in section 1.

Dynamic Key Management

The ISO Host Interface process uses working keys to protect the transaction data moving through the BASE24 network. The installation and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management.

Dynamic key management allows an external processor to automatically change working keys based on a number of factors. Thresholds can be established based on time, number of transactions, number of errors, number of consecutive errors, or any combination of these factors. One or more working keys can be changed when any of the defined thresholds is surpassed. Working keys can also be changed using a network control command.

Dynamic key management provided by the ISO Host Interface process is based on the ANSI X9.17 standard, ***Financial Institution Key Management (Wholesale)*** and the ISO 8732 standard, ***Banking—Key Management (Wholesale)***.

Dynamic key management requires the use of security modules to perform PIN encryption, PIN verification, and message authentication, and is not supported when these functions are performed in software.

Working Keys

The ISO Host Interface process uses four working keys, which are described in the paragraphs that follow.

Inbound PIN Encryption Keys — If a PIN is encrypted coming from the external processor, this is the key under which it is encrypted. If a PIN is in the clear coming from the external processor and must be encrypted by BASE24 Remote Banking, this is the key under which it is to be encrypted.

Outbound PIN Encryption Keys — If PIN encryption is required by the external processor, this is the key under which the PIN is to be encrypted by BASE24 Remote Banking.

Inbound Message Authentication Keys — If inbound messages from the external processor contain a message authentication code (MAC), this is the key the ISO Host Interface process uses to verify the MAC.

Outbound Message Authentication Keys — If outbound messages for the external processor contain a MAC, this is the key the ISO Host Interface process uses to generate the MAC.

Key Exchange Keys

Dynamic key management requires key exchange keys (KEKs) in addition to the working keys. Working keys must be encrypted before being exchanged between BASE24 Remote Banking and the external processor. Two KEKs are used to encrypt the working keys: one for PIN encryption working keys and one for message authentication working keys. These KEKs must be manually exchanged and entered in the BASE24 database before dynamic key management can occur.

Atalla uses the term *key exchange key* (KEK) to describe the keys used for exchanging PIN keys and MAC keys. Racal-Guardata uses the term *zone master key* (ZMK) to describe the key used for exchanging PIN keys and MAC keys. References to the key exchange key (KEK) in BASE24 documentation include both the Atalla and Racal-Guardata keys.

Key Management Processors

Key management involves two parties: the ISO Host Interface process and the external processor. Each party has specific key management responsibilities.

The processor type assigned to each party identifies its responsibilities when performing key management, whether key management is performed manually or automatically. There are three possible processor types, each of which is described below.

Master Key Processor — The master key processor controls all key generation. When one key management party is the master, the other party must be the slave. The master can generate a key itself or in response to a request from the slave.

Slave Key Processor — The slave key processor has no responsibility for key generation. When one key management party is the slave, the other party must be the master. The slave cannot generate a key, but can initiate a key change by requesting the master to generate a new key. The slave can, however, accept or reject a key generated by the master.

Co-Network Key Processor — A co-network key processor shares key management responsibilities with another co-network key processor. When one key management party is a co-network, the other party must also be a co-network.

The distinction between inbound and outbound keys is important in this configuration. Each party is responsible for generating the key used as its outbound key. The key that serves as the inbound key for one co-network serves as the outbound key for the other co-network, as shown below.

In this example, co-network 1 (BASE24 Remote Banking) is responsible for generating key A because co-network 1 uses it for outbound messages. Co-network 2 (external processor) is responsible for generating key B because co-network 2 uses it for outbound messages.

Key	Co-network 1 (BASE24 Remote Banking)	Co-network 2 (external processor)
A	Outbound key	Inbound key
B	Inbound key	Outbound key

Note: The term co-network typically applies to the BIC Interface external processor. When defining key processing responsibilities, however, the term co-network applies to BASE24 Remote Banking and the DPC as well.

Key Management Messages

Key management messages make up a subset of network management (0800) messages. There are four key management messages, each of which is described below.

Note: The change key and new key messages can be generated automatically as the result of thresholds being surpassed. Thresholds are described in the topic “Dynamic Key Management Thresholds” presented later in this section. In addition, all four of these messages can be initiated by an operator issuing a network control command. For more information on using network control commands to generate key management messages, and for complete syntax of the commands involved, refer to the ***BASE24 Text Command Reference Manual***.

Change Key Message — A change key message requests that a processor generate a new PIN encryption or message authentication key. The key-generating processor sends a response to the change key message and then, in a separate action, generates the new key and sends it to the requesting processor in a new key message.

The requesting processor sends this type of message to a key-generating processor when the requesting processor has determined that the key needs to be changed. The requesting processor can make that determination based on a threshold being passed or in response to a Key Change command (the Key Change command has the same effect as passing a threshold).

New Key Message — A new key message carries a new PIN encryption or message authentication key between processors. The key-using processor verifies and begins using the new key and returns a response to the key-generating processor indicating that the new key has been implemented.

A key-generating processor sends this type of message to a key-using processor when the key-generating processor has either determined that the key needs to be changed or been requested by the key-using processor to change or repeat the key. This can occur as a result of a threshold being passed, notification from the key-using processor (using a change key or repeat key message), or in response to a Key Change command (the Key Change command has the same effect as passing a threshold).

Repeat Key Message — A repeat key message requests that a key-generating processor resend a key that was generated earlier. Upon receipt of a repeat key message, the key-generating processor obtains the requested key from its database and sends it to the key-using processor in a new key message. The key itself is not changed, nor are any threshold counters affected.

The ISO Host Interface process that is the key-using processor generates the repeat key message only as the result of receiving a Key Repeat command. The ISO Host Interface process that is the key-generating processor responds to a repeat key message from the key-using processor with a new key message carrying the requested key.

Verify Key Message — A verify key message requests that a key-generating processor verify the key by comparing check digits. The verify key message contains check digits for the key in question, and the processor receiving the message compares the check digits in the message with the check digits it has on file for the particular key. The key-generating processor then responds to the verify key message with the results of the check digit comparison. (If the check digits do not match, the key-using processor that sent the verify key message should use a change key message to request a new key.)

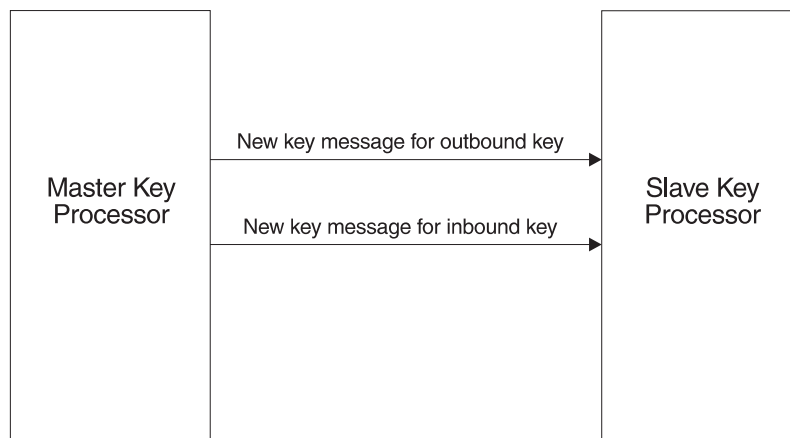
The ISO Host Interface process that is the key-using processor generates the verify key message only as the result of receiving a Key Verify command. The ISO Host Interface process that is the key-generating processor responds to a verify key message from the key-using processor by comparing check digits for the specified key and returning the results.

Processing Key Management Messages

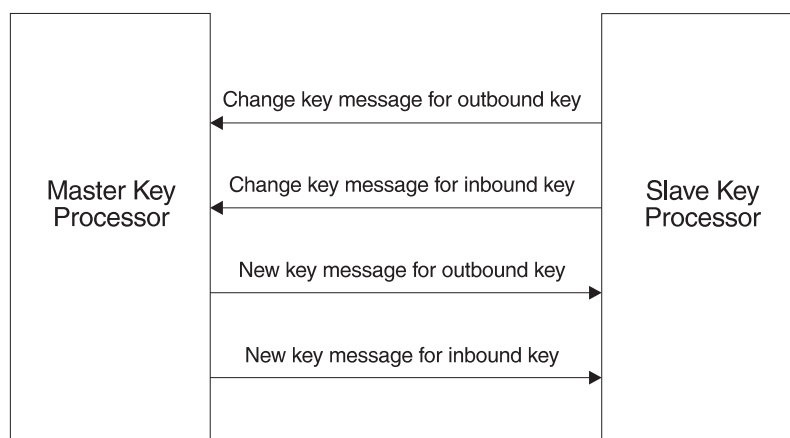
All four key management messages can be requested for the inbound key only, the outbound key only, or both keys. However, the key management messages generated by the interface processes contain only one key, so a request affecting both keys results in two key management messages being generated if the inbound and outbound keys are separate and unique. If the keys are the same, as set up in the NUMBER OF KEYS field on KEYF screen 1 and the KEY PROCESSING TYPE field on KEYF screen 3, both the outbound and inbound keys are replaced with one key. If the external processor is a co-network, the inbound and outbound keys must be separate and unique. The NUMBER OF KEYS and KEY PROCESSING TYPE fields in the KEYF must be set in a consistent manner.

This processing is demonstrated in the following example. In this case, an interface process receives a Key Change command for both (inbound and outbound) keys.

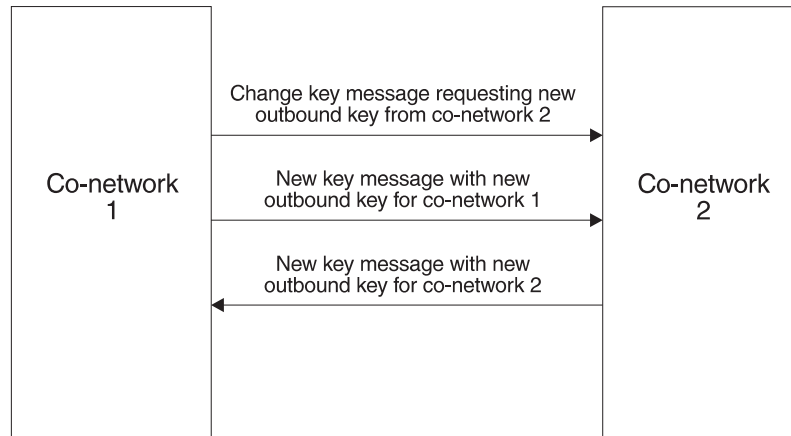
When the ISO Host Interface process is the master key processor and is configured for two keys, it sends two new key messages to the slave key processor as shown below: one for the key used by the master key processor as its outbound key and one for the key used by the master key processor as its inbound key.



When the ISO Host Interface process is the slave key processor and is configured for two keys, it sends two change key messages to the master key processor as shown below: one for the key used by the slave key processor as its outbound key and one for the key used by the slave key processor as its inbound key. The master key processor responds by sending a new key message for each key.



When the ISO Host Interface process is a co-network key processor and is configured for two keys, it sends a change key message and a new key message to the second co-network key processor as shown below. The change key message is a request to change the key used as an outbound key by the second co-network key processor. The new key message contains the new key to be used as an outbound key by the first co-network key processor. The second co-network receives the change key message and responds by sending a new key message that contains the new outbound key used by the second co-network key processor.



Network Management Messages

Key management messages are network management messages. Each key management message is a request or a response. For ISO messages, the request is an 0800 message and the response is an 0810 message. The following subsections describe the ISO message elements used to carry the data required for key management messages.

For more information on ISO messages, refer to the following manuals:

- ***BASE24 ISO Host Interface Manual***
- ***BASE24 External Message Manual***

ISO Messages

This subsection describes the ISO data elements used in 0800 or 0810 key management messages.

Data Element	Description
Network Management Information Code (S-70)	Identifies the type of key management messages being sent. Valid values are as follows: 161 = Change key 162 = New key 163 = Repeat key 164 = Verify key
Response Code (P-39)	Identifies whether the key management process is successful. Valid values are as follows: 00 = Approved 05 = Declined 12 = Declined—invalid check digits
Network Management Message Additional Data (P-48)	Includes the following five fields: Number of Keys — Indicates the number of keys supported by the interface process. Valid values are as follows: 1 = Combined keys (inbound and outbound keys are equal) 0, or 2 = Separate keys (0 means blank) Key Length — Indicates the type of key management supported by the interface process. Valid values are as follows: 0 or 1 = Single-length keys 2 = Double-length keys (inbound and outbound keys are equal) Key Processor Type — Indicates if the interface process is the master, slave, or co-network key processor. Valid values are as follows: 0 or N = None or not applicable C = Co-network M = Master S = Slave

Continued on the following page

Data Element	Description
Network Management Message Additional Data (P-48) <i>continued</i>	MAC Type — Indicates the level of MACs supported by the interface process. Valid values are as follows: 0 or 0 = No MAC support (0 means blank) 1 = Hardware MAC Support 2 = Software MAC support MAC Data Type — Indicates the character set in which the data will be authenticated. Valid values are as follows: 0 or 0 = ASCII (0 means blank) 1 = EBCDIC
Security-Related Control Information (P-53)	Includes the following two fields: Key Type — Identifies whether the key being exchanged is used for message authentication or PIN encryption. Valid values are as follows: 00 = PIN key 01 = MAC key Key Direction — Identifies whether the key being exchanged is used for inbound messages, outbound messages, or both. Valid values are as follows: 00 = Outbound messages only 02 = Inbound messages only 03 = Both inbound and outbound messages
BASE24 Key Management (S-120)	Contains the check digits for the key being exchanged.
Cryptographic Service Message (S-123)	Contains the Cryptographic Service Message (CSM), which is used for moving key management data between processors. This message is discussed later in this section.
Primary Message Authentication Code (P-64)	Contains the MAC if message authentication is performed and secondary data elements are not included in the message.
Secondary Message Authentication Code (S-128)	Contains the MAC if message authentication is performed and secondary data elements are included in the message.

Cryptographic Service Message

The ANSI X9.17 standard, *Financial Institution Key Management (Wholesale)*, establishes standards for key management. This standard also defines the Cryptographic Service Message (CSM) used for moving key management data between processors when the keys are distributed automatically. CSM information is contained in the Cryptographic Service Message data element (S-123) of the ISO external message.

Key Distribution Environment

BASE24 Remote Banking uses the point-to-point key distribution environment. In this environment, two parties share a key encryption key so that further data keys can be exchanged and one or both of the parties has the capability to generate or otherwise acquire keys.

Message Classes

BASE24 Remote Banking supports the following classes of CSMs used in point-to-point environments:

- Request Service Initiation Message (RSI)
- Key Service Message (KSM)
- Response Service Message (RSM)
- Error Service Message (ESM)

Message Fields and Subfields

Each CSM class contains several fields, with many of the fields containing subfields. Some fields and subfields are mandatory while others are optional. Because of this flexibility, each field begins with a unique 2- or 3-character identifier and ends with a blank. The CSM itself always begins with the literal CSM, and the message contents are always carried between a pair of parentheses.

The fields shown in the following table are the only ones that appear in the CSMs created by BASE24 Remote Banking. There are five CSM formats because the ESM class can have two formats, depending on the type of error being reported. Each of the CSM formats is described in greater detail, following the table.

CSM Class	RSI	KSM	RSM	ESM	
				With Counts	Without Counts
CSM Fields	MCL RCV ORG SVR	MCL RCV ORG KD CTP	MCL RCV ORG	MCL RCV ORG CTP CTR ERF	MCL RCV ORG ERF

When the ISO Host interface process receives CSMs generated by an external processor, it searches only for the fields shown in the preceding table. It is recommended, but not mandatory, that the fields in the message be kept in the order shown in the table.

Message Formats

BASE24 Remote Banking dynamic key management processing uses the CSM to move key information between BASE24 and the external processor in three of the four key management messages described earlier in this section: Change Key messages, New Key messages, and Repeat Key messages. The Verify Key message does not use the CSM because it only moves check digits and does not need the additional information contained in the CSM.

The following describes and presents examples of the BASE24 Remote Banking supported CSM classes:

Request Service Initiation Message (RSI) — Used when a key change request is being passed between processors. An example of a CSM containing the RSI format is shown below:

CSM(MCL/RSI| RCV/1234567890123456| ORG/1234567890123456| SVR/I)

Each field in the message is preceded by one of the following identifiers. The identifiers are shown in boldface type. The blanks identified by the character **␣** in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

- MCL/** = Message class (RSI)
Field Length: 3 alphabetic characters
- RCV/** = Receiver of the message
Field Length: 4–16 alphanumeric characters
- ORG/** = Originator of the message
Field Length: 4–16 alphanumeric characters
- SVR/** = Service request
Field Length: 0

Note: The absence of a value following this identifier implies a request for one data key.

Key Service Message (KSM) — Used when a new key is passed between processors. An example of a CSM containing the KSM format is shown below:

```
CSM(MCL/KSM␣RCV/1234567890123456␣ORG/1234567890123456␣  
KD/1234567890123456␣CTP/12345678901234␣)
```

The information in this example is actually a continuous line. It has been broken here due to space constraints. Each field in the message is preceded by one of the following identifiers. The identifiers are shown in boldface type. The blanks identified by the character **␣** in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

- MCL/** = Message class (KSM)
Field Length: 3 alphabetic characters
- RCV/** = Receiver of the message
Field Length: 4–16 alphanumeric characters
- ORG/** = Originator of the message
Field Length: 4–16 alphanumeric characters

KD/ = New key value, generated by the security module for a new key request and taken from the database for a repeat key message
Field Length: 16 hexadecimal characters

CTP/ = Hexadecimal key counter
Field Length: 1–14 hexadecimal characters

Response Service Message (RSM) — Used to respond to a KSM that is processed successfully. An example of a CSM containing the RSM format is shown below:

CSM(**MCL**/RSM|**RCV**/1234567890123456|**ORG**/1234567890123456|)

Each field in the message is preceded by one of the following identifiers. The identifiers are shown in boldface type. The blanks identified by the character | in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

MCL/ = Message class (RSM)
Field Length: 3 alphabetic characters

RCV/ = Receiver of the message
Field Length: 4–16 alphanumeric characters

ORG/ = Originator of the message
Field Length: 4–16 alphanumeric characters

Error Service Message (ESM) — Used to respond to a Request Service Initiation Message (RSI) or a Key Service Message (KSM) that cannot be processed successfully. The ESM can have two formats, depending on the type of error being reported. The two counter fields are only included in the message when the expected key count (CTP field) and the key count received from the message originator (CTR field) do not match. The error code (ERF field) in the examples below can contain only alphabetic characters. Numbers are used to demonstrate field length.

An example of a CSM containing the ESM format with the key counters is shown below:

CSM(**MCL**/ESM|**RCV**/1234567890123456|**ORG**/1234567890123456|
CTP/12345678901234|**CTR**/12345678901234|**ERF**/1234567890123456|)

An example of a CSM containing the ESM format without the key counters is shown below:

CSM(**MCL**/ESM|**RCV**/1234567890123456|**ORG**/1234567890123456|
ERF/1234567890123456|)

The information in each of these examples is actually a continuous line. It has been broken here due to space constraints. Each field in the messages is preceded by one of the following identifiers. The identifiers are shown in boldface type. The blanks identified by the character | in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

- MCL**/ = Message class (ESM)
Field Length: 3 alphabetic characters
- RCV**/ = Receiver of the message
Field Length: 4–16 alphanumeric characters
- ORG**/ = Originator of the message
Field Length: 4–16 alphanumeric characters
- CTP**/ = Hexadecimal key count expected by the receiver of the KSM. This identifier and field are only included in the message when error code P is returned.
Field Length: 1–14 hexadecimal characters
- CTR**/ = Hexadecimal key count originally sent in the KSM by the message originator. This identifier and field are only included in the message when error code P is returned.
Field Length: 1–14 hexadecimal characters
- ERF**/ = The following codes identify the errors that were detected during processing.
- C = Cannot process (general error)
 - F = Format error
 - H = Invalid receiver ID or originator ID
 - P = The value in the CTP field of the KSM does not match the expected count
- Field Length: 1-16 alphabetic characters

Message Length

The use of multiple CSM formats makes the CSM a variable-length element in the ISO external message. The CSM can also be a fixed-length element in the ISO external message.

The ISO Host Interface process uses the values in the MESSAGE FORMAT fields on all screens of the Host Configuration File (HCF). If any of these fields indicates that fixed-length messages are used, the fixed-length format is used for the CSM. If all of these fields indicate that variable-length messages are used, the variable-length format is used for the CSM.

Fixed-Length Messages

When the CSM is used as a fixed-length element, the data is left-justified within each field of the CSM format being used and each field is blank filled, if necessary, to obtain its maximum length. The CSM format is then left-justified in the external message data element and the data element is blank filled to reach the specified length.

In the fixed-length ISO message, the CSM (S-123) has a length of 150 bytes, excluding the 3-position field length indicator. Although the total length of the CSM is 150 bytes, the longest CSM is currently 114 bytes. Therefore, positions 115 through 150 of the CSM always contain blanks.

The ESM can have two lengths, depending on whether it includes the two key count fields (CTP and CTR). The key count fields are only included when the error code field (ERF) contains a P (counts do not match). Neither key count field is included when the error code is a value other than P.

CSM Format	RSI	KSM	RSM	ESM	
				With Counts	Without Counts
CSM Data Length	60	94	55	114	76

Variable-Length Messages

When the CSM is used as a variable-length element, the length of each field in the CSM is determined by the data it carries. The message format field descriptions presented earlier in this section include the range of possible lengths for each field.

All fields are required except for the key count fields (CTP and CTR) in the ESM. The key count fields are conditional in the ESM, depending on whether the error code field (ERF) contains a P (counts do not match). Neither key count field is included when the error code is a value other than P. Data element S-123 can have the following lengths (excluding the 3-position field length indicator), depending on the CSM format it contains.

CSM Format	RSI	KSM	RSM	ESM	
				With Counts	Without Counts
Minimum Length	36	57	31	49	37
Maximum Length	60	94	55	114	76

Examples

The following examples show an RSM when the receiver and originator are less than the maximum length of 16 alphanumeric characters. The first example illustrates a variable-length format and the second example illustrates a fixed-length format. The variable-length RSM shown has a length of 36 characters while the fixed-length RSM shown maintains its length of 55 characters because each field is padded with blanks. The field identifiers are shown in boldface type, while blanks are identified by the character `b` in the examples.

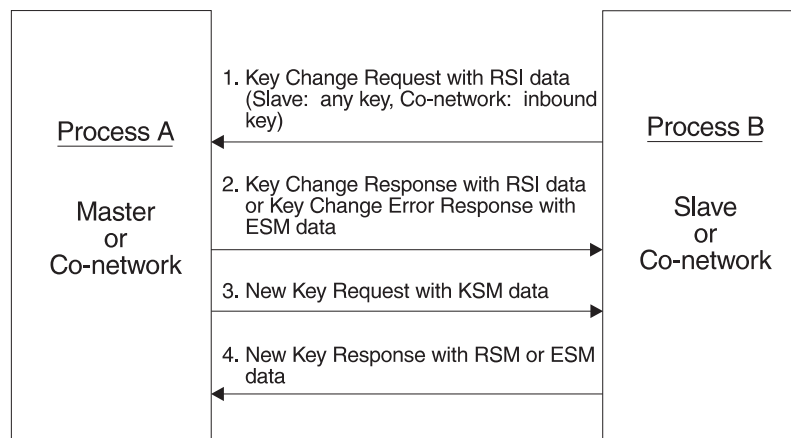
CSM(~~MCL/RSM/RCV/ABC123/ORG/DEFG456~~)
CSM(~~MCL/RSM/RCV/ABC123~~~~bbbbbbbbbb~~~~ORG/DEFG456~~~~bbbbbbbbbb~~)

Message Flow Diagrams

The following subsections present message flow diagrams for key change, new key, and repeat key transactions.

Key Change

The following diagram illustrates how messages containing CSM formats are used in a key change request.

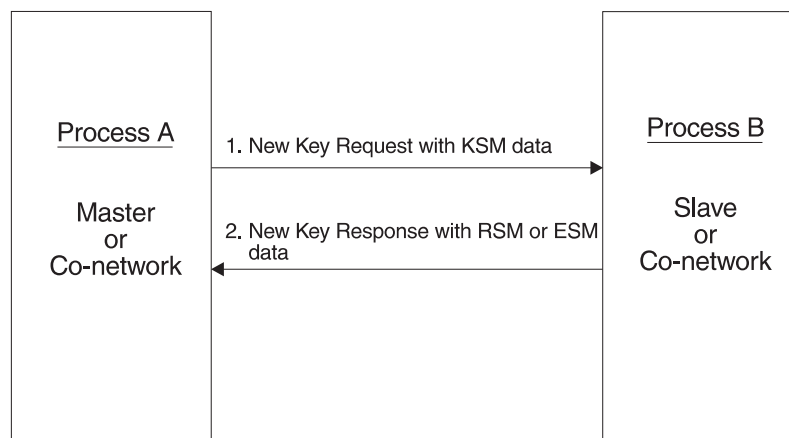


1. The process requesting a key change (process B) sends a request message containing an RSI to the process responsible for generating the key (process A).

2. Process A responds to process B. This response message notifies process B that the key change request has been received, but does not return the new key. The CSM in the response contains the RSI echoed from the original key change request when no errors are detected in the request and the new key can be generated. The CSM in the response contains an ESM when errors are found in the request and the new key cannot be generated.
3. Process A sends a request message containing a KSM with the new key information to process B.
4. Process B responds to process A. This response message contains an RSM when no errors are detected in the KSM or an ESM when errors are found in the KSM.

New Key

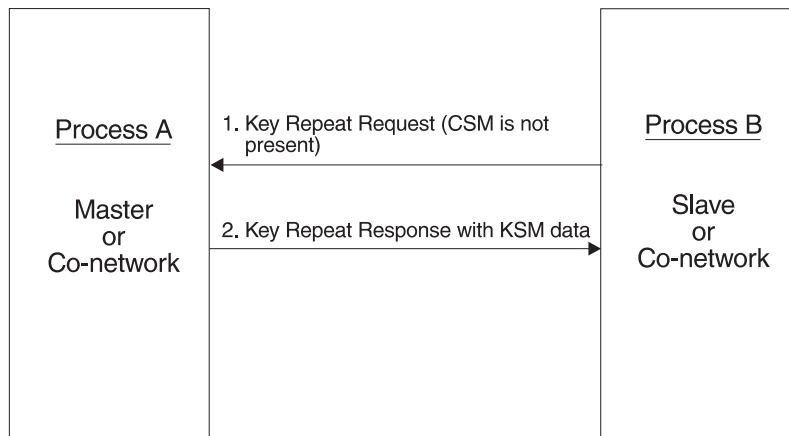
The following diagram illustrates how messages containing CSM formats are used when a new key is passed between processors.



1. The process generating the key (process A) sends a request message containing a KSM with the new key information to the receiving process (process B) without receiving a request from process B.
2. Process B responds to process A. This response message contains an RSM when no errors are detected in the KSM or an ESM when errors are found in the KSM.

Repeat Key

The following diagram illustrates how messages containing CSM formats are used when a key is repeated.



1. The process that originally received the key (process B) sends a key repeat request message to the process that originally generated the key (process A). The CSM is not present in this message unless this element is configured in the External Message File (EMF) as a mandatory element.
2. Process A sends a response message containing a KSM with the existing key information to process B. Process A does not change the key or any key-related counters before sending the response. Since the KSM is in a response, process B cannot return an RSM or ESM.

Note: This command may not work on systems using Racal-Guardata security modules. The repeat key command uses the GC transaction code. In some Racal-Guardata security module firmware versions, the GC transaction code may be available only when the security module is in an authorized state. The security module should not be in an authorized state during normal online processing. Check with your Racal-Guardata customer support representative for more information regarding the capabilities of your security module and its firmware.

Dynamic Key Management Thresholds

BASE24 Remote Banking dynamic key management supports four types of thresholds for changing PIN encryption keys and five types of thresholds for changing message authentication keys. One or more of the thresholds can be established for each type of key; however, dynamic key management only occurs when at least one threshold is established for each type of key. One set of thresholds is used for both inbound and outbound keys and counters are maintained for each. Therefore, if the keys are separate (that is, the NUMBER OF KEYS field on KEYF screen 1 is set to 2), two sets of counters are maintained. When the first threshold is exceeded, the appropriate key (that is, PIN key or MAC key) is generated. The first threshold can be exceeded by the slave process or by the co-network and its key will be generated. These thresholds are set on KEYF screen 3.

BASE24 Remote Banking dynamic key management thresholds are based on the following information:

- Length of time (used with both key types)
- Number of transactions (used with both key types)
- Number of errors (used with both key types)
- Number of consecutive errors (used with both key types)
- Number of synchronization errors (used with message authentication keys only)

The length-of-time thresholds use timers to track the amount of time that has elapsed since the last key exchange. The remaining thresholds use counters to track the number of transactions or errors that have occurred since the last key exchange. The circumstances that cause a timer or counter to be increased or reset depend on the threshold being monitored. These thresholds are described in the following subsections.

Note: When new PIN encryption and message authentication working keys are both being generated, the message authentication working keys must be generated first so they can be available for generating the MAC in the new PIN key messages.

Time Threshold

The length-of-time threshold is based solely on the amount of time that has elapsed since the timer was set. The time is not updated based on transaction activity and this threshold is not checked during transaction processing. When this timer expires, the interface process sends a key change message or a new key message for the key in question to the external processor. A timer can be maintained for each of the four working keys (that is, inbound PIN encryption, outbound PIN encryption, inbound message authentication, and outbound message authentication) or, if the inbound and outbound keys are the same, one timer can be maintained for PIN encryption keys and one for message authentication keys.

A timer is reset in the following situations:

- When the interface process is initialized or receives a warmboot command. A check is performed at logon and logoff to determine if the time limit has changed and if the timer should be deleted, added, or remain as is (not actually reset).
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.

Transaction Thresholds

Each of the transaction thresholds is based on the number of transactions that have occurred since its corresponding transaction counter was set. Each time a counter is increased, it is compared to its threshold. When the counter exceeds its threshold, the interface process sends a key change message or a new key message for the key in question to the external processor. For each threshold, a counter can be maintained for each of the four working keys or, if the inbound and outbound keys are the same, one counter is maintained for PIN encryption keys and one for message authentication keys.

Number of Transactions

The message content and direction determine which transaction counter is increased by the interface process:

- The inbound PIN transaction counter is increased each time an external message is received from an external processor containing an internal or external encrypted PIN.
- The outbound PIN transaction counter is increased each time an internal message is received from the Integrated Authorization Server process containing an internal or external encrypted PIN.
- The inbound message authentication transaction counter is increased each time the external message received from an external processor requires MAC verification.
- The outbound message authentication transaction counter is increased each time a MAC must be generated before the external message is sent to an external processor.

A transaction counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key change message is sent.

Number of Errors

The message content and direction determine which total error counter is increased by the interface process:

- The inbound PIN total error counter is increased each time an error occurs while attempting to process an encrypted PIN contained in the external message received from an external processor. This occurs when the external PIN is encrypted and the internal PIN is in the clear.
- The outbound PIN total error counter is increased each time an error occurs while attempting to generate an encrypted PIN from the internal message to the external message to send to the external processor.

- The inbound message authentication total error counter is increased each time the external message received from an external processor requires MAC verification and the verification attempt fails, and the check digits received in the response from the security module match the check digits maintained in the BASE24 database.

This counter is also increased when the NUMBER OF KEYS field on KEYF screen 1 is set to 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 197, and the message is not a MAC key change message.

- The outbound message authentication total error counter is increased when the NUMBER OF KEYS field on KEYF screen 1 is set to 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 197, and the message is not a MAC key change message.

A total error counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key change message is sent.

Number of Consecutive Errors

The message content and direction determine which consecutive error counter is increased by the interface process:

- The inbound PIN consecutive error counter is increased each time an error occurs while attempting to process an encrypted PIN contained in the external message received from an external processor. This occurs when the external PIN is encrypted and the internal PIN is in the clear.
- The outbound PIN consecutive error counter is increased each time an error occurs while attempting to process an encrypted PIN contained in the external message received from an external processor.
- The inbound message authentication consecutive error counter is increased each time the external message received from an external processor requires MAC verification and the verification attempt fails. The consecutive error

counter is increased without regard to whether the check digits received in the response from the security module match the check digits maintained in the BASE24 database.

This counter is also increased when the NUMBER OF KEYS field on KEYF screen 1 is set to 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 197, and the message is not a MAC key change message.

- The outbound message authentication consecutive error counter is increased when the NUMBER OF KEYS field on KEYF screen 1 is set to 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 196 or 197, and the message is not a MAC key change message.

A consecutive error counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key change message is sent.
- When the key it is associated with is used successfully. For outbound MAC messages, this is when a response is received.
- When an approved response is received from a request that contained a hardware-encrypted PIN.

Number of Synchronization Errors

The message content and direction determine which message authentication key (KMAC) error counter is increased by the interface process:

- The inbound KMAC error counter is increased each time the external message received from an external processor requires MAC verification, the verification attempt fails, and the check digits received in the response from the security module do not match the check digits maintained in the BASE24 database.

This counter is also increased when the NUMBER OF KEYS field on KEYF screen 1 is set to 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 196, and the message is not a MAC key change message.

- The outbound KMAC error counter is increased when the NUMBER OF KEYS field on KEYF screen 1 is set to 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 196, and the message is not a MAC key change message.

A KMAC error counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and processes it successfully.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key exchange message is sent.

Setting Up the BASE24 Database

Settings in the Key File (KEYF), External Message File (EMF), and Host Configuration File (HCF) must be considered when using dynamic key management. Appendix A contains illustrations of the EMF, HCF, and KEYF screens presented in this section.

For more information on KEYF, EMF, and HCF screens, refer to the ***BASE24 Base Files Maintenance Manual***.

Key File Settings

The following settings are needed in the Key File (KEYF) for each external processor using dynamic key management.

KEYF Screen 1

KEYF screen 1 contains PIN encryption, PIN translation, and message authentication parameters, intermediate keys, PIN key exchange keys, and message authentication code (MAC) key exchange keys. The following fields are required when using key management.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor.

- If any of the PIN key timer fields on KEYF screen 3 are set to a nonzero value, this field must be set to 1 (security module PIN management).
- If the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 contains a nonzero value or the KEY PROCESSING TYPE field on KEYF screen 3 contains a value other than N (N indicates no dynamic key management is to be performed), this field or the MAC ENCRYPT TYPE field must be set to 1.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages.

- If dynamic key management is used, which requires the use of security modules for PIN encryption, this field must be set to a value of 1 (ANSI PIN block) or 3 (PIN/PAD PIN block).
- If dynamic key management is not used for PIN keys, this field can be set to any valid value.

MAC ENCRYPT TYPE — Specifies how message authentication will be performed for messages between the external processor and the interface process.

- If any of the MAC key timer fields on KEYF screen 3 are set to a nonzero value, this field must be set to 1 (security module MAC processing).
- If the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 contains a nonzero value or the KEY PROCESSING TYPE field on KEYF screen 3 contains a value other than N (N indicates no dynamic key management is to be performed), this field or the ENCRYPT TYPE field must be set to 1.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 are combined or separate. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If the same key is used for inbound and outbound messages, this field must be set to 1.
- If different keys are used for inbound and outbound messages or the KEY PROCESSING TYPE field on KEYF screen 3 is set to C (co-network), this field must be set to 2.

KEY LENGTH — Identifies whether single- or double-length key exchange keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged using dynamic key management. The PIN KEK and MAC KEK must be the same length. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If single-length KEKs are used for key exchanges, this field must be set to 1.
- If double-length KEKs are used for key exchanges, this field must be set to 2.

EXCHANGE KEYS

The following fields must be set as indicated to create keys that BASE24 Remote Banking uses when MAC keys are automatically exchanged between BASE24 and the external processor.

PIN KEY — The security module-encrypted form of the key exchange key (KEK) used when inbound and outbound PIN keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. Valid values for each character in these fields are 0 through 9 and A through F. Both fields must contain 16 valid characters. If the value in the KEY LENGTH field is 1, the second set of characters must be zeros.

PIN KEY CHECK DIGITS — The check digits associated with the PIN KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with PIN KEKs.

MAC KEY — The security module-encrypted form of the key exchange key (KEK) used when inbound and outbound MAC keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. Valid values for each character in these fields are 0 through 9 and A through F. Both fields must contain 16 valid characters. If the value in the KEY LENGTH field is 1, the second set of characters must be zeros.

MAC KEY CHECK DIGITS — The check digits associated with the MAC KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with MAC KEKs.

KEYF Screen 2

KEYF screen 2 consists of inbound and outbound working keys for PINs and MACs. Descriptions of the fields required are as follows.

PIN KEY INFORMATION

The fields in the PIN KEY INFORMATION section on KEYF screen 2 contain working keys for PINs.

OUTBOUND KEYS

The following fields contain the PIN keys used for outbound transactions (that is, transactions sent by the interface from BASE24 to the external processor).

PIN KEY1 — The key used for the encryption or translation of PINs in transactions sent to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS1 — The check digits corresponding to the value in the PIN KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

PIN KEY2 — The key used for the encryption or translation of PINs in transactions sent to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS2 — The check digits corresponding to the value in the PIN KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

KEY COUNTER — The number of times the outbound PIN key has been changed by dynamic key management processing.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is, transactions sent by the external processor to BASE24).

PIN KEY1 — The key used by the external processor to encrypt PINs in the transaction sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS1 — The check digits corresponding to the value in the PIN KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

CURRENT INDEX — Indicates which inbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

PIN KEY2 — The key used by the external processor to encrypt PINs in the transaction sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS2 — The check digits corresponding to the value in the PIN KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

KEY COUNTER — The number of times the inbound PIN key has been changed by dynamic key management processing.

MAC KEY INFORMATION

The fields in the MAC KEY INFORMATION section on KEYF screen 2 contain working keys for message authentication codes (MACs).

OUTBOUND KEYS

The following fields contain working keys for message authentication codes (MACs).

Note: If an external processor supports automatic key changes, the MAC KEY1, MAC KEY2, CURRENT INDEX, CHECK DIGITS1, and CHECK DIGITS2 fields will automatically be set by the interface process when it is first started.

MAC KEY1 — The key used by the ISO Host Interface process to generate MACs in messages from BASE24 to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS1 — The check digits corresponding to the value in the MAC KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

CURRENT INDEX — Indicates which outbound MAC key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the MAC KEY1 field
- 2 = Value in the MAC KEY2 field

MAC KEY2 — The key used by the ISO Host Interface process to generate MACs in messages from BASE24 to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS2 — The check digits corresponding to the value in the MAC KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

KEY COUNTER — The number of times the outbound MAC key has been changed by dynamic key management processing.

INBOUND KEYS

The following fields contain the MAC keys used in inbound transactions (that is, transactions sent by an external processor to BASE24).

Note: If an external processor supports automatic key changes, the MAC KEY1, MAC KEY2, CURRENT INDEX, CHECK DIGITS1, and CHECK DIGITS2 fields will automatically be set by the interface process when it is first started.

MAC KEY1 — The key used by the ISO Host Interface process to verify MACs generated by the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS1 — The check digits corresponding to the value in the MAC KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

CURRENT INDEX — Indicates which inbound MAC key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the MAC KEY1 field
- 2 = Value in the MAC KEY2 field

MAC KEY2 — The key used by the ISO Host Interface process to verify MACs generated by the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS2 — The check digits corresponding to the value in the MAC KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMcom and RSMcom utilities.

KEY COUNTER — The number of times the inbound MAC key has been changed by dynamic key management processing.

KEYF Screen 3

KEYF screen 3 contains the fields used to set the thresholds for dynamic key management. Descriptions of the fields required are as follows. If all limits on KEYF screen 3 are set to zero, the interface process using the KEYF record does not initiate key changes automatically based on timers or counters, so the external processor must be responsible for initiating any key changes if dynamic key management is to occur. At least one PIN key limit on KEYF screen 3 must be set to a nonzero value for the ISO Host Interface process to initiate dynamic key management for PINs and at least one MAC key limit on KEYF screen 3 must be set to a nonzero value for the ISO Host Interface process to initiate dynamic key management for MACs.

PIN KEY VARIANT — Specifies the key variant to be applied to the PIN key exchange key (KEK) when translating PIN keys.

- If the ISO Host Interface process and the external processor use Atalla hardware security modules (ASM or NSP), this field must be set to 1 (variant 1).
- If the ISO Host Interface process uses an Atalla hardware security module (ASM or NSP) and the external processor uses a Racal-Guardata hardware security module, this field must be set to 1 (variant 1).
- If the ISO Host Interface process uses a Racal-Guardata hardware security module and the external processor uses an Atalla security module (ASM or NSP), this field must be set to 1 (variant 1).
- If the ISO Host Interface process and the external processor use Racal-Guardata hardware security modules, this field can be set to 0 (variant 0) or 1 (variant 1); however, the ISO Host Interface process and the external processor must use the same setting.

MAC KEY VARIANT — Specifies the key variant to be applied to the MAC key exchange key (KEK) when translating MAC keys. Regardless of the hardware security module used by BASE24 Remote Banking or the external processor this field must be set to 3 (variant 3).

LIMITS

The following fields are used to set the parameters for dynamic key management for this interface. The parameters specify when the key used for PIN encryption and translation should be changed and when the key used for message authentication should be changed. Both types of keys can be changed based on time intervals, transaction activity, the number of errors since the last key change, or the number of consecutive errors. The MAC key can also be changed based on the number of key synchronization errors since the last key change.

PIN KEY TIMER VALUE — Specifies the maximum length of time that a PIN working key should be used, if a time limit is being used to determine how often the key is changed.

The time period can be expressed in minutes, hours, or days, depending on the value in the PIN KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid values are as follows:

- 0, 5–1500 If the value in the PIN KEY TIMER INTERVAL field is M (minutes)
- 0–1000 If the value in the PIN KEY TIMER INTERVAL field is H (hours)
- 0–180 If the value in the PIN KEY TIMER INTERVAL field is D (days)

BASE24 Remote Banking checks the values in the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields when the KEYF record is added or updated. If the time is evenly divisible by a longer interval, BASE24 Remote Banking changes the value in the PIN KEY TIMER INTERVAL field to the longer interval and redisplay the value in the PIN KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

The interface process does not use this parameter if the value in this field is set to 0. If the ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to 0.

MAC KEY TIMER VALUE — Specifies the maximum length of time that a MAC working key should be used, if a time limit is being used to determine how often the key is changed.

The time period can be expressed in minutes, hours, or days, depending on the value in the MAC KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid values are as follows:

- 0, 5–1500 If the value in the MAC KEY TIMER INTERVAL field is
M (minutes)
- 0–1000 If the value in the MAC KEY TIMER INTERVAL field is H (hours)
- 0–180 If the value in the MAC KEY TIMER INTERVAL field is D (days)

BASE24 Remote Banking checks the values in the MAC KEY TIMER VALUE and MAC KEY TIMER INTERVAL fields when the KEYF record is added or updated. If the time is evenly divisible by a longer interval, BASE24 Remote Banking changes the value in the MAC KEY TIMER INTERVAL field to the longer interval and redisplay the value in the MAC KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is

automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

The interface process does not use this parameter if the value in this field is set to 0. If the MAC ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to 0.

PIN KEY TIMER INTERVAL — A code specifying the unit of time used with the value in the PIN KEY TIMER VALUE field to set the maximum length of time that a PIN working key should be used. BASE24 Remote Banking changes the value in the PIN KEY TIMER INTERVAL field to the largest possible unit of time when the KEYF record is added or updated. Refer to the PIN KEY TIMER VALUE field for additional information. Valid values are as follows:

D = Days
H = Hours
M = Minutes

MAC KEY TIMER INTERVAL — A code specifying the unit of time used with the value in the MAC KEY TIMER VALUE field to set the maximum length of time that a MAC working key should be used. BASE24 Remote Banking changes the value in the MAC KEY TIMER INTERVAL field to the largest possible unit of time when the KEYF record is added or updated. Refer to the MAC KEY TIMER VALUE field for additional information. Valid values are as follows:

D = Days
H = Hours
M = Minutes

PIN KEY TRAN — The maximum number of transactions that can be performed before changing the current PIN working key.

- If a total transaction limit is being used, this field must contain a value in the range of 50–100000.
- If a total transaction limit is not being used, this field must contain a 0. If the ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to 0.

MAC KEY TRAN — The maximum number of transactions that can be performed before changing the current MAC working key.

- If a total transaction limit is being used, this field must contain a value in the range of 50–100000.
- If a total transaction limit is not being used, this field must contain a 0. If the MAC ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to 0.

PIN KEY ERROR — The maximum number of PIN key-related errors that can occur with the present PIN working key before changing the key.

- If a PIN key error limit is being used, this field must contain a value in the range of 5–9999.
- If a PIN key error limit is not being used, this field must contain a 0. If the ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to 0.

MAC KEY ERROR — The maximum number of MAC generation and verification errors that can occur with the present MAC working key before changing the key.

- If a MAC generation and verification error limit is being used, this field must contain a value in the range of 5–9999.
- If a MAC generation and verification error limit is not being used, this field must contain a 0. If the MAC ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to 0.

CONSECUTIVE PIN KEY ERROR — The maximum number of consecutive PIN key-related errors that can occur with the present PIN working key before changing the key.

- If a consecutive PIN key error limit is being used, this field must contain a value in the range of 5–9999.
- If a consecutive PIN key error limit is not being used, this field must contain a 0. If the ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to 0.

CONSECUTIVE MAC KEY ERROR — The maximum number of consecutive MAC generation and verification errors that can occur with the present MAC working key before changing the key.

- If a consecutive MAC generation and verification error limit is being used, this field must contain a value in the range of 5–9999.
- If a consecutive MAC generation and verification error limit is not being used, this field must contain a 0. If the MAC ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to 0.

KMAC SYNCHRONIZATION ERROR — The maximum number of MAC working key synchronization errors that can occur with the present MAC working key before changing the key. Key synchronization is the process of comparing the check digits returned from a security module for the present MAC working key with the check digits stored in the KEYF record for the present MAC working key. These KEYF MAC working key check digit values are set in the CHECK DIGITS1 and CHECK DIGITS2 fields on KEYF screen 2.

- If a MAC working key synchronization error limit is being used, this field must contain a value in the range of 3–9999.
- If a MAC working key synchronization error limit is not being used, this field must contain a 0. If the MAC ENCRYPT TYPE field on KEYF screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to 0.

CLEAR OLD KEY TIMER VALUE — The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the interface process.

The INBOUND PIN KEY, OUTBOUND PIN KEY, INBOUND MAC KEY, and OUTBOUND MAC KEY fields on KEYF screen 2 can contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key (1 or 2) is currently in use. At the time a new key is placed in service, the other key becomes the old key. For example, if inbound PIN key number 1 is currently in use when the ISO Host Interface process receives a new key, the new key is stored as inbound PIN key number 2. Inbound PIN key number 2 becomes the current key and inbound PIN key number 1 becomes the old key. (If inbound PIN key number 2 is the current key, it becomes the old key and the new key is stored as inbound PIN key number 1.)

When processing a transaction, BASE24 Remote Banking uses the new key for PIN processing or message authentication. If the security device used by BASE24 Remote Banking for PIN processing or message authentication detects an error because the external processor is still using the old key, it can repeat the PIN processing or message authentication using the old key before BASE24 Remote Banking declines the transaction. The value in this field controls how long the old key is available. If the old key is not available, a second PIN processing or message authentication attempt is not made.

- If the old key is to be cleared, this field must contain a value in the range of 1–9999 to specify the time, in seconds, that an old key can be used before being cleared.
- If the old key is not to be cleared, this field must contain a value of 0. If the ENCRYPT TYPE field on KEYF screen 1 and the MAC ENCRYPT TYPE field on KEYF screen 1 both are set to values other than 1, this field must be set to 0.

ORIGINATING ID — An identification code that will be placed in the originator field of the Cryptographic Service Message (CSM) in a message originated by BASE24. The values in the ORIGINATING ID and RECEIVING ID fields must be agreed on by the sender and receiver. The sender's originating ID is the receiver's receiving ID and the sender's receiving ID is the receiver's originating ID. These values do not need to match any specific values in the BASE24 database.

RECEIVING ID — An identification code that will be placed in the receiver field of the Cryptographic Service Message (CSM) in a message originated by BASE24. The values in the ORIGINATING ID and RECEIVING ID fields must be agreed on by the sender and receiver. The sender's originating ID is the receiver's receiving ID and the sender's receiving ID is the receiver's originating ID. These values do not need to match any specific values in the BASE24 database.

KEY PROCESSING TYPE — A code identifying the type of dynamic key management processing this interface process can perform.

- If each interface process is responsible for generating its outbound working keys and can request its inbound working keys from the other interface process in the co-network, this field must be set to C (co-network).
- If this interface process is responsible for all key generation, this field must be set to M (master).
- If this interface process does not generate or receive keys, this field must be set to N (none).

- If this interface process can request the master process to generate keys but does not generate keys itself, this field must be set to S (slave).

Dynamic key management requires hardware security devices for PIN management and message authentication. If the ENCRYPT TYPE and MAC ENCRYPT TYPE fields on KEYF screen 1 both are set to values other than 1, meaning security devices are not being used, the value in this field must be set to N (none).

Dynamic key management in a co-network configuration requires the use of separate keys for inbound and outbound messages. This is because each co-network process must generate the key used for its outbound messages and two processes cannot share the responsibility for generating one key. If the NUMBER OF KEYS field on KEYF screen 1 is set to 1, the value in this field cannot be set to C (co-network).

NOTARIZATION SUPPORTED — A code identifying whether this interface process supports key notarization. This field must be set to 0 (not supported) because BASE24 Remote Banking does not currently support key notarization.

External Message File Settings

The following field on EMF screen 1 must be considered for network management messages when dynamic key management is used.

MESSAGE ELEMENTS — Identifies the elements contained in ISO messages.

- If default values are being used, an EMF record is not needed for network management (0800 and 0810) messages.
- If default values are not being used, the following elements must be defined as conditional in the EMF record for network management (0800 and 0810) messages:
 - ❑ Response code (P-39)
 - ❑ PIN/MAC flag and inbound/outbound/both flag (P-53)
 - ❑ Key management message type (S-70)
 - ❑ Check digits for working key (S-120)
 - ❑ Cryptographic service message (CSM) (S-123)

Host Configuration File Settings

The following fields on HCF screen 1 must be considered for network management messages when dynamic key management is used.

These fields must be set regardless of the setting of the NMM ENABLED PROCESSING FLAG field on HCF screen 1. Dynamic key management messages are considered a type of network management message, but are not affected by the value in the NMM ENABLED PROCESSING FLAG field because the working keys they are used to manage are necessary for financial transactions.

TIMER LIMITS

The following fields are used to configure timer limits that are not product-specific.

NETWORK MANAGEMENT — Identifies the number of seconds that the ISO Host Interface process waits for a response after transmitting a network management message to an external processor station. Network management messages include echo-test messages, logons, logoffs, and dynamic key management messages.

If the time indicated in this field expires before a response message is received, the ISO Host Interface process marks the station as down. Valid values are 0 through 9999.

EXTENDED NETWORK — Identifies the number of seconds the ISO Host Interface process is to wait, once the initial message has timed out, before resending dynamic key management messages (that is, new key and change key).

Applied Communications, Inc.

Section 13

Security Module Support

This section describes the use of security modules with BASE24 Remote Banking and the functions of each that are supported by BASE24 Remote Banking. The information a user requires for configuring specific security modules for use with BASE24 Remote Banking is found in appendixes E and F of this manual.

BASE24 Remote Banking Security Module Usage

BASE24 Remote Banking allows for integration of different security modules into its routine transaction processing. It can use security modules to perform the following functions.

PIN Verification — Verifying PINs with a security module involves sending prescribed data from a transaction and the BASE24 database to the security module and allowing the module to verify the PIN. The Integrated Authorization Server process can be configured to send the appropriate data required for PIN verification to the security module and accept the results returned by the module.

PIN Translation — Translating PINs with a security module involves the translation of PINs from encryption under one key to encryption under another key by the security module. The ISO Host Interface process can be set up to use security modules for PIN translation in cases where BASE24 Remote Banking handles PINs under different keys than the external processor with which it communicates. Integrated Authorization Server processes can also be set up to use security modules for PIN translation in cases where they handle PINs under different keys than the remote banking endpoint device with which they communicate.

PIN Changes — Changing PINs with a security module involves receiving old and new PINs entered by the customer at a remote banking endpoint device or customer service representative terminal, verifying the old PIN, and generating PIN verification digits for the new PIN.

PIN verification digits include the following:

- The PIN offset for the IBM 3624-format and Diebold PIN verification methods
- The PIN verification value (PVV) for the Visa PVV PIN verification method
- The PIN verification number (PVN) for the Identikey PIN verification method

The BASE24 database is updated by replacing the PIN verification digits for the old PIN with the PIN verification digits for the new PIN. A specific transaction, the PIN change transaction, is supported by some remote banking endpoint devices and customer service representative terminals for this feature. Similar to PIN

verification, the Integrated Authorization Server process can be set up to use security modules to verify the existing PIN and generate PIN verification digits for the new PIN.

Message Authentication — Performing message authentication with a security module involves a two-step process. The first step is generating a message authentication code (MAC) based on the contents of a message. This MAC is added to the message before it is sent. The second step is verifying the MAC when the message is received. The MAC calculation is repeated and the result is compared to the value received in the message. The ISO Host Interface process can use security modules to generate the MACs placed in outgoing messages and verify the MACs received in incoming messages. Message authentication is not supported between BASE24 and remote banking terminals at this time.

Note: BASE24 Remote Banking does not support message authentication using the Atalla Security Module model A1. However, the Atalla Security Module model A1 can be used for dynamic key management in environments where message authentication is not being performed.

Key Management — BASE24 Remote Banking requires key management by a security module in two situations.

- Generating new working keys.
 - For an RBSI-compliant device, working keys are always provided by the BASE24 system. A security module is used to generate the working keys. The inbound PIN encryption keys carried in the Interface Security File (ISF) for RBSI-compliant devices are working keys.
 - For an external processor, working keys can be provided by the BASE24 system or by an external processor. A security module is used to generate the working keys when the BASE24 system is responsible for providing them. The inbound and outbound keys carried in the Key File (KEYF) for external processors are working keys.
- Translating working keys. When the BASE24 system and an external processor support online key changes, the ISO Host Interface process must take a working key encrypted under the key exchange key (also known as a zone master key (ZMK)) for the external processor and obtain the working key encrypted under the security module master key. Key exchange keys for the external processor are carried in the PIN KEY EXCHANGE KEY and MAC KEY EXCHANGE KEY fields of the KEYF.

BASE24 Remote Banking dynamic key management support automates the generation and translation of new working keys based on parameters such as length of time, number of transactions, number of errors, and number of consecutive errors.

Supported Security Modules

BASE24 Remote Banking supports several models of the following security modules in its processing:

- Atalla Security Module (ASM)
- Atalla Network Security Processor (NSP)
- Racal-Guardata Host Security Module (RSM)

Each security module supports a subset of BASE24 Remote Banking transaction security processing. Specific support information has been outlined in appendixes to this manual.

While any of the above security modules can be used with BASE24 Remote Banking, any one BASE24 Remote Banking process or transaction path can only use one type of security module. For example, an NSP and an RSM cannot both be used by the same Integrated Authorization Server process, and all Integrated Authorization Server processes, Interface Master processes, and ISO Host Interface processes must use the same type of security module (that is, ASM, NSP, or RSM).

In a configuration with more than one BASE24 product (for example, BASE24-atm and BASE24 Remote Banking), one product can use one type of security module while the other product uses a different type; however, ACI recommends that, whenever possible, one type of security module be used throughout the BASE24 system configuration.

Number of Security Modules

If a security module is to be used by BASE24 Remote Banking for any transaction security processing, ACI recommends that at least two physical devices be configured for the system. This is particularly true in a production system where the malfunction of a security module could halt transaction processing if no backup is available.

The number of security modules supported by a BASE24 Remote Banking process can vary according to the type of process that accesses them. These differences are explained in the following paragraphs.

Integrated Authorization Server and Interface Master Processes

The Integrated Authorization Server process and Interface Master process can support an essentially unlimited number of security modules. Each security module is assigned to an Integrated Authorization Server process and Interface Master process using the SECURE-DEVICE n assign in the LCONF, where n is a value from 1 to 32767. The SECURE-DEVICE-TYPE param in the LCONF identifies the type of security module specified in the SECURE-DEVICE n assigns.

When one of these processes requires the services of a security module, it uses the next available security module. The n in the assign name identifies the order in which the process accesses the security modules. For example, these processes access the security module identified by the SECURE-DEVICE1 assign before they access the security module identified by the SECURE-DEVICE2 assign.

The SECURE-DEVICE-COUNT LCONF param identifies the number of security modules being used by an Integrated Authorization Server process or Interface Master process. For example, if the SECURE-DEVICE-COUNT param is set to 5, the Integrated Authorization Server process or Interface Master process accesses five security modules. When the process accesses the security module identified by the SECURE-DEVICE5 assign, it then starts over with the security module identified by the SECURE-DEVICE1 assign.

Both the Integrated Authorization Server process and Interface Master process maintain the status of all their security modules in an internal table. If an error occurs while attempting to access a security module, the process will attempt to process the request with the same security module up to two more times. If these retry attempts also fail, the process marks the module as “device failed” in this table and attempts to access the next available module to process the request. Again, the process will allow for up to three attempts with the second security module before it marks the module as “device failed.” Failed modules are taken out of service until they are successfully reinitialized by the Integrated Authorization Server process.

Note: Although the number of security modules is essentially unlimited, the Integrated Authorization Server process and Interface Master process never attempt to use more than two security modules to process a particular command message to avoid the possibility of a transaction timeout.

You can obtain the status of all security modules used by an Integrated Authorization Server process or Interface Master process by sending the STATUS command to the Transaction Security object of the process. The current status of each device (that is, open, closed, or failed), along with the failure count for each device, total number of security devices configured, and total number of security devices available are displayed in an Event Management Service (EMS) event. The STATUS command is entered on the Status (STAT) screen. Refer to the *NET24 ITS Kernel Files Maintenance Manual* for a description of the Status screen.

ISO Host Interface Process

Each BASE24 Remote Banking ISO Host Interface process requires two security modules to be assigned to each process, with one security module designated as the primary module and the other designated as the secondary module. The primary and secondary security modules are assigned to each ISO Host Interface process using the SECURE-DEV1 and SECURE-DEV2 LCONF assigns, respectively. The SECURE-DEV-TYP param in the LCONF identifies the type of security modules specified in the SECURE-DEV1 and SECURE-DEV2 assigns.

While each BASE24 Remote Banking ISO Host Interface process requires a primary and secondary module, processes do not require exclusive access to a pair of security modules. For example, if an institution has three security modules and three processes that require a security module for transaction processing, the configuration could be set up as shown below:

Process	Primary	Secondary
BASE24-atm Device Handler process	Module C	Module A
BASE24 Remote Banking ISO Host Interface process	Module B	Module C
BASE24-atm Interchange Interface process	Module C	Module B

Note: In this example, an Integrated Authorization Server process or Interface Master process could use all three security modules (A, B, and C), since these processes do not use the primary and secondary designation.

When a transaction is received by an ISO Host Interface process that requires the use of a security module, the process first attempts to communicate with its primary security module. If the request to the primary security module times out or the security module is considered inoperative, the ISO Host Interface process attempts to communicate with its secondary security module to process the transaction.

BASE24 processes that use primary and secondary security modules can use the SECURE-DEV-MONITOR-THRESHOLD LCONF param to provide the following two options involving security modules:

- Checking the secondary security module at specified intervals
- Balancing transactions between the primary and secondary security modules

Because the Integrated Authorization Server process and Interface Master process do not use primary and secondary security module designations, these options do not apply to Integrated Authorization Server and Interface Master processes.

Checking the Secondary Security Module

To minimize the possibility of both security modules becoming inoperative, BASE24 processes can be configured to occasionally perform one transaction using the secondary security module instead of the primary security module. The transaction is performed using the secondary security module at intervals based on the number of transactions performed by the primary security module.

Each process using a security device maintains its own transaction counter. The value specified in the SECURE-DEV-MONITOR-THRESHOLD param is the number of transactions processed by an individual process using the primary security module before it sends the next request to the secondary security module. For example, if this param was set to a value of 1000, the ISO Host Interface process would send 1000 requests to the primary security module and then send the next request to the secondary security module. This would be repeated for every 1000 requests processed by the primary security module for the process.

Occasionally using the secondary security module permits the identification and resolution of any problems with the secondary security module while the primary security module is functioning properly.

Balancing Transactions Between Security Modules

The SECURE-DEV-MONITOR-THRESHOLD param can also be used to balance the number of transactions processed by the primary and secondary security modules. For example, by setting this param to 1, each ISO Host Interface process alternates between the primary and secondary security modules. Each ISO Host Interface process uses the primary security module for one transaction, uses the secondary security module for one transaction, and then repeats the cycle.

Security Device Failures and Timeouts

The user can configure the maximum time, in hundredths of a second, an Integrated Authorization Server process, Interface Master process, or ISO Host Interface process waits for a response from a security module before the request is considered timed out using the SECURE-DEV-TIM-LMT param in the LCONF. After three timeouts, BASE24 Remote Banking processes consider the security module to be inoperative.

Should two security modules for an Integrated Authorization Server process or Interface Master process become inoperative, or both the primary and secondary security modules for an ISO Host Interface process, BASE24 Remote Banking generates event messages to notify network operators of the failure.

For security modules used by ISO Host Interface processes, the SECURE-DEV-FAIL-NOTIFY param in the LCONF specifies the interval at which critical messages are logged when the security modules fail.

For security modules used by Integrated Authorization Server and Interface Master processes, an error event message is logged when the module fails and the module is removed from rotation in the internal device table. The module is not used in subsequent processing until an Assign Warmboot or LCONF Warmboot command is successfully executed on the process.

Configuring Security Modules

Although the processing within each security module is proprietary to the vendor manufacturing the device, most such devices perform similar functions and are integrated into BASE24 Remote Banking processing in the same way.

All security modules used by BASE24 Remote Banking must be defined to the Tandem system configuration and the BASE24 logical network. Depending on the version level of the Tandem operating system, modules are defined to the Tandem system configuration using the Tandem SYSGEN configuration programs or the Tandem Dynamic System Configuration (DSC) utility. Modules are defined to each BASE24 logical network using the Logical Network Configuration File (LCONF).

Note: All BASE24 application processes interface with security devices directly. Therefore, security modules need not be specified as processes or stations to the XPNET process.

For detailed information regarding configuration of specific security devices, refer to the following sources:

- The Tandem configuration manual specific to the method desired for the appropriate operating system used.
- The configuration manual provided by the device vendor.
- Appendix E, “Atalla Security Modules.”
- Appendix F, “Racal-Guardata Host Security Modules.”

Note: Appendixes E and F define only the LCONF assigns and params required specifically on behalf of security modules. The Integrated Authorization Server process, Interface Master process, and ISO Host Interface process require additional LCONF assigns to support transaction security processing in general. These assigns include the INTFEMT, ISF, KEYAEMT, and ISECEMT-SWAPVOL assigns for the Integrated Authorization Server process and the KEYF assign for the ISO Host Interface process. The Interface Master process requires the same additional assigns as the Integrated Authorization Server process, except for the KEYAEMT assign. Refer to the ***BASE24 Logical Network Configuration File Manual*** for detailed descriptions of these assigns.

Applied Communications, Inc.

Appendix A

File and Table Maintenance Screens

This appendix contains illustrations of the following file and table maintenance screens.

- Customer Table (CSTT) screen 1
- Host Configuration File (HCF) screen 1
- Interface Security File (ISF) screens 1 and 2
- Institution Definition File (IDF) screens 2, 4, and 40
- Key Authorization File (KEYA) screens 1 through 5
- Key File (KEYF) screens 1 through 3
- VRU Configuration Data (VCD) screens 1, 2, 4, and 18

These screen illustrations are provided for your convenience as you review the instructions presented throughout this manual for setting up the BASE24 database. These illustrations are identical to the ones provided in the ***BASE24 Base Files Maintenance Manual*** and the ***BASE24 Core Files and Tables Maintenance Manual***.

Customer Table (CSTT) Screen 1

CSTT screen 1 defines authorization parameters and usage accumulation information for BASE24 Remote Banking customers. For a complete discussion of the fields on CSTT screen 1, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

```
BASE24-CORE CUSTOMER CONFIGURATION  LLLL      YY/MM/DD  hh:mm  01 OF 03
CUSTOMER ID:

      FIID:      BRANCH ID:      PROFILE:      TYPE: C
      BEGIN DATE:      (YYYYMMDD)      END DATE:      (YYYYMMDD)
VERIFY STATUS: V      (LAST UPDATE:      )
      PIN VERIFICATION DIGITS:      (LAST UPDATE:      )
      PIN VERIFICATION KEY INDEX:      BAD PIN COUNT: 0000
      PIN CHANGE REQUIRED: N (Y/N/S)      LAST RESET DATE:
      MAX HISTORY RECORDS: 0
      DEFAULT ACCT TYPE: NONE      DEFAULT ACCT NUM:

CUST INFO:
      ALTERNATE CONTACT:
      PAYMENT HIGH LIMIT:      BILLPAY BILLING GROUP:
      PACKET STATUS:      (LAST UPDATE: 00000000000000 )
      BILLPAY BILLING TYPE:      (LAST UPDATE: 00000000000000 )
      SERVICE FEE ACCT TYPE: NONE      SERVICE FEE ACCT NUM:

RECORD LAST CHANGED:      BY USER:      ,
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F12-HELP
```


Host Configuration File (HCF) Screen 1

HCF screen 1 contains timer limits, processing flags, and data prefix characters. For a complete discussion of the fields on HCF screen 1, refer to the ***BASE24 Base Files Maintenance Manual***.

```

BASE24-BASE  HOST CONFIGURATION      LLLL      YY/MM/DD  hh:mm  01 OF 25
              DPC NUMBER: 0000      HISF NAME: P1A^HISF1
                                      TOKEN GROUP: ****
              TIMER LIMITS          PROCESSING FLAGS

NETWORK MANAGEMENT:  30 (SEC)      STORE AND FORWARD: 0  (INTERPERSE)
EXTENDED NETWORK:    60 (SEC)      DPC TYPE: 0  (MULTITHREADED)
STORE AND FORWARD:   30 (SEC)      ACK TO DPC: N (Y/N)
WAIT FOR TRAFFIC:    60 (SEC)      ACK FROM DPC: N (Y/N)
PERFORMANCE PERIOD:  20 (MIN)      MAX TIMEOUTS: 2
                                      NMM ENABLED: Y (Y/N)
MAXIMUM OUTSTANDING SAFS: 1      MAX SAF RETRY: 0
MAXIMUM OUTSTANDING REQUESTS      PROTOCOL TYPE: 00  (NUCLEUS MGMT)
OUTBOUND:  30  INBOUND:  30      MESSAGE SEQUENCE FLAG: 0  (BY PROCESS)
                                      TIME DISCREPANCY CHECK: N (Y/N)
RELEASE INDICATOR: 01  (CURRENT REL)  MESSAGE FORMAT: 01 (VARIABLE FORMAT)
              DATA PREFIX CHARACTERS
              1.      2.      3.      4.      5.      6.      7.      8.      9.
VALID ENTRIES FOR DATA PREFIX CHARACTERS ARE 0 THRU 9 AND A THRU F.

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
              F12-HELP

```

Institution Definition File (IDF) Screen 2

IDF screen 2 enables an institution to set up transaction security parameters at the institution level. For a complete discussion of the fields on IDF screen 2, refer to the ***BASE24 Base Files Maintenance Manual***.

```
BASE24-BASE  INSTITUTION FILE          LLLL          YY/MM/DD  hh:mm  02 OF 45

          FIID:          FI-NAME:

          CARD PARAMETERS
CHECK IF HOST ONLINE  LIMITS: N          PIN: N
          CARD STATUS: N          EXP DATE: N

      MAX PIN TRIES:    1          BAD PIN ACTION: 0  (RETURN CARD)

      POFST/PVV LOC: 0  (NO OFFSET)          PIN CHECK TYPE: 00  (NO VERIFICATION)

      ALGO NUMBER LOC: 0  (NOT REQUIRED)          CARDHOLDER PIN SELECT: N  (Y/N)

      EXP CHECK DISP: 0  (RETURN CARD)

      SHARING GROUP:

      PIN TRIES RESET OPTION: 0  (RESET EACH CAF/UAF EACH USAGE PERIOD)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Institution Definition File (IDF) Screen 4

IDF screen 4 enables an institution to specify its withdrawal period parameters. For a complete discussion of the fields on IDF screen 4, refer to the ***BASE24 Base Files Maintenance Manual***.

BASE24-BASE INSTITUTION FILE LLLL YY/MM/DD hh:mm 04 OF 45

FIID:

FI-NAME:

WITHDRAWAL PERIOD PARAMETERS

BEGINNING DATE: (YYMMDD)

NEXT BEGINNING DATE:

PERIOD LENGTH: 1 DAYS IN PERIOD

WORK DAY CODE: 0 (NO HOLIDAYS)

HOLIDAYS

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:
F12-HELP

NEW LOGICAL NETWORK ID:

Institution Definition File (IDF) Screen 40

IDF screen 40 contains BASE24 Remote Banking processing information. For a complete discussion of the fields on IDF screen 40, refer to the ***BASE24 Base Files Maintenance Manual***.

```
BASE24-TB      INSTITUTION FILE      LLLL      YY/MM/DD  hh:mm  40 OF 45

              FIID:                  FI-NAME:

              TELEBANKING IDF INFORMATION

              ROUTE PROFILE:
              PIN VERIFICATION GROUP:
              CUSTOMER BALANCE INFO: 4 (BOTH, PREFER AVAIL BALANCE)
MULTIPLE ACCOUNT SELECT DISPLAY: 1 (ACCOUNT DESCRIPTIONS)
DISCARD NON-FINANCIAL REVERSALS: N (Y/N)
              CURRENT BUSINESS DATE: 000000 (YYMMDD)
              REPORT BUSINESS DATE: 000000 (YYMMDD)
              CUTOVER END: 00:00

              CUSTOMER SERVICE INTERFACE CONTROL PARAMETERS
              PIN REQUIRED: Y (Y/N/OVERRIDE)
              PIN CHANGE ALLOWED: N (Y/N)

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
              F12-HELP                SF7-BILLPAY BANK TABLE
```

Interface Security File (ISF) Screen 1

Interface Security File (ISF) screen 1 contains PIN encryption parameters, message authentication parameters, and key transport key information. For a complete discussion of the fields on ISF screen 1, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

```
BASE24-CORE  INTERFACE SECURITY FILE  LLLL          YY/MM/DD  hh:mm  01 OF 04
                INTERFACE NAME:

EXTRN PIN ENCRYPT TYPE: H (HARDWARE)      INTRN PIN ENCRYPT TYPE: H (HARDWARE)
      PIN BLOCK FORMAT: 1 (ANSI)          PIN PAD CHARACTER: F
      PIN ENCRYPT LEVEL: 1 (SINGLE)        OLD KEY TIMER LIMIT:    0 (SEC)
      MAC ALGO: 0 (ANSI X9.19)           MAC ENCRYPT TYPE: N (NONE)

                KEY TRANSPORT KEY INFORMATION
KEY LENGTH: 1 (SINGLE)
KKT-PT: 0000000000000000                VVK: 0000      VARIANT: 0 (KKT)
KKT-AM: 0000000000000000                VVK: 0000      VARIANT: 0 (KKT)
KKT-DT: 0000000000000000                VVK: 0000      VARIANT: 0 (KKT)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Interface Security File (ISF) Screen 2

ISF screen 2 contains PIN transport key information. For a complete discussion of the fields on ISF screen 2, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

```
BASE24-CORE  INTERFACE SECURITY FILE  LLLL          YY/MM/DD  hh:mm  02 OF 04
                INTERFACE NAME:

                PIN TRANSPORT KEY INFORMATION

INTERM SFTWR KEY: 0000000000000000
INTERM HDWR KEY: 0000000000000000                VVK: 0000

INBOUND KEYS:
    KPT1: 0000000000000000                VVK1: 0000
    KPT2: 0000000000000000                VVK2: 0000
    CURRENT INDEX: 0                KEY COUNTER: 000000

OUTBOUND KEYS:
    KPT1: 0000000000000000                VVK1: 0000
    KPT2: 0000000000000000                VVK2: 0000
    CURRENT INDEX: 0                KEY COUNTER: 000000

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HEL
```

Key Authorization File (KEYA) Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. For a complete discussion of the fields on KEYA screen 1, refer to the ***BASE24 Base Files Maintenance Manual***.

BASE24-BASE KEY AUTH FILE LLLL YY/MM/DD hh:mm 01 OF 06

GRP: BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID:

RECORD TYPE: 00 (01)-IBM DES PIN VERIFICATION
(02)-DIEBOLD PIN VERIFICATION
(03)-IDENTIKEY PIN VERIFICATION
(04)-VISA PVV PIN VERIFICATION
(11)-CARD VERIFICATION

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

Key Authorization File (KEYA) Screen 2

KEYA screen 2 contains information for the IBM 3624-format PIN verification method. For a complete discussion of the fields on KEYA screen 2, refer to the ***BASE24 Base Files Maintenance Manual***.

BASE24-BASE KEY AUTH FILE LLLL YY/MM/DD hh:mm 02 OF 06

GRP: BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID:

DES (IBM 3624) PIN VERIFICATION

DECIMALIZATION TABLE: 0000000000000000

PAN VERIFY OFFSET: 00

CLEAR KEY: 0000000000000000

PAN VERIFY LENGTH: 00

ENCRYPTED KEY: 0000000000000000

PAN PAD CHARACTER: F

CHECK DIGITS: 0000

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:

NEW LOGICAL NETWORK ID:

F12-HELP

Key Authorization File (KEYA) Screen 3

KEYA screen 3 contains Diebold PIN verification information. For a complete discussion of the fields on KEYA screen 3, refer to the **BASE24 Base Files Maintenance Manual**.

BASE24-BASE KEY AUTH FILE LLLL YY/MM/DD hh:mm 03 OF 06

GRP: BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID:

DIEBOLD VERIFICATION DATA

NUMBER TABLE RELATIVE LOCATION: 0 ALGO NUMBER: 00
CLEAR KEY: 0000000000000000 ENCRYPTED KEY: 0000000000000000

DIEBOLD TABLE:

00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

Key Authorization File (KEYA) Screen 4

KEYA screen 4 contains Identikkey PIN verification information. For a complete discussion of the fields on KEYA screen 4, refer to the ***BASE24 Base Files Maintenance Manual***.

BASE24-BASE KEY AUTH FILE LLLL YY/MM/DD hh:mm 04 OF 06

GRP: BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID:

IDENTIKEY PIN VERIFICATION

INSTITUTION IDENTIFIER: 00000000 ID LENGTH: 0
PARTIAL PAN OFFSET: 00 PARTIAL PAN LENGTH: 4
COMPARE INDICATOR:

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

Key Authorization File (KEYA) Screen 5

KEYA screen 5 contains Visa PVV PIN verification information. For a complete discussion of the fields on KEYA screen 5, refer to the ***BASE24 Base Files Maintenance Manual***.

```
BASE24-BASE  KEY AUTH FILE          LLLL      YY/MM/DD  hh:mm  05 OF 06

GRP:      BEGIN DATE (YYYYMM): ***** END DATE (YYYYMM): ***** FIID:

                VISA PVV KEYS
          CLEAR      ENCRYPT      CLEAR      ENCRYPT
1 0000000000000000 0000000000000000 0000000000000000 0000000000000000
   CHECK DIGITS: 0000                      CHECK DIGITS: 0000
2 0000000000000000 0000000000000000 0000000000000000 0000000000000000
   CHECK DIGITS: 0000                      CHECK DIGITS: 0000
3 0000000000000000 0000000000000000 0000000000000000 0000000000000000
   CHECK DIGITS: 0000                      CHECK DIGITS: 0000
4 0000000000000000 0000000000000000 0000000000000000 0000000000000000
   CHECK DIGITS: 0000                      CHECK DIGITS: 0000
5 0000000000000000 0000000000000000 0000000000000000 0000000000000000
   CHECK DIGITS: 0000                      CHECK DIGITS: 0000
6 0000000000000000 0000000000000000 0000000000000000 0000000000000000
   CHECK DIGITS: 0000                      CHECK DIGITS: 0000

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```

Key File (KEYF) Screen 1

KEYF screen 1 contains PIN encryption, PIN translation, and message authentication parameters, intermediate keys, PIN key exchange keys, and message authentication code (MAC) key exchange keys. For a complete discussion of the fields on KEYF screen 1, refer to the ***BASE24 Base Files Maintenance Manual***.

```
BASE24-BASE KEY FILE          LLLL      YY/MM/DD  hh:mm  01 OF 03
DPC/FIID:                     INTERFACE PROCESS:

  ENCRYPT TYPE: 0 (NO ENCRYPTION)  BASE24 ENCRYPT TYPE: 0 (NO ENCRYPTION)
PIN BLOCK FORMAT: 0 (CLEAR)      ANSI PAN FORMAT: 0 (12 RIGHT/NO CHK)
MAC ENCRYPT TYPE: 0 (NO PROCESSING) PIN PAD CHARACTER: F
  MAC DATA TYPE: 0 (ASCII)      NUMBER OF KEYS: 1 (COMBINED)
    KEY LENGTH: 1 (SINGLE)      FULL MESSAGE MAC: N (SELECTED FIELDS)

                        INTERMEDIATE KEYS

      CLEAR: 0000000000000000      CHECK DIGITS: 0000
    ENCRYPTED: 0000000000000000

                        EXCHANGE KEYS

      PIN KEY: 0000000000000000 0000000000000000 CHECK DIGITS: 0000
      MAC KEY: 0000000000000000 0000000000000000 CHECK DIGITS: 0000

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Key File (KEYF) Screen 2

KEYF screen 2 consists of inbound and outbound working keys for PINs and MACs. For a complete discussion of the fields on KEYA screen 2, refer to the ***BASE24 Base Files Maintenance Manual***.

```
BASE24-BASE KEY FILE          LLLL      YY/MM/DD  hh:mm  02 OF 03
DPC/FIID:                     INTERFACE PROCESS:

                                PIN KEY INFORMATION
OUTBOUND KEYS:
  PIN KEY1: 0000000000000000  CHECK DIGITS1: 0000  CURRENT INDEX: 1
  PIN KEY2: 0000000000000000  CHECK DIGITS2: 0000  KEY COUNTER: 000000
INBOUND KEYS:
  PIN KEY1: 0000000000000000  CHECK DIGITS1: 0000  CURRENT INDEX: 1
  PIN KEY2: 0000000000000000  CHECK DIGITS2: 0000  KEY COUNTER: 000000

                                MAC KEY INFORMATION
OUTBOUND KEYS:
  MAC KEY1: 0000000000000000  CHECK DIGITS1: 0000  CURRENT INDEX: 1
  MAC KEY2: 0000000000000000  CHECK DIGITS2: 0000  KEY COUNTER: 000000
INBOUND KEYS:
  MAC KEY1: 0000000000000000  CHECK DIGITS1: 0000  CURRENT INDEX: 1
  MAC KEY2: 0000000000000000  CHECK DIGITS2: 0000  KEY COUNTER: 000000

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                   F12-HELP
```

Key File (KEYF) Screen 3

KEYF screen 3 contains dynamic key management parameters. For a complete discussion of the fields on KEYA screen 3, refer to the **BASE24 Base Files Maintenance Manual**.

Note: If all limits on this screen are set to zero, the interface process using this KEYF record will not initiate key changes automatically based on timers or counters, so the external processor must be responsible for initiating any key changes if dynamic key management is to occur. At least one PIN key limit on this screen must be set to a nonzero value for the ISO Host Interface process to initiate dynamic key management for PINs, and at least one MAC key limit on this screen must be set to a nonzero value for the ISO Host Interface process to initiate dynamic key management for MACs.

```
BASE24-BASE KEY FILE          LLLL      YY/MM/DD  hh:mm  03 OF 03
DPC/FIID:                    INTERFACE PROCESS:

    PIN KEY VARIANT: 0                      MAC KEY VARIANT: 0

                                LIMITS
    PIN KEY TIMER VALUE:      0              MAC KEY TIMER VALUE:      0
    PIN KEY TIMER INTERVAL: M (MINUTES)      MAC KEY TIMER INTERVAL: M (MINUTES)
    PIN KEY TRAN:            0              MAC KEY TRAN:            0
    PIN KEY ERROR:           0              MAC KEY ERROR:           0
    CONSECUTIVE PIN KEY ERROR: 0              CONSECUTIVE MAC KEY ERROR: 0
                                           KMAC SYNCHRONIZATION ERROR: 0

CLEAR OLD KEY TIMER VALUE:      0 (SECONDS)

    ORIGINATING ID:
    RECEIVING ID:

    KEY PROCESSING TYPE: N (NONE)
    NOTARIZATION SUPPORTED: 0 (NOT SUPPORTED)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                   F12-HELP
```

VRU Configuration Data (VCD) Screen 1

VCD screen 1 defines the processing options associated with a particular interface instance. For a complete discussion of the fields on VCD screen 1, refer to the *BASE24 Core Files and Tables Maintenance Manual*.

BASE24-CORE VRU CONFIGURATION

INTERFACE NAME:

PROCESSING OPTIONS

DE-SAF LIMIT: 1

MAX SAF RETRY: 0

NMM ENABLED: Y (Y/N)

AUTO LOGON: Y (Y/N)

KEY MGMT TYPE: N (NONE)

MAC TYPE: N (NONE)

MAC NMM: N (Y/N)

MAC DATA TYPE: A (ASCII)

IMP SYMBOLIC NAME:

PRODUCT INDICATOR TABLE (Y/N)

Y BASE

LLLL YY/MM/DD hh:mm 01 OF 45

INTERFACE TYPE:

FIID:

NAME:

ADDRESS 1:

ADDRESS 2:

POSTAL CODE:

COUNTRY:

UNKNOWN ROUTE CODE:

SAF GROUP:

DESTINATION GROUP:

MEASURE COUNTER NAME:

IAS QUEUE:

Y TB

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:

F12-HELP

NEW LOGICAL NETWORK ID:

VRU Configuration Data (VCD) Screen 2

VCD screen 2 defines additional processing options associated with a particular interface instance. For a complete discussion of the fields on VCD screen 2, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

```
BASE24-CORE VRU CONFIGURATION      LLLL      YY/MM/DD  hh:mm  02 OF 45
INTERFACE NAME:                     INTERFACE TYPE:

                                TIMER LIMITS
NETWORK MANAGEMENT:    30 (SEC)      EXTENDED NETWORK:    60 (SEC)
STORE AND FORWARD:    60 (SEC)      WAIT FOR TRAFFIC:    120 (SEC)
OVERLOAD:              30 (SEC)

THRESHOLD MAX OUTSTANDING REQUESTS: 0      MAX TIMEOUTS:      0
STAN UPDATE INTERVAL:  5000      STATION SELECT METHOD: R (ROTARY)
STAN UNIQUE INDICATOR: D (DESTINATION GROUP)
CURRENCY CODE: 840
CURRENCY CONVERSION GROUP:
BALANCE INFO OPTION: 1 (LEDGER BALANCE ONLY)

                                MESSAGE CLASS WILD CARD OPTIONS (Y/N)
SUB FIELD 1: N      SUB FIELD 2: N      SUB FIELD 3: N      SUB FIELD 4: N

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```


VRU Configuration Data (VCD) Screen 4

VCD screen 4 defines the dynamic key management processing options associated with a particular interface instance. For a complete discussion of the fields on VCD screen 4, refer to the *BASE24 Core Files and Tables Maintenance Manual*.

BASE24-CORE VRU CONFIGURATION
INTERFACE NAME: LLLL YY/MM/DD hh:mm 04 OF 45
INTERFACE TYPE:

LIMITS

	KPT	KAM	KDT
KEY SYNC ERRORS:	0		
TIMER VALUE:	0 (MIN)	0 (MIN)	0 (MIN)
TRANSACTIONS:	0	0	0
ERRORS:	0	0	0
CONSECUTIVE ERRORS:	0	0	0
SYNC ERRORS:		0	

KEY SYNC ERROR ACTION: 0 (EXCHANGE KEYS AND CONTINUE)
KPT CONSECUTIVE ERROR ACTION: 0 (EXCHANGE KEYS AND CONTINUE)
KAM CONSECUTIVE ERROR ACTION: 0 (EXCHANGE KEYS AND CONTINUE)
KDT CONSECUTIVE ERROR ACTION: 0 (EXCHANGE KEYS AND CONTINUE)
NUMBER OF KEYS: 1 (COMBINED)
ORIGINATING ID: RECEIVING ID:

RECORD LAST CHANGED: BY USER: /
***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

VRU Configuration Data (VCD) Screen 18

VCD screen 18 defines various BASE24 Remote Banking processing controls associated with an interface instance. For a complete discussion of the fields on VCD screen 18, refer to the ***BASE24 Core Files and Tables Maintenance Manual***.

```
BASE24-CORE VRU CONFIGURATION          LLLL      YY/MM/DD  hh:mm  18 OF 45
INTERFACE NAME:                        INTERFACE TYPE:

                                TELEBANKING INFORMATION
                                TIMER LIMITS
                                AUTH REQUEST:    15
                                MULTIPLE ACCOUNT SELECT: 60
                                QUEUE LIMIT:    5
                                DEFAULT SOURCE CODE:
                                DEFAULT POINT OF SVC DATA CODE: 61050061510C
                                MULTIPLE ACCOUNT SELECT MAX COUNT: 0
                                LAST TRANSACTION MAX COUNT: 0
                                CUSTOMER VENDOR MAX COUNT: 0
                                HIST INQ PMNT/XFER MAX COUNT: 0
                                HIST INQ ALL TRANS MAX COUNT: 0
                                MASTER VENDOR INQ MAX COUNT: 0
                                BANK LIST INQ MAX COUNT: 0
                                MISCELLANEOUS REQ MAX COUNT: 0
                                APPROVAL CODE LENGTH: 6
                                LOCAL PIN VERIFICATION: Y (Y/N)
                                DEFAULT TERMINAL ID:
                                DEFAULT ACQUIRER ID:
                                LOG ID: 0000
                                SCHEDULED TRANSFER MAX COUNT: 0
                                SCHEDULED PAYMENT MAX COUNT: 0
                                SCHEDULED TRANSACTIONS MAX COUNT: 0
                                ACCOUNT LIST MAX COUNT: 0
                                MULTIPLE ACCT BAL MAX COUNT: 0
                                BANK BRANCH LIST INQ MAX COUNT: 0
                                CUST VEND MASK INQ MAX COUNT: 0

RECORD LAST CHANGED:                  BY USER:
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:    NEW LOGICAL NETWORK ID:
                                F12-HELP
```

Appendix B

BASE24 Supercrypt Utility

BASE24 Remote Banking uses the BASE24 Supercrypt utility to encrypt certain keys for storage in the BASE24 database.

The BASE24 Supercrypt utility accepts clear keys as input and provides encrypted key values for entry in the BASE24 database. The keys encrypted in this manner are carried in the INBOUND PIN KEYS and OUTBOUND PIN KEYS fields on KEYF screen 2 and in the INBOUND KEYS:KPT1 and INBOUND KEYS:KPT2 fields on ISF screen 2; however, they only need to be encrypted in this manner when BASE24 Remote Banking requires clear PINs for its own processing and the external processor is using encrypted PINs.

Using the Supercrypt Utility

The remainder of this appendix describes how to use the BASE24 Supercrypt utility.

Starting the Supercrypt Utility

The BASE24 Supercrypt utility is a line mode utility that generates prompts and displays lines. This utility is run from a TACL prompt on a Tandem computer. The syntax for running the Supercrypt utility is described below.

Syntax

[RUN] [*\$volume.subvolume.*] SKCRYPT

volume — The name of the Tandem volume on which the Supercrypt utility is located. An operator must either move to the volume and subvolume on which the utility can be found, prior to entering the command, or fully qualify the name of the utility.

subvolume — The name of the Tandem subvolume on which the Supercrypt utility is located. An operator must either move to the volume and subvolume on which the utility can be found, prior to entering the command, or fully qualify the name of the utility.

Exiting the Supercrypt Utility

An operator can exit the utility by pressing the **Return** key or the **Control-Y** keys in answer to any of its prompts.

Accessing the Supercrypt Utility

The Supercrypt utility requests data for encryption immediately following startup, and repeats the request until the operator exits the utility. No menu is necessary to operate the Supercrypt utility because it is a single-function utility.

Using the Supercrypt utility involves the following steps:

1. Enter the 16-digit number to be encrypted (for example, 1234567890123456) at the “Enter data to be ENCRYPTED” prompt. Then press the **Return** key.

The utility replies with the following message, where the variable xxxxxxxxxxxxxxxx indicates the encrypted value (for example, F1843DF1FBEC273A).

Your super-key ENCRYPTED input is xxxxxxxxxxxxxxxx

The utility then displays another “Enter data to be ENCRYPTED” prompt.

2. Enter another 16-digit number to be encrypted or press the **Return** key or the **Control-Y** keys to exit the utility.

Error Messages

The Supercrypt utility displays the following possible error messages:

- If alphabetic characters other than A through F are entered, the following message is displayed:

Error - must be valid HEX digits.

The “Enter data to be ENCRYPTED” prompt is then displayed again.

- If fewer than 16 digits are entered before the **Return** key is pressed, the following message is displayed:

Must be 16 digits, try again

Enter the data to be encrypted again.

Applied Communications, Inc.

Appendix C

ASMcom Utility

This appendix describes the ASMcom utility provided by ACI for communicating with the Atalla Security Modules (ASMs) and the Atalla Network Security Processors (NSPs). The ASMcom utility is a line mode utility that generates prompts and displays lines on a terminal attached to a Tandem computer.

This appendix also provides operating procedures for each of the ASMcom utility options:

- Translating PIN working keys from encryption under the key exchange key (KEK) to encryption under the master file key (MFK) of the ASM or NSP
- Translating MAC working keys from encryption under the KEK to encryption under the MFK of the ASM or NSP
- Translating BASE24 keys from encryption under one MFK to encryption under a new MFK
- Validating check digits for BASE24 keys
- Generating check digits for BASE24 keys
- Selecting the security module device name
- Selecting the Logical Network Configuration File (LCONF)
- Selecting the ASMcom utility report output location

The material in this appendix is not meant to be a tutorial on the ASM or NSP; it presumes a familiarity with the device capabilities and nomenclature as discussed in documentation published by Atalla. This appendix also contains references to BASE24 products other than BASE24 Remote Banking.

Note: The check digit generation, validation, and BASE24 key translation options of the ASMcom utility result in the modification of security-related information in the BASE24 database. The use of this utility by anyone other than qualified, trusted network security staff is discouraged, and should not be attempted without a full understanding of the processing involved. These procedures do not currently

work with the Interface Security File (ISF). For the ISF, these procedures must be performed without the aid of the ASMcom utility with the results manually entered into the ISF.

BASE24 Keys

Key translation and check digit generation or validation can be performed for the following keys located in BASE24 files. The MFK variant also is shown for each key. These variants are used to generate and validate check digits.

Key	MFK Variant	Generate Check Digits	Validate Check Digits	Translate MFK to MFK	Translate KEK to MFK
Interface Security File (ISF)					
Intermediate Hardware	1				
PIN Inbound 1 and 2	1				✓
Key Authorization File (KEYA)					
IBM 3624-Format PIN Verification	4	✓	✓	✓	
Diebold PIN Verification	0			✓	
Visa PVV Verification	4	✓	✓	✓	
Key File (KEYF)					
Intermediate Encrypted	1	✓	✓	✓	
PIN Exchange	0	✓	✓	✓	
PIN Inbound 1 and 2	1	✓	✓	✓	✓
PIN Outbound 1 and 2	1	✓	✓	✓	✓
MAC Exchange	0	✓	✓	✓	
MAC Inbound 1 and 2	3	✓	✓	✓	✓
MAC Outbound 1 and 2	3	✓	✓	✓	✓

The ASMcom utility also processes keys located in the Terminal Data File (TDF), POS Terminal Data File (PTDF), and Teller Terminal Data File (TTDF). BASE24 Remote Banking products do not use the keys in these files.

Translating Keys

The ASMcom utility translates keys for use with Atalla ASM and NSP models.

The Atalla Security Initialization Unit (SIU) performs offline key management and the programming of the KEK or MFK into the Atalla Proprietary Part (APP) used by the ASM. Programming the KEK or MFK into the APP is also known as injecting the KEK or MFK. Injecting the KEK is required with ASM models A2 and A3. The ASMcom utility requires that the APP in all three models contain the KEK and MFK before the utility is started.

The NSP does not use APPs; however, the KEK and MFK still must be loaded in the security device before the ASMcom utility is started. The Key Loading Module (KLM) and the KLMKEY program are used to perform offline key management and the programming of the KEK and MFK directly into the NSP.

Changing Master File Keys

The ASMcom utility can translate hardware-encrypted keys in the BASE24 database from encryption under one MFK to another MFK, automating what otherwise is a manual process. These keys are listed at the start of this appendix. The network must be shut down while this function is performed, so its use requires careful planning and should only be performed by authorized personnel.

Old and new MFKs are required in the ASM or NSP to translate the encrypted keys from encryption under one MFK to another MFK. ASMs must have the old and new MFKs injected in the APP prior to starting this function. Normally, the APP contains the MFK and KEK. However, this function requires the APP to contain the new MFK in place of the MFK and the old MFK in place of the KEK. The previous KEK must be returned to the APP once this function is complete. NSPs must contain the old MFK as the current MFK and contain the new MFK as the pending MFK.

Note: This procedure does not currently work for the Interface Security File (ISF).

Variants

The security module used by the external processor affects the KEK used by the ASMcom utility. If the working key provided by the external processor has been encrypted with the KEK using a non-Atalla security module, an extra step must be taken in setting up the KEK for ASMcom utility use.

This extra step instructs the SIU or KLM to treat the constant for variant 1 or variant 3 as a component part in addition to all of the KEK component parts received from the external processor. The Atalla variant 1 constant (hexadecimal 0800 0000 0000 0000) is required when the KEK is being used for PIN keys. The Atalla variant 3 constant (hexadecimal 1800 0000 0000 0000) is required when the KEK is being used for message authentication code (MAC) keys. When all of the KEK component parts have been entered at the SIU or KLM, either for injection or for cryptogram display, the appropriate Atalla variant constant must be entered as the final component part.

The following table illustrates how the KEK for PIN keys is entered when the external processor is using Atalla equipment and when the external processor is using non-Atalla equipment. This illustration uses an MFK of 0123456789ABCDEF. The constant (hexadecimal 0800 0000 0000 0000) is for Atalla variant 1 because this KEK is for PIN keys.

SIU Entry	Atalla Equipment	Non-Atalla Equipment
Number of key parts	2	3
Part 1 (KEK part 1)	2222 2222 2222 2222	2222 2222 2222 2222
Part 2 (KEK part 2)	1212 1212 1212 1212	1212 1212 1212 1212
Part 3 (constant)	No entry	0800 0000 0000 0000
Cryptogram	F393 9F9C 3037 E8A5	841C CA39 C99D 6484

Using the ASMcom Utility

The information below provides operating instructions for the ASMcom utility.

Starting the ASMcom Utility

The BASE24 ASMcom utility is a line mode utility that generates prompts and displays lines. This utility is run from a TACL prompt on a Tandem computer. The syntax for running the ASMcom utility is described below.

Syntax

```
[ RUN ] [ $volume.subvolume. ] ASMCOM [ / IN device-name ]  
      [ ,OUT report-location/ ]
```

volume — The name of the Tandem volume on which the ASMcom utility is located. You must either move to the volume and subvolume on which the utility can be found, prior to entering the command, or fully qualify the name of the utility.

subvolume — The name of the Tandem subvolume on which the ASMcom utility is located. You must either move to the volume and subvolume on which the utility can be found, prior to entering the command, or fully qualify the name of the utility.

IN *device-name* — The name of the port to which the ASM or NSP is cabled (for example, \$ASM1). This name is configured using the Tandem SYSGEN utility. If the *device-name* portion of the command is omitted and the ASMcom utility needs to access the ASM or NSP, you are prompted to enter this value.

OUT *report-location* — The name of the spooler or disk file to which reports are sent. If the *report-location* portion of the command is omitted, you are prompted to enter this value after you select any ASMcom utility Main Menu option that includes reports.

Specialized Use of the Return Key

The use of the **Return** key to enter the input values on the ASMcom utility console display varies from field to field as follows:

Return Key	Condition
Required	If the input value is less than the maximum length allowed for the following: • The SYSGEN device name. • The report output location. • The LCONF name. • The final character if the keys are entered as 16-character string.
Not required	If the keys are entered with spaces.
	If selecting a menu option.
	If responding to questions that request single character responses.
Optional	If entering the KEK and working key because these keys can be entered in 16-character strings or in four groups of four characters each separated by spaces, according to operator preference.

ASMcom Utility Variables

The ASMcom utility documentation in this appendix uses the following variables:

device-name — The SYSGEN device name (for example, \$ASM1) for the ASM or NSP. You can enter this value when you start the utility or by selecting option 5 from the Main Menu. The ASMcom utility prompts you for this value if the processing option you select requires it and you have not yet entered it.

device-subtype — The code that identifies the security device subtype (that is, 1 for ASM and 2 for NSP).

encrypted-kek — The encrypted KEK used by the ASMcom utility to translate a PIN encryption or MAC encryption working key if the KEK has not been injected at the SIU.

lconf-name — The fully qualified name of the LCONF used by the logical network (for example, \sys.\$volume.subvolume.L1CONF) to which the ASM or NSP is attached. You can enter this value by selecting option 6 from the Main Menu. However, you cannot enter this value when you start the utility. The ASMcom utility prompts you for this value if the processing option you select requires it and you have not yet entered it.

output — The output location (for example, \$S.#PRTR for a spooler or \sys.\$volume.subvolume.PRTRFILE for a disk file). You can enter this value when you start the utility or by selecting option 7 from the Main Menu. The ASMcom utility prompts you for this value if the processing option you select requires it and you have not yet entered it.

working-key — The PIN encryption or MAC encryption working key being translated from encryption under the KEK to encryption under the MFK.

Main Menu

Upon startup, the ASMcom utility displays the following Main Menu:

```
Atalla Security Module Utility Program - ASMCOM - (DDMMYY)

-- Main Menu --

1)  Validate/Generate Check Digits
2)  Translate Security Keys
3)  Translate KEK-encrypted PIN Working Key for Local Storage
4)  Translate KEK-encrypted MAC Working Key for Local Storage
5)  Alter Security Module Device Name
6)  Alter LCONF Location
7)  Alter Summary Report Output Location

Enter selection code (1 - 7, "H" for Help, or "X" to Exit):
```

Enter one of the selection code values and the utility begins the selected function. The Main Menu reappears on completion of each function, prompting you to select the next action.

Validate/Generate Check Digits

Selection 1 on the Main Menu generates check digits for keys that do not have them or validates the check digits contained in one or more BASE24 files.

Note: This procedure does not currently work for the Interface Security File (ISF).

Check Digits Processing

Validating or generating check digits using the ASMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 1 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1-7, "H" for Help, or "X" to Exit)
Entry: 1
2. Enter the code that identifies the appropriate device subtype.
Prompt: Enter security device subtype (1 (A1/A2/A3 or 2 (NSP))
Entry: *device-subtype*
3. Enter the SYSGEN device name for the ASM or NSP. (The ASMcom utility skips this step if a valid device name has already been entered.)
Prompt: Enter ASM device name
Entry: *device-name*
4. Enter the report output location. (The ASMcom utility skips this step if a valid report output location has already been entered.)
Prompt: Enter Report File Name
Entry: *output*
5. Enter the LCONF name. (The ASMcom utility skips this step if a valid LCONF name has already been entered.)
Prompt: Enter LCONF File Name
Entry: *lconf-name*

The ASMcom utility at this point presents the Check Digits Menu, which is described under the topic "Check Digits Menu" in the following subsection.

6. Enter the code for the desired check digit processing at the selection code prompt on the Check Digits Menu. Valid values are as follows:

- 1 = Generate Check Digits; All BASE24 security keys
- 2 = Validate Check Digits; All BASE24 security keys
- 3 = Validate Check Digits; BASE24 KEYA security keys
- 4 = Validate Check Digits; BASE24 KEYF security keys
- 5 = Validate Check Digits; BASE24-atm TDF security keys
- 6 = Validate Check Digits; BASE24-pos PTDF security keys
- 7 = Validate Check Digits; BASE24-teller TTDF security keys

Note: Values 5–7 are not used for BASE24 Remote Banking.

Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, or 7, depending on processing to be performed

The ASMcom utility performs the requested processing, displays processing status messages if appropriate, and generates the report.

7. Enter another code at the selection code prompt on the Check Digits Menu to perform additional check digit processing or enter X to return to the Main Menu.

Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, 7, or X, depending on processing to be performed

If you entered 1 through 7, the utility performs the requested processing. If you entered X, make your next Main Menu selection.

Check Digits Menu

The Check Digits Menu is displayed in response to selecting 1 from the Main Menu and successfully entering the required information as described in steps 1 through 5 in the topic “Check Digits Processing.” The Check Digits Menu is shown below followed by descriptions of its fields.

```
Atalla Security Module Utility Program - ASMCOM - (DDMMYY)

-- Check Digits Menu --

1) Generate Check Digits; All BASE24 Security Keys
2) Validate Check Digits; All BASE24 Security Keys
3) Validate Check Digits; BASE24 KEYA Security Keys
4) Validate Check Digits; BASE24 KEYF Security Keys
5) Validate Check Digits; BASE24-atm TDF Security Keys
6) Validate Check Digits; BASE24-pos PTDF Security Keys
7) Validate Check Digits; BASE24-teller TTDF Security Keys

Enter selection code (1 - 7, "H" for Help, or "X" to Exit):
```

Generate Check Digits; All BASE24 Security Keys — Generates check digits for all keys currently in use that have check digit entries of zeros. Check digits are generated for keys in all files that the ASMcom utility can process.

Validate Check Digits; All BASE24 Security Keys — Validates the nonzero check digit entries for the keys that are currently in use. With this command check digit entries are validated in all files as if selections 3 through 7 were chosen sequentially.

Validate Check Digits; BASE24 KEYA Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the KEYA.

Validate Check Digits; BASE24 KEYF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the KEYF.

Validate Check Digits; BASE24-atm TDF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the TDF. This entry appears only if BASE24-atm support is bound into the ASMcom utility source file.

Validate Check Digits; BASE24-pos PTDF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the PTDF. This entry appears only if BASE24-pos support is bound into the ASMcom utility source file.

Validate Check Digits; BASE24-teller TTDF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the TTDF. This entry appears only if BASE24-teller support is bound into the ASMcom utility source file.

Using the Check Digits Menu

Enter one of the selection code values and the utility begins the selected function. The Check Digits Menu also appears on completion of each check digits function, containing any operator status messages and prompting you to make another check digits selection or return to the ASMcom utility Main Menu.

The BASE24 files that can be updated by the ASMcom utility check digits processing and the keys contained in each file are listed at the beginning of this appendix.

Translate Security Keys

Selection 2 on the Main Menu passes the keys contained in BASE24 files encrypted under one MFK to the ASM or NSP to be translated to encryption under a new MFK. The BASE24 files are then updated to contain the keys encrypted under the new MFK.

Note: This procedure does not currently work for the Interface Security File (ISF).

Key Translation Processing

Translating security keys using the ASMcom utility involves the following steps. The variables in *italics* were described earlier in this section.

1. Enter 2 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)
Entry: 2
2. Enter the code that identifies the appropriate device subtype.
Prompt: Enter security device subtype (1 (A1/A2/A3) or 2 (NSP))
Entry: *device-subtype*
3. Enter the SYSGEN device name for the ASM or NSP. (The ASMcom utility skips this step if a valid device name has already been entered.)
Prompt: Enter ASM device name
Entry: *device-name*
4. Enter the report output location. (The ASMcom utility skips this step if a valid report output location has already been entered.)
Prompt: Enter Report File Name
Entry: *output*
5. Enter the LCONF name. (The ASMcom utility skips this step if a valid LCONF name has already been entered.)
Prompt: Enter LCONF File Name
Entry: *lconf-name*

The ASMcom utility at this point presents, the Key Translation Menu, which is described under the topic “Key Translation Menu” in this section.

6. Enter the code for the desired key translation processing at the selection code prompt on the Key Translation Menu. Valid values are as follows:

- 1 = Translate all BASE24 security keys
- 2 = Translate BASE24 KEYA security keys
- 3 = Translate BASE24 KEYF security keys
- 4 = Translate BASE24-atm TDF security keys
- 5 = Translate BASE24-pos PTDF security keys
- 6 = Translate BASE24-teller TTDF security keys

Note: Values 4–6 are not used for BASE24 Remote Banking.

Prompt: Enter selection code (1–6, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, or 6, depending on processing to be performed

The ASMcom utility performs the requested processing, displays processing status messages if appropriate, and generates the report.

7. Enter another code at the selection code prompt on the Key Translation Menu to perform additional key translation processing or enter X to return to the Main Menu.

Prompt: Enter selection code (1–6, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, or X, depending on processing to be performed

If you entered 1 through 6, the utility performs the requested processing. If you entered X, make your next Main Menu selection.

Key Translation Menu

The Key Translation Menu is displayed in response to selecting 2 from the Main Menu and successfully entering the required information as described in steps 1 through 5 in the topic “Key Translation Processing.” The Key Translation Menu is shown below followed by descriptions of its fields.

```
Atalla Security Module Utility Program - ASMCOM - (DDMMYY)
```

```
-- Key Translation Menu --
```

- 1) Translate All BASE24 Security Keys
- 2) Translate BASE24 KEYA Security Keys
- 3) Translate BASE24 KEYF Security Keys
- 4) Translate BASE24-atm TDF Security Keys
- 5) Translate BASE24-pos PTDF Security Keys
- 6) Translate BASE24-teller TTDF Security Keys

```
Enter selection code (1 - 6, "H" for Help, or "X" to Exit):
```

Translate All BASE24 Security Keys — Translates the hardware-encrypted keys that are currently in use. This command translates keys in all files as if selections 2 through 6 were chosen sequentially.

Translate BASE24 KEYA Security Keys — Translates the hardware-encrypted keys that are currently in use in the KEYA.

Translate BASE24 KEYF Security Keys — Translates the hardware-encrypted keys that are currently in use in the KEYF or ISF.

Translate BASE24-atm TDF Security Keys — Translates the hardware-encrypted keys that are currently in use in the TDF. This entry appears only if BASE24-atm support is bound into the ASMcom utility source file.

Translate BASE24-pos PTDF Security Keys — Translates the hardware-encrypted keys that are currently in use in the PTDF. This entry appears only if BASE24-pos support is bound into the ASMcom utility source file.

Translate BASE24-teller TTDF Security Keys — Translates the hardware-encrypted keys that are currently in use in the TTDF. This entry appears only if BASE24-teller support is bound into the ASMcom utility source file.

Using the Key Translation Menu

Enter one of the selection code values and the utility begins the selected function. The Key Translation Menu also appears on completion of each key translation function, containing any operator status messages and prompting you to make another key translation selection or return to the ASMcom utility Main Menu.

The BASE24 files that can be updated by the ASMcom utility key translation processing and the keys contained in each file are listed at the beginning of this appendix. For more information on special key requirements for key translation reference the topic “Changing Master File Keys” earlier in this appendix.

Note: The network must be shut down while this function is performed, so its use requires special planning and should be performed only by authorized personnel.

Translate KEK-Encrypted PIN Working Key for Local Storage

Selection 3 on the Main Menu translates a KEK-encrypted PIN encryption working key manually sent from a host or interchange to encryption under the ASM or NSP MFK for manual entry on KEYF screen 2 or ISF screen 2. This ASMcom utility selection does not update any BASE24 files. This function is performed by selecting 3 on the ASMcom Main Menu.

The utility prompts for the KEK-encrypted PIN working key, then accesses the ASM or NSP to translate the working key from encryption under the KEK to encryption under the MFK. This translated key is then displayed, and can be copied down for entry into the KEYF or ISF.

Translating a KEK-encrypted PIN working key using the ASMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 3 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)
Entry: 3
2. Enter the SYSGEN device name for the ASM or NSP. (The ASMcom utility skips this step if a valid device name has already been entered.)
Prompt: Enter ASM device name
Entry: *device-name*
3. Enter a code to identify whether the KEK has been injected in the APP or must be manually entered.
Prompt: Has the KEK been injected in the APP?
Entry: Y or N
If you entered Y, go to step 5. If you entered N, go to step 4.
4. Enter the encrypted KEK to be used by the ASMcom utility. (The ASMcom utility skips this step if the KEK has been injected at the SIU.)
Prompt: Enter encrypted KEK
Entry: *encrypted-kek*
5. Enter the PIN working key to be translated.
Prompt: Enter KEK-encrypted PIN working key
Entry: *working-key*

6. Obtain the version of the key to be stored in the KEYF or ISF.

Prompt: Translated working key is “NNNN NNNN NNNN NNNN”
(nnnn nnnn nnnn nnnn)
(Copy this key down for entry into the Key File (KEYF))

Press RETURN to continue, or enter “X” to return to menu)

Note: The two values shown on the screen are uppercase and lowercase displays of the same key.

Entry: **Return** key to translate another PIN encryption working key or to return to the Main Menu.

If you pressed the **Return** key, go to step 3. If you entered X, make your next Main Menu selection.

Translate KEK-Encrypted MAC Working Key for Local Storage

Selection 4 on the Main Menu translates a KEK-encrypted MAC encryption working key manually sent from a host or interchange to encryption under the ASM or NSP MFK for manual entry on KEYF screen 2. This ASMcom utility selection does not update any BASE24 files. This function is performed by selecting 4 on the ASMcom Main Menu.

The utility prompts for the KEK-encrypted MAC working key, then accesses the ASM or NSP to translate the working key from encryption under the KEK to encryption under the MFK. This translated key is then displayed, and can be copied down for entry into the KEYF.

Translating a KEK-encrypted MAC working key using the ASMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 4 at the selection code prompt on the Main Menu.

Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)

Entry: 4

2. Enter the SYSGEN device name for the ASM or NSP. (The ASMcom utility skips this step if a valid device name has already been entered.)

Prompt: Enter ASM device name

Entry: *device-name*

3. Enter a code to identify whether the KEK has been injected in the APP or must be manually entered.

Prompt: Has the KEK been injected in the APP? (Y/N)

Entry: Y or N

If you entered Y, go to step 5. If you entered N, go to step 4.

4. Enter the encrypted KEK to be used by the ASMcom utility. (The ASMcom utility skips this step if the KEK has been injected at the SIU.)

Prompt: Enter encrypted KEK

Entry: *encrypted-kek*

5. Enter the working key to be translated.

Prompt: Enter KEK-encrypted MAC working key

Entry: *working-key*

6. Obtain the version of the key to be stored in the KEYF.

Prompt: Translated working key is “NNNN NNNN NNNN NNNN”
(nnnn nnnn nnnn nnnn)

(Copy this key down for entry into the Key File (KEYF))

Press RETURN to continue, or enter “X” to return to menu)

Note: The two values shown on the screen are uppercase and lowercase displays of the same key.

Entry: **Return** key to translate another MAC encryption working key or enter X to return to the Main Menu.

If you pressed the **Return** key, go to step 3. If you entered X following the prompt displayed in step 6, make your next Main Menu selection.

Alter Security Module Device Name

Selection 5 on the Main Menu allows you to specify or change the ASM or NSP being accessed by the utility. It can be used to switch to a different ASM or NSP device in the event of an error.

Altering a security module device name using the ASMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 5 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)
Entry: 5
2. Enter the code that identifies the appropriate security device subtype.
Prompt: Enter security device subtype (1 (A1/A2/A3) or 2 (NSP))
Entry: *device-subtype*
3. Enter the new SYSGEN device name for the ASM or NSP.
Prompt: Enter ASM device name
Entry: *device-name*
4. Make your next Main Menu selection.
Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)
Entry: 1, 2, 3, 4, 5, 6, 7, H, or X

Alter LCONF Location

Selection 6 on the Main Menu allows you to specify or change the LCONF accessed by the Validate/Generate Check Digits and Translate Security Keys functions (selections 1 and 2). Assigns in the LCONF identify the files that contain the keys included in ASMcom utility processing.

Altering an LCONF location using the ASMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 6 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)
Entry: 6
2. Enter the new LCONF name.
Prompt: Enter LCONF File Name
Entry: *lconf-name*
3. Make your next Main Menu selection.
Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)
Entry: 1, 2, 3, 4, 5, 6, 7, H, or X

Alter Summary Report Output Location

Selection 7 on the Main Menu allows you to change the output location for reports generated by the ASMcom utility. It also allows you to specify the report output location if you did not specify one during the ASMcom utility startup.

Altering a summary report output location using the ASMcom utility involves the following steps. The variables in *italics* were described earlier in this section.

1. Enter 7 at the selection code prompt on the Main Menu.

Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)

Entry: 7

2. Enter the new report output location.

Prompt: Enter Report File Name

Entry: *output*

3. Make your next Main Menu selection.

Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, 7, H, or X

Help Function

The ASMcom utility Help selection provides an online description of the ASMcom utility selections and gives instructions on how to exit the utility. This selection is accessed by pressing H any time one of the menus or prompts is displayed. Help presents information requirements for each topic on the ASMcom utility Main Menu.

Status Messages

The ASMcom utility displays status messages on the computer terminal screen and includes them on reports. These messages indicate processing results and identify the cause of any processing problems encountered.

These messages are documented in the ATALMSGs file located on the BA53ATAL subvolume (the subvolume that contains source code for the ASMcom utility). A Tandem editor can be used to read this file.

Exiting the ASMcom Utility

You can exit the utility by pressing X or the **Control-Y** keys in response to the Main Menu prompt. Pressing either of these keys in response to any other prompt displays the next higher level menu.

Reports

The ASMcom utility generates three reports:

- Check digit generation report
- Check digit validation report
- Key translation report

The following files can appear on reports generated by the ASMcom utility:

- Key Authorization File (KEYA)
- Key File (KEYF)
- Terminal Data File (TDF)
- POS Terminal Data File (PTDF)
- Teller Terminal Data File (TTDF)

These reports are illustrated on the following pages.

Check Digit Generation Report

A Check Digit Generation Report identifies the keys processed under the ASMcom utility Generate Check Digits option. The report shown below is a sample of the one generated for keys in the KEYA. All the keys listed may not be processed for each KEYA record because keys with zeros in the key field are not processed.

ASMCOM CHECK DIGIT GENERATION REPORT FOR PR01		MM/DD/YY	PAGE 1
		HH:MM:SS	
\$vol.subvol.KEYA			
(any KEYA processing errors appear here)			
TOTAL RECORDS PROCESSED	0	CHECK DIGITS	
KEY TYPES	GENERATED	IN ERROR	
IBM-DES	0	0	
VISA PVV	0	0	
CV	0	0	
TOTAL KEYA CHECK DIGITS PROCESSED	0	0	

Check Digit Validation Report

The Check Digit Validation Report identifies the keys processed under the ASMcom utility Validate Check Digits option. The report shown below is a sample of the one generated for keys in the KEYF. All the keys listed may not be processed for each KEYF record because keys with zeros in the key field are not processed.

ASMCOM CHECK DIGIT VALIDATION REPORT FOR PR01		MM/DD/YY	PAGE 1
		HH:MM:SS	
\$vol.subvol.KEYF			
(any KEYF processing errors appear here)			
TOTAL RECORDS PROCESSED		0	
KEY TYPES	CHECK DIGITS		
	VALIDATED	IN ERROR	
INTERM.KEY-ENCRYPT	0	0	
EXCHNG-KEY	0	0	
OUTBOUND.PIN.KEY1	0	0	
OUTBOUND.PIN.KEY2	0	0	
INBOUND.PIN.KEY1	0	0	
INBOUND.PIN.KEY2	0	0	
OUTBOUND.MAC.KEY1	0	0	
OUTBOUND.MAC.KEY2	0	0	
INBOUND.MAC.KEY1	0	0	
INBOUND.MAC.KEY2	0	0	
MAC-EXCHNG-KEY	0	0	
TOTAL KEYF CHECK DIGITS PROCESSED		0	0

Key Translation Report

The Key Translation Report identifies the keys processed under the ASMcom utility Translate Security Keys option. The report shown below is a sample of the one generated for keys in the KEYF. All the keys listed may not be processed for each KEYF record because keys that are not in use are not processed.

ASCOM KEY TRANSLATION REPORT FOR PR01		MM/DD/YY	PAGE 1
		HH:MM:SS	
\$vol.subvol.KEYF			
(any KEYF processing errors appear here)			
TOTAL RECORDS PROCESSED		0	
KEY TYPES	KEYS TRANSLATED	KEY ERRORS	
INTERM.KEY-ENCRYPT	0	0	
EXCHNG-KEY	0	0	
OUTBOUND.PIN.KEY1	0	0	
OUTBOUND.PIN.KEY2	0	0	
INBOUND.PIN.KEY1	0	0	
INBOUND.PIN.KEY2	0	0	
OUTBOUND.MAC.KEY1	0	0	
OUTBOUND.MAC.KEY2	0	0	
INBOUND.MAC.KEY1	0	0	
INBOUND.MAC.KEY2	0	0	
MAC-EXCHNG-KEY	0	0	
TOTAL KEYF KEYS PROCESSED		0	0

Applied Communications, Inc.

Appendix D

RSMcom Utility

This appendix describes the RSMcom utility provided by ACI for communicating with the Racal-Guardata Host Security Module (RSM). The RSMcom utility is a line mode program that generates prompts and displays lines on a terminal attached to a Tandem computer.

The RSMcom utility was formerly called the VSMcom utility; the name change coincides with the Visa Security Module (VSM) becoming the Racal-Guardata Host Security Module (RSM). BASE24 DDLs and configuration information may still contain references to VSM, and such references are equally applicable to the RSM.

This appendix also provides operating procedures for each of the RSMcom utility options:

- Generating clear intermediate keys
- Translating PIN working keys encrypted under the zone master key (ZMK) to encryption under the local master key (LMK) of the RSM
- Translating MAC working keys encrypted under the ZMK to encryption under the LMK of the RSM
- Translating BASE24 keys from encryption under one LMK to encryption under a new LMK
- Validating check digits for BASE24 keys
- Generating check digits for BASE24 keys
- Selecting the security module device name
- Selecting the Logical Network Configuration File (LCONF)
- Selecting the RSMcom utility report output location
- Uploading the encrypted Diebold number table (DNT) to the Key Authorization File (KEYA)

The material in this appendix is not meant to be a tutorial on the RSM; it presumes a familiarity with the device capabilities and nomenclature as discussed in documentation published by Racal-Guardata. This appendix also contains references to BASE24 products other than BASE24 Remote Banking.

Note: The check digit generation, validation, and BASE24 key translation options of the RSMcom utility result in the modification of security-related information in the BASE24 database. The use of this utility by anyone other than qualified, trusted network security staff is discouraged, and should not be attempted without a full understanding of the processing involved. These procedures do not currently work with the Interface Security File (ISF). For the ISF, these procedures must be performed without the aid of the RSMcom utility with the results manually entered into the ISF.

BASE24 Keys

Key translation and check digit generation or validation can be performed for the following keys located in BASE24 files.

Key	LMK Pair	Generate Check Digits	Validate Check Digits	Translate LMK to LMK	Translate ZMK to LMK
Interface Security File (ISF)					
Intermediate Hardware	06–07				
PIN Inbound 1 and 2	06–07				✓
Key Authorization File (KEYA)					
IBM 3624-Format Verification	14–15	✓	✓	✓	
Visa PVV Verification	14–15	✓	✓	✓	
Key File (KEYF)					
Intermediate Encrypted	06–07	✓	✓	✓	
PIN Exchange	04–05	✓	✓	✓	
PIN Inbound 1 and 2	06–07	✓	✓	✓	✓
PIN Outbound 1 and 2	06–07	✓	✓	✓	✓
MAC Exchange	04–05	✓	✓	✓	
MAC Inbound 1 and 2	16–17	✓	✓	✓	✓
MAC Outbound 1 and 2	16–17	✓	✓	✓	✓

The RSMcom utility also processes keys located in the Terminal Data File (TDF), POS Terminal Data File (PTDF), and Teller Terminal Data File (TTDF). BASE24 Remote Banking products do not use the keys in these tables.

Key Pairs

The RSM used by the RSMcom utility contains a local master key (LMK) that consists of 24 keys. These keys are arranged in pairs to provide 12 double-length keys. Each key pair, also called an LMK pair, is designated for use with a certain type of data. The list of keys at the start of this appendix shows the LMK pair used for each BASE24 key that can be processed by the RSMcom utility.

Changing Local Master Key Pairs

The relationship between BASE24 keys and LMK pairs becomes especially important when the RSMcom utility is used to translate BASE24 keys between the old and new values of an LMK pair (the RSMcom utility Main Menu selection 4). When the value of an LMK pair changes, all BASE24 keys using that LMK pair must be translated from encryption under the old value of the LMK pair to the new value of the LMK pair before they can be used. This selection must be processed for each possible combination of BASE24 file and LMK pair. After specifying the BASE24 file to be processed, you must specify the LMK pair or pairs being changed.

Note: This procedure does not currently work for the Interface Security File (ISF).

The RSM can translate only one LMK pair at a time, regardless of the number of LMK pairs that need to be translated. As a result, it is important to know which BASE24 files must be processed when a particular LMK pair changes. LMK pairs affect the following BASE24 files:

LMK Pair	KEYA	KEYF	PTDF	TDF	TTDF
LMK pair 04-05		✓			
LMK pair 06-07		✓			
LMK pair 14-15	✓		✓	✓	✓
LMK pair 16-17		✓	✓	✓	✓

The RSMcom utility Main Menu selection 4 provides the option of translating keys under a specific LMK pair or under all four possible LMK pairs. When you select the option to translate keys under all four possible LMK pairs, you must load the old and new values for LMK pair 04–05 in the RSM before you start processing even if the BASE24 files selected do not contain keys encrypted under

LMK pair 04–05. After all keys in the selected BASE24 files that use LMK pair 04–05 have been translated, the RSMcom utility instructs you to load the old and new values for LMK pair 06–07 in the RSM, then pauses until you load the LMK pair 06–07 values. All keys in the selected BASE24 files that use LMK pair 06–07 are then translated. This process is repeated for LMK pairs 14–15 and 16–17.

The old and new values for the LMK pair are required in the RSM to translate the encrypted keys from encryption under one value of an LMK pair to encryption under another value of the same LMK pair. The old value of the LMK pair must be transferred to the key change storage area of the RSM, then the new value of the LMK pair must overlay the old value of the LMK pair. The RSM offline console command T is used to transfer the old value to the key change storage area within the RSM, and the RSM offline console command K is used to load the new value.

The BASE24 network must be shut down while this RSMcom utility function is performed, so its use requires careful planning and should only be performed by authorized personnel.

Changing the Diebold Number Table Key Pair

The Diebold number table (DNT) in the KEYA uses LMK pair 10–11 and must be updated whenever LMK pair 10–11 is changed. This LMK pair is not included in the processing performed by the RSMcom utility Main Menu selection 5 (translating BASE24 keys between encryption under old and new LMK pair values). Translation involves two encrypted values. Since the RSM contains a clear DNT and the KEYA contains an encrypted DNT, translation does not work to update the DNT in the KEYA.

The RSMcom utility Main Menu selection 9 (upload DNT to KEYA) uploads the DNT to the KEYA. The DNT must be entered at the RSM console before the RSMcom utility Main Menu selection 9 is used to upload it.

Using the RSMcom Utility

The information below provides operating instructions for the RSMcom utility.

Starting the RSMcom Utility

The BASE24 RSMcom utility is a line mode program that generates prompts and displays lines. This utility is run from a TACL prompt on a Tandem computer. The syntax for running the RSMcom utility is described below.

Syntax

```
[ RUN ] [ $volume.subvolume. ] RSMCOM [ / IN device-name ]  
      [ ,OUT report-location/ ]
```

volume — The name of the Tandem volume on which the RSMcom program is located. You must either move to the volume and subvolume on which the program can be found, prior to entering the command, or fully qualify the name of the program.

subvolume — The name of the Tandem subvolume on which the RSMcom program is located. You must either move to the volume and subvolume on which the program can be found, prior to entering the command, or fully qualify the name of the program.

IN *device-name* — The name of the port to which the RSM is cabled (for example, \$RSM1). This name is configured using the Tandem SYSGEN utility. If the *device-name* portion of the command is omitted and the RSMcom utility needs to access the RSM, you are prompted to enter this value.

OUT *report-location* — The name of the spooler or disk file to which reports are sent. If the *report-location* portion of the command is omitted, you are prompted to enter this value after you select any RSMcom utility Main Menu option that includes reports.

Specialized Use of the Return Key

The use of the **Return** key to enter the input values on the RSMcom utility console display varies from field to field as follows:

Return Key	Condition
Required	If the input value is less than the maximum length allowed for the following: • The SYSGEN device name. • The report output location. • The LCONF name. • The DNT relative location. • The final character if the keys are entered as 16-character string.
Not required	If the keys are entered with spaces.
	If selecting a menu option.
	If responding to questions that request single character responses.
Optional	If entering the ZMK, ZMK components, and the working key because these keys can be entered in 16-character strings or in four groups of four characters each separated by spaces, according to operator preference.

Documentation of the RSMcom Utility Variables

The RSMcom utility documentation in this appendix uses the following variables:

device-name — The SYSGEN device name (for example, \$RSM1) for the RSM. You can enter this value when you start the utility or by selecting option 6 from the Main Menu. The RSMcom utility prompts you for this value if the processing option you select requires it and you have not yet entered it.

dnt-location — The relative location of the Diebold number table (DNT) within the RSM. The RSMcom utility calculates the Racal user storage address from this number. This value must not exceed the total number of DNTs actually being stored in the RSM.

keya-group — The FIID or card prefix for which a DNT has been created. (The KEYA record for this group ID must already exist. The RSMcom utility does not add it.)

lconf-name — The fully qualified name of the LCONF used by the logical network (for example, \sys.\$volume.subvolume.L1CONF) to which the RSM is attached. You can enter this value by selecting option 7 from the Main Menu. However, you cannot enter this value when you start the utility. The RSMcom utility prompts you for this value if the processing option you select requires it and you have not yet entered it.

lmk-pair — The code identifying the LMK pair that must be loaded before key translation processing can continue.

mac-working-key — The MAC encryption working key being translated from encryption under the ZMK to encryption under LMK pair 16–17.

output — The output location (for example, \$S.#PRTR for a spooler or \sys.\$volume.subvolume.PRTRFILE for a disk file). You can enter this value when you start the utility or by selecting option 8 from the Main Menu. The RSMcom utility prompts you for this value if the processing option you select requires it and you have not yet entered it.

pin-working-key — The PIN encryption or MAC encryption working key being translated from encryption under the ZMK to encryption under the LMK pair 06-07.

zmk — An encrypted zone master key. The ZMK is used by a host or interchange to encrypt a manually sent PIN or MAC working key.

zmk-component — One of three clear values that are combined to form a key that is used to decrypt the intermediate key. A number (that is, 1, 2, 3) follows this variable name to uniquely identify each component.

zmk-intermediate-key — An intermediate key that is encrypted under the ZMK.

Main Menu

Upon startup, the RSMcom utility displays the following Main Menu:

```
Racal Security Module Utility Program - RSMCOM - (DDMMYY)
```

```
-- Main Menu --
```

- 1) Generate Clear Intermediate Working Key
- 2) Translate PIN Working Key from ZMK to LMK Encryption
- 3) Translate MAC Working Key from ZMK to LMK Encryption
- 4) Validate/Generate Check Digits
- 5) Translate Security Keys
- 6) Alter Security Module Device Name
- 7) Alter LCONF Location
- 8) Alter Summary Report Output Location
- 9) Upload Diebold Number Table to KEYA

```
Enter selection code (1 - 9, "H" for Help, or "X" to Exit):
```

Enter one of the selection code values and the program begins the selected function. The Main Menu reappears on completion of each function, prompting you to select the next action.

Generate Clear Intermediate Working Key

Selection 1 on the Main Menu decrypts a ZMK-encrypted intermediate key, generated by the RSM console command B, to obtain a clear intermediate key for storage in the KEYF or ISF.

The program prompts for the three clear ZMK components (which can be chosen or can be generated by the RSM console command F), and then prompts for the ZMK-encrypted intermediate working key (generated by the RSM console command B).

The ZMK components are combined to form the key that is used to decrypt the intermediate key. The resulting clear intermediate key is then displayed and can be copied down for entry in the INTERMEDIATE KEYS: CLEAR field on KEYF screen 1 or the INTERM SFTWR KEY on ISF screen 2.

Note: Selection 2, Translate ZMK-encrypted PIN working key, must be used following selection 1 to obtain the value entered in the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1 or the INTERM HDWR KEY on ISF screen 2.

Clear Intermediate Working Keys Processing

Decrypting a ZMK-encrypted intermediate key using the RSMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 1 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)
Entry: 1
2. Enter the 16 hexadecimal characters for the first clear ZMK component.
Prompt: Enter clear-ZMK component number 1
(from console command “F”)
Entry: *zmk-component-1*
3. Enter the 16 hexadecimal characters for the second clear ZMK component.
Prompt: Enter clear-ZMK component number 2
(from console command “F”)
Entry: *zmk-component-2*

4. Enter the 16 hexadecimal characters for the third clear ZMK component.

Prompt: Enter clear-ZMK component number 3
(from console command “F”)

Entry: *zmk-component-3*
5. Enter the 16 hexadecimal characters for the ZMK-encrypted intermediate working key (key to be decrypted).

Prompt: Enter ZMK-encrypted work key
(WK ENCRYPTED FOR TRANSMISSION)

Entry: *zmk-intermediate-key*
6. Obtain the clear intermediate working key.

Prompt: Clear working key is “NNNN NNNN NNNN NNNN”
(*nnnn nnnn nnnn nnnn*)
(Copy this key down for entry into the Key File (KEYF))

Press RETURN to continue, or enter “X” to return to menu)

Note: The two values shown on the screen are uppercase and lowercase displays of the same key.

Entry: **Return** key to generate another key or X to return to the Main Menu.

If you pressed X, make your next Main Menu selection. If you pressed the **Return** key, go to step 7.
7. Enter a code to identify whether to retain or modify the ZMK.

Prompt: Shall I use the current clear ZMK? (Y/N)

Entry: Y or N

If you entered Y, go to step 5. If you entered N, go to step 2.

Translate PIN Working Key from ZMK to LMK Encryption

Selection 2 on the Main Menu translates a ZMK-encrypted PIN working key manually sent from a host or interchange to encryption under the RSM LMK pair 06–07 for entry in the PIN INBOUND KEY or PIN OUTBOUND KEY field on KEYF screen 2 or for entry in the INBOUND KPT1 or INBOUND KPT2 fields on ISF screen 2. This selection also provides the encrypted version of an intermediate key for entry in the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1 or for entry in the INTERM HDWR KEY field on ISF screen 2.

The program prompts for the encrypted Zone Master Key and the ZMK-encrypted PIN working key, then accesses the RSM to translate the working key from encryption under the ZMK to encryption under the RSM LMK pair 06–07. This translated key is then displayed, and can be copied down for entry in the appropriate field on KEYF screen 1 or 2 or ISF screen 2.

Translating a ZMK-encrypted PIN working key using the RSMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 2 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)
Entry: 2
2. Enter the SYSGEN device name for the RSM. (The RSMcom utility skips this step if a valid device name has already been entered.)
Prompt: Enter RSM device name
Entry: *device-name*
3. Enter the 16 hexadecimal characters of the encrypted ZMK to be used by the RSMcom utility.
Prompt: Enter encrypted ZMK (from console command “D”)
Entry: *zmk*
4. Enter the 16 hexadecimal characters of the ZMK-encrypted working key to be translated.
Prompt: Enter ZMK-encrypted PIN working key
(WK ENCRYPTED FOR TRANSMISSION)
Entry: *pin-working-key*
5. Obtain the version of the key to be stored in the KEYF or ISF.
Prompt: Translated working key is “NNNN NNNN NNNN NNNN”
(*nnnn nnnn nnnn nnnn*)
(Copy this key down for entry into the Key File (KEYF)

Press RETURN to continue, or enter “X” to return to menu)
Note: The two values shown on the screen are uppercase and lowercase displays of the same key.
Entry: **Return** key to translate another key or enter X to return to the Main Menu.

If you entered X, make your next Main Menu selection. If you pressed the **Return** key, go to step 6.

6. Enter a code to identify whether to retain or modify the ZMK.

Prompt: Shall I use the current ZMK? (Y/N)

Entry: Y or N

If you entered Y, go to step 4. If you entered N, go to step 3.

Translate MAC Working Key from ZMK to LMK Encryption

Selection 3 on the Main Menu translates a ZMK-encrypted MAC working key manually sent from a host or interchange to encryption under the RSM LMK pair 16–17 for entry in the MAC INBOUND KEY or MAC OUTBOUND KEY field on KEYF screen 2.

The program prompts for the encrypted zone master key and the ZMK-encrypted working key, then accesses the RSM to translate the MAC working key from encryption under the ZMK to encryption under the RSM LMK pair 16–17. This translated key is then displayed, and can be copied down for entry in the appropriate field on KEYF screen 2.

Translating a ZMK-encrypted MAC working key using the RSMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 3 at the selection code prompt on the Main Menu.

Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)

Entry: 3

2. Enter the SYSGEN device name for the RSM. (The RSMcom utility skips this step if a valid device name has already been entered.)

Prompt: Enter RSM device name

Entry: *device-name*

3. Enter the 16 hexadecimal characters of the encrypted ZMK to be used by the RSMcom utility.

Prompt: Enter encrypted ZMK (from console command “D”)

Entry: *zmk*

4. Enter the 16 hexadecimal characters of the ZMK-encrypted MAC working key to be translated.

Prompt: Enter ZMK-encrypted MAC working key
(WK ENCRYPTED FOR TRANSMISSION)

Entry: *mac-working-key*

5. Obtain the version of the key to be stored in the KEYF.

Prompt: Translated working key is “NNNN NNNN NNNN NNNN”
(nnnn nnnn nnnn nnnn)
(Copy this key down for entry into the Key File (KEYF))

Press RETURN to continue, or enter “X” to return to menu)

Note: The two values shown on the screen are uppercase and lowercase displays of the same key.

Entry: **Return** key to translate another key or enter X to return to the Main Menu.

If you entered X, make your next Main Menu selection. If you pressed the **Return** key, go to step 6.

6. Enter a code to identify whether to retain or modify the ZMK.

Prompt: Shall I use the current ZMK? (Y/N)

Entry: Y or N

If you entered Y, go to step 4. If you entered N, go to step 3.

Validate/Generate Check Digits

Selection 4 on the Main Menu generates check digits for keys that do not have them or validates the check digits contained in one or more BASE24 files.

Note: This procedure does not currently work for the Interface Security File (ISF).

Check Digits Processing

Validating or generating check digits using the RSMcom utility involves the following steps. The variables in *italics* were described earlier in this section.

1. Enter 4 at the selection code prompt on the Main Menu.

Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)

Entry: 4

2. Enter the SYSGEN device name for the RSM. (The RSMcom utility skips this step if a valid device name has already been entered.)

Prompt: Enter RSM device name

Entry: *device-name*

3. Enter the LCONF name. (The RSMcom utility skips this step if a valid LCONF name has already been entered.)

Prompt: Enter LCONF File Name

Entry: *lconf-name*

4. Enter the report output location. (The RSMcom utility skips this step if a valid report output location has already been entered.)

Prompt: Enter report file name

Entry: *output*

The RSMcom utility at this point presents, the Check Digits Menu, which is described under the topic “Check Digits Menu” in the following subsection.

5. Enter the code for the desired check digit processing at the selection code prompt on the Check Digits Menu. Valid values are as follows:

1 = Generate Check Digits; All BASE24 security keys

2 = Validate Check Digits; All BASE24 security keys

3 = Validate Check Digits; BASE24 KEYA security keys

4 = Validate Check Digits; BASE24 KEYF security keys

5 = Validate Check Digits; BASE24-atm TDF security keys

6 = Validate Check Digits; BASE24-pos PTDF security keys

7 = Validate Check Digits; BASE24-teller TTDF security keys

Note: Values 5–7 are not used for BASE24 Remote Banking.

Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, or 7, depending on processing to be performed

The RSMcom utility performs the requested processing, displays status messages, if appropriate, and generates reports.

6. Enter another code at the selection code prompt on the Check Digits Menu to perform additional check digit processing or enter X to return to the Main Menu.

Prompt: Enter selection code (1–7, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, 7, or X, depending on processing to be performed

If you entered 1 through 7, the utility performs the requested processing. If you entered X, make your next Main Menu selection.

Check Digits Menu

The Check Digits Menu is displayed in response to selecting 4 from the Main Menu and entering the required information as described in steps 1 through 4 in the topic “Check Digits Processing.” The Check Digits Menu is shown below, followed by descriptions of its fields.

```
Racal Security Module Utility Program - RSMCOM - (DDMMYY)

-- Check Digits Menu --

1) Generate Check Digits; All BASE24 Security Keys
2) Validate Check Digits; All BASE24 Security Keys
3) Validate Check Digits; BASE24 KEYA Security Keys
4) Validate Check Digits; BASE24 KEYF Security Keys
5) Validate Check Digits; BASE24-atm TDF Security Keys
6) Validate Check Digits; BASE24-pos PTDF Security Keys
7) Validate Check Digits; BASE24-teller TTDF Security Keys

Enter selection code (1 - 7, "H" for Help, or "X" to Exit):
```

Generate Check Digits; All BASE24 Security Keys — Generates check digits for all keys currently in use that do not have check digits. Check digits are generated for keys in all files that the ASMcom utility can process.

Validate Check Digits; All BASE24 Security Keys — Validates the nonzero check digit entries for the keys that are currently in use. With this command check digit entries are validated in all files as if selections 3 through 7 were chosen sequentially.

Validate Check Digits; BASE24 KEYA Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the KEYA.

Validate Check Digits; BASE24 KEYF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the KEYF.

Validate Check Digits; BASE24-atm TDF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the TDF. This entry appears only if BASE24-atm support is bound into the ASMcom utility source file.

Validate Check Digits; BASE24-pos PTDF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the PTDF. This entry appears only if BASE24-pos support is bound into the ASMcom utility source file.

Validate Check Digits; BASE24-teller TTDF Security Keys — Validates the nonzero check digit entries for the keys that are currently in use in the TTDF. This entry appears only if BASE24-teller support is bound into the ASMcom utility source file.

Using the Check Digits Menu

Enter one of the selection code values and the utility begins the selected function. The Check Digits Menu also appears on completion of each check digits function, containing any operator status messages and prompting you to make another check digits selection or return to the RSMcom utility Main Menu.

The BASE24 files that can be updated by the RSMcom utility check digits processing and the keys contained in each file are listed at the beginning of this appendix. For all selections, a key with a check digit entry of zeros is considered the same as a key with no check digit entry.

Translate Security Keys

Selection 5 on the Main Menu passes the keys contained in BASE24 files encrypted under an LMK pair to the RSM to be translated to encryption under a new value of the same LMK pair. The BASE24 files are then updated to contain the keys encrypted under the new value of the LMK pair. This selection involves two menus: a Key Translation Menu and a Key Translation Submenu.

Note: The network must be shut down while this selection is performed, so its use requires special planning and should only be performed by authorized personnel. In addition, this procedure does not currently work for the Interface Security File (ISF).

Key Translation Processing

Translating security keys using the RSMcom utility involves the following steps. The variables in italics were described earlier in this section.

Note: With the complexity of different LMK pair combinations for each BASE24 file and the requirement of manually changing the LMK pair information in the RSM for each LMK pair, special planning is required to make the appropriate selections on the Key Translation Menu and the Key Translation Submenu.

1. Enter 5 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)
Entry: 5
2. Enter the SYSGEN device name for the RSM. (The RSMcom utility skips this step if a valid device name has already been entered.)
Prompt: Enter RSM device name
Entry: *device-name*
3. Enter the LCONF name. (The RSMcom utility skips this step if a valid LCONF name has already been entered.)
Prompt: Enter LCONF File Name
Entry: *lconf-name*
4. Enter the report output location. (The RSMcom utility skips this step if a valid report output location has already been entered.)
Prompt: Enter report file name
Entry: *output*

The RSMcom utility at this point presents the Key Translation Menu, which is described under the topic “Key Translation Menu” in the following subsection.

5. Enter the code for the files containing the keys to be translated. Valid values are as follows:
 - 1 = Translate all BASE24 security keys
 - 2 = Translate BASE24 KEYA security keys
 - 3 = Translate BASE24 KEYF security keys
 - 4 = Translate BASE24-atm TDF security keys
 - 5 = Translate BASE24-pos PTDF security keys
 - 6 = Translate BASE24-teller TTDF security keys

Note: Values 4–6 are not used for BASE24 Remote Banking.

Prompt: Enter selection code (1–6, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, or 6, depending on which LMK pairs are being changed

The RSMcom utility at this point presents the Key Translation Submenu, which is described under the topic “Key Translation Submenu” in the following subsection.

6. Enter the code for the LMK pairs being changed. Valid values are as follows:
 - 1 = Translate security keys; all
 - 2 = Translate security keys; ZMK (LMK pairs 04-05)
 - 3 = Translate security keys; ZWK (LMK pairs 06-07)
 - 4 = Translate security keys; CVK, TMK, TPK, PVK (LMK pairs 14-15)
 - 5 = Translate security keys; TAK (LMK pairs 16-17)

Prompt: Validating or generating check digits using the ASMcom utility.

Entry: 1, 2, 3, 4, or 5, depending on which LMK pairs are being changed

If you entered 1 (change all LMK pairs) at the Key Translation Submenu, go to step 7.

If you entered selections 2 through 5 (changing one LMK pair), remain at step 6 to enter another code at the selection code prompt on the Key Translation Submenu to change additional key pairs or enter X to return to the Key Translation Menu and go to step 8.

7. If you entered 1 (change all LMK pairs) in step 6, enter the RSM key change storage information for each LMK pair as it is requested by the RSMcom utility.

Prompt: Load RSM key change storage for key pair *lmk-pair*
Press "X" to exit or press RETURN to continue when ready

Entry: X or the **Return** key, depending on processing to be performed

Continue to load the RSM key change storage information until all requested information has been completed and go to step 8.

8. The utility performs the requested processing, displays status messages, if appropriate, and generates the reports.

Enter another code at the selection code prompt on the Key Translation Menu to perform key translation processing for additional BASE24 files or enter X to return to the Main Menu.

Prompt: Enter selection code (1–6, "H" for Help, or "X" to Exit)

Entry: 1, 2, 3, 4, 5, or 6, depending on processing to be performed

If you entered 1 through 6, go to step 2. If you entered X, make your next Main Menu selection.

Key Translation Menu

The Key Translation Menu is displayed in response to selecting 5 from the Main Menu and successfully entering the required information as described in steps 1 through 5 in the topic “Key Translation Processing.” The Key Translation Menu is shown below, followed by descriptions of its fields.

```
Racal Security Module Utility Program - RSMCOM - (DDMMYY)

-- Key Translation Menu --

1) Translate All BASE24 Security Keys
2) Translate BASE24 KEYA Security Keys
3) Translate BASE24 KEYF Security Keys
4) Translate BASE24-atm TDF Security Keys
5) Translate BASE24-pos PTDF Security Keys
6) Translate BASE24-teller TTDF Security Keys

Enter selection code (1 - 6, "H" for Help, or "X" to Exit):
```

Translate All BASE24 Security Keys — Translates the hardware-encrypted keys that are currently in use. This command translates keys in all files as if selections 2 through 6 were chosen sequentially.

Translate BASE24 KEYA Security Keys — Translates the hardware-encrypted keys that are currently in use in the KEYA.

Translate BASE24 KEYF Security Keys — Translates the hardware-encrypted keys that are currently in use in the KEYF.

Translate BASE24-atm TDF Security Keys — Translates the hardware-encrypted keys that are currently in use in the TDF. This entry appears only if BASE24-atm support is bound into the ASMcom utility source file.

Translate BASE24-pos PTDF Security Keys — Translates the hardware-encrypted keys that are currently in use in the PTDF. This entry appears only if BASE24-pos support is bound into the ASMcom utility source file.

Translate BASE24-teller TTDF Security Keys — Translates the hardware-encrypted keys that are currently in use in the TTDF. This entry appears only if BASE24-teller support is bound into the ASMcom utility source file.

Using the Key Translation Menu

Enter one of the selection code values and the Key Translation Submenu is displayed. When you exit the Key Translation Submenu, the Key Translation Menu is redisplayed, from which you can select another file for key translation or return to the RSMcom utility Main Menu.

The BASE24 files that can be updated by the RSMcom utility key translation processing and the keys contained in each file are listed at the beginning of this appendix.

Key Translation Submenu

The RSM can translate keys for only one LMK pair at a time. The old and new values of this LMK pair must be loaded in the RSM prior to starting this function. The LMK pair used by each BASE24 key is listed at the start of this appendix. The LMK pair that has been loaded in the RSM must be identified on the Key Translation Submenu. This menu is displayed immediately after a selection has been made on the Key Translation Menu. The Key Translation Submenu is shown below, followed by descriptions of its fields.

Note: With the complexity of different LMK pair combinations for each BASE24 file and the requirement of manually changing the LMK pair information in the RSM for each LMK pair, special planning is required to make the appropriate selections on the Key Translation Menu and the Key Translation Submenu.

```
Racal Security Module Utility Program - RSMCOM - (DDMMYY)

-- Key Translation Sub-Menu --

1) Translate Security Keys; all
2) Translate Security Keys; ZMK                (LMK pairs 04-05)
3) Translate Security Keys; ZWK                (LMK pairs 06-07)
4) Translate Security Keys; CVK, TMK, TPK, PVK (LMK pairs 14-15)
5) Translate Security Keys; TAK                (LMK pairs 16-17)

Enter desired key pair (1 - 5, "H" for Help, or "X" to Exit):
```

Translate Security Keys; all — Specifies the LMK pairs to be used during BASE24 key translation. This command translates keys in the selected files using the four LMK pairs as if selections 2 through 5 were chosen sequentially.

The old and new values for LMK pair 04–05 must be loaded in the RSM before processing begins. This selection expects to start with LMK pair 04–05 even if the BASE24 file being processed does not contain any keys encrypted under that LMK pair.

After all BASE24 keys using LMK pair 04–05 in the files selected on the Key Translation Menu have been translated, the RSMcom utility displays a message instructing you to place the old and new values for LMK pair 06–07 in the RSM. The RSMcom utility pauses until you change the LMK pair values, then translates the BASE24 keys that use LMK pair 06–07. This process is repeated for LMK pair 14–15 and LMK pair 16–17. Selection 1 should only be used when multiple LMK pairs are being changed.

Translate Security Keys; ZMK (LMK pairs 04-05) — Specifies individual LMK pairs to be used during BASE24 key translation. The old and new values for the LMK pair must be loaded in the RSM before processing begins.

Translate Security Keys; ZWK (LMK pairs 06-07) — Specifies individual LMK pairs to be used during BASE24 key translation. The old and new values for the LMK pair must be loaded in the RSM before processing begins.

Translate Security Keys; CVK, TMK, TPK, PVK (LMK pairs 14-15) — Specifies individual LMK pairs to be used during BASE24 key translation. The old and new values for the LMK pair must be loaded in the RSM before processing begins.

Translate Security Keys; TAK (LMK pairs 16-17) — Specifies individual LMK pairs to be used during BASE24 key translation. The old and new values for the LMK pair must be loaded in the RSM before processing begins.

Using the Key Translation Submenu

Enter one of the selection code values and the utility begins the selected function. The Key Translation Submenu also appears after translating the keys applicable to the selected LMK pair, prompting you to select another LMK pair or return to the Key Translation Menu.

Alter Security Module Device Name

Selection 6 on the Main Menu allows you to specify or change the RSM being accessed by the utility. It can be used to switch to a different RSM device in the event of an error.

Altering the security module device name using the RSMcom utility involves the following steps. The variables in *italics* were described earlier in this section.

1. Enter 6 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)
Entry: 6
2. Enter the SYSGEN device name for the RSM to be specified.
Prompt: Enter RSM device name
Entry: *device-name*
3. Make your next Main Menu selection.
Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)
Entry: 1, 2, 3, 4, 5, 6, 7, 8, 9, H, or X

Alter LCONF Location

Selection 7 on the Main Menu allows you to specify or change the LCONF accessed by the RSMcom utility. Assigns in the LCONF identify the files that contain the keys included in RSMcom utility processing.

Altering an LCONF location using the RSMcom utility involves the following steps. The variables in *italics* were described earlier in this section.

1. Enter 7 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)
Entry: 7
2. Enter the LCONF name to be specified or the new LCONF name if the LCONF is to be changed.
Prompt: Enter LCONF File Name
Entry: *lconf-name*

3. Make your next Main Menu selection.

Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, 7, 8, 9, H, or X

Alter Summary Report Output Location

Selection 8 on the Main Menu allows you to change the output location for reports generated by the RSMcom utility. It also allows you to specify the report output location if you did not specify one during RSMcom utility startup.

Altering the summary report output location using the RSMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 8 at the selection code prompt on the Main Menu.

Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)

Entry: 8

2. Enter the report output location.

Prompt: Enter report file name

Entry: *output*

3. Make your next Main Menu selection.

Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, 7, 8, 9, H, or X

Upload Diebold Number Table to KEYA

Selection 9 on the Main Menu obtains the encrypted Diebold number table (DNT) from the RSM user storage for entry in the KEYA for an institution or a card prefix.

The program prompts you for the RSM device name (if it was not provided previously), the name of the KEYA, the ID of the group (FIID or card prefix) for which the DNT was created, and the relative location of the DNT within the RSM.

The RSMcom utility uses RSM command LE to transfer the encrypted DNT from the RSM to the program. It then accesses the KEYA using the group ID as the key and updates the DIEBOLD TABLE and NUMBER TABLE RELATIVE LOCATION fields.

Obtaining the encrypted DNT from the RSM using the RSMcom utility involves the following steps. The variables in italics were described earlier in this section.

1. Enter 9 at the selection code prompt on the Main Menu.
Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)
Entry: 9
2. Enter the SYSGEN device name. (The RSMcom utility skips this step if a valid device name has previously been entered.)
Prompt: Enter RSM device name
Entry: *device-name*
3. Enter the report output location. (The RSMcom utility skips this step if a valid report output location has previously been entered.)
Prompt: Enter report file name
Entry: *output*
4. Enter the LCONF name. (The RSMcom utility skips this step if a valid LCONF name has previously been entered.)
Prompt: Enter LCONF file name
Entry: *lconf-name*
5. Enter the ID of the group (FIID or card prefix) for which the DNT is being placed in the KEYA.
Prompt: Enter KEYA Group
Entry: *keya-group*
6. Enter the beginning date for the KEYA record in which the DNT is being placed.
Prompt: Enter KEYA Begin Date
Entry: *yymmdd* or ******* (Asterisks mean that the KEYA record does not have beginning and ending dates.)
7. Enter the ending date for the KEYA record in which the DNT is being placed, if a KEYA beginning date was entered in step 6. The RSMcom utility skips this step if asterisks were entered in step 6.
Prompt: Enter KEYA End Date
Entry: *yymmdd*
8. Enter the relative location of the DNT in the KEYA.
Prompt: Enter DNT RELATIVE LOCATION
Entry: *dnt-location*

9. Make your next Main Menu selection.

Prompt: Enter selection code (1–9, “H” for Help, or “X” to Exit)

Entry: 1, 2, 3, 4, 5, 6, 7, 8, 9, H, or X

Help Function

The RSMcom utility Help selection provides an online description of the RSMcom utility Main Menu selections and gives instructions on how to exit the program. The RSMcom utility Help Function is accessed by entering H any time one of the menus or prompts is displayed. Help presents information requirements for each topic on the ASMcom utility Main Menu.

Status Messages

The RSMcom utility displays status messages on the computer terminal screen and includes them on reports. These messages indicate processing results and identify the cause of any processing problems encountered.

These messages are documented in the STATMSGs file located on the BA53RACL subvolume (the subvolume that contains source code for the RSMcom utility). A Tandem editor can be used to read this file.

Exiting the RSMcom Utility

You can exit the program by pressing X or the **Control-Y** keys in response to the Main Menu prompt. Pressing either of these keys in response to any other prompt displays the next higher level menu.

Reports

The RSMcom utility generates four reports:

- Check Digit Generation Report
- Check Digit Validation Report
- Key Translation Report
- DNT Upload Report

The following files can appear on reports generated by the RSMcom utility:

- Key Authorization File (KEYA)
- Key File (KEYF)
- Terminal Data File (TDF)
- POS Terminal Data File (PTDF)
- Teller Terminal Data File (TTDF)

These reports are illustrated on the following pages.

Check Digit Generation Report

The Check Digit Generation Report identifies the keys processed under the RSMcom utility Generate Check Digits option. The report shown below is a sample of the one generated for keys in the KEYA. All the keys listed may not be processed for each KEYA record because keys with zeros in the key field are not processed.

RSMCOM CHECK DIGIT GENERATION REPORT FOR PR01		MM/DD/YY	PAGE 1
		HH:MM:SS	
\$vol.subvol.KEYA			
(any KEYA processing errors appear here)			
TOTAL RECORDS PROCESSED	0	CHECK DIGITS	
KEY TYPES	GENERATED	IN ERROR	
IBM-DES	0	0	
VISA PVV	0	0	
TOTAL KEYA CHECK DIGITS PROCESSED	0	0	

Check Digit Validation Report

The Check Digit Validation Report identifies the keys processed under the RSMcom utility Validate Check Digits option. The report shown below is a sample of the one generated for keys in the KEYF. All the keys listed may not be processed for each KEYF record because keys with zeros in the key field are not processed.

RSMCOM CHECK DIGIT VALIDATION REPORT FOR PR01		MM/DD/YY	PAGE 1
		HH:MM:SS	
\$vol.subvol.KEYF			
(any KEYF processing errors appear here)			
TOTAL RECORDS PROCESSED		0	
KEY TYPES	CHECK DIGITS		
	VALIDATED	IN ERROR	
INTERM.KEY-ENCRYPT	0	0	
EXCHNG-KEY	0	0	
OUTBOUND.PIN.KEY1	0	0	
OUTBOUND.PIN.KEY2	0	0	
INBOUND.PIN.KEY1	0	0	
INBOUND.PIN.KEY2	0	0	
OUTBOUND.MAC.KEY1	0	0	
OUTBOUND.MAC.KEY2	0	0	
INBOUND.MAC.KEY1	0	0	
INBOUND.MAC.KEY2	0	0	
MAC-EXCHNG-KEY	0	0	
TOTAL KEYF CHECK DIGITS PROCESSED		0	0

Key Translation Report

The Key Translation Report identifies the keys processed under the RSMcom utility Translate Security Keys option. The report shown below is a sample of the one generated for keys in the KEYF. All the keys listed may not be processed for each KEYF record because keys that are not in use are not processed.

RSMCOM KEY TRANSLATION REPORT FOR PR01		MM/DD/YY	PAGE 1
		HH:MM:SS	
\$vol.subvol.KEYF			
(any KEYF processing errors appear here)			
TOTAL RECORDS PROCESSED		0	
KEY TYPES	KEYS TRANSLATED	KEY ERRORS	
INTERM.KEY-ENCRYPT	0	0	
EXCHNG-KEY	0	0	
OUTBOUND.PIN.KEY1	0	0	
OUTBOUND.PIN.KEY2	0	0	
INBOUND.PIN.KEY1	0	0	
INBOUND.PIN.KEY2	0	0	
OUTBOUND.MAC.KEY1	0	0	
OUTBOUND.MAC.KEY2	0	0	
INBOUND.MAC.KEY1	0	0	
INBOUND.MAC.KEY2	0	0	
MAC-EXCHNG-KEY	0	0	
TOTAL KEYF KEYS PROCESSED		0	0

DNT Upload Summary Report

The DNT Upload Summary Report identifies the DNTs processed under the RSMcom utility Upload Diebold Number Table to KEYA option.

RSMCOM DNT UPLOAD SUMMARY REPORT FOR PR01		MM/DD/YY	PAGE 1
		HH:MM:SS	
\$vol.subvol.KEYA			
KEY SELECTION CRITERIA			
KEYA GROUP:	BEGIN DATE: YYYYMM	END DATE: YYYYMM	DNT LOC:
(any processing errors appear here)			
SUCCESSFUL UPLOADS	IN ERROR		
-----	-----		
0	0		

Applied Communications, Inc.

Appendix E

Atalla Security Modules

This appendix describes how Atalla security modules must be configured to work with BASE24 products, including BASE24 Remote Banking. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 and the information a user will require for configuring the modules for use by BASE24.

BASE24 Remote Banking supports the following Atalla security modules in processing:

- Atalla Switch Security Module (ASM) model A1
- Atalla Node Security Module (ASM) model A2
- Atalla Node Security Module (ASM) model A3
- Atalla Network Security Processor (NSP) model A5000
- Atalla Network Security Processor (NSP) model A6000
- Atalla Network Security Processor (NSP) model A7000
- Atalla Network Security Processor (NSP) model A10000E

In reviewing the information in this appendix, note that the terms *SYSGEN utility* and *DSC utility* refer to the Tandem System Generation (SYSGEN) and the Tandem Dynamic System Configuration (DSC) utilities.

Security Processing Support

Each security module supports a subset of BASE24 transaction security processing. The processing limitations may be the result of the security module features or BASE24 product options.

In the following table, a check mark (✓) indicates processing that is supported by each security module. In addition to the various security modules, the table also indicates transaction security processing that can be performed in software (SW).

Function	SW	ASM			NSP
		A1	A2	A3	
PIN encryption	✓	✓	✓	✓	✓
PIN verification—IBM 3624-format	✓	✓	✓	✓	✓
PIN verification—Diebold	✓	✓	✓	✓	✓
PIN verification—Identikey		✓	✓	✓	✓
PIN verification—Visa PVV	✓	✓	✓	✓	✓
Cardholder PIN select—IBM 3624-format	✓				
Cardholder PIN select—Diebold	✓				
Cardholder PIN select—Identikey	Not supported				
Cardholder PIN select—Visa PVV	Not supported				
PIN change—IBM 3624-format	✓				✓
PIN change—Diebold	✓				✓
PIN change—Identikey					✓
PIN change—Visa PVV	✓				✓
Card verification	✓	✓	✓	✓	✓
Message authentication	✓				✓
Dynamic key management		✓	✓*	✓*	✓

- * ASM models A2 and A3 can be used for dynamic key management; however, they support only one key exchange key. Two key exchange keys are necessary when PIN encryption and message authentication working keys are being managed.

Configuring Atalla Security Devices

This section presents configuration information for each of the BASE24 supported Atalla security devices. The LCONF assigns and params used in their configuration with BASE24 are also discussed.

Atalla Security Modules (ASMs) and Network Security Processors (NSPs) can be configured using the SYSGEN utility or the Tandem DSC utility. The DSC utility supports online reconfiguration of controllers, devices, and some system operational parameters. The DSC changes made remain in effect through all system loads or reloads and remain in effect until the next SYSGEN. To make DSC changes permanent, they can be integrated into the CONFTEXT file for the next SYSGEN.

Atalla Security Modules

BASE24 communicates online with Atalla Switch Security Modules (model A1) or Node Security Modules (models A2 and A3) using an asynchronous protocol and ASCII data transmission.

Atalla Security Modules must be configured using the SYSGEN utility or Dynamic System Configuration (DSC) utility as 6401 point-to-point asynchronous terminals. The SYSGEN utility or DSC utility definition can use the 6401 default modifiers, but the three modifiers shown in the following example must be changed from the defaults.

The following is an example of a SYSGEN utility or DSC utility definition for a security module named \$ASM1 connected to the controller ASYNCF on port number 25:

<i>device-name</i>	<i>controller.port</i>	<i>product-number</i>	<i>modifier</i>	<i>modifier</i>	<i>modifier</i>
\$ASM1	ASYNCF.25	6401	BAUD9600,	NOECHO,	EIA

The device names configured using the SYSGEN utility or DSC utility must be specified in the SECURE-DEVICE n assigns in the LCONF for the Integrated Authorization Server process and Interface Master process or in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF for the ISO Host Interface process. These assigns are described later in this section.

Atalla A5000 Network Security Processors

BASE24 communicates online with the Atalla A5000 Network Security Processor (formerly known as the Atalla High Performance Security Module) using a Front End Board (FEB) controller that is installed in the I/O bay of a Tandem computer system.

Each FEB controller can be configured using the SYSGEN utility or the DSC utility. The DSC utility supports online reconfiguration of controllers, devices, and some system operational parameters. The DSC utility changes made remain in effect through all system loads or reloads and remain in effect until the next SYSGEN. To make DSC utility changes permanent, they can be integrated into the CONFTXT file for the next SYSGEN.

The FEB controller must be configured as a dummy controller (product number 6999). The controller definition also specifies the attached CPU numbers and the device address the FEB controller will use. The following is an example of a SYSGEN utility or DSC utility definition for a FEB controller named FEB1:

```
controller-name    product-number    processors    subchannels
      FEB1              6999              02, 03        %320;
```

Each A5000 device must be configured as a peripheral device. The device definition includes three logical unit names for the FEB, the name of the macro containing definition default values, and entries to modify the three default modifiers shown in the following example.

The definition macro is a part of the additional software that is required to install and operate the A5000 device. Refer to the appropriate Atalla device manual for information about this software.

The following is an example of a SYSGEN utility or DSC utility definition for an A5000 device named \$ATALLA that is attached to FEB1:

```
device-name        logical-unit-names
      $ATALLA        FEB1.0, FEB1.1, FEB1.2    HPSMA12    macro-name
                                     TYPE 15,    modifier,
                                     SUBTYPE 3,    modifier,
                                     RSIZE 1000;    modifier;
```

The macro and modifiers shown above are required when defining A5000 devices for use with BASE24 products. With the HPSMA12 macro, the A5000 device returns the proper length message to BASE24. With the HPSMA11 macro,

another available macro, BASE24 cannot process the message received from an A5000 device because the message contains carriage return and line feed characters.

The device names configured using the SYSGEN utility or DSC utility must be specified in the SECURE-DEVICE n assigns in the LCONF for the Integrated Authorization Server process and Interface Master process or in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF for the ISO Host Interface process. These assigns are described later in this section.

The SECURE-DEV-CONF-TXT param in the LCONF can cause the Integrated Authorization Server or ISO Host Interface processes that use the A5000 device to send a command that configures the A5000 device to check for a minimum PIN length and specifies the response code the A5000 device returns for a PIN length error. This param also can cause these processes that use the A5000 device to send a command that configures the A5000 device to exclude carriage return and line feed characters from its response messages, permitting the SYSGEN utility or DSC definition for the A5000 device to use the HPSMA11 macro. More information is provided on the SECURE-DEV-CONF-TXT param later in this section.

Some of the A5000 device commands used in BASE24 processing may need to be enabled before transaction processing begins. The SECURE-DEV-CONF-CMD100 param in the LCONF defines the text that an Integrated Authorization Server or ISO Host Interface process sends to an A5000 device in a 100 command during initialization or warmboot. A 100 command enables commands that an A5000 device contains in disabled form. The text contained in the 100 command must be obtained directly from Atalla. The commands to be enabled may already exist in the A5000 device or may have to be obtained from Atalla and installed in the A5000 device. More information is provided on the SECURE-DEV-CONF-CMD100 param later in this section.

Atalla A6000 Network Security Processors

BASE24 processes communicate online with the Atalla A6000 Network Security Processor using a 3605 bit-synchronous communications board that is installed in the I/O bay of a Tandem computer system.

The A6000 device uses an interface module (IFM) provided by Atalla to communicate with BASE24. The A6000 device name is not specified in the LCONF. Instead, the process name assigned to the A6000 interface module at

startup is used as the device name in the SECURE-DEVICE n or SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF. These assigns are described later in this section.

Hardware Configuration

Each 3605 controller must be configured using the SYSGEN utility or DSC utility. The controller definition also specifies the attached CPU numbers and the device address the 3605 controller will use. The following is an example of a SYSGEN utility or DSC definition for a 3605 controller named SD1:

```
controller-name      product-number    processors      subchannels
      SD1                3605                02, 03          %320;
```

A path name must also be configured using the SYSGEN utility or DSC utility for each 3605 controller. The path name is then used in the device definition. An example of a SYSGEN utility or DSC path entry for controller SD1 is as follows:

```
path-name           controller-name
      SDP1                SD1;
```

Each A6000 device must be configured as a peripheral device. The SYSGEN utility or DSC utility device definition includes the path name for the 3605 controller, the name of the macro containing definition default values, and entries to modify several of the default modifiers. The following is an example of a SYSGEN utility or DSC definition for an A6000 device named \$ATALLA that is attached to clip 0 of controller SD1:

```
device-name          path.clip
      $ATALLA          SDP1.0
                        SDLCXF      macro-name
                        TYPE 15,    modifier,
                        SUBTYPE 4,  modifier,
                        RSIZE 268,  modifier,
                        FULL,      modifier,
                        PRIMARY,   modifier,
                        INTERRUPT BS^6100^INTERRUPT^XF#; modifier;
```

Interface Module

Each A6000 device requires a special interface to communicate with BASE24. The A6000 interface module (IFM) is a NonStop process provided by Atalla that performs high-level SDLC protocol handling for messages between BASE24 and the A6000 device. Refer to the appropriate Atalla device manual for configuration and startup information for the A6000 device IFM.

BASE24 cannot process the message it receives from an A6000 device if the message contains carriage return and line feed characters. The SECURE-DEV-CONF-TXT param in the LCONF can cause BASE24 processes that use the A6000 device to send a command that configures the A6000 device to omit carriage return and line feed characters from the message it returns to BASE24. The SECURE-DEV-CONF-TXT param can also cause BASE24 processes to send a command that configures the A6000 device to check for a minimum PIN length and specifies the response code the A6000 device returns for a PIN length error. More information is provided on the SECURE-DEV-CONF-TXT param later in this section.

Some of the A6000 device commands used in BASE24 processing may need to be enabled before transaction processing begins. The SECURE-DEV-CONF-CMD100 param in the LCONF defines the text that BASE24 sends to an A6000 device in a 100 command during initialization or warmboot. A 100 command enables commands that an A6000 device contains in disabled form. The text contained in the 100 command must be obtained directly from Atalla. The commands to be enabled may already exist in the A6000 device or may have to be obtained from Atalla and installed in the A6000 device. More information is provided on the SECURE-DEV-CONF-CMD100 param later in this section.

Note: The A6000 interface module must be running when BASE24 processes that use the A6000 device are initialized or warmbooted. If the A6000 interface module is stopped, BASE24 processes that use the A6000 device must be warmbooted after the interface module is restarted.

Atalla A7000 Network Security Processors

BASE24 communicates online with the Atalla A7000 Network Security Processor using high speed small computer system interface (SCSI) communications through a 3214 communications controller that is installed in the I/O bay of a Tandem computer system. The A7000 NSP itself appears to the Tandem system as if it is a 5170 tape drive. This is accomplished using the SYSGEN utility or DSC utility.

The A7000 device uses a NonStop system application interface called Boxcar to communicate with BASE24. For the A7000 device, it is the Boxcar process name (for example, \$BOX) configured using the SYSGEN utility or the DSC utility that is specified in the SECURE-DEVICEn, or SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF. Configuration requirements for the Boxcar interface module are provided later in this section.

Hardware Configuration

Because the A7000 NSP is a SCSI device, it must connect to a 3214 tape controller defined using the SYSGEN utility or DSC utility. The following is an example of a SYSGEN utility or DSC definition for a 3214 controller named A7000X:

```
controller-name    product-number    processors    subchannels
      A7000X              3214              0, 1              %130;
```

A path name must also be configured for each 3214 controller. The path name is then used in the device definition. The following is an example of a SYSGEN utility or DSC path entry for a controller named A7000X:

```
path-name          controller-name
      A7P1              A7000X;
```

Each A7000 device must be configured as a peripheral device. The SYSGEN utility or DSC device definition includes the path name for the 3214 controller, the name of the macro containing definition default values, and entries to modify several of the default modifiers. The following is an example of a SYSGEN utility definition for an A7000 device named \$A7000 that is attached to clip 5 of controller A7000X:

```
device-name          path.clip          SDLCXF          macro-name
      $A7000              A7P1.5              TYPE 15,          modifier,
                                SUBTYPE 4,          modifier,
                                RSIZE 268,          modifier,
                                FULL,          modifier,
                                PRIMARY,          modifier,
                                INTERRUPT BS^6100^INTERRUPT^XF#;          modifier;
```

Atalla A10000E Network Security Processors

The Atalla A10000E Network Security Processor communicates with the Tandem using Transmission Control Protocol/Internet Protocol (TCP/IP) over an Ethernet connection. Thus, the A10000E NSP requires the configuration and startup of an Ethernet controller, Ethernet port, and the TCP/IP process.

The A10000E device uses a NonStop system application interface called Boxcar to communicate with BASE24. For the A10000E device, it is the BOXCAR process name (i.e., \$BOX) that is specified in the SECURE-DEVICEn, or SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF. Configuration requirements for the Boxcar interface module are provided later in this section.

Hardware Configuration for K-Series Servers

Because the A10000E is a TCP/IP device, each Ethernet controller (e.g., 3613, 3615) must be configured using the SYSGEN utility or DSC utility. For detailed information on configuring an Ethernet controller and line, refer to Tandem's ***TLAM Configuration and Management Manual***. The following is an example of a SYSGEN utility definition example for a 3615 controller named A10000x:

```
controller-name product-number processor-numbers channel addresses
A10000X          3615          0,1          %030;
```

The following is an example of the SYSGEN utility definition that shows the minimum parameters required to configure a 3615 controller (line):

```
device-name      controller-name  macro  modifiers
$LAM1           LAM1.0, LAM1.1    MLAM   TYPE 56,
                                     SUBTYPE 0,
                                     RSIZE 32000,
                                     READBUFFERSIZE 8192,
                                     INTERRUPT
                                     IOP^INTERRUPT^HANDLER#
```

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the A10000E device and the Tandem server. Refer to the "Software Configuration" discussion on the following pages.

Hardware Configuration for S-Series Servers

Because the A10000E is a TCP/IP device, it must communicate with the S-series server through an Ethernet port on either a 3861 Ethernet 4 ServerNet Adapter (E4SA) controller or through a Processor MultiFunction Unit (PMF). The configuration of a E4SA controller on an S-series server requires the use of the Tandem Subsystem Control Facility (SCF) to configure and start objects associated with the E4SA controller. These objects include the adapter, the ServerNet addressable controllers (SACs), the logical interfaces (LIFs), and the physical interfaces (PIFs). For additional information on configuring the 3681 (E4SA) controller, refer to Tandem's ***Himalaya S-Series Hardware Installation Guide***.

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the A10000E device and the Tandem server. Refer to the "Software Configuration" discussion on the following pages.

Software Configuration

The software configuration is very similar for the K-series and S-series servers. Examples for both types of servers are provided below. In general, the TCP/IP process must be started against a previously configured port and the Boxcar interface module must be started. Some of the values in these examples, most importantly the IP addresses, are site-specific and must be modified accordingly.

K-Series Server. The following is an example TACL macro for starting the TCP/IP process on a K-series server.

```
?tacl macro
#frame
delete define=tcPIP^host^file
add define=tcPIP^host^file, file $system.ztcPIP.hosts
$system.ztcPIP.tcPIP.lib $system.zqiolib.qiolib,name $ZTCO,&
    nowait, pri 180, cpu 0/1
#push scfin
#append scfin allow all errors
#append scfin assume process $ZTCO
#append scfin alter subnet, #loop0, ipaddress 127.1
#append scfin add subnet #sn0, type ethernet, devicename $LAN1,&
    ipaddress 130.130.63.16, forceqio on
#append scfin add route #r80, destination 0.0.0.0,&
    gateway 130.130.5.1
#append scfin start route *
#append scfin start subnet *
scfin/inv scfin/
#unframe
```

S-Series Server. The following are examples for starting the TCP/IP process on an S-series server.

The first example starts the TCP/IP process using an SCF obey file.

```
=====
= Configure & Start $ZB110 TCPIP process for ADAPTER E1151,LIFL100 ==
=====
ASSUME PROCESS $ZB110
ADD SUBNET #SN1, TYPE ETHERNET, IPADDRESS 192.231.036.99&
DEVICENAME L110
ADD ROUTE #GW, DESTINATION 0, GATEWAY 192.231.036.017,&
DESTTYPE BROADCAST
ALTER SUBNET #LOOP0, IPADDRESS 127.1
START SUBNET *
START ROUTE *
```


The next example starts the TCP/IP process using a TACL macro.

```
TCP/IP /name $ZB110,NOWAIT, CPU 0, PRI 160, TERM $ZHOME/1
```

Starting the Boxcar Process. After the TCP/IP process is started, the Boxcar process must be started using one of the Atalla-supplied utilities (i.e., A10KWARM, A10KCOLD). The following is an example of the A10KWARM utility.

Note: The BOXCAR-DEVICE-NAME param must be set to the IP address of the A10000E device.

```
COMMENT A10KWARM restarts boxcar without downloading the A10000 image
```

```
CLEAR ALL PARAM
PARAM BOXCAR-WCOUNT 100
COMMENT PARAM BOXCAR-NO-INIT Y
COMMENT PARAM BOXCAR-NUMOPENS 2046
COMMENT PARAM BOXCAR-DEVICE-TYPE 03C3
```

```
PARAM BOXCAR-LOGFILE BOXLOG
PARAM BOXCAR-TRACE-FILE TRACELOG
PARAM BOXCAR-TRACE-RECORDS 0
PARAM BOXCAR-BACKUPCPU 3
PARAM BOXCAR-DEVICE-NAME 172.17.11.82
PARAM BOXCAR-A10KE-PORT 7000
PARAM BOXCAR-MAXSOCKET-OPENS 14
PARAM BOXCAR-NO-INIT Y
```

```
purge boxlog
purge tracelog
fup create boxlog,type e,rec 200,block 4096,maxextents 100,ext 100
fup create tracelog,type e,rec 200,block 4096,maxextents 100,ext 100
```

```
run boxcaro /name $box1,nowait,cpu 3,highpin on/
```

Boxcar Interface Module

Each A7000 and A10000E device requires a special interface to communicate with the BASE24. The Boxcar interface module is a NonStop process provided on the Tandem System Utility Tape (SUT) that runs in the Tandem Guardian environment. The Boxcar interface module is designed to improve overall efficiency of communication between application processes and an Atalla NSP.

The Boxcar interface module increases the efficiency of I/O by grouping together a collection of commands from multiple requesting application processes and sending this block of commands to an Atalla NSP as a single I/O event. The Atalla NSP then separates and processes the individual commands and formats them as a block of responses that is returned to the Boxcar interface module. The Boxcar interface module separates the individual responses and sends replies to the appropriate individual requesters. Refer to the appropriate Atalla device manual for configuration and startup information for the Boxcar process.

BASE24 cannot process the message it receives from an Atalla NSP if the message contains carriage return and line feed characters. The SECURE-DEV-CONF-TXT param in the LCONF causes BASE24 processes that use an Atalla NSP to send a command that configures the NSP to omit carriage return and line feed characters from the message it returns to BASE24. The SECURE-DEV-CONF-TXT param can also cause BASE24 processes to send a command that configures the Atalla NSP to check for a minimum PIN length and specifies the response code the NSP returns for a PIN length error. More information is provided on the SECURE-DEV-CONF-TXT param later in this section.

Some of the Atalla NSP commands used in BASE24 processing may need to be enabled before transaction processing begins. The SECURE-DEV-CONF-CMD100 param in the LCONF defines the text that BASE24 sends to an Atalla NSP in a 100 command during initialization or warmboot. A 100 command enables commands that an Atalla NSP contains in disabled form. The text contained in the 100 command must be obtained directly from Atalla. The commands to be enabled may already exist in the Atalla NSP or may have to be obtained from Atalla and installed. More information is provided on the SECURE-DEV-CONF-CMD100 param later in this section.

The process name assigned to the Boxcar interface module at startup is used as the device name in the SECURE-DEVICEn, SECURE-DEV1, and SECURE-DEV2 assigns in the LCONF.

Note: The Boxcar interface module must be running when BASE24 processes that use an Atalla NSP are initialized or warmbooted. If the Boxcar interface module is stopped, BASE24 processes that use an Atalla NSP must be warmbooted after the interface module is restarted.

Logical Network Configuration File (LCONF)

The following section presents the Logical Network Configuration File (LCONF) information required for configuration of the Atalla security devices.

LCONF Assigns

The following assigns are required in the LCONF to configure an Atalla security device. The SECURE-DEV1 and SECURE-DEV2 assigns are required for ISO Host Interface processes, while the SECURE-DEVICEn assigns are required for Integrated Authorization Server and Interface Master processes.

SECURE-DEV1 — Specifies the name of the primary security device or the name of the interface module used with the primary security device. For security devices other than the Atalla A6000, A7000, or A10000E security devices, this assign specifies the SYSGEN utility or DSC name of the primary security device. For Atalla A6000, A7000, or A10000E security devices, this assign specifies the name given to the A6000, A7000, or A10000E interface module used with the primary security device.

There can be multiple SECURE-DEV1 assigns in the LCONF, depending on whether a primary device is to be specified individually for each process requiring the use of a security device (in which case, each process has its own SECURE-DEV1 assign in the LCONF).

To associate a primary security device with a specific process, the symbolic name of the process must be placed in the Process Name field on the LCONF Assign screen. If a specific primary security device need not be associated with a specific process, the Process Name field can be filled in with asterisks.

When determining its primary security device, a process first reads through the LCONF for a SECURE-DEV1 assign with its own name in the Process Name field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV1 assign with asterisks in the Process Name field.

Note: Each BASE24 process uses the value in the User Field field on the LCONF Assign screen to identify the Atalla security device being used. For ASM security modules, the User Field field must be set to the module's subtype: A1, A2, or A3. For NSP security devices, the User Field field must be set to NSP.

SECURE-DEV2 — Specifies the name of the secondary security device or the name of the interface module used with the secondary security device. For security devices other than the Atalla A6000, A7000, or A10000E security devices, this assign specifies the SYSGEN or DSC name of the secondary security device. For Atalla A6000, A7000, or A10000E security devices, this assign specifies the name given to the A6000, A7000, or A10000E interface module used with the secondary security device.

There can be multiple SECURE-DEV2 assigns in the LCONF, depending on whether a secondary device is to be specified individually for each process requiring the use of a security device (in which case each process has its own SECURE-DEV2 assign in the LCONF).

To associate a secondary security device with a specific process, the symbolic name of the process must be placed in the Process Name field on the LCONF Assign screen. If a specific secondary security device need not be associated with a specific process, the Process Name field can be filled in with asterisks.

When determining its secondary security device, a process first reads through the LCONF for a SECURE-DEV2 assign with its own name in the Process Name field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV2 assign with asterisks in the Process Name field.

Note: Each BASE24 process uses the value in the User Field field on the LCONF Assign screen to identify the Atalla security device being used. For ASM security modules, the User Field field must be set to the module's subtype: A1, A2, or A3. For NSP security devices, the User Field field must be set to NSP.

SECURE-DEVICE n — The fully qualified name of a security module.

For Atalla security modules and Atalla Network Security Processors model A5000, this assign specifies the Tandem SYSGEN device name. For Atalla Network Security Processors (NSPs) other than model A5000, this assign specifies the PPD name given to the NSP interface module used with the security device. The device name or interface module name must be compatible with the Tandem Expand product.

The Integrated Authorization Server process and Interface Master process use this assign. Each security module used by an Integrated Authorization Server process or Interface Master process requires its own SECURE-DEVICE n assign.

When the Integrated Authorization Server process or Interface Master process requires the services of a security module, they use the next available security module. The n in the assign name identifies the order in which each process accesses the security modules. For example, each process accesses the security module identified by the SECURE-DEVICE1 assign before it accesses the security module identified by the SECURE-DEVICE2 assign.

The SECURE-DEVICE-COUNT param identifies the number of security modules being used. For example, if the SECURE-DEVICE-COUNT param is set to 5, the Integrated Authorization Server process and Interface Master process access five security modules. When a process accesses the security module identified by the SECURE-DEVICE5 assign, it then starts over with the security module identified by the SECURE-DEVICE1 assign.

LCONF Params

The following params are required in the LCONF to configure an Atalla security device. Some params are required by Integrated Authorization Server, Interface Master, and ISO Host Interface processes while others are required by only one type of process and not the other. The params required by the Integrated Authorization Server process and Interface Master process are the same. The params required by each type of process are depicted in the following table. In this table, IAS stands for Integrated Authorization Server process, IMP stands for Interface Master process, and HISO stands for ISO Host Interface process.

Param Name	IAS/IMP	HISO
SECURE-DEV-CONF-CMD100	✓	✓
SECURE-DEV-CONF-TXT	✓	✓
SECURE-DEV-FAIL-NOTIFY		✓
SECURE-DEVICE-COUNT	✓	
SECURE-DEVICE-TYPE	✓	
SECURE-DEV-MONITOR-THRESHOLD		✓
SECURE-DEV-TIM-LMT	✓	✓
SECURE-DEV-TYP		✓
TSEC-OBJ-MASTER	✓	

SECURE-DEV-CONF-CMD100 — Defines the text that a BASE24 process sends to an Atalla Network Security Processor (NSP) model A5000, A6000, A7000, or A10000E in a 100 command during initialization or warmboot. A 100 command enables commands that an NSP contains in disabled form. The text contained in the 100 command must be obtained directly from Atalla. The commands to be enabled may already exist in the NSP. The text to be contained in the 100 command must be entered in the Text field of the Param screen in the following format:

100#Text-obtained-from-Atalla

The following NSP commands must be enabled with a 100 command before they can be used:

- 30 = Encrypt Clear PIN. This command is needed if a clear PIN is passed to the NSP for encryption.
- 37 = PIN Change Transaction. This command is needed if the BASE24-atm or BASE24 Remote Banking PIN change transaction is offered and the NSP is used to generate the new PIN verification digits. Option 066 (verify old PIN in a separate call to the device) must also be enabled with NSP command 37.
- 5E = Verify Card Verification Value. This command is needed if card verification is performed and the NSP is used to verify the MasterCard card validation code (CVC) or the Visa card verification value (CVV).
- 11D = Generate MAC Key. This command is needed if message authentication is performed and the NSP is used to generate a new MAC transaction key for a terminal. This command is not needed to generate a new MAC key for an interface process.
- D0 = Verify Clear PIN. This command is needed if a clear PIN is passed to the NSP for verification.

This param is needed only if any of the commands listed above are used. A single SECURE-DEV-CONF-CMD100 param can be defined in the LCONF with the Process Name field on the LCONF Param screen set to asterisks (default value). The BASE24 security utilities do not send this param to non-NSP security devices.

Each 100 command received by an NSP can override the settings established by any prior 100 commands if the commands do not contain the same text. If multiple processes (BASE24 or non-BASE24) issue 100 commands to an NSP, care must be taken to ensure that all 100 commands contain the same text.

SECURE-DEV-CONF-TXT — Defines the configuration text sent to an Atalla Network Security Processor (NSP) model A5000, A6000, A7000, or A10000E in a 101 command during initialization or warmboot. A 101 command can modify three of the default configuration settings used by the NSP. The text to be contained in the 101 command must be entered in the Text field of the Param screen and can consist of one or more four-character option words.

Each option word contains a three-digit option ID followed by a one-character action flag. If more than one option word is defined, the option words must be separated by commas. Each of the available option IDs can only be used once in each SECURE-DEV-CONF-TXT param. The valid option IDs are as follows:

023 = Specifies whether the NSP should remove carriage return and line feed characters from all response messages returned to the BASE24. This option ID is required for the A6000, A7000, or A10000E device and is optional for the A5000 device. (The HPSMA12 macro, if used, configures the A5000 device to remove carriage return and line feed characters.) This option ID has the following valid action flag value:

E = Enable. Remove carriage return and line feed characters from all response messages returned to BASE24 by the NSP.

024 = Specifies whether the NSP should identify PIN length errors by returning a unique code in response messages to BASE24. This option ID has the following action flag values:

D = Disable. Return an S error response for PIN length and sanity check errors. BASE24 processes an S error response as a sanity check error.

E = Enable. Return an L error response for a PIN length error and return an S error response for a sanity check error. BASE24 processes an L error response as an invalid PIN and processes an S error response as a sanity check error.

040 = Sets the minimum valid PIN length used by the NSP when performing PIN length checks. This option ID has the following valid action flag values:

0 = Perform the PIN length check using a minimum PIN length of zero. This value has the effect of disabling the PIN length check because any PIN entered passes.

1–9, A–C = Perform the PIN length check using this value as the minimum PIN length. A, B, and C represent lengths of 10, 11, and 12, respectively.

As an example, to remove carriage return and line feed characters from all response messages returned to BASE24, return an L response code for a PIN length error, and set a minimum PIN length of 4 characters, the following information would be entered in the Text field on the Param screen:

023E, 024E, 0404

This param is needed only if any of these option IDs (023, 024, or 040) are used. Once a configuration value is changed using one of these option IDs, the new value remains in effect until the NSP is reset or the configuration value is changed by receiving the option ID in another 101 command. To avoid possible inconsistencies caused by one process changing a configuration value and another process expecting the default configuration value, ACI recommends the use of this param with all three option words for all BASE24 processes using NSP security modules. A single SECURE-DEV-CONF-TXT param can be defined in the LCONF with the Process Name field on the LCONF Param screen set to asterisks (that is, default value). The BASE24 security utilities do not send this param to non-NSP security devices.

Each 101 command received by an NSP can override the settings established by any prior 101 commands if the commands do not contain the same text. If multiple processes (BASE24 or non-BASE24) issue 101 commands to an NSP, care must be taken to ensure that all 101 commands contain the same text.

SECURE-DEV-FAIL-NOTIFY — The interval at which critical event messages are generated due to the failure of a primary or secondary security device for an ISO Host Interface process. Valid values are 10 to 1000; the default value is 20.

If the primary and secondary security devices cannot be initialized during process startup and this param is set to the default of 20, every twentieth transaction that an ISO Host Interface process must decline because the security devices are inoperative causes a message to be generated indicating the security devices have failed. (Note that a critical event message is always generated immediately upon failure of both devices.)

If a primary security device is marked down and this param is set to the default of 20, every twentieth transaction that is directed to the primary security device causes a message to be generated even though the transactions are being handled properly by the secondary security device. Transactions routed directly to the secondary security device do not affect the primary security device transaction counter. The SECURE-DEV-MONITOR-THRESHOLD param controls when a transaction is directed to the secondary security module instead of the primary security device.

If the primary and secondary security devices fail at some time after initialization, a separate failed transaction counter is maintained for each security device. A transaction that is sent to one security device and then the other increments both counters. An event message is generated each time either counter reaches the value established in this param.

SECURE-DEVICE-COUNT — The number of security devices configured in this logical network. The value of this param should match the number of SECURE-DEVICE n assigns in this LCONF. The Integrated Authorization Server process and Interface Master process use this param. Valid values are 1 through 32767.

SECURE-DEVICE-TYPE — A code identifying the type of security module specified in the SECURE-DEVICE n assigns. The Integrated Authorization Server process and Interface Master process use this param. Valid values are as follows:

ASM = Atalla Security Module
NSP = Atalla Network Security Processor
N/A = No security module required

If this param is not configured, it defaults to a value of N/A.

Note: The Integrated Authorization Server process, Interface Master process, and ISO Host Interface process must use the same type of security device. For example, if the Integrated Authorization Server process uses an ASM, the Interface Master process or ISO Host Interface process cannot use an NSP or Racal-Guardata Security Module (RSM).

SECURE-DEV-MONITOR-THRESHOLD — The interval, expressed in the number of transactions processed by the security device defined in the SECURE-DEV1 assign, for checking the security device defined by the SECURE-DEV2 assign. Valid values are 0–32000. The default value of 0 indicates no check is performed.

This param can be used to monitor the secondary security device or to balance the load between the primary and secondary security devices. Each ISO Host Interface process using a security device maintains its own transaction counter. This means the interval specified in the SECURE-DEV-MONITOR-THRESHOLD param needs to be set for the number of transactions processed by individual processes using the security device, not the total number of transactions processed by the security device. For example, if five BASE24 processes use a security device and the param is set to 1000, up to 5000 transactions (1000 by each

of the five processes) could be processed by the primary security device before the secondary security device is used. Setting this param to 1 causes each of the BASE24 processes to alternate between the two security devices.

If all processes use the same threshold for monitoring the secondary security device, a single SECURE-DEV-MONITOR-THRESHOLD param can be defined in the LCONF. In this case, the Process Name field on the LCONF Param screen must be set to asterisks (default value).

If different secondary security device monitoring thresholds are used, SECURE-DEV-MONITOR-THRESHOLD params must be defined in the LCONF for each threshold and the Process Name field can be set to identify the processes using that threshold. If a majority of processes use a particular threshold, an alternative to identifying all of its processes can be making that threshold the default by placing asterisks in the Process Name field. Then, only processes using a different threshold need to be defined specifically.

SECURE-DEV-TIM-LMT — The maximum time, in hundredths of a second, to wait for a response from the security device before the request is considered to be timed out. Valid values are 1 to 6000; the default value is 500 (that is, 5 seconds). The same time limit applies to all processes so there is only one SECURE-DEV-TIM-LMT param in the LCONF, and the Process Name field must be set to asterisks.

SECURE-DEV-TYP — Specifies the type of security device being used by ISO Host Interface processes. Valid values are as follows:

ASM = Atalla Security Module (ASM) or Atalla Network Security Processor (NSP)
N/A = No security module required

If no security devices are used by ISO Host Interface processes, a SECURE-DEV-TYP param must still be defined in the LCONF. The Process Name field on the LCONF Param screen must be set to asterisks (default value) and the field on the LCONF Param screen must be set to N/A (no security module required).

Note: The ISO Host Interface process, Integrated Authorization Server process, and Interface Master process must use the same type of security device. For example, if the ISO Host Interface process uses an ASM, the Integrated Authorization Server process and Interface Master process cannot use an NSP or Racal-Guardata Security Module (RSM).

TSEC-OBJ-MASTER — The symbolic name of the Integrated Authorization Server process containing the Transaction Security object responsible for performing the following initialization steps:

- Downloading the Diebold number table (DNT) to the hardware security modules used by the process.
- Sending the command text defined in the SECURE-DEV-CONF-CMD100 param to the hardware security modules used by the process.
- Sending the command text defined in the SECURE-DEV-CONF-TXT param to the hardware security modules used by the process.

These initialization steps need to be performed only once. Without this param, unnecessary processing would be performed because each Transaction Security object in each Integrated Authorization Server process or Interface Master process would repeat these initialization steps.

Atalla Security Device Support

Summaries of Atalla Security Module (ASM) and Atalla Network Security Processor (NSP) functions supported by BASE24 products presented in this section include the following:

- The ASM and NSP specifications required for generating, encrypting, and translating keys for storage in the BASE24 database.
- The ASM and NSP online functions supported by BASE24 products.
- The online functions supported by each model of the ASM and NSP.

This material is not meant to be a tutorial on the ASM or NSP; it presumes a familiarity with the capabilities and nomenclature as discussed in Atalla documentation.

Preparing Data for the BASE24 Files

The Atalla Security Initialization Unit (SIU) is used to perform offline key management and the programming of the Master File Key (MFK) directly into the Atalla Proprietary Parts (APP) used by the ASM. The Key Loading Module (KLM) and the KLMKEY program are used to perform offline key management and the programming of the MFK directly into the NSP. The BASE24 ASMcom utility can also be used to perform offline key management of the KEYA and KEYF for the ASM and NSP. Refer to appendix C for information regarding the ASMcom utility.

Note: The ASMcom utility does not currently modify the Interface Security File (ISF) in any way. For the ISF, only procedures in which you must copy down the ASMcom utility output and manually enter it, can be used with the ISF.

The SIU and KLM can be used to encrypt or translate the keys under the proper MFK variant for storage in the BASE24 files, as shown in the following table. The table identifies the BASE24 field name, BASE24 file maintenance screen, MFK variant, and key type. The BASE24 files maintenance screens listed in this table are illustrated in appendix A.

Field Name in BASE24 File	Screen	Key Variant	Key Type
KKT-PT	ISF1	0	Key exchange key
INTERM HDWR KEY	ISF 2	1	PIN encryption key
INBOUND KPT1	ISF 2	1	PIN encryption key
INBOUND KPT2	ISF 2	1	PIN encryption key
PIN KEY EXCHANGE KEY	KEYF 1	0	Key exchange key
MAC KEY EXCHANGE KEY	KEYF 1	0	Key exchange key
INTERMEDIATE KEY ENCRYPTED	KEYF 1	1	PIN encryption key
INBOUND PIN KEY 1	KEYF 2	1	PIN encryption key
INBOUND PIN KEY 2	KEYF 2	1	PIN encryption key
OUTBOUND PIN KEY 1	KEYF 2	1	PIN encryption key
OUTBOUND PIN KEY 2	KEYF 2	1	PIN encryption key
INBOUND MAC KEY 1	KEYF 2	3	Message authentication key
INBOUND MAC KEY 2	KEYF 2	3	Message authentication key
OUTBOUND MAC KEY 1	KEYF 2	3	Message authentication key
OUTBOUND MAC KEY 2	KEYF 2	3	Message authentication key
DES (IBM 3624) ENCRYPTED KEY	KEYA 2	4	PIN verification key
DIEBOLD ENCRYPTED KEY	KEYA 3	0	Key exchange key
DIEBOLD TABLE*	KEYA 3	—	
VISA PVV KEYS ENCRYPTED	KEYA 5	4	PIN verification key

* The Diebold number table must be placed in the KEYA in the clear. A BASE24 process will encrypt it for sending to the ASM or NSP.

ASM and NSP Online Functions Supported by BASE24

The table below summarizes the ASM message IDs and functions used by the Integrated Authorization Server process, Interface Master process, ISO Host Interface process, and the ASMcom utility. Refer to appendix C for information regarding the ASMcom utility.

Message ID	Function	Integrated Authorization Server	Interface Master Process	ISO Host Interface Process	ASMcom Utility*
10	Generate Working Key Type J	N	Y	Y	N
11	Translate Working Key for Distribution	N	N	Y	N
12	Translate Working Key for Local Storage	N	N	Y	Y
13	Translate Working Key for Local Storage (switch-to-switch)	N	N	Y	Y
15	Change Transaction Key	N	N	N	N
17	Translate Working Key from Master File Key encryption to APP-resident Key Exchange Key encryption Note: The ASMcom utility uses this message ID to generate check digits.	N	N	Y	Y [†]
30	Clear PIN Encryption	N	N	Y	N
31	Translate PIN	Y	N	Y	N
32, 1	Verify PIN—Identikey	Y	N	N	N
32, 2	Verify PIN—IBM 3624-format	Y	N	N	N
32, 3	Verify PIN—Visa PVV	Y	N	N	N
32, 5	Verify PIN—Diebold	Y	N	N	N
33	Translate PIN (differing PAN digits)	Y	N	Y	N

Message ID	Function	Integrated Authorization Server	Interface Master Process	ISO Host Interface Process	ASMcom Utility [*]
37, 1	Change PIN—Identikey [‡]	Y	N	N	N
37, 2	Change PIN—IBM 3624-format [‡]	Y	N	N	N
37, 3	Change PIN—Visa PVV [‡]	Y	N	N	N
37, 5	Change PIN—Diebold [‡]	Y	N	N	N
5E, 3	Verify CVV, Track 2, Online	N	N	N	N
74	Load Diebold Number Table [§]	Y	N	N	N
98	Generate MAC	N	N	Y	N
99	Verify MAC	N	N	Y	N
9E	Translate Working Key from Current MFK to Pending MFK	N	N	N	Y [†]
BD	Translate PIN and Generate MAC	N	N	Y	N
D0, 1	Verify PIN—Clear PIN Identikey	N	N	N	N
100	Security Processor Configuration Command	Y	N	Y	N
101	Configure Security Processor Options	Y	N	Y	N
11D	Generate MAC or Data Encryption Key	Y	N	N	N

^{*} Some of the ASMcom utility functions will work for the Key File (KEYF) only; the ASMcom utility does not currently modify the Interface Security File (ISF) in any way. For the ISF, only procedures in which you must copy down the ASMcom utility output and manually enter it, can be used with the ISF.

[†] These functions update the BASE24 database automatically, and, therefore, cannot be used with the Interface Security File (ISF).

- ‡ The 37 message ID can be configured by Atalla to be performed with or without the old PIN, depending on the value used in command 100 when this command is activated. For a BASE24 process, it must be configured to be performed without the old PIN. This configuration is called option 66.
- § The Integrated Authorization Server process sends Diebold number tables to the ASM at initialization using a 74 message ID.

Security Module Capabilities

The table below summarizes the ASM and NSP message IDs and functions supported by each of the Atalla security modules: A1 (ASM), A2 (ASM), A3 (ASM), A5000 (NSP), A6000 (NSP), A7000 (NSP), and A10000E (NSP).

The NSP column contains two values for each message ID. The first value indicates whether the message ID is a standard or optional feature and the level of NSP firmware required to support the feature. There are two levels of NSP firmware involved in the optional features listed. The second value indicates whether the feature is shipped to the customer in enabled or disabled form. Atalla must be contacted directly to obtain the ACI-specific release of NSP firmware and enable any disabled features that are required for BASE24 processing. BASE24 processes can pass the text of the command to enable the required features to the NSP during initialization or warmboot by specifying the text in the SECURE-DEV-CONF-CMD100 param in the LCONF.

The following is a key to the values displayed in the third through seventh columns of the table.

Key:

- A = Optional feature for the security module with the ACI-specific release of firmware (NSP only).
- D = Optional NSP feature shipped in disabled form.
- E = Optional NSP feature shipped in enabled form.
- O = Optional feature for the security module (ASM only) or the security module without the ACI-specific release of firmware (NSP only).
- X = Standard feature for the security module (ASM or NSP).

Note: Modifications to standard product (RPQs) developed for the A1 security module do not necessarily function in the same manner with the A5000, A6000, A7000, and A10000E security modules.

Msg ID	Function	NSP		A1	A2	A3
		Std/Op	Shipped Enabled/Disabled			
10	Generate Working Key Type J	X		X	X	
11	Translate Working Key for Distribution	X		X		
12	Translate Working Key for Local Storage	X		X	X	X
13	Translate Working Key for Local Storage (switch-to-switch)	X		X		
15	Change Transaction Key	X		X	X	
17	Translate Working Key from Master File Key encryption to APP-resident Key Exchange Key encryption	X		X	X	X
30	Clear PIN Encryption	O	D	X	X	X
31	Translate PIN	X		X	X	X
32, 1	Verify PIN—Identikey	X		X	X	X
32, 2	Verify PIN—IBM 3624-format	X		X	X	X
32, 3	Verify PIN—Visa PVV	X		X	X	X
32, 5	Verify PIN—Diebold	X		X	X	X
33	Translate PIN (differing PAN digits)	X		O	O	O
37, 1	Change PIN—Identikey	A	D			
37, 2	Change PIN—IBM 3624-format	A	D			
37, 3	Change PIN—Visa PVV	A	D			
37, 5	Change PIN—Diebold	A	D			
5E, 3	Verify CVV, Track 2, Online	O	D	O	O	O

Msg ID	Function	NSP		A1	A2	A3
		Std/Op	Shipped Enabled/ Disabled			
74	Load Diebold Number Table	X		X	X	X
98	Generate MAC	X				
99	Verify MAC	X				
9E	Translate Working Key from Current MFK to Pending MFK	X				
BD	Translate PIN and Generate MAC	A	E			
D0, 1	Verify PIN—Clear PIN Identikey	O	D	O	O	O
100	Security Processor Configuration Command	X				
101	Configure Security Processor Options	X				
11D	Generate MAC or Data Encryption Key	A	D			

Appendix F

Racal-Guardata Host Security Modules

This appendix describes how Racal-Guardata Host Security Modules must be configured to work with BASE24 products, including BASE24 Remote Banking. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 and the information a user will require for configuring the modules for use by BASE24.

BASE24 supports the following Racal-Guardata Host Security Module (RSM) models:

- RG6000
- RG7100 and RG7110
- RG7300 and RG7310
- RG7400

Note: The RG7110 and RG7310 models are high performance versions of the RG7100 and RG7300 models, respectively. The configuration for these high performance models is the same as that of their lower performance counterparts.

In reviewing the information in this appendix, note that the terms *SYSGEN utility* and *DSC utility* refer to the Tandem System Generation (SYSGEN) and the Tandem Dynamic System Configuration (DSC) utilities.

Security Processing Support

The RSM security module supports a subset of BASE24 transaction security processing. The processing limitations may be the result of the security module features or BASE24 options. In the following table, a check mark (✓) indicates processing that is supported by the RSM security module. In addition, the table also indicates the transaction security processing that can be performed in software (SW).

Function	SW	RSM
PIN encryption	✓	✓
PIN verification—IBM 3624-format	✓	✓
PIN verification—Diebold	✓	✓
PIN verification—Identikey		
PIN verification—Visa PVV	✓	✓
Cardholder PIN select—IBM 3624-format	✓	
Cardholder PIN select—Diebold	✓	
Cardholder PIN select—Identikey	Not supported	
Cardholder PIN select—Visa PVV	Not supported	
PIN change—IBM 3624-format	✓	✓
PIN change—Diebold	✓	✓
PIN change—Identikey		
PIN change—Visa PVV	✓	✓
Card verification	✓	✓
Message authentication	✓	✓
Dynamic key management		✓

Configuring Racal-Guardata Security Devices

This section presents configuration information for the BASE24 supported Racal-Guardata security devices.

Racal-Guardata Host Security Modules (RSMs) can be configured using the SYSGEN utility or the Tandem DSC utility. The DSC utility supports online reconfiguration of controllers, devices, and some system operational parameters. The DSC changes made remain in effect through all system loads or reloads and remain in effect until the next SYSGEN. To make DSC changes permanent, they can be integrated into the CONTEXT file for the next SYSGEN.

Data Communications Options

BASE24 supports different data communications modes for each of the RSM models as shown in the following table. The first column of the table lists each of the supported RSM models. The second through fourth columns of the table indicate the data communications modes available for each model. A check mark (✓) in columns two through four indicates that the data communications mode is available for the model. Each of the three available data communications modes and their configuration requirements are explained below and on the following pages.

RSM Model	ASYNCH	SDLC	TCP/IP
RG6000	✓		
RG7100	✓		✓
RG7300	✓	✓	
RG7400	✓		

Asynchronous Emulation (ASYNCH)

Asynchronous emulation mode is full duplex, allowing the Tandem server to send commands to the RSM independently of reading the associated responses from the RSM. With this mode, there is no inherent flow control; the RSM returns a response as soon as it has finished processing a command. The following paragraphs discuss the configuration requirements for BASE24-supported Racal-Guardata security devices using asynchronous emulation mode.

Using asynchronous emulation, BASE24 communicates online with RSMs using an asynchronous protocol and ASCII data transmission.

With this communications mode, RSMs must be configured using the SYSGEN utility as 6603 point-to-point asynchronous terminals. The SYSGEN utility definition can use the 6603 default modifiers, but four modifiers must be changed from the defaults.

The following is an example of a SYSGEN utility definition for a security module named \$RSM1 connected to the controller ASYNCF on port number 25:

<i>device-name</i>	<i>controller.port</i>	<i>product-number</i>	<i>modifier</i>	<i>modifier</i>	<i>modifier</i>	<i>modifier</i>
\$RSM1	ASYNCF.25	6603	BAUD9600,	NOECHO,	PAGE,	PAGETERM %3

Note: Using asynchronous emulation, the Racal-Guardata Host Security Module (RSM) must be configured to operate in nontransparent mode. The RSM will not function properly with BASE24 when configured in transparent mode. Refer to the appropriate Racal-Guardata documentation for RSM configuration procedures.

SDLC Emulation

Synchronous data-link control (SDLC) emulation mode allows the Tandem server to view the RSM as data communications equipment (DCE) operating as a non-switched, point-to-point, half duplex device. The electrical interface conforms to the RS-449 standard without the secondary channel.

BASE24 communicates online with RG7300 Racal-Guardata Host Security Modules using a 3604 bit-synchronous communications board, or any controller capable of supporting Tandem's EnvoyACP/XF product, that is installed in the I/O bay of a Tandem computer system.

The RG7300 device also uses a NonStop system application interface process called the HSM Interface Process (HIP) to communicate with BASE24. For the RG7300 device, it is the HIP process name (i.e., \$HIP) configured using the SYSGEN utility or the DSC utility that is specified in the SECURE-DEVICE*n*,

SECURE-DEV1, and SECURE-DEV2 assigns in the LCONF. Refer to the appropriate Racal-Guardata documentation for HIP installation and user procedures.

The HIP requires the use of Tandem's EnvoyACP/XF product using bit synchronous SDLC protocol. One HIP is required for each RG7300 device connected to the Tandem. Each HIP accepts requests from one or more BASE24 processes. BASE24 processes are responsible for all queue and error handling since the HIP can be considered to be a single-threaded SDLC device driver.

Ethernet (TCP/IP)

The RSM acts as a TCP server supporting connections of up to seven TCP ports or sockets. Applications establish connections to the RSM ports by first connecting to the Well-Known-Port at the IP address. The port number and IP address are defined for the RSM during configuration.

Since the RSM communicates with the Tandem server using TCP/IP over an Ethernet connection, the configuration and startup of an appropriate Ethernet controller, Ethernet port, and TCP/IP process are required.

The RSM uses an ACI-supplied interface process called the Remote Application Manager (RAM) process to communicate with BASE24. For the RSM, it is the RAM process name (i.e., \$RAM) that is specified in the SECURE-DEVICEn, SECURE-DEV1, and SECURE-DEV2 assigns in the LCONF. Configuration requirements for the RAM interface process are provided later in this section.

Hardware Configuration for K-Series Servers

Because the RSM is a TCP/IP device, each Ethernet controller (e.g., 3613, 3615) must be configured using the SYSGEN utility or DSC utility. For detailed information on configuring an Ethernet controller and line, refer to Tandem's ***TLAM Configuration and Management Manual***. The following is an example of a SYSGEN utility definition example for a 3615 controller named RG7100x:

```
controller-name product-number processor-numbers channel addresses
  RG7100X          3615          0,1          %030;
```

The following is an example of the SYSGEN utility definition that shows the minimum parameters required to configure a 3615 controller (line):

<i>device-name</i>	<i>controller-name</i>	<i>macro</i>	<i>modifiers</i>
\$LAM1	LAM1.0, LAM1.1	MLAM	TYPE 56, SUBTYPE 0, RSIZE 32000, READBUFFERSIZE 8192, INTERRUPT IOP^INTERRUPT^HANDLER#

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the RG7100 device and the Tandem server. Refer to the “Software Configuration” discussion below.

Hardware Configuration for S-Series Servers

Because the RSM is a TCP/IP device, it must communicate with the S-series server through an Ethernet port on either a 3861 Ethernet 4 ServerNet Adapter (E4SA) controller or through a Processor MultiFunction Unit (PMF). The configuration of a E4SA controller on an S-series server requires the use of the Tandem Subsystem Control Facility (SCF) to configure and start objects associated with the E4SA controller. These objects include the adapter, the ServerNet addressable controllers (SACs), the logical interfaces (LIFs), and the physical interfaces (PIFs). For additional information on configuring the 3681 (E4SA) controller, refer to Tandem’s *Himalaya S-Series Hardware Installation Guide*.

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the RG7100 device and the Tandem server. Refer to the “Software Configuration” discussion below.

Software Configuration

The software configuration is very similar for the K-series and S-series servers. Examples for both types of servers are provided below. In general, the TCP/IP process must be started against a previously configured port and the RAM interface process must be started. Some of the values in these examples, most importantly the IP addresses, are site-specific and must be modified accordingly.

K-Series Server. The following is an example TACL macro for starting the TCP/IP process on a K-series server.

```
?tacl macro
#frame
delete define=tcPIP^host^file
add define=tcPIP^host^file, file $system.ztcPIP.hosts
$system.ztcPIP.tcPIP.lib $system.zqiolib.qiolib,name $ZTC0,&
    nowait, pri 180, cpu 0/1
#push scfin
#append scfin allow all errors
#append scfin assume process $ZTC0
#append scfin alter subnet, #loop0, ipaddress 127.1
#append scfin add subnet #sn0, type ethernet, devicename $LAN1,&
    ipaddress 130.130.63.16, forceqio on
#append scfin add route #r80, destination 0.0.0.0,&
    gateway 130.130.5.1
#append scfin start route *
#append scfin start subnet *
scfin/inv scfin/
#unframe
```

S-Series Server. The following are examples for starting the TCP/IP process on an S-series server.

The first example starts the TCP/IP process using an SCF obey file.

```
=====
= Configure & Start $ZB110 TCPIP process for ADAPTER E1151,LIFL100 ==
=====
ASSUME PROCESS $ZB110
ADD SUBNET #SN1, TYPE ETHERNET, IPADDRESS 192.231.036.99&
DEVICENAME L110
ADD ROUTE #GW, DESTINATION 0, GATEWAY 192.231.036.017,&
DESTTYPE BROADCAST
ALTER SUBNET #LOOP0, IPADDRESS 127.1
START SUBNET *
START ROUTE *
```

The next example starts the TCP/IP process using a TACL macro.

```
TCP/IP /name $ZB110,NOWAIT, CPU 0, PRI 160, TERM $ZHOME/1
```

Remote Application Manager (RAM) Process. The RAM process runs under control of the XPNET process, and, therefore, must be configured to the XPNET system. For detailed information on configuring and starting the RAM process, refer to the following informal document available from ACI: ***Module Specification - Remote Application Manager***.

Logical Network Configuration File (LCONF)

The following section presents the Logical Network Configuration File (LCONF) information required for configuration of the Racal-Guardata Host Security Module (RSM).

LCONF Assigns

The following assigns are required in the LCONF to configure a Racal-Guardata Host Security Module (RSM). The SECURE-DEV1 and SECURE-DEV2 assigns are required for ISO Host Interface processes, while the SECURE-DEVICEn assigns are required for Integrated Authorization Server and Interface Master processes.

SECURE-DEV1 — Specifies the name of the primary security device or the name of the interface module used with the primary security device. For security devices other than the RG7100 security device using TCP/IP or the RG7300 security device using SDLC, this assign specifies the SYSGEN utility or DSC name of the primary security device. For an RG7100 device using TCP/IP, this assign specifies the process name given to the Remote Application Manager process (e.g., \$RAM). For an RG7300 device using SDLC, this assign specifies the process name given to the HSM Interface process (e.g., \$HIP).

There can be multiple SECURE-DEV1 assigns in the LCONF, depending on whether a primary device is to be specified individually for each ISO Host Interface process requiring the use of a security device (in which case each process has its own SECURE-DEV1 assign in the LCONF).

To associate a primary security device with a specific process, the symbolic name of the process must be placed in the Process Name field on the LCONF Assign screen. If a specific primary security device need not be associated with a specific process, the Process Name field can be filled in with asterisks.

When determining its primary security device, a process first reads through the LCONF for a SECURE-DEV1 assign with its own name in the Process Name field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV1 assign with asterisks in the Process Name field.

SECURE-DEV2 — Specifies the name of the secondary security device or the name of the interface module used with the secondary security device. For security devices other than the RG7100 security device using TCP/IP or the RG7300 security device using SDLC, this assign specifies the SYSGEN utility or DSC name of the secondary security device. For an RG7100 device using TCP/IP, this assign specifies the process name given to the Remote Application Manager process (e.g., \$RAM). For an RG7300 device using SDLC, this assign specifies the process name given to the HSM Interface process (e.g., \$HIP).

There can be multiple SECURE-DEV2 assigns in the LCONF, depending on whether a secondary device is to be specified individually for each ISO Host Interface process requiring the use of a security device (in which case each process has its own SECURE-DEV2 assign in the LCONF).

To associate a secondary security device with a specific process, the symbolic name of the process must be placed in the Process Name field on the LCONF Assign screen. If a specific secondary security device need not be associated with a specific process, the Process Name field can be filled in with asterisks.

When determining its secondary security device, a process first reads through the LCONF for a SECURE-DEV2 assign with its own name in the Process Name field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV2 assign with asterisks in the Process Name field.

SECURE-DEVICEn — Specifies the fully qualified name of the security device or of the interface module used with the security device. For security devices other than the RG7100 security device using TCP/IP or the RG7300 security device using SDLC, this assign specifies the SYSGEN utility or DSC name of the primary security device. For an RG7100 device using TCP/IP, this assign specifies the process name given to the Remote Application Manager process (e.g., \$RAM). For an RG7300 device using SDLC, this assign specifies the process name given to the HSM Interface process (e.g., \$HIP).

The device name or interface module name must be compatible with the Tandem Expand product.

The Integrated Authorization Server and Interface Master processes use this assign. Each security module used by an Integrated Authorization Server or Interface Master process requires its own SECURE-DEVICEn assign.

When an Integrated Authorization Server or Interface Master process requires the services of a security module, it uses the next available security module. The *n* in the assign name identifies the order in which the process accesses the security

modules. For example, the process accesses the security module identified by the SECURE-DEVICE1 assign before it accesses the security module identified by the SECURE-DEVICE2 assign.

The SECURE-DEVICE-COUNT param identifies the number of security modules being used. For example, if the SECURE-DEVICE-COUNT param is set to 5, the Integrated Authorization Server and Interface Master process access five security modules. When a process accesses the security module identified by the SECURE-DEVICE5 assign, it then starts over with the security module identified by the SECURE-DEVICE1 assign.

LCONF Params

The following params are required in the LCONF to configure a Racal-Guardata Host Security Module (RSM). Some params are required by Integrated Authorization Server processes, Interface Master processes, and ISO Host Interface processes while others are required by one type of process and not the other. The params required by the Integrated Authorization Server process and Interface Master process are the same. The params required by each type of process are depicted in the following table. In this table, IAS stands for Integrated Authorization Server process, IMP stands for Interface Master process, and HISO stands for ISO Host Interface process.

Param Name	IAS/IMP	HISO
RSM-TERM-MASTER-KEY		✓
SECURE-DEV-FAIL-NOTIFY		✓
SECURE-DEVICE-COUNT	✓	
SECURE-DEVICE-TYPE	✓	
SECURE-DEV-MONITOR-THRESHOLD		✓
SECURE-DEV-TIM-LMT	✓	✓
SECURE-DEV-TYP		✓
TSEC-OBJ-MASTER	✓	

RSM-TERM-MASTER-KEY — The encrypted terminal master key used with Racal-Guardata security devices. This key is required when the ISO Host Interface process uses Racal-Guardata security devices to generate message authentication code (MAC) encryption keys. This param is not required when Racal-Guardata security devices are not used to generate MAC encryption keys. Since the Integrated Authorization Server process and Interface Master process do not currently support message authentication, this param is not required for these processes.

The ISO Host Interface process uses this key in an intermediate step that is performed when BASE24 is responsible for generating and distributing MAC encryption keys. The key in this param does not coincide with any other keys being used by the external processor. For ease of maintenance, ACI recommends using one RSM-TERM-MASTER-KEY param for a network.

When this param is required, the default value of zeros must be changed. The Racal-Guardata console command K can be used to obtain the encrypted key for this param. Refer to the appropriate Racal-Guardata manual for information regarding the console command.

SECURE-DEV-FAIL-NOTIFY — The interval at which critical event messages are generated due to the failure of a primary or secondary security device for an ISO Host Interface process. Valid values are 10 to 1000; the default value is 20.

If the primary and secondary security devices cannot be initialized during ISO Host Interface process startup and this param is set to the default of 20, every twentieth transaction that a process must decline because the security devices are inoperative causes a message to be generated indicating the security devices have failed. (Note that a critical event message is always generated immediately upon failure of both devices.)

If a primary security device is marked down and this param is set to the default of 20, every twentieth transaction that is directed to the primary security device causes a message to be generated even though the transactions are being handled properly by the secondary security device. Transactions routed directly to the secondary security device do not affect the primary security device transaction counter. The SECURE-DEV-MONITOR-THRESHOLD param controls when a transaction is directed to the secondary security module instead of the primary security device.

If the primary and secondary security devices fail at some time after initialization, a separate failed transaction counter is maintained for each security device. A transaction that is sent to one security device and then the other increments both counters. An event message is generated each time either counter reaches the value established in this param.

SECURE-DEVICE-COUNT — The number of security devices configured in this logical network. The value of this param should match the number of SECURE-DEVICE n assigns in this LCONF. The Integrated Authorization Server and Interface Master process use this param. Valid values are 1 through 32767.

SECURE-DEVICE-TYPE — A code identifying the type of security module specified in the SECURE-DEVICE n assigns. The Integrated Authorization Server and Interface Master process use this param. Valid values are as follows:

RSM = Racal-Guardata Host Security Module (same as VSM)
N/A = No security module required

If this param is not configured, it defaults to a value of N/A.

Note: The Integrated Authorization Server process, Interface Master process, and ISO Host Interface process must use the same type of security device. For example, if the Integrated Authorization Server process uses an RSM, the Interface Master process and the ISO Host Interface process cannot use an Atalla Security Module (ASM) or Atalla Network Security Processor (NSP).

SECURE-DEV-MONITOR-THRESHOLD — The interval, expressed in the number of transactions processed by the security device defined in the SECURE-DEV1 assign, for checking the security device defined by the SECURE-DEV2 assign. Valid values are 0–32000. The default value of 0 indicates no check is performed.

This param can be used to monitor the secondary security device or to balance the load between the primary and secondary security devices. Each ISO Host Interface process using a security device maintains its own transaction counter. This means the interval specified in the SECURE-DEV-MONITOR-THRESHOLD param needs to be set for the number of transactions processed by individual processes using the security device, not the total number of transactions processed by the security device. For example, if five BASE24 processes use a security device and the param is set to 1000, up to 5000 transactions (1000 by each of the five processes) could be processed by the primary security device before the secondary security device is used. Setting this param to 1 causes each of the BASE24 processes to alternate between the two security devices.

If all processes use the same threshold for monitoring the secondary security device, a single SECURE-DEV-MONITOR-THRESHOLD param can be defined in the LCONF. In this case, the Process Name field on the LCONF Param screen must be set to asterisks (default value).

If different secondary security device monitoring thresholds are used, SECURE-DEV-MONITOR-THRESHOLD params must be defined in the LCONF for each threshold and the Process Name field can be set to identify the processes using that threshold. If a majority of processes use a particular threshold, an alternative to identifying all of its processes can be making that threshold the default by placing asterisks in the Process Name field. Then, only processes using a different threshold need to be defined specifically.

SECURE-DEV-TIM-LMT — The maximum time, in hundredths of a second, to wait for a response from the security device before the request is considered to be timed out. Valid values are 1 to 6000; the default value is 500 (that is, 5 seconds). The same time limit applies to all processes so there is only one SECURE-DEV-TIM-LMT param in the LCONF, and the Process Name field must be set to asterisks.

SECURE-DEV-TYP — Specifies the type of security device being used by ISO Host Interface processes. Valid values are as follows:

RSM = Racal-Guardata Host Security Module (same as VSM)
VSM = Racal-Guardata Host Security Module (same as RSM)
N/A = No security module required

If no security devices are used by ISO Host Interface processes, a SECURE-DEV-TYP param must still be defined in the LCONF. The Process Name field on the LCONF Param screen must be set to asterisks (default value) and the field on the LCONF Param screen must be set to N/A (no security module required).

Note: The ISO Host Interface process, Integrated Authorization Server process, and Interface Master process must use the same type of security device. For example, if the ISO Host Interface process uses an RSM, the Integrated Authorization Server process and Interface Master process cannot use an Atalla Security Module (ASM) or Atalla Network Security Processor (NSP).

TSEC-OBJ-MASTER — The symbolic name of the Integrated Authorization Server process containing the Transaction Security object responsible for downloading the Diebold number table (DNT) to the hardware security modules used by the process.

This initialization step needs to be performed only once. Without this param, unnecessary processing would be performed because each Transaction Security object in each Integrated Authorization Server process or Interface Master process would repeat these initialization steps.

Racal-Guardata Host Security Module Support

Summaries of Racal-Guardata Host Security Module (RSM) functions supported by BASE24 presented in this section include the following:

- The RSM commands required for generating, encrypting, and translating data for storage in the BASE24 database.
- The RSM online functions supported by BASE24.

This material is not meant to be a tutorial on the RSM; it presumes a familiarity with the RSM's capabilities and nomenclature as discussed in the Racal-Guardata documentation. Specifically, it does not address the creation and maintenance of the RSM local master keys (LMKs), which also are discussed in the Racal-Guardata documentation.

Note: The RSM was formerly known as the Visa Security Module (VSM). DDLs and other BASE24 documentation may still reference the VSM. Such references should be considered to apply to the RSM instead. The Visa PVV PIN verification method and the Visa Security Module are not related. The verification method name remains unchanged.

Preparing Data for the BASE24 Files

Offline RSM master key loading and key management are accomplished using a user-supplied asynchronous ASCII terminal connected to the RSM asynchronous port. This asynchronous ASCII terminal is known as the RSM console. The BASE24 RSMcom utility can also be used to perform offline key management. Refer to appendix D for information regarding the RSMcom utility.

The table on the following pages summarizes the RSM console commands used to create, encrypt, and translate keys for storage in the BASE24 database and to load and encrypt the Diebold number table. Note that the RSM master keys must have already been loaded and the RSM placed in the authorized state before the commands can be performed. The BASE24 files maintenance screens listed in this table are illustrated in appendix A.

Field Name in BASE24 File	Screen	Master Key Pair	RSM Console Command
INTERM HDWR KEY	ISF 2	06–07	B = Generate and encrypt a working key [*]
INBOUND KPT1	ISF 2	06–07	B = Generate and encrypt a working key [‡]
INBOUND KPT2	ISF 2		
KKT-PT	ISF 1	04–05	F = Generate and encrypt a ZMK component Z = Encrypt a user-supplied ZMK component D = Combine 3 encrypted ZMK components [†]
INTERMEDIATE KEY ENCRYPTED	KEYF 1	06–07	B = Generate and encrypt a working key [*]
PIN KEY EXCHANGE KEY	KEYF 1	04–05	F = Generate and encrypt a ZMK component Z = Encrypt a user-supplied ZMK component D = Combine 3 encrypted ZMK components [†]
MAC KEY EXCHANGE KEY	KEYF 1	04–05	F = Generate and encrypt a ZMK component Z = Encrypt a user-supplied ZMK component D = Combine 3 encrypted ZMK components [†]
INBOUND PIN KEY 1	KEYF 2	06–07	B = Generate and encrypt a working key [‡]
INBOUND PIN KEY 2			

Field Name in BASE24 File	Screen	Master Key Pair	RSM Console Command
OUTBOUND PIN KEY 1	KEYF 2	06–07	B = Generate and encrypt a working key [‡]
OUTBOUND PIN KEY 2			
INBOUND MAC KEY 1	KEYF 2	16–17	O = Encrypt a terminal authentication key [§]
INBOUND MAC KEY 2			
OUTBOUND MAC KEY 1	KEYF 2	16–17	O = Encrypt a terminal authentication key [§]
OUTBOUND MAC KEY 2			
DES (IBM 3624) ENCRYPTED KEY	KEYA 2	14–15	K = Encrypt a PIN verification key
DIEBOLD TABLE	KEYA 3	10–11	R = Load and encrypt a clear Diebold table ^{**}
VISA PVV KEYS ENCRYPTED	KEYA 5	14–15	K = Encrypt a PIN verification key

* **Handling the intermediate key** — The intermediate key encrypted under the RSM master key pair 06–07 must be stored in the ISF record for a remote banking device or in the KEYF record for the external processor if a mixed clear and encrypted PIN interface environment is required. The ZMK-encrypted intermediate key stored in the KEYF or ISF can be used in the BASE24 RSMcom utility (described in appendix D) to obtain the clear intermediate key.

† **Handling key exchange keys** — A key exchange key is the key under which working keys are encrypted for exchange between BASE24 and an RBSI-compliant device or an external processor. It equates to Racal-Guardata's zone master key (ZMK).

An encrypted ZMK need not be placed in the KKT-PT field of the ISF or the KEY EXCHANGE KEY fields of the KEYF unless automatic key exchange is supported. (BASE24 only supports automatic key exchange with RBSI-compliant devices, DPCs, and certain interchanges.)

To generate a ZMK, the console command F or Z must first be utilized to obtain three encrypted ZMK components. Console command F is used if the RSM is to randomly generate the ZMK components. Console command Z is used if the components are to be supplied by the BASE24 system administrator (which is the case when the key has already been determined by another party).

Once the three encrypted ZMK components have been derived, console command D can be used to combine the three components to form the encrypted ZMK for storage in the KKT-PT or KEY EXCHANGE KEY fields.

- ‡ **Handling inbound and outbound keys** — If the BASE24 operations staff is responsible for generating the inbound and outbound keys for PIN encryption, console command B can be used to generate and encrypt these keys under the RSM master key pair 06–07. These keys can then be placed in the ISF and KEYF, and the ZMK-encrypted forms of these keys can be conveyed to the remote banking device or external processor.

If the inbound and outbound working keys are generated by an external processor, the keys can be distributed automatically or manually. If these keys are distributed automatically when the interface process is first brought up, inbound and outbound entries in the KEYF must be left as zeros. The ZMK must be provided in the KEY EXCHANGE KEY fields in the KEYF (refer to note 2).

Inbound encryption keys cannot be generated by an RBSI-compliant device.

When inbound and outbound working keys are distributed manually, the BASE24 operations staff will receive ZMK-encrypted versions of these keys. These ZMK-encrypted versions of the keys must be used to obtain versions encrypted under the RSM master key pair 06–07 for storage in the the ISF and KEYF. The BASE24 RSMcom utility (described in appendix D) can be used to obtain the master key-encrypted version of the PIN key for the KEYF and ISF. Since the inbound and outbound keys must be encrypted or decrypted using the ZMK, depending on whether the keys are being sent to or received from an external processor or sent to an RBSI-compliant device, the ZMK must be provided in either console command B or the RSMcom utility.

- § **Generating zone authentication keys** — There is no RSM console command to generate a zone authentication key (ZAK). If authentication keys are not distributed automatically, they must be generated manually and encrypted using the RSM console O command.

**** Handling the Diebold Number Table** — Initially, the clear Diebold number table (DNT) must be manually entered using the RSM console command R. This command causes the clear DNT to be encrypted and loaded into the RSM. It also returns an encrypted version of the DNT to the BASE24 system administrator.

The BASE24 system administrator can then either copy the encrypted form of the DNT and place it in the KEYA manually or use the BASE24 RSMcom utility (described in appendix D) to place the RSM-encrypted DNT in the KEYA automatically.

RSM Online Functions Supported by BASE24

The following table summarizes the RSM transaction codes and functions used by the Integrated Authorization Server and ISO Host Interface processes, and the RSMcom utility.

Note: The RSMcom utility does not currently modify the Interface Security File (ISF) in any way. For the ISF, only procedures in which you must copy down the RSMcom utility output and manually enter it, can be used with the ISF.

Transaction Code	Function	Integrated Authorization Server	Interface Master Process	ISO Host Interface Process	RSMcom Utility
AA	Translate terminal master key, terminal PIN key, or PIN verification key (old LMK pair 14–15 to new LMK pair 14–15)	N	N	N	Y
AC	Translate MAC transaction key or working key (encryption under old LMK pair 16–17 to encryption under new LMK pair 16–17)	N	N	N	Y
AY	Translate card verification key pair (encryption under variant of old LMK pair 14–15 to encryption under variant of new LMK pair 14–15)	N	N	N	Y

Transaction Code	Function	Integrated Authorization Server	Interface Master Process	ISO Host Interface Process	RSMcom Utility
CA	Translate PIN (encryption under LMK pair 14–15 to encryption under LMK pair 06–07)	Y	N	Y	N
CC	Translate PIN (encryption under LMK pair 06–07 to encryption under LMK pair 06–07)	Y	N	Y	N
CE	Generate Diebold PIN Offset	Y	N	N	N
CG	Verify PIN from terminal—Diebold	Y	N	N	N
CY	Verify a Visa CVV	N	N	N	N
DA	Verify PIN from terminal—IBM 3624-format	Y	N	N	N
DC	Verify PIN from terminal—Visa PVV	Y	N	N	N
DE	Generate IBM 3624 PIN Offset	Y	N	N	N
DG	Generate Visa PVV	Y	N	N	N
EA	Interchange PIN Verify—IBM 3624-format	Y	N	N	N
EC	Interchange PIN Verify—Visa PVV	Y	N	N	N
EG	Interchange PIN Verify—Diebold	Y	N	N	N
FA	Interchange Key Translate (encryption under ZMK to encryption under LMK pair 06–07)	N	N	N	N
GC	Interchange Key Translate (encryption under LMK pair 06–07 to encryption under ZMK)*	N	N	N	N
GE	Translate ZMK (encryption under old LMK pair 04–05 to encryption under new LMK pair 04–05)	N	N	N	Y
HA	Generate TAK under TMK, TPK, or LMK	N	N	Y	N
HC	Generate a New Terminal Master Key or PIN key	N	Y	N	N

Transaction Code	Function	Integrated Authorization Server	Interface Master Process	ISO Host Interface Process	RSMcom Utility
IA	Interchange Key Generate (encryption under ZMK and LMK pair 06–07)	Y	N	N	N
IE	Encrypt Docutel Test Pattern under a Terminal Master Key	N	N	N	N
JC	Translate PIN from TMK or TPK to LMK	Y	N	N	N
JE	Translate PIN from ZWK to LMK	Y	N	N	N
KA	Generate a key check value	Y	N	N	Y
KC	Translate ZWK from old LMK pair 06–07 to new LMK pair 06–07	N	N	N	Y
LA	Load Encrypted Diebold number table to RSM [†]	Y	N	N	N
LC	Verify Diebold number table held in RSM [†]	Y	N	N	N
LE	Upload Diebold number table held in RSM to the RSMcom utility [†]	N	N	N	Y
MG	Translate TAK from LMK to ZMK	N	N	Y	N
MI	Translate TAK from ZMK to LMK	N	N	Y	N
MK	Generate MAC [‡]	N	N	Y	N
MM	Verify MAC [‡]	N	N	Y	N

* In some RSM firmware versions, the GC transaction code may be available only when the RSM is in the authorized state. As a result, the Repeat Key command for dynamic key management may not work. Check with your Racal-Guardata customer support representative.

† If Diebold PIN verification is required, the Integrated Authorization Server process restores or verifies the encrypted DNT in the RSM. If the DNT in the KEYA contains any values other than zeros, the Integrated Authorization Server

process uses the LA message to restore the DNT in the RSM from the KEYA. If the DNT in the KEYA is zeros (that is, it was never loaded into the KEYA), the Integrated Authorization Server process uses the LC message to verify the DNT in the RSM.

- ‡ The MK and MM transaction codes are not supported in the standard Racal-Guardata product. They must be obtained from Racal-Guardata.

Applied Communications, Inc.

Index

A

- A1, A2, and A3 security modules, configuration of, [E-4](#)
- A10000E network security processor
 - configuration of, [E-10](#)
 - hardware configuration, K-series servers, [E-11](#)
 - hardware configuration, S-series servers, [E-11](#)
 - software configuration, [E-12](#)
- A5000 network security processor, configuration of, [E-6](#)
- A6000 network security processor
 - configuration of, [E-7](#)
 - hardware configuration, [E-8](#)
 - interface module, [E-8](#)
- A7000 network security processor
 - boxcar interface process, [E-9](#)
 - configuration of, [E-9](#)
 - hardware configuration, [E-10](#)
- Algorithm number, Diebold PIN verification, [8-4](#)
- ANSI standards
 - PIN block, [2-3](#)
 - X3.92 standard, [1-9](#)
 - X9.17 standard, cryptographic service message, [11-9](#)
 - X9.17 standard, cryptographic service message, [12-12](#)
 - X9.17 standard, dynamic key management, [1-23](#), [11-2](#), [12-2](#)
 - X9.17 standard, message authentication, [1-15](#)
 - X9.19 standard, [A-7](#)
 - X9.8 standard, [2-3](#)
 - X9.9 standard, [1-15](#), [5-6](#)
- ASM
 - see* Atalla Security Module (ASM)
- ASMcom utility
 - changing master file keys, [C-4](#)
 - Check Digit Generation Report, [C-25](#)
 - Check Digit Validation Report, [C-26](#)
 - Check Digits Menu, [C-12](#)
 - check digits processing, [C-10](#)
 - description of, [1-20](#)
 - device name, altering, [C-20](#)
 - exiting, [C-23](#)
 - help function, [C-22](#)
 - Key Translation Menu, [C-16](#)
 - key translation processing, [C-14](#)
 - Key Translation Report, [C-27](#)
 - LCONF location, altering, [C-21](#)
 - Main Menu, [C-9](#)
 - report location, altering, [C-22](#)
 - reports, [C-24](#)

- starting, [C-6](#)
 - status messages, [C-22](#)
 - translating a KEK-encrypted MAC working key, [C-19](#)
 - translating a KEK-encrypted PIN working key, [C-18](#)
 - translating keys, [C-4](#)
 - variants, [C-4](#)
- Asynchronous emulation
 - Atalla security modules, [E-4](#)
 - Racal-Guardata host security modules, [F-3](#)
- Atalla Network Security Processor (NSP)
 - A6000 interface module, [E-8](#)
 - Boxcar interface module, [E-9](#), [E-10](#), [E-13](#)
 - configuring the A10000E NSP, [E-10](#)
 - configuring the A5000 NSP, [E-6](#)
 - configuring the A6000 NSP, [E-7](#)
 - configuring the A7000 NSP, [E-9](#)
 - functions, BASE24 supported, [E-26](#)
 - module capabilities, by message ID, [E-28](#)
 - processing support, [E-2](#)
- Atalla Security Module (ASM)
 - configuring the A1, A2, or A3 ASM, [E-4](#)
 - functions, BASE24 supported, [E-26](#)
 - module capabilities, by message ID, [E-28](#)
 - processing support, [E-2](#)

B

- Bank ID, [9-4](#)
- BASE24 database
 - master key loading from RSM, [F-16](#)
 - settings for interface processes, [4-9](#), [4-15](#), [4-27](#)
 - settings for PIN change transaction, [6-20](#)
- BASE24-telebanking Standard Internal Message (BSTM), [2-10](#)
- Billpay Server process, [2-10](#)
- Boxcar interface module, [E-13](#)
- BSTM
 - see* BASE24-telebanking Standard Internal Message
- BSTM field information, [2-10](#)

C

- Change key message, [12-6](#)
- Check Digit Generation Report, [C-25](#)
- Check Digit Validation Report, [C-26](#)
- Check digits, definition of, [1-18](#)

Compare indicator, [9-5](#)

Co-network key processor, definition of, [12-4](#)

Cryptographic Service Message (CSM)

- fixed-length messages, [12-17](#)
- keys, [11-9](#), [12-12](#), [12-19](#)
- message classes, [11-9](#), [12-12](#)
- message fields, [11-9](#), [12-12](#)
- message flow diagrams, [12-19](#)
- message formats, [11-10](#), [12-13](#)
- variable-length messages, [11-13](#), [12-18](#)

Customer ID

- Diebold PIN verification, [8-4](#), [8-15](#)
- IBM 3624-format PIN verification, [7-3](#), [7-16](#)
- Identikey PIN verification, [9-3](#), [9-13](#)
- PIN verification support, [6-12](#)
- Visa PVV PIN verification, [10-3](#), [10-15](#)

Customer PIN verification parameters, [6-12](#)

Customer Table (CSTT)

- screen 1, [A-2](#)
- screen 1 field descriptions, [6-12](#)

D

Data encryption and decryption

- definition of, [1-8](#)
- security of, [1-10](#)

Data Encryption Standard (DES), [1-8](#)

Data processing center, [1-2](#)

Decimalization table, [7-5](#)

DES

- see Data Encryption Standard (DES)

DES algorithm, [1-9](#)

Diebold number table (DNT)

- Atalla security devices, [E-23](#)
- definition of, [8-3](#)
- hardware module requirements, [8-7](#)
- Racal-Guardata Host Security Module (RSM), [F-14](#), [F-20](#)

Diebold PIN verification

- algorithm number, [8-4](#)
- Diebold number table (DNT), [8-3](#)
- hardware module requirements, [8-7](#)
- input parameters, [8-3](#)
- PIN, [8-5](#)
- PIN offset, [8-5](#)
- primary account number (PAN), [8-4](#)
- processing steps, [8-2](#)
- setting up the BASE24 database, [8-7](#)

DNT

- see Diebold number table (DNT)

Dynamic key management

- see Key management, dynamic

Dynamic System Configuration (DSC) utility

- configuring security modules, [13-9](#)
- defined, [E-4](#), [F-3](#)

E

Enhanced telebanking application, [1-4](#)

Error Service Message (ESM)

- external processor, [12-15](#)
- remote banking standard interface, [11-12](#)

Ethernet communications mode

- A10000E NSP configuration requirements, [E-10](#)
- RSM configuration requirements, [F-5](#)

External processors, definition of, [1-2](#)

H

Hardware module requirements

- Diebold PIN verification, [8-7](#)
- dynamic key management, [11-2](#), [12-2](#)
- IBM 3624-format PIN verification, [7-8](#)
- Identikey PIN verification, [9-6](#)
- supported security modules, [13-4](#)
- Visa PVV PIN verification, [10-7](#)

HIP

- see HSM Interface process (HIP)

Host Configuration File (HCF) screen 1, [A-3](#)

HSM Interface process (HIP), [F-4](#)

I

IBM 3624-format PIN verification

- customer ID, [7-3](#)
- decimalization table, [7-5](#)
- hardware module requirements, [7-8](#)
- input parameters, [7-3](#)
- PIN, [7-7](#)
- PIN offset, [7-6](#)
- primary account number (PAN), [7-3](#)
- processing steps, [7-2](#)
- setting up the BASE24 database, [7-8](#)
- validation data, [7-3](#)

Identikey PIN verification

- bank ID, [9-4](#)
- compare indicator, [9-5](#)
- customer ID, [9-3](#)
- hardware module requirements, [9-6](#)
- input parameters, [9-3](#)
- PIN, [9-4](#)
- PIN verification number (PVN), [9-5](#)
- primary account number (PAN), [9-3](#)
- setting up the BASE24 database, [9-6](#)

Institution Definition File (IDF)

- screen 2, [A-4](#)
- screen 4, [A-5](#)
- screen 40, [A-6](#)
- settings for Diebold PIN verification, [8-8](#)
- settings for IBM 3624-format PIN verification, [7-8](#)
- settings for Identikey PIN verification, [9-6](#)
- settings for Visa PVV PIN verification, [10-8](#)

- Integrated Authorization Server process
 - internal transaction data usage, [2-10](#)
 - LCONF assign requirements, [13-9](#), [E-15](#), [F-9](#)
 - LCONF param requirements, [E-17](#), [F-11](#)
 - PIN encryption support, remote banking devices, [3-2](#)
 - transaction security processing, [1-6](#)
 - Interface Configuration File (ICF), [11-27](#)
 - Interface Master process
 - LCONF assign requirements, [13-9](#), [E-15](#), [F-9](#)
 - LCONF param requirements, [E-17](#), [F-11](#)
 - transaction security processing, [1-6](#)
 - Interface Security File (ISF)
 - screen 1, [A-7](#)
 - screen 1 field descriptions, [3-11](#), [11-23](#)
 - screen 2, [A-8](#)
 - screen 2 field descriptions, [3-13](#), [11-25](#)
 - Intermediate keys
 - hardware, [3-4](#)
 - processing, [3-4](#)
 - Racal-Guardata Host Security Module (RSM)
 - support, [F-18](#)
 - software, [3-4](#)
 - Intermediate PIN
 - Diebold PIN verification, [8-5](#)
 - IBM 3624-format PIN verification, [7-6](#)
 - Internal Transaction Data (ITD), [2-10](#)
 - ISO Host Interface process
 - internal message usage, [2-10](#)
 - LCONF assign requirements, [13-9](#), [E-15](#), [F-9](#)
 - LCONF param requirements, [E-17](#), [F-11](#)
 - PIN encryption support, external processors, [4-2](#)
 - transaction security processing, [1-7](#)
 - ISO standards
 - 8583, 1987, [4-2](#)
 - 8583, 1993, [3-1](#), [6-18](#), [11-1](#)
 - 8732 standard, [1-23](#), [11-2](#), [12-2](#)
 - 9564 standard, [2-3](#)
 - key management message data elements, [12-10](#)
 - ITD
 - see Internal Transaction Data (ITD)
 - ITD field information, [2-10](#)
- ## K
- Key Authorization File (KEYA)
 - screen 1, [A-9](#)
 - screen 2, [A-10](#)
 - screen 3, [A-11](#)
 - screen 4, [A-12](#)
 - screen 5, [A-13](#)
 - settings for Diebold PIN verification, [8-17](#), [8-18](#)
 - settings for IBM 3624-format PIN verification, [7-17](#)
 - settings for Identikv PIN verification, [9-15](#)
 - settings for Visa PVV verification, [10-17](#)
 - Key Change command, [12-6](#)
 - Key exchange key
 - definition of, [11-3](#), [12-3](#)
 - Racal-Guardata Host Security Module, [F-18](#)
 - Key File (KEYF)
 - screen 1, [A-14](#)
 - screen 2, [A-15](#)
 - screen 3, [A-16](#)
 - settings for interfaces, [4-9](#), [4-15](#), [4-27](#)
 - Key Loading Module (KLM)
 - ASMcom utility, [C-4](#)
 - Atalla security modules, [E-24](#)
 - Key management, dynamic
 - change key, [12-6](#)
 - definition of, [1-22](#)
 - external processor interface support, [12-1](#)
 - ICF settings, [11-27](#)
 - ISF settings, [11-23](#)
 - ISO message data elements, [12-10](#)
 - key change processing, [12-19](#)
 - key exchange keys, [11-3](#), [12-3](#)
 - KEYF settings, [12-28](#)
 - message processing, [12-7](#)
 - messages, defined, [11-5](#), [12-6](#)
 - network management messages, [12-9](#)
 - network management request message, [11-5](#)
 - network management response message, [11-6](#)
 - new key, [12-6](#)
 - new key message, [11-5](#)
 - new key processing, [11-15](#), [12-20](#)
 - number of consecutive errors, [11-21](#), [12-25](#)
 - number of errors, [11-21](#), [12-24](#)
 - number of synchronization errors, [11-22](#), [12-26](#)
 - number of transactions, [11-21](#), [12-24](#)
 - processors, defined, [11-4](#), [12-4](#)
 - RBSI support, [11-1](#)
 - RBSI thresholds, [11-19](#)
 - repeat key, [12-7](#), [12-21](#)
 - setting up the BASE24 database, [11-23](#), [12-28](#)
 - thresholds, [12-22](#)
 - tools provided, [1-20](#)
 - verify key, [12-7](#)
 - working keys, [11-2](#), [12-2](#)
 - Key Repeat command, [12-7](#)
 - Key Service Message (KSM)
 - external processor, [12-14](#)
 - remote banking standard interface, [11-10](#)
 - Key Translation Report, [C-27](#)
 - Key Verify command, [12-7](#)
 - Keys
 - definition of, [1-9](#)
 - Diebold PIN verification, [8-3](#)
 - dynamic key management, [4-6](#), [11-2](#), [12-2](#)
 - encryption, inbound, [3-3](#), [4-3](#)
 - encryption, outbound, [4-4](#)
 - IBM 3624-format PIN verification, [7-4](#)
 - intermediate, [3-3](#), [4-4](#)
 - message authentication, [5-5](#)
 - message authentication, inbound, [12-2](#)
 - message authentication, outbound, [12-3](#)
 - PIN encryption, inbound, [11-2](#), [12-2](#)
 - PIN encryption, outbound, [12-2](#)
 - PIN key exchange, [4-6](#)
 - previous, [4-5](#)

Keys *continued*

- remote banking device, [3-3](#)
- security module usage, [13-3](#)
- translation, [1-11](#)
- Visa PVV PIN verification, [10-4](#)
- working, [3-3](#), [4-3](#), [4-4](#), [11-2](#), [12-2](#)

KLMKEY program, [C-4](#), [E-24](#)

L

Logical Network Configuration File (LCONF) assigns

- INTFEMT, [13-9](#)
- ISECEMT-SWAPVOL, [13-9](#)
- ISF, [13-9](#)
- KEYAEMT, [13-9](#)
- KEYF, [13-9](#)
- SECURE-DEV1, [E-15](#), [F-9](#)
- SECURE-DEV2, [E-16](#), [F-10](#)
- SECURE-DEVICEn, [E-16](#), [F-10](#)
- security module configuration, process requirements, [E-15](#), [F-9](#)

Logical Network Configuration File (LCONF) params

- RSM-TERM-MASTER-KEY, [F-12](#)
- SECURE-DEV-CONF-CMD100, [E-18](#)
- SECURE-DEV-CONF-TXT, [F-19](#)
- SECURE-DEV-FAIL-NOTIFY, [E-20](#), [F-12](#)
- SECURE-DEVICE-COUNT, [E-21](#), [F-13](#)
- SECURE-DEVICE-TYPE, [E-21](#), [F-13](#)
- SECURE-DEV-MONITOR-THRESHOLD, [E-21](#), [F-13](#)
- SECURE-DEV-TIM-LMT, [E-22](#), [F-14](#)
- SECURE-DEV-TYP, [E-22](#), [F-14](#)
- security module configuration, process requirements, [E-17](#), [F-11](#)
- TSEC-OBJ-MASTER, [E-23](#), [F-14](#)

M

MAC

see Message authentication code (MAC)

Master file key (MFK), [E-24](#)

Master key processor definition of

- external processor, [12-4](#)
- remote banking standard interface, [11-4](#)

Message authentication

- authentication algorithm, [5-6](#)
- authentication input parameters, defined, [5-5](#)
- BASE24 Remote Banking support, [5-2](#)
- data to be authenticated, [5-5](#)
- definition of, [1-15](#)
- hardware and software authentication, [5-4](#)
- key, [5-5](#)
- MAC length, [5-6](#)
- MAC location, [5-6](#)
- message authentication code (MAC), [5-2](#)
- transaction flow, [5-3](#)

Message authentication code (MAC), [5-2](#)

Message processing

- incoming PIN encryption support, [3-8](#), [4-9](#), [4-13](#), [4-26](#), [4-34](#)
- outgoing PIN encryption support, [4-9](#), [4-11](#), [4-32](#)

N

Network control command, changing working

- keys, [11-2](#), [12-2](#)

Network management messages

- defined, [12-9](#)
- ISO, [12-10](#)

Network management request message, [11-5](#)

Network management response message, [11-6](#)

New key message

- external processor, [12-6](#)
- remote banking standard interface, [11-5](#)

NSP

see Atalla Network Security Processor (NSP)

P

PAN

- defined, Diebold PIN verification, [8-4](#)
- defined, IBM 3624-format PIN verification, [7-3](#)
- defined, Identikey PIN verification, [9-3](#)
- defined, Visa PVV PIN verification, [10-3](#)
- length, [9-3](#)
- offset value, [9-3](#)
- PAN string within PIN/PAN PIN blocks, [2-4](#)
- see also Primary account number (PAN) and Customer ID

Personal Identification Number

see PIN

PIN

- definition of, [1-12](#)
- management, [1-12](#)
- PIN string within PIN/PAN PIN block, [2-3](#)

PIN block

- ANSI, [2-3](#)
- definition of, [2-3](#)
- PIN/PAD, [2-7](#)
- PIN/PAN, [2-3](#)

PIN change

- database settings, [6-20](#)
- Diebold PIN verification transaction, [8-5](#)
- IBM 3624-format PIN verification transaction, [7-7](#)
- Identikey PIN verification transaction, [9-4](#)
- option, definition of, [6-20](#)
- processing, [6-21](#)
- Visa PVV PIN verification transaction, [10-4](#)

PIN check number, [7-6](#)

PIN encryption

- BASE24 clear and external processor clear option, [4-9](#)
- BASE24 clear and external processor encrypted option, [4-11](#)

- PIN encryption *continued*
 - BASE24 encrypted and external processor clear option, [4-24](#)
 - BASE24 encrypted and external processor encrypted option, [4-32](#)
 - definition of, [2-2](#)
 - options, [3-6](#), [4-7](#)
 - PIN offset
 - Diebold PIN verification, [8-5](#)
 - IBM 3624-format PIN verification, [7-6](#)
 - PIN select option
 - definition of, [6-19](#)
 - processing of, [6-21](#)
 - PIN verification
 - BASE24 Remote Banking support, [6-2](#)
 - Diebold, definition of, [8-2](#)
 - IBM 3624-format decimalization table, [7-5](#)
 - IBM 3624-format, definition of, [7-2](#)
 - Identikey, definition of, [9-2](#)
 - institution parameters, [6-3](#)
 - internal messages, [6-15](#)
 - local, BASE24 Remote Banking, [6-18](#)
 - PIN change option, definition of, [6-20](#)
 - PIN select option, definition of, [6-19](#)
 - setting up Diebold PIN verification, [8-7](#)
 - setting up IBM 3624-format PIN verification, [7-8](#)
 - setting up Identikey PIN verification, [9-6](#)
 - setting up Visa PVV PIN verification, [10-7](#)
 - transaction processing, Integrated Authorization Server process, [6-16](#)
 - Visa PVV, definition of, [10-2](#)
 - PIN verification key (PVK)
 - IBM 3624-format PIN verification, [7-4](#)
 - Visa PVV PIN verification, [10-4](#)
 - PIN verification key indicator (PVKI), [10-5](#)
 - PIN verification number (PVN), [9-5](#)
 - PIN verification value (PVV)
 - control, [10-5](#)
 - definition of, [10-5](#)
 - transaction, [10-5](#)
 - Primary account number (PAN)
 - definition of, [1-3](#)
 - see also* PAN and Customer ID
- ## R
- Racal-Guardata Host Security Module (RSM)
 - asynchronous emulation, configuration of, [F-3](#)
 - asynchronous SYSGEN definition, [F-4](#)
 - BASE24 support, [F-16](#)
 - console, [F-16](#)
 - console command, [F-16](#), [F-20](#)
 - data communications options, [F-3](#)
 - Ethernet communications mode, configuration of, [F-5](#)
 - IBM 3624-format PIN verification, [2-8](#)
 - PIN blocks, special requirements, [2-7](#), [4-33](#), [4-36](#)
 - PIN translation, [2-8](#)
 - processing support, [F-2](#)
 - SDLC emulation, configuration of, [F-4](#)
 - transaction codes used by BASE24, [F-20](#)
 - utility program, [D-1](#)
 - Visa PIN verification value (PVV), [2-8](#)
 - RAM
 - see* Remote Application Manager process
 - Remote Application Manager (RAM) process, [F-8](#)
 - Remote banking endpoint device, definition of, [1-2](#)
 - Repeat key message, [12-7](#)
 - Request Service Initiation Message (RSI), [12-13](#)
 - Response Service Message (RSM)
 - external processor, [12-15](#)
 - remote banking standard interface, [11-11](#)
 - RG6000 host security module, data communications options, [F-3](#)
 - RG7100 host security module
 - data communications options, [F-3](#)
 - Ethernet software configuration requirements, [F-6](#)
 - hardware configuration, K-series server, [F-5](#)
 - hardware configuration, S-series server, [F-6](#)
 - Remote Application Manager process, [F-8](#)
 - RG7110 host security module, [F-1](#)
 - RG7300 host security module
 - data communications options, [F-3](#)
 - HSM Interface process, [F-4](#)
 - SDLC emulation configuration requirements, [F-4](#)
 - RG7310 host security module, [F-1](#)
 - RG7400 host security module, data communications options, [F-3](#)
 - RSM
 - see* Racal-Guardata Host Security Module
 - RSMcom utility
 - Check Digit Generation Report, [D-30](#)
 - check digit processing, [D-14](#)
 - Check Digit Validation Report, [D-31](#)
 - Check Digits Menu, [D-16](#)
 - description of, [1-21](#)
 - device name, altering, [D-25](#)
 - DNT Upload Summary Report, [D-33](#)
 - exiting, [D-28](#)
 - help function, [D-28](#)
 - Key Translation Menu, [D-21](#)
 - key translation processing, [D-18](#)
 - Key Translation Report, [D-32](#)
 - LCONF location, altering, [D-25](#)
 - report location, altering, [D-26](#)
 - reports, [D-29](#)
 - starting, [D-6](#)
 - status messages, [D-28](#)
 - uploading Diebold number table (DNT), [D-26](#)
 - working key translation processing, [D-13](#)
- ## S
- SDLC
 - see* Synchronous data-link control (SDLC)
 - emulation, [F-4](#)

Security modules

- Atalla, [E-1](#)
- balancing transactions between, [13-8](#)
- BASE24 files, preparing data for, [E-24](#)
- BASE24 Remote Banking support, [13-4](#)
- checking the secondary, [13-7](#)
- configuring, [13-9](#), [E-4](#), [E-6](#)
- definition of, [1-17](#)
- functions, BASE24 supported, [E-26](#)
- number of, [13-4](#)
- primary and secondary, [13-6](#)
- Racal-Guardata Host Security Module (RSM), [F-1](#)

Slave key processor, definition of

- external processor, [12-4](#)
- remote banking standard interface, [11-4](#)

STATUS command, [13-6](#)

Supercrypt utility

- accessing, [B-3](#)
- description of, [1-22](#)
- error messages, [B-3](#)
- exiting, [B-2](#)
- starting, [B-2](#)

Synchronous data-link control (SDLC) emulation,
configuration requirements, [F-4](#)**T**Transaction Security object, [13-6](#)

Transaction security processing software

- Integrated Authorization Server process, [1-6](#)
- Interface Master process, [1-6](#)
- ISO Host Interface process, [1-7](#)

U

Utilities

- ASMcom, [C-1](#)
- RSMcom, [D-1](#)
- Supercrypt, [B-1](#)

VVerify key message, [12-7](#)

Visa PVV PIN verification

- customer ID, [10-3](#)
- hardware module requirements, [10-7](#)
- input parameters, [10-3](#)
- PIN, [10-3](#)
- PIN change transaction, [10-4](#)
- PIN verification key (PVK) pair, [10-4](#)
- PIN verification key indicator (PVKI), [10-5](#)
- PIN verification value (PVV), [10-5](#)
- primary account number (PAN), [10-3](#)
- setting up the BASE24 database, [10-7](#)

VRU Configuration Data (VCD)

- screen 1, [A-17](#)
- screen 1 field descriptions, [11-27](#)
- screen 18, [A-20](#)
- screen 18 field descriptions, [6-18](#)
- screen 2, [A-18](#)
- screen 2 field descriptions, [11-28](#)
- screen 4, [A-19](#)
- screen 4 field descriptions, [11-28](#)

ZZone master key (ZMK), [F-18](#)

Index by Field Name

A

ALGO NUMBER, [8-18](#)
ALGO NUMBER LOC
IDF screen 2, Diebold PIN verification, [8-9](#)
IDF screen 2, IAS PIN verification, [6-6](#)
IDF screen 2, IBM 3624-format PIN verification, [7-10](#)
IDF screen 2, Identikey PIN verification, [9-7](#)
IDF screen 2, Visa PVV PIN verification, [10-9](#)
ANSI PAN FORMAT, [4-10](#), [4-16](#), [4-28](#), [4-36](#)

B

BAD PIN ACTION, [6-5](#)
BAD PIN COUNT
CSTT screen 1, [6-13](#)
CSTT screen 1, Diebold PIN verification, [8-16](#)
CSTT screen 1, IBM 3624-format PIN verification, [7-16](#)
CSTT screen 1, Identikey PIN verification, [9-14](#)
CSTT screen 1, Visa PVV PIN verification, [10-15](#)
BASE24 ENCRYPT TYPE, [4-9](#), [4-16](#), [4-27](#), [4-35](#)
BEGIN DATE
KEYA screen 1, Diebold PIN verification, [8-17](#)
KEYA screen 1, IAS PIN verification, [6-3](#)
KEYA screen 1, IBM 3624-format PIN verification, [7-18](#)
KEYA screen 1, Identikey PIN verification, [9-15](#)
KEYA screen 1, Visa PVV PIN verification, [10-17](#)
BEGINNING DATE
IDF screen 4, Diebold PIN verification, [8-11](#)
IDF screen 4, IAS PIN verification, [6-7](#)
IDF screen 4, IBM 3624-format PIN verification, [7-11](#)
IDF screen 4, Identikey PIN verification, [9-9](#)
IDF screen 4, Visa PVV PIN verification, [10-10](#)

C

CARDHOLDER PIN SELECT, [6-6](#)
CHECK DIGITS
KEYA screen 2, IBM 3624-format PIN verification, [7-19](#)
KEYA screen 5, Visa PVV PIN verification, [10-19](#)
KEYF screen 1, [4-17](#), [4-18](#), [4-29](#), [4-37](#)

CHECK DIGITS1
KEYF screen 2, key management, inbound keys, [12-32](#), [12-34](#)
KEYF screen 2, key management, outbound keys, [12-31](#), [12-33](#)
KEYF screen 2, PIN encryption, inbound keys, [4-22](#), [4-30](#), [4-41](#)
KEYF screen 2, PIN encryption, outbound keys, [4-19](#), [4-38](#)
CHECK DIGITS2
KEYF screen 2, key management, inbound keys, [12-33](#), [12-35](#)
KEYF screen 2, key management, outbound keys, [12-32](#), [12-34](#)
KEYF screen 2, PIN encryption, inbound keys, [4-23](#), [4-31](#), [4-42](#)
KEYF screen 2, PIN encryption, outbound keys, [4-20](#), [4-39](#)
CLEAR
KEYA screen 5, Visa PVV PIN verification, [10-18](#)
KEYF screen 1, [4-17](#), [4-29](#)
CLEAR KEY
KEYA screen 2, IBM 3624-format PIN verification, [7-19](#)
KEYA screen 3, Diebold PIN verification, [8-18](#)
CLEAR OLD KEY TIMER VALUE, [12-40](#)
CONSECUTIVE ERRORS, [11-29](#)
CONSECUTIVE MAC KEY ERROR, [12-40](#)
CONSECUTIVE PIN KEY ERROR, [12-39](#)
CURRENT INDEX
ISF screen 2, [3-15](#)
ISF screen 2, PIN transport keys, inbound, [11-26](#)
KEYF screen 2, key management, inbound keys, [12-32](#), [12-35](#)
KEYF screen 2, key management, outbound keys, [12-31](#), [12-34](#)
KEYF screen 2, PIN encryption, inbound keys, [4-22](#), [4-30](#), [4-41](#)
KEYF screen 2, PIN encryption, outbound keys, [4-20](#), [4-39](#)
CUSTOMER ID
CSTT screen 1, Diebold PIN verification, [8-15](#)
CSTT screen 1, IAS PIN verification, [6-12](#)
CSTT screen 1, IBM 3624-format PIN verification, [7-15](#)
CSTT screen 1, Identikey PIN verification, [9-13](#)
CSTT screen 1, Visa PVV PIN verification, [10-14](#)

CUSTOMER SERVICE INTERFACE CONTROL PARAMETERS

- IDF screen 40, Diebold PIN verification, [8-14](#)
- IDF screen 40, IBM 3624-format PIN verification, [7-15](#)
- IDF screen 40, Identikkey PIN verification, [9-12](#)
- IDF screen 40, Visa PVV PIN verification, [10-14](#)

D

DECIMALIZATION TABLE, [7-18](#)

DIEBOLD TABLE, [8-19](#)

E

ENCRYPT, [10-18](#)

ENCRYPT KEY, [8-18](#)

ENCRYPT TYPE

- KEYF screen 1, key management, [12-28](#)
- KEYF screen 1, PIN encryption, [4-9](#), [4-15](#), [4-27](#), [4-35](#)

ENCRYPTED, [4-17](#), [4-29](#)

ENCRYPTED KEY, [7-19](#)

END DATE

- KEYA screen 1, Diebold PIN verification, [8-17](#)
- KEYA screen 1, IAS PIN verification, [6-3](#)
- KEYA screen 1, IBM 3624-format PIN verification, [7-18](#)
- KEYA screen 1, Identikkey PIN verification, [9-15](#)
- KEYA screen 1, Visa PVV PIN verification, [10-17](#)

ERRORS, [11-29](#)

EXCHANGE KEYS

- KEYF screen 1, key management, [12-30](#)
- KEYF screen 1, PIN encryption, [4-37](#)
- KEYF screen1, PIN encryption, [4-18](#)

EXTENDED NETWORK, [12-43](#)

EXTRN PIN ENCRYPT TYPE, [3-11](#), [11-23](#)

F

FIID, [6-12](#)

G**GRP**

- KEYA screen 1, Diebold PIN verification, [8-17](#)
- KEYA screen 1, IAS PIN verification, [6-3](#)
- KEYA screen 1, IBM 3624-format PIN verification, [7-18](#)
- KEYA screen 1, Identikkey PIN verification, [9-15](#)
- KEYA screen 1, Visa PVV PIN verification, [10-17](#)

H**HOLIDAYS**

- IDF screen 4, Diebold PIN verification, [8-13](#)
- IDF screen 4, IAS PIN verification, [6-10](#)
- IDF screen 4, IBM 3624-format PIN verification, [7-14](#)
- IDF screen 4, Identikkey PIN verification, [9-11](#)
- IDF screen 4, Visa PVV PIN verification, [10-13](#)

I

ID LENGTH, [9-16](#)

IMP SYMBOLIC NAME, [11-28](#)

INBOUND KEYS

- ISF screen 2, [3-14](#), [11-26](#)
- KEYF screen 1, key management, [12-32](#)
- KEYF screen 2, key management, [12-34](#)
- KEYF screen 2, PIN encryption, [4-21](#), [4-29](#), [4-40](#)

INSTITUTION IDENTIFIER, [9-16](#)

INTERFACE NAME, [3-11](#)

INTERM HDWR KEY, [3-14](#)

INTERM SFTWR KEY, [3-13](#)

INTERMEDIATE KEYS, [4-17](#), [4-29](#)

INTRN PIN ENCRYPT TYPE, [3-11](#), [11-23](#)

K**KEY COUNTER**

- ISF screen 2, PIN transport keys, inbound, [11-26](#)
- KEYF screen 2, key management, inbound keys, [12-33](#), [12-35](#)
- KEYF screen 2, key management, outbound keys, [12-32](#), [12-34](#)
- KEYF screen 2, PIN encryption, inbound keys, [4-23](#)
- KEYF screen 2, PIN encryption, outbound keys, [4-21](#), [4-40](#)

KEY LENGTH, [11-25](#)

- KEYF screen 1, key management, [12-29](#)
- KEYF screen 1, PIN encryption, [4-10](#), [4-17](#), [4-28](#), [4-37](#)

KEY MGMT TYPE, [11-27](#)

KEY PROCESSING TYPE, [12-41](#)

KEY SYNC ERROR ACTION, [11-30](#)

KEY SYNC ERRORS, [11-29](#)

KEY TRANSPORT KEY INFORMATION, [11-25](#)

KKT-PT, [11-25](#)

KMAC SYNCHRONIZATION ERROR, [12-40](#)

KPT CONSECUTIVE ERROR ACTION, [11-30](#)

KPT1, [3-14](#)

- ISF screen 2, PIN transport keys, inbound, [11-26](#)

KPT2, [3-14](#)

- ISF screen 2, PIN transport keys, inbound, [11-26](#)

L**LAST RESET DATE**

- CSTT screen 1, Diebold PIN verification, [8-16](#)
- CSTT screen 1, IAS PIN verification, [6-13](#)
- CSTT screen 1, IBM 3624-format PIN verification, [7-17](#)
- CSTT screen 1, Identikey PIN verification, [9-14](#)
- CSTT screen 1, Visa PVV PIN verification, [10-16](#)

LAST UPDATE

- CSTT screen 1, Diebold PIN verification, [8-16](#)
- CSTT screen 1, IAS PIN verification, [6-12](#)
- CSTT screen 1, IBM 3624-format PIN verification, [7-16](#)
- CSTT screen 1, Identikey PIN verification, [9-14](#)
- CSTT screen 1, Visa PVV PIN verification, [10-15](#)

LIMITS, [11-28](#), [12-36](#)**M****MAC ENCRYPT TYPE, [12-29](#)****MAC KEY, [12-30](#)****MAC KEY CHECK DIGITS, [12-30](#)****MAC KEY ERROR, [12-39](#)****MAC KEY INFORMATION, [12-33](#)****MAC KEY TIMER INTERVAL, [12-38](#)****MAC KEY TIMER VALUE, [12-37](#)****MAC KEY TRAN, [12-39](#)****MAC KEY VARIANT, [12-36](#)****MAC KEY1, [12-33](#), [12-34](#)****MAC KEY2, [12-34](#), [12-35](#)****MAX PIN TRIES**

- IDF screen 2, Diebold PIN verification, [8-9](#)
- IDF screen 2, IAS PIN verification, [6-5](#)
- IDF screen 2, IBM 3624-format PIN verification, [7-9](#)
- IDF screen 2, Identikey PIN verification, [9-7](#)
- IDF screen 2, Visa PVV PIN verification, [10-8](#)

MESSAGE ELEMENTS, [12-42](#)**N****NETWORK MANAGEMENT, [11-28](#), [12-43](#)****NEXT BEGINNING DATE**

- IDF screen 4, Diebold PIN verification, [8-11](#)
- IDF screen 4, IAS PIN verification, [6-8](#)
- IDF screen 4, IBM 3624-format PIN verification, [7-12](#)
- IDF screen 4, Identikey PIN verification, [9-9](#)
- IDF screen 4, Visa PVV PIN verification, [10-11](#)

NMM ENABLED, [11-27](#)**NOTARIZATION SUPPORTED, [12-42](#)****NUMBER OF KEYS, [11-30](#)**

- KEYF screen 1, key management, [12-29](#)
- KEYF screen 1, PIN encryption, [4-10](#), [4-16](#), [4-28](#), [4-36](#)

NUMBER TABLE RELATIVE LOCATION, [8-18](#)**O****OLD KEY TIMER LIMIT, [3-12](#), [11-24](#)****ORIGINATING ID, [11-30](#), [12-41](#)****OUTBOUND KEYS**

- KEYF screen 1, key management, [12-31](#)
- KEYF screen 2, key management, [12-33](#)
- KEYF screen 2, PIN encryption, [4-19](#), [4-38](#)

P**PAN PAD CHARACTER, [7-19](#)****PAN VERIFY LENGTH, [7-19](#)****PAN VERIFY OFFSET, [7-18](#)****PARTIAL PAN LENGTH, [9-16](#)****PARTIAL PAN OFFSET, [9-16](#)****PERIOD LENGTH**

- IDF screen 4, Diebold PIN verification, [8-12](#)
- IDF screen 4, IAS PIN verification, [6-8](#)
- IDF screen 4, IBM 3624-format PIN verification, [7-12](#)
- IDF screen 4, Identikey PIN verification, [9-10](#)
- IDF screen 4, Visa PVV PIN verification, [10-11](#)

PIN

- IDF screen 2, [6-4](#)
- IDF screen 2, Diebold PIN verification, [8-8](#)
- IDF screen 2, IBM 3624-format PIN verification, [7-9](#)
- IDF screen 2, Identikey PIN verification, [9-7](#)
- IDF screen 2, Visa PVV PIN verification, [10-8](#)

PIN BLOCK FORMAT, [11-23](#)

- ISF screen 1, [3-12](#)
- KEYF screen 1, key management, [12-29](#)
- KEYF screen 1, PIN encryption, [4-10](#), [4-16](#), [4-28](#), [4-36](#)

PIN CHANGE ALLOWED

- IDF screen 40, Diebold PIN verification, [8-15](#)
- IDF screen 40, IAS PIN verification, [6-11](#), [6-23](#)
- IDF screen 40, IBM 3624-format PIN verification, [7-15](#)
- IDF screen 40, Identikey PIN verification, [9-13](#)
- IDF screen 40, Visa PVV PIN verification, [10-14](#)

PIN CHANGE REQUIRED

- CSTT screen 1, Diebold PIN verification, [8-16](#)
- CSTT screen 1, IAS PIN verification, [6-13](#), [6-20](#)
- CSTT screen 1, IBM 3624-format PIN verification, [7-17](#)
- CSTT screen 1, Identikey PIN verification, [9-14](#)
- CSTT screen 1, Visa PVV PIN verification, [10-16](#)

PIN CHECK TYPE

- IDF screen 2, Diebold PIN verification, [8-9](#)
- IDF screen 2, IAS PIN verification, [6-5](#)
- IDF screen 2, IBM 3624-format PIN verification, [7-10](#)
- IDF screen 2, Identikey PIN verification, [9-7](#)
- IDF screen 2, Visa PVV PIN verification, [10-9](#)

PIN ENCRYPT LEVEL, [3-12](#), [11-24](#)

PIN KEY

- KEYF screen 1, exchange keys, [4-37](#)
- KEYF screen 1, key management, [12-30](#)
- KEYF screen 1, PIN encryption, [4-18](#)

PIN KEY CHECK DIGITS, [12-30](#)PIN KEY ERROR, [12-39](#)

PIN KEY INFORMATION

- KEYF screen 1, key management, [12-31](#)
- KEYF screen 2, PIN encryption, [4-19](#)

PIN KEY TIMER INTERVAL, [12-38](#)PIN KEY TIMER VALUE, [12-36](#)PIN KEY TRAN, [12-38](#)PIN KEY VARIANT, [12-36](#)

PIN KEY1

- KEYF screen 2, inbound keys, [4-21](#), [4-30](#), [4-40](#)
- KEYF screen 2, key management, inbound keys, [12-32](#)
- KEYF screen 2, key management, outbound keys, [12-31](#)
- KEYF screen 2, outbound keys, [4-19](#), [4-38](#)

PIN KEY2

- KEYF screen 2, inbound keys, [4-22](#), [4-41](#)
- KEYF screen 2, key management, inbound keys, [12-32](#)
- KEYF screen 2, key management, outbound keys, [12-31](#)
- KEYF screen 2, outbound keys, [4-20](#), [4-39](#)
- KEYF screen 2, PIN encryption, inbound keys, [4-31](#)

PIN PAD CHARACTER

- ISF screen 1, [3-12](#), [11-24](#)
- KEYF screen 1, [4-10](#), [4-16](#), [4-28](#), [4-36](#)

PIN REQUIRED

- IDF screen 40, [6-11](#), [6-23](#)
- IDF screen 40, Diebold PIN verification, [8-14](#)
- IDF screen 40, IBM 3624-format PIN verification, [7-15](#)
- IDF screen 40, Identikay PIN verification, [9-12](#)
- IDF screen 40, Visa PVV PIN verification, [10-14](#)

PIN TRANSPORT KEY INFORMATION, [3-13](#), [11-25](#)

PIN TRIES RESET OPTION

- IDF screen 2, Diebold PIN verification, [8-9](#)
- IDF screen 2, IAS PIN verification, [6-6](#)
- IDF screen 2, IBM 3624-format PIN verification, [7-10](#)
- IDF screen 2, Identikay PIN verification, [9-8](#)
- IDF screen 2, Visa PVV PIN verification, [10-9](#)

PIN VERIFICATION DIGITS

- CSTT screen 1, Diebold PIN verification, [8-15](#)
- CSTT screen 1, IAS PIN verification, [6-12](#)
- CSTT screen 1, IBM 3624-format PIN verification, [7-16](#)
- CSTT screen 1, Identikay PIN verification, [9-13](#)
- CSTT screen 1, Visa PVV PIN verification, [10-15](#)

PIN VERIFICATION GROUP

- IDF screen 40, Diebold PIN verification, [8-14](#)
- IDF screen 40, IAS PIN verification, [6-10](#)
- IDF screen 40, IBM 3624-format PIN verification, [7-14](#)
- IDF screen 40, Identikay PIN verification, [9-12](#)
- IDF screen 40, Visa PVV PIN verification, [10-13](#)

PIN VERIFICATION KEY INDEX

- CSTT screen 1, Diebold PIN verification, [8-16](#)
- CSTT screen 1, IAS PIN verification, [6-13](#)
- CSTT screen 1, IBM 3624-format PIN verification, [7-16](#)
- CSTT screen 1, Identikay PIN verification, [9-14](#)
- CSTT screen 1, Visa PVV PIN verification, [10-15](#)

POFST/PVV LOC

- IDF screen 2, Diebold PIN verification, [8-9](#)
- IDF screen 2, IAS PIN verification, [6-5](#)
- IDF screen 2, IBM 3624-format PIN verification, [7-9](#)
- IDF screen 2, Identikay PIN verification, [9-7](#)
- IDF screen 2, Visa PVV PIN verification, [10-8](#)

R

RECEIVING ID

- KEYF screen 3, [12-41](#)
- VCD screen 4, [11-31](#)

RECORD TYPE

- KEYA screen 1, Diebold PIN verification, [8-17](#)
- KEYA screen 1, IAS PIN verification, [6-3](#)
- KEYA screen 1, IBM 3624-format PIN verification, [7-18](#)
- KEYA screen 1, Identikay PIN verification, [9-15](#)
- KEYA screen 1, Visa PVV PIN verification, [10-17](#)

T

TELEBANKING IDF INFORMATION

- IDF screen 40, Diebold PIN verification, [8-14](#)
- IDF screen 40, IBM 3624-format PIN verification, [7-14](#)
- IDF screen 40, Identikay PIN verification, [9-12](#)
- IDF screen 40, Visa PVV PIN verification, [10-13](#)

TIMER LIMITS, [12-43](#)TIMER VALUE, [11-29](#)TRANSACTIONS, [11-29](#)**V**

VARIANT

- ISF screen 1, key transport keys, KKT-PT, [11-25](#)

VVK, [3-14](#)

- ISF screen 1, key transport keys, KKT-PT, [11-25](#)

VVK1, [3-14](#)

- ISF screen 2, PIN transport keys, inbound, [11-26](#)

VVK2, [3-15](#)

- ISF screen 2, PIN transport keys, inbound, [11-26](#)

W**WITHDRAWAL PERIOD PARAMETERS**

- IBM 3624-format PIN verification, [7-11](#)
- IDF screen 4, Diebold PIN verification, [8-11](#)
- IDF screen 4, IAS PIN verification, [6-7](#)
- IDF screen 4, IBM 3624-format PIN verification, [7-11](#)
- IDF screen 4, Identikey PIN verification, [9-9](#)
- IDF screen 4, Visa PVV PIN verification, [10-10](#)

WORK DAY CODE

- IDF screen 4, Diebold PIN verification, [8-12](#)
- IDF screen 4, IAS PIN verification, [6-9](#)
- IDF screen 4, IBM 3624-format PIN verification, [7-13](#)
- IDF screen 4, Identikey PIN verification, [9-10](#)
- IDF screen 4, Visa PVV PIN verification, [10-12](#)

Applied Communications, Inc.